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PSoC 4100S Plus 256KB

PSoC 4 Registers Technical Reference Manual (TRM)

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20.1.5	ROMTABLE_PID6	531
20.1.6	ROMTABLE_PID7	532
20.1.7	ROMTABLE_PID0	533
20.1.8	ROMTABLE_PID1	534
20.1.9	ROMTABLE_PID2	535
20.1.10	ROMTABLE_PID3	536
20.1.11	ROMTABLE_CID0	537
20.1.12	ROMTABLE_CID1	538
20.1.13	ROMTABLE_CID2	539
20.1.14	ROMTABLE_CID3	540

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Register Mapping



The Register Mapping section discusses the PSoC 4100S Plus 256KB PSoC 4 registers and lists all the registers in mapping tables, in address order. For Architecture details, see [PSoC 4100S Plus 256KB: PSoC 4 Architecture Technical Reference Manual \(TRM\)](#).

Note that memory mapped I/O (MMIO) registers support only 32-bit access, unless otherwise specified in the register description.

Register General Conventions

The register conventions specific to this section and the Register Reference chapter are listed in this table.

Convention	Description	Explanation
RW	Read/Write	These bits can be both read and written.
R	Read only	These bits can only be read. Writing has no effect on the bit value.
W	Write only	These bits can only be written. Reading the bit returns the reset value.
RW1C	Read/Write '1' to clear	These bits can be read as well as cleared by writing '1'. Writing '0' has no effect on the bit value.
RW0C	Read/Write '0' to clear	These bits can be read as well as cleared by writing '0'. Writing '1' has no effect on the bit value.
RW1S	Read/Write '1' to set	These bits can be read as well as set by writing '1'. Writing '0' has no effect on the bit value.
A	Alias	This register is an additional alias for another register. This convention is used when a physical register is mapped to multiple addresses, typically for firmware debug purposes. See the corresponding register descriptions for more details.
None / Reserved	Reserved bits	Keep these bits at the default value
'x' in a register /bit field name	Multiple instances	Multiple instances/address ranges of the same register/bit field

Acronyms

This table lists the acronyms used in this document.

Table 1-1. Acronyms

Symbol	Unit of Measure
ABUS	analog output bus
AC	alternating current
ADC	analog-to-digital converter
ADV	advertising
AES	Advanced Encryption Standard
AHB	AMBA (advanced microcontroller bus architecture) high-performance bus, an ARM data transfer bus
API	application programming interface
APOR	analog power-on reset
BC	broadcast clock
BLE	Bluetooth Low Energy (Bluetooth Smart)
BLESS	BLE subsystem

Table 1-1. Acronyms

Symbol	Unit of Measure
BOM	bill of materials
BR	bit rate
BRA	bus request acknowledge
BRQ	bus request
CAN	controller area network
CI	carry in
CMP	compare
CO	carry out
CPU	central processing unit
CRC	cyclic redundancy check
CSD	CapSense sigma delta
CT	continuous time
CTBm	continuous time block-mini
DAC	digital-to-analog converter
DC	direct current
DI	digital or data input
DMA	direct memory access
DNL	differential nonlinearity
DO	digital or data output
DSI	digital signal interface
DSM	deep-sleep mode
ECO	external crystal oscillator
EEPROM	electrically erasable programmable read only memory
EMIF	external memory interface
FB	feedback
FIFO	first in first out
FSR	full scale range
GAP	generic access profile
GATT	generic attribute profile
GFSK	Gaussian frequency-shift keying
GPIO	general purpose I/O
HCI	host-controller interface
HFCLK	high-frequency clock
I ² C	inter-integrated circuit
IDE	integrated development environment
ILO	internal low-speed oscillator
IMO	internal main oscillator
INL	integral nonlinearity
I/O	input/output
IOR	I/O read
IOW	I/O write

Table 1-1. Acronyms

Symbol	Unit of Measure
IRES	initial power on reset
IRA	interrupt request acknowledge
IRQ	interrupt request
ISR	interrupt service routine
IVR	interrupt vector read
L2CAP	logical link control and adaptation protocol
LPCOMP	low-power comparator
LRb	last received bit
LRB	last received byte
LSb	least significant bit
LSB	least significant byte
LUT	lookup table
MISO	master-in-slave-out
MMIO	memory mapped input/output
MOSI	master-out-slave-in
MSb	most significant bit
MSB	most significant byte
PC	program counter
PCH	program counter high
PCL	program counter low
PD	power down
PDU	protocol data unit
PGA	programmable gain amplifier
PHY	physical layer
PM	power management
PMA	PSoC memory arbiter
POR	power-on reset
PPOR	precision power-on reset
PRS	pseudo random sequence
PSoC®	Programmable System-on-Chip
PSRR	power supply rejection ratio
PSSDC	power system sleep duty cycle
PWM	pulse width modulator
RAM	random-access memory
RETI	return from interrupt
RF	radio frequency
ROM	read only memory
RW	read/write
SAR	successive approximation register
SC	switched capacitor
SCB	serial communication block

Table 1-1. Acronyms

Symbol	Unit of Measure
SIE	serial interface engine
SIO	special I/O
SE0	single-ended zero
SNR	signal-to-noise ratio
SOF	start of frame
SOI	start of instruction
SP	stack pointer
SPD	sequential phase detector
SPI	serial peripheral interconnect
SPIM	serial peripheral interconnect master
SPIS	serial peripheral interconnect slave
SRAM	static random-access memory
SRSS	system resources sub-system
SROM	supervisory read only memory
SSADC	single slope ADC
SSC	supervisory system call
SYSCLK	system clock
SWD	single wire debug
TC	terminal count
TD	transaction descriptors
UART	universal asynchronous receiver/transmitter
UDB	universal digital block
USB	universal serial bus
USBIO	USB I/O
WCO	watch crystal oscillator
WDT	watchdog timer
WDR	watchdog reset
XRES	external reset
XRES_N	external reset, active low

1 Supervisory Flash Registers



This section discusses the Supervisory Flash registers. It lists all the registers in mapping tables, in address order.

1.1 Register Details

Register	Address	Description
SFLASH_SILICON_ID	0x0FFFF244	Silicon ID
SFLASH_HIB_KEY_DELAY	0x0FFFF250	Hibernate wakeup value for PWR_KEY_DELAY
SFLASH_DPSLP_KEY_DELAY	0x0FFFF252	DeepSleep wakeup value for PWR_KEY_DELAY
SFLASH_SWD_CONFIG	0x0FFFF254	SWD pinout selector (not present in TSG4/TSG5-M)
SFLASH_SWD_LISTEN	0x0FFFF258	Listen Window Length
SFLASH_FLASH_START	0x0FFFF25C	Flash Image Start Address
SFLASH_CSDV2_CSD0_ADC_TRIM1	0x0FFFF260	CSDV2 CSD0 ADC TRIM 1
SFLASH_CSDV2_CSD0_ADC_TRIM2	0x0FFFF261	CSDV2 CSD0 ADC TRIM2
SFLASH_SAR_TEMP_MULTIPLIER	0x0FFFF264	SAR Temperature Sensor Multiplication Factor
SFLASH_SAR_TEMP_OFFSET	0x0FFFF266	SAR Temperature Sensor Offset
SFLASH_IMO_TRIM_USBMODE_24	0x0FFFF33E	USB IMO TRIM 24MHz
SFLASH_IMO_TRIM_USBMODE_48	0x0FFFF33F	USB IMO TRIM 48MHz
SFLASH_IMO_TCTRIM_LT0	0x0FFFF34C	IMO TempCo Trim Register (SRSS-Lite)
SFLASH_IMO_TCTRIM_LT1	0x0FFFF34D	IMO TempCo Trim Register (SRSS-Lite). See SFLASH_IMO_TCTRIM_LT0 for the details of bit fields.
SFLASH_IMO_TCTRIM_LT2	0x0FFFF34E	IMO TempCo Trim Register (SRSS-Lite). See SFLASH_IMO_TCTRIM_LT0 for the details of bit fields.
SFLASH_IMO_TCTRIM_LT3	0x0FFFF34F	IMO TempCo Trim Register (SRSS-Lite). See SFLASH_IMO_TCTRIM_LT0 for the details of bit fields.
SFLASH_IMO_TCTRIM_LT4	0x0FFFF350	IMO TempCo Trim Register (SRSS-Lite). See SFLASH_IMO_TCTRIM_LT0 for the details of bit fields.
SFLASH_IMO_TCTRIM_LT5	0x0FFFF351	IMO TempCo Trim Register (SRSS-Lite). See SFLASH_IMO_TCTRIM_LT0 for the details of bit fields.
SFLASH_IMO_TCTRIM_LT6	0x0FFFF352	IMO TempCo Trim Register (SRSS-Lite). See SFLASH_IMO_TCTRIM_LT0 for the details of bit fields.
SFLASH_IMO_TCTRIM_LT7	0x0FFFF353	IMO TempCo Trim Register (SRSS-Lite). See SFLASH_IMO_TCTRIM_LT0 for the details of bit fields.
SFLASH_IMO_TCTRIM_LT8	0x0FFFF354	IMO TempCo Trim Register (SRSS-Lite). See SFLASH_IMO_TCTRIM_LT0 for the details of bit fields.
SFLASH_IMO_TCTRIM_LT9	0x0FFFF355	IMO TempCo Trim Register (SRSS-Lite). See SFLASH_IMO_TCTRIM_LT0 for the details of bit fields.
SFLASH_IMO_TCTRIM_LT10	0x0FFFF356	IMO TempCo Trim Register (SRSS-Lite). See SFLASH_IMO_TCTRIM_LT0 for the details of bit fields.

Register	Address	Description
SFLASH_IMO_TCTRIM_LT11	0x0FFF357	IMO TempCo Trim Register (SRSS-Lite). See SFLASH_IMO_TCTRIM_LT0 for the details of bit fields.
SFLASH_IMO_TCTRIM_LT12	0x0FFF358	IMO TempCo Trim Register (SRSS-Lite). See SFLASH_IMO_TCTRIM_LT0 for the details of bit fields.
SFLASH_IMO_TCTRIM_LT13	0x0FFF359	IMO TempCo Trim Register (SRSS-Lite). See SFLASH_IMO_TCTRIM_LT0 for the details of bit fields.
SFLASH_IMO_TCTRIM_LT14	0x0FFF35A	IMO TempCo Trim Register (SRSS-Lite). See SFLASH_IMO_TCTRIM_LT0 for the details of bit fields.
SFLASH_IMO_TCTRIM_LT15	0x0FFF35B	IMO TempCo Trim Register (SRSS-Lite). See SFLASH_IMO_TCTRIM_LT0 for the details of bit fields.
SFLASH_IMO_TCTRIM_LT16	0x0FFF35C	IMO TempCo Trim Register (SRSS-Lite). See SFLASH_IMO_TCTRIM_LT0 for the details of bit fields.
SFLASH_IMO_TCTRIM_LT17	0x0FFF35D	IMO TempCo Trim Register (SRSS-Lite). See SFLASH_IMO_TCTRIM_LT0 for the details of bit fields.
SFLASH_IMO_TCTRIM_LT18	0x0FFF35E	IMO TempCo Trim Register (SRSS-Lite). See SFLASH_IMO_TCTRIM_LT0 for the details of bit fields.
SFLASH_IMO_TCTRIM_LT19	0x0FFF35F	IMO TempCo Trim Register (SRSS-Lite). See SFLASH_IMO_TCTRIM_LT0 for the details of bit fields.
SFLASH_IMO_TCTRIM_LT20	0x0FFF360	IMO TempCo Trim Register (SRSS-Lite). See SFLASH_IMO_TCTRIM_LT0 for the details of bit fields.
SFLASH_IMO_TCTRIM_LT21	0x0FFF361	IMO TempCo Trim Register (SRSS-Lite). See SFLASH_IMO_TCTRIM_LT0 for the details of bit fields.
SFLASH_IMO_TCTRIM_LT22	0x0FFF362	IMO TempCo Trim Register (SRSS-Lite). See SFLASH_IMO_TCTRIM_LT0 for the details of bit fields.
SFLASH_IMO_TCTRIM_LT23	0x0FFF363	IMO TempCo Trim Register (SRSS-Lite). See SFLASH_IMO_TCTRIM_LT0 for the details of bit fields.
SFLASH_IMO_TCTRIM_LT24	0x0FFF364	IMO TempCo Trim Register (SRSS-Lite). See SFLASH_IMO_TCTRIM_LT0 for the details of bit fields.
SFLASH_IMO_TRIM_LT0	0x0FFF365	IMO Frequency Trim Register (SRSS-Lite)
SFLASH_IMO_TRIM_LT1	0x0FFF366	IMO Frequency Trim Register (SRSS-Lite). See SFLASH_IMO_TRIM_LT0 for the details of bit fields.
SFLASH_IMO_TRIM_LT2	0x0FFF367	IMO Frequency Trim Register (SRSS-Lite). See SFLASH_IMO_TRIM_LT0 for the details of bit fields.
SFLASH_IMO_TRIM_LT3	0x0FFF368	IMO Frequency Trim Register (SRSS-Lite). See SFLASH_IMO_TRIM_LT0 for the details of bit fields.
SFLASH_IMO_TRIM_LT4	0x0FFF369	IMO Frequency Trim Register (SRSS-Lite). See SFLASH_IMO_TRIM_LT0 for the details of bit fields.
SFLASH_IMO_TRIM_LT5	0x0FFF36A	IMO Frequency Trim Register (SRSS-Lite). See SFLASH_IMO_TRIM_LT0 for the details of bit fields.
SFLASH_IMO_TRIM_LT6	0x0FFF36B	IMO Frequency Trim Register (SRSS-Lite). See SFLASH_IMO_TRIM_LT0 for the details of bit fields.
SFLASH_IMO_TRIM_LT7	0x0FFF36C	IMO Frequency Trim Register (SRSS-Lite). See SFLASH_IMO_TRIM_LT0 for the details of bit fields.
SFLASH_IMO_TRIM_LT8	0x0FFF36D	IMO Frequency Trim Register (SRSS-Lite). See SFLASH_IMO_TRIM_LT0 for the details of bit fields.
SFLASH_IMO_TRIM_LT9	0x0FFF36E	IMO Frequency Trim Register (SRSS-Lite). See SFLASH_IMO_TRIM_LT0 for the details of bit fields.
SFLASH_IMO_TRIM_LT10	0x0FFF36F	IMO Frequency Trim Register (SRSS-Lite). See SFLASH_IMO_TRIM_LT0 for the details of bit fields.
SFLASH_IMO_TRIM_LT11	0x0FFF370	IMO Frequency Trim Register (SRSS-Lite). See SFLASH_IMO_TRIM_LT0 for the details of bit fields.
SFLASH_IMO_TRIM_LT12	0x0FFF371	IMO Frequency Trim Register (SRSS-Lite). See SFLASH_IMO_TRIM_LT0 for the details of bit fields.
SFLASH_IMO_TRIM_LT13	0x0FFF372	IMO Frequency Trim Register (SRSS-Lite). See SFLASH_IMO_TRIM_LT0 for the details of bit fields.
SFLASH_IMO_TRIM_LT14	0x0FFF373	IMO Frequency Trim Register (SRSS-Lite). See SFLASH_IMO_TRIM_LT0 for the details of bit fields.
SFLASH_IMO_TRIM_LT15	0x0FFF374	IMO Frequency Trim Register (SRSS-Lite). See SFLASH_IMO_TRIM_LT0 for the details of bit fields.

Register	Address	Description
SFLASH_IMO_TRIM_LT16	0x0FFFF375	IMO Frequency Trim Register (SRSS-Lite). See SFLASH_IMO_TRIM_LT0 for the details of bit fields.
SFLASH_IMO_TRIM_LT17	0x0FFFF376	IMO Frequency Trim Register (SRSS-Lite). See SFLASH_IMO_TRIM_LT0 for the details of bit fields.
SFLASH_IMO_TRIM_LT18	0x0FFFF377	IMO Frequency Trim Register (SRSS-Lite). See SFLASH_IMO_TRIM_LT0 for the details of bit fields.
SFLASH_IMO_TRIM_LT19	0x0FFFF378	IMO Frequency Trim Register (SRSS-Lite). See SFLASH_IMO_TRIM_LT0 for the details of bit fields.
SFLASH_IMO_TRIM_LT20	0x0FFFF379	IMO Frequency Trim Register (SRSS-Lite). See SFLASH_IMO_TRIM_LT0 for the details of bit fields.
SFLASH_IMO_TRIM_LT21	0x0FFFF37A	IMO Frequency Trim Register (SRSS-Lite). See SFLASH_IMO_TRIM_LT0 for the details of bit fields.
SFLASH_IMO_TRIM_LT22	0x0FFFF37B	IMO Frequency Trim Register (SRSS-Lite). See SFLASH_IMO_TRIM_LT0 for the details of bit fields.
SFLASH_IMO_TRIM_LT23	0x0FFFF37C	IMO Frequency Trim Register (SRSS-Lite). See SFLASH_IMO_TRIM_LT0 for the details of bit fields.
SFLASH_IMO_TRIM_LT24	0x0FFFF37D	IMO Frequency Trim Register (SRSS-Lite). See SFLASH_IMO_TRIM_LT0 for the details of bit fields.

1.1.1 SFLASH_SILICON_ID

Silicon ID

Address: 0x0FFFF244

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access								
Name	ID [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	RW							
HW Access								
Name	ID [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
15 : 0	ID	Silicon ID Default Value: X

1.1.2 SFLASH_HIB_KEY_DELAY

Hibernate wakeup value for PWR_KEY_DELAY

Address: 0x0FFFF250

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access	R							
Name	WAKEUP_HOLDOFF [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	None						RW	
HW Access	None						R	
Name	None [15:10]						WAKEUP_HOLDOFF [9:8]	

Bits	Name	Description
9 : 0	WAKEUP_HOLDOFF	Delay (in 12MHz IMO clock cycles) to wait for references to settle on wakeup from hibernate/deepsleep. PBOD is ignored and system does not resume until this delay expires. Note that the same delay on POR is hard-coded. Default Value: X

1.1.3 SFLASH_DPSLP_KEY_DELAY

DeepSleep wakeup value for PWR_KEY_DELAY

Address: 0x0FFFF252

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access	R							
Name	WAKEUP_HOLDOFF [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	None						RW	
HW Access	None						R	
Name	None [15:10]						WAKEUP_HOLDOFF [9:8]	

Bits	Name	Description
9 : 0	WAKEUP_HOLDOFF	Delay (in 12MHz IMO clock cycles) to wait for references to settle on wakeup from hibernate/deepsleep. PBOD is ignored and system does not resume until this delay expires. Note that the same delay on POR is hard-coded. Default Value: X

1.1.4 SFLASH_SWD_CONFIG

SWD pinout selector (not present in TSG4/TSG5-M)

Address: 0x0FFFF254

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None							RW
HW Access	None							
Name	None [7:1]							SWD_SELECT

Bits	Name	Description
0	SWD_SELECT	0: Use Primary SWD location 1: Use Alternate SWD location Default Value: X

1.1.5 SFLASH_SWD_LISTEN

Listen Window Length

Address: 0x0FFFF258

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access								
Name	CYCLES [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	RW							
HW Access								
Name	CYCLES [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	RW							
HW Access								
Name	CYCLES [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	RW							
HW Access								
Name	CYCLES [31:24]							

Bits	Name	Description
31 : 0	CYCLES	Number of clock cycles Default Value: X

1.1.6 SFLASH_FLASH_START

Flash Image Start Address

Address: 0x0FFFF25C

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access								
Name	ADDRESS [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	RW							
HW Access								
Name	ADDRESS [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	RW							
HW Access								
Name	ADDRESS [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	RW							
HW Access								
Name	ADDRESS [31:24]							

Bits	Name	Description
31 : 0	ADDRESS	Start Address Default Value: X

1.1.7 SFLASH_CSDV2_CSD0_ADC_TRIM1

CSDV2 CSD0 ADC TRIM 1

Address: 0x0FFFF260

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access								
Name	CSD_ADC_CAL_LSB [7:0]							

Bits	Name	Description
7 : 0	CSD_ADC_CAL_LSB	Low byte of CSDv2 Calibration Default Value: X

1.1.8 SFLASH_CSDV2_CSD0_ADC_TRIM2

CSDV2 CSD0 ADC TRIM2

Address: 0x0FFFF261

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access								
Name	CSD_ADC_CAL_MSB [7:0]							

Bits	Name	Description
7 : 0	CSD_ADC_CAL_MSB	High byte of CSDv2 Calibration Default Value: X

1.1.9 SFLASH_SAR_TEMP_MULTIPLIER

SAR Temperature Sensor Multiplication Factor

Address: 0x0FFFF264

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access								
Name	TEMP_MULTIPLIER [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	RW							
HW Access								
Name	TEMP_MULTIPLIER [15:8]							

Bits	Name	Description
15 : 0	TEMP_MULTIPLIER	Multiplier value for SAR temperature sensor in fixed point 0.16 format. Note: this field exists in products that contain SAR (m0s8sar) only. Default Value: X

1.1.10 SFLASH_SAR_TEMP_OFFSET

SAR Temperature Sensor Offset

Address: 0x0FFFF266

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access								
Name	TEMP_OFFSET [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	RW							
HW Access								
Name	TEMP_OFFSET [15:8]							

Bits	Name	Description
15 : 0	TEMP_OFFSET	Offset value for SAR temperature sensor in fixed point 10.6 format. Note: this field exists in products that contain SAR (m0s8sar) only. Default Value: X

1.1.11 SFLASH_IMO_TRIM_USBMODE_24

USB IMO TRIM 24MHz

Address: 0x0FFF33E

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access								
Name	TRIM_24 [7:0]							

Bits	Name	Description
7 : 0	TRIM_24	TRIM value for IMO with USB at 24MHz Default Value: X

1.1.12 SFLASH_IMO_TRIM_USBMODE_48

USB IMO TRIM 48MHz

Address: 0x0FFF33F

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access								
Name	TRIM_24 [7:0]							

Bits	Name	Description
7 : 0	TRIM_24	TRIM value for IMO with USB at 24MHz Default Value: X

1.1.13 SFLASH_IMO_TCTRIM_LT0

IMO TempCo Trim Register (SRSS-Lite)

Address: 0x0FFFF34C

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None	RW		RW				
HW Access	None	R		R				
Name	None	TCTRIM [6:5]		STEP_SIZE [4:0]				

Bits	Name	Description
6 : 5	TCTRIM	IMO temperature compensation trim. These bits are determined at manufacturing time to adjust for temperature dependence. These bits are dependent on frequency and need to be changed using the Cypress provided frequency change algorithm. Default Value: 2
4 : 0	STEP_SIZE	IMO trim stepsize bits. These bits are determined at manufacturing time to adjust for process variation. They are used to tune the stepsize of the FSOFFSET and OFFSET trims. Default Value: 16

1.1.14 SFLASH_IMO_TRIM_LT0

IMO Frequency Trim Register (SRSS-Lite)

Address: 0x0FFF365

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access								
Name	OFFSET [7:0]							

Bits	Name	Description
7 : 0	OFFSET	Frequency trim bits. These bits are determined at manufacturing time for each FREQ setting (IMO_TRIM2) and stored in SFLASH. Default Value: X

2 Peripheral Registers



This section discusses the Peripheral registers. It lists all the registers in mapping tables, in address order.

2.1 Register Details

Register	Address	Description
PERI_DIV_CMD	0x40010000	Divider command register
PERI_PCLK_CTL0	0x40010100	Programmable clock control register
PERI_PCLK_CTL1	0x40010104	Programmable clock control register. See PERI_PCLK_CTL0 for the details of bit fields.
PERI_PCLK_CTL2	0x40010108	Programmable clock control register. See PERI_PCLK_CTL0 for the details of bit fields.
PERI_PCLK_CTL3	0x4001010C	Programmable clock control register. See PERI_PCLK_CTL0 for the details of bit fields.
PERI_PCLK_CTL4	0x40010110	Programmable clock control register. See PERI_PCLK_CTL0 for the details of bit fields.
PERI_PCLK_CTL5	0x40010114	Programmable clock control register. See PERI_PCLK_CTL0 for the details of bit fields.
PERI_PCLK_CTL6	0x40010118	Programmable clock control register. See PERI_PCLK_CTL0 for the details of bit fields.
PERI_PCLK_CTL7	0x4001011C	Programmable clock control register. See PERI_PCLK_CTL0 for the details of bit fields.
PERI_PCLK_CTL8	0x40010120	Programmable clock control register. See PERI_PCLK_CTL0 for the details of bit fields.
PERI_PCLK_CTL9	0x40010124	Programmable clock control register. See PERI_PCLK_CTL0 for the details of bit fields.
PERI_PCLK_CTL10	0x40010128	Programmable clock control register. See PERI_PCLK_CTL0 for the details of bit fields.
PERI_PCLK_CTL11	0x4001012C	Programmable clock control register. See PERI_PCLK_CTL0 for the details of bit fields.
PERI_PCLK_CTL12	0x40010130	Programmable clock control register. See PERI_PCLK_CTL0 for the details of bit fields.
PERI_PCLK_CTL13	0x40010134	Programmable clock control register. See PERI_PCLK_CTL0 for the details of bit fields.
PERI_PCLK_CTL14	0x40010138	Programmable clock control register. See PERI_PCLK_CTL0 for the details of bit fields.
PERI_PCLK_CTL15	0x4001013C	Programmable clock control register. See PERI_PCLK_CTL0 for the details of bit fields.
PERI_PCLK_CTL16	0x40010140	Programmable clock control register. See PERI_PCLK_CTL0 for the details of bit fields.
PERI_PCLK_CTL17	0x40010144	Programmable clock control register. See PERI_PCLK_CTL0 for the details of bit fields.
PERI_PCLK_CTL18	0x40010148	Programmable clock control register. See PERI_PCLK_CTL0 for the details of bit fields.
PERI_DIV_16_CTL0	0x40010300	Divider control register (for 16.0 divider)
PERI_DIV_16_CTL1	0x40010304	Divider control register (for 16.0 divider). See PERI_DIV_16_CTL0 for the details of bit fields.
PERI_DIV_16_CTL2	0x40010308	Divider control register (for 16.0 divider). See PERI_DIV_16_CTL0 for the details of bit fields.
PERI_DIV_16_CTL3	0x4001030C	Divider control register (for 16.0 divider). See PERI_DIV_16_CTL0 for the details of bit fields.
PERI_DIV_16_CTL4	0x40010310	Divider control register (for 16.0 divider). See PERI_DIV_16_CTL0 for the details of bit fields.
PERI_DIV_16_CTL5	0x40010314	Divider control register (for 16.0 divider). See PERI_DIV_16_CTL0 for the details of bit fields.
PERI_DIV_16_CTL6	0x40010318	Divider control register (for 16.0 divider). See PERI_DIV_16_CTL0 for the details of bit fields.

Register	Address	Description
PERI_DIV_16_CTL7	0x4001031C	Divider control register (for 16.0 divider). See PERI_DIV_16_CTL0 for the details of bit fields.
PERI_DIV_16_CTL8	0x40010320	Divider control register (for 16.0 divider). See PERI_DIV_16_CTL0 for the details of bit fields.
PERI_DIV_16_CTL9	0x40010324	Divider control register (for 16.0 divider). See PERI_DIV_16_CTL0 for the details of bit fields.
PERI_DIV_16_CTL10	0x40010328	Divider control register (for 16.0 divider). See PERI_DIV_16_CTL0 for the details of bit fields.
PERI_DIV_16_CTL11	0x4001032C	Divider control register (for 16.0 divider). See PERI_DIV_16_CTL0 for the details of bit fields.
PERI_DIV_16_5_CTL0	0x40010400	Divider control register (for 16.5 divider)
PERI_DIV_16_5_CTL1	0x40010404	Divider control register (for 16.5 divider). See PERI_DIV_16_5_CTL0 for the details of bit fields.
PERI_DIV_16_5_CTL2	0x40010408	Divider control register (for 16.5 divider). See PERI_DIV_16_5_CTL0 for the details of bit fields.
PERI_DIV_16_5_CTL3	0x4001040C	Divider control register (for 16.5 divider). See PERI_DIV_16_5_CTL0 for the details of bit fields.
PERI_DIV_16_5_CTL4	0x40010410	Divider control register (for 16.5 divider). See PERI_DIV_16_5_CTL0 for the details of bit fields.
PERI_DIV_24_5_CTL	0x40010500	Divider control register (for 24.5 divider)
PERI_TR_CTL	0x40010600	Trigger control register
PERI_TR_GROUP0_TR_OUT_CTL0	0x40012000	Trigger control register
PERI_TR_GROUP0_TR_OUT_CTL1	0x40012004	Trigger control register. See PERI_TR_GROUP0_TR_OUT_CTL0 for the details of bit fields.
PERI_TR_GROUP0_TR_OUT_CTL2	0x40012008	Trigger control register. See PERI_TR_GROUP0_TR_OUT_CTL0 for the details of bit fields.
PERI_TR_GROUP0_TR_OUT_CTL3	0x4001200C	Trigger control register. See PERI_TR_GROUP0_TR_OUT_CTL0 for the details of bit fields.
PERI_TR_GROUP0_TR_OUT_CTL4	0x40012010	Trigger control register. See PERI_TR_GROUP0_TR_OUT_CTL0 for the details of bit fields.
PERI_TR_GROUP0_TR_OUT_CTL5	0x40012014	Trigger control register. See PERI_TR_GROUP0_TR_OUT_CTL0 for the details of bit fields.
PERI_TR_GROUP0_TR_OUT_CTL6	0x40012018	Trigger control register. See PERI_TR_GROUP0_TR_OUT_CTL0 for the details of bit fields.
PERI_TR_GROUP0_TR_OUT_CTL7	0x4001201C	Trigger control register. See PERI_TR_GROUP0_TR_OUT_CTL0 for the details of bit fields.
PERI_TR_GROUP1_TR_OUT_CTL0	0x40012200	Trigger control register
PERI_TR_GROUP1_TR_OUT_CTL1	0x40012204	Trigger control register. See PERI_TR_GROUP1_TR_OUT_CTL0 for the details of bit fields.
PERI_TR_GROUP1_TR_OUT_CTL2	0x40012208	Trigger control register. See PERI_TR_GROUP1_TR_OUT_CTL0 for the details of bit fields.
PERI_TR_GROUP1_TR_OUT_CTL3	0x4001220C	Trigger control register. See PERI_TR_GROUP1_TR_OUT_CTL0 for the details of bit fields.
PERI_TR_GROUP1_TR_OUT_CTL4	0x40012210	Trigger control register. See PERI_TR_GROUP1_TR_OUT_CTL0 for the details of bit fields.
PERI_TR_GROUP1_TR_OUT_CTL5	0x40012214	Trigger control register. See PERI_TR_GROUP1_TR_OUT_CTL0 for the details of bit fields.
PERI_TR_GROUP1_TR_OUT_CTL6	0x40012218	Trigger control register. See PERI_TR_GROUP1_TR_OUT_CTL0 for the details of bit fields.
PERI_TR_GROUP2_TR_OUT_CTL	0x40012400	Trigger control register
PERI_TR_GROUP3_TR_OUT_CTL	0x40012600	Trigger control register
PERI_TR_GROUP4_TR_OUT_CTL	0x40012800	Trigger control register. See PERI_TR_GROUP2_TR_OUT_CTL for the details of bit fields.

2.1.1 PERI_DIV_CMD

Divider command register

Address: 0x40010000

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW		RW					
HW Access	R		R					
Name	SEL_TYPE [7:6]		SEL_DIV [5:0]					
Bits	15	14	13	12	11	10	9	8
SW Access	RW		RW					
HW Access	R		R					
Name	PA_SEL_TYPE [15:14]		PA_SEL_DIV [13:8]					
Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							
Bits	31	30	29	28	27	26	25	24
SW Access	RW	RW	None					
HW Access	RW1C	RW1C	None					
Name	ENABLE	DISABLE	None [29:24]					

Bits	Name	Description
31	ENABLE	Clock divider enable command (mutually exclusive with DISABLE). Typically, SW sets this field to '1' to enable a divider and HW sets this field to '0' to indicate that divider enabling has completed. When a divider is enabled, its integer and fractional (if present) counters are initialized to "0". If a divider is to be re-enabled using different integer and fractional divider values, the SW should follow these steps: 0: Disable the divider using the DIV_CMD.DISABLE field. 1: Configure the divider's DIV_XXX_CTL register. 2: Enable the divider using the DIV_CMD_ENABLE field. The SEL_DIV and SEL_TYPE fields specify which divider is to be enabled. The enabled divider may be phase aligned to either "clk_hf" (typical usage) or to ANY enabled divider. The PA_SEL_DIV and P_SEL_TYPE fields specify the reference divider. The HW sets the ENABLE field to '0' when the enabling is performed and the HW set the DIV_XXX_CTL.EN field of the divider to '1' when the enabling is performed. Note that enabling with phase alignment to a low frequency divider takes time. E.g. To align to a divider that generates a clock of "clk_hf"/n (with n being the integer divider value INT_DIV+1), up to n cycles may be required to perform alignment. Phase alignment to "clk_hf" takes affect immediately. SW can set this field to '0' during phase alignment to abort the enabling process. Default Value: 0

2.1.1 PERI_DIV_CMD (continued)

30	DISABLE	Clock divider disable command (mutually exclusive with ENABLE). SW sets this field to '1' and HW sets this field to '0'. The SEL_DIV and SEL_TYPE fields specify which divider is to be disabled. The HW sets the DISABLE field to '0' immediately and the HW sets the DIV_XXX_CTL.EN field of the divider to '0' immediately. Default Value: 0
15 : 14	PA_SEL_TYPE	Specifies the divider type of the divider to which phase alignment is performed for the clock enable command: 0: 8.0 (integer) clock dividers. 1: 16.0 (integer) clock dividers. 2: 16.5 (fractional) clock dividers. 3: 24.5 (fractional) clock dividers. Default Value: 3
13 : 8	PA_SEL_DIV	(PA_SEL_TYPE, PA_SEL_DIV) specifies the divider to which phase alignment is performed for the clock enable command. Any enabled divider can be used as reference. This allows all dividers to be aligned with each other, even when they are enabled at different times. If PA_SEL_DIV is "63" and "PA_SEL_TYPE" is "3", "clk_hf" is used as reference. Default Value: 63
7 : 6	SEL_TYPE	Specifies the divider type of the divider on which the command is performed: 0: 8.0 (integer) clock dividers. 1: 16.0 (integer) clock dividers. 2: 16.5 (fractional) clock dividers. 3: 24.5 (fractional) clock dividers. Default Value: 3
5 : 0	SEL_DIV	(SEL_TYPE, SEL_DIV) specifies the divider on which the command (DISABLE/ENABLE) is performed. If SEL_DIV is "63" and "SEL_TYPE" is "3" (default/reset value), no divider is specified and no clock signal(s) are generated. Default Value: 63

2.1.2 PERI_PCLK_CTL0

Programmable clock control register

Address: 0x40010100

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW		None		RW			
HW Access	R		None		R			
Name	SEL_TYPE [7:6]		None [5:4]		SEL_DIV [3:0]			

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
7 : 6	SEL_TYPE	Specifies divider type: 0: 8.0 (integer) clock dividers. 1: 16.0 (integer) clock dividers. 2: 16.5 (fractional) clock dividers. 3: 24.5 (fractional) clock dividers. Default Value: 3
3 : 0	SEL_DIV	Specifies one of the dividers of the divider type specified by SEL_TYPE. If SEL_DIV is "63" and "SEL_TYPE" is "3" (default/reset value), no divider is specified and no clock control signal(s) are generated. When transitioning a clock between two out of phase dividers, spurious clock control signals may be generated for one "clk_hf" cycle during this transition. These clock control signals may cause a single clock period that is smaller than any of the two divider periods. To prevent these spurious clock signals, the clock multiplexer can be disconnected (SEL_DIV is "63" and "SEL_TYPE" is "3") for a transition time that is larger than the smaller of the two divider periods. Default Value: 15

2.1.3 PERI_DIV_16_CTL0

Divider control register (for 16.0 divider)

Address: 0x40010300

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None							R
HW Access	None							RW
Name	None [7:1]							EN

Bits	15	14	13	12	11	10	9	8
SW Access	RW							
HW Access	R							
Name	INT16_DIV [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	RW							
HW Access	R							
Name	INT16_DIV [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
23 : 8	INT16_DIV	Integer division by (1+INT16_DIV). Allows for integer divisions in the range [1, 65,536]. Note: this type of divider does NOT allow for a fractional division. For the generation of a divided clock, the integer division range is restricted to [2, 65,536]. For the generation of a 50/50% duty cycle digital divided clock, the integer division range is restricted to even numbers in the range [2, 65,536]. The generation of a 50/50 % duty cycle analog divided clock has no restrictions. Note that this field is retained. However, the counter that is used to implement the division is not and will be initialized by HW to "0" when transitioning from DeepSleep to Active power mode. Default Value: 0
0	EN	Divider enabled. HW sets this field to '1' as a result of an ENABLE command. HW sets this field to '0' as a result on a DISABLE command. Note that this field is retained. As a result, the divider does NOT have to be re-enabled after transitioning from DeepSleep to Active power mode. Default Value: 0

2.1.4 PERI_DIV_16_5_CTL0

Divider control register (for 16.5 divider)

Address: 0x40010400

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW					None		R
HW Access	R					None		RW
Name	FRAC5_DIV [7:3]					None [2:1]		EN

Bits	15	14	13	12	11	10	9	8
SW Access	RW							
HW Access	R							
Name	INT16_DIV [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	RW							
HW Access	R							
Name	INT16_DIV [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
23 : 8	INT16_DIV	Integer division by (1+INT16_DIV). Allows for integer divisions in the range [1, 65,536]. Note: combined with fractional division, this divider type allows for a division in the range [1, 65,536 31/ 32] in 1/32 increments. For the generation of a divided clock, the division range is restricted to [2, 65,536 31/32]. For the generation of a 50/50% duty cycle divided clock, the division range is restricted to [2, 65,536]. Note that this field is retained. However, the counter that is used to implement the division is not and will be initialized by HW to "0" when transitioning from DeepSleep to Active power mode. Default Value: 0
7 : 3	FRAC5_DIV	Fractional division by (FRAC5_DIV/32). Allows for fractional divisions in the range [0, 31/32]. Note that fractional division results in clock jitter as some clock periods may be 1 "clk_hf" cycle longer than other clock periods. Note that this field is retained. However, the counter that is used to implement the division is not and will be initialized by HW to "0" when transitioning from DeepSleep to Active power mode. Default Value: 0

2.1.4 PERI_DIV_16_5_CTL0 (continued)

0	EN	Divider enabled. HW sets this field to '1' as a result of an ENABLE command. HW sets this field to '0' as a result on a DISABLE command. Note that this field is retained. As a result, the divider does NOT have to be re-enabled after transitioning from DeepSleep to Active power mode. Default Value: 0
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2.1.5 PERI_DIV_24_5_CTL

Divider control register (for 24.5 divider)

Address: 0x40010500

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW					None		R
HW Access	R					None		RW
Name	FRAC5_DIV [7:3]					None [2:1]		EN

Bits	15	14	13	12	11	10	9	8
SW Access	RW							
HW Access	R							
Name	INT24_DIV [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	RW							
HW Access	R							
Name	INT24_DIV [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	RW							
HW Access	R							
Name	INT24_DIV [31:24]							

Bits	Name	Description
31 : 8	INT24_DIV	Integer division by (1+INT24_DIV). Allows for integer divisions in the range [1, 16,777,216]. Note: combined with fractional division, this divider type allows for a division in the range [1, 16,777,216 31/32] in 1/32 increments. For the generation of a divided clock, the integer division range is restricted to [2, 16,777,216 31/32]. For the generation of a 50/50% duty cycle divided clock, the division range is restricted to [2, 16,777,216]. Note that this field is retained. However, the counter that is used to implement the division is not and will be initialized by HW to "0" when transitioning from DeepSleep to Active power mode. Default Value: 0
7 : 3	FRAC5_DIV	Fractional division by (FRAC5_DIV/32). Allows for fractional divisions in the range [0, 31/32]. Note that fractional division results in clock jitter as some clock periods may be 1 "clk_hf" cycle longer than other clock periods. Note that this field is retained. However, the counter that is used to implement the division is not and will be initialized by HW to "0" when transitioning from DeepSleep to Active power mode. Default Value: 0

2.1.5 PERI_DIV_24_5_CTL (continued)

0	EN	Divider enabled. HW sets this field to '1' as a result of an ENABLE command. HW sets this field to '0' as a result on a DISABLE command. Note that this field is retained. As a result, the divider does NOT have to be re-enabled after transitioning from DeepSleep to Active power mode. Default Value: 0
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2.1.6 PERI_TR_CTL

Trigger control register

Address: 0x40010600

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None	RW						
HW Access	None	R						
Name	None	TR_SEL [6:0]						

Bits	15	14	13	12	11	10	9	8
SW Access	None				RW			
HW Access	None				R			
Name	None [15:12]				TR_GROUP [11:8]			

Bits	23	22	21	20	19	18	17	16
SW Access	RW							
HW Access	RW							
Name	TR_COUNT [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	RW	RW	None					
HW Access	RW1C	R	None					
Name	TR_ACT	TR_OUT	None [29:24]					

Bits	Name	Description
31	TR_ACT	SW sets this field to '1' by to activate (set to '1') a trigger as identified by TR_SEL and TR_OUT for TR_COUNT cycles. HW sets this field to '0' when the cycle counter is decremented to "0". Note: a TR_COUNT value of 255 is a special case and trigger activation is under direct control of the TR_ACT field (the counter is not decremented). Default Value: 0
30	TR_OUT	Specifies whether trigger activation is for a specific input or output trigger of the trigger multiplexer. Activation of a specific input trigger, will result in activation of all output triggers that have the specific input trigger selected through their TR_OUT_CTL.SEL field. Activation of a specific output trigger, will result in activation of the specified TR_SEL output trigger only. '0': TR_SEL selection and trigger activation is for an input trigger to the trigger multiplexer. '1': TR_SEL selection and trigger activation is for an output trigger from the trigger multiplexer. Default Value: 0
23 : 16	TR_COUNT	Amount of cycles a specific trigger is activated. During activation (TR_ACT is '1'), HW decrements this field to "0" using a cycle counter. During activation, SW should not modify this register field. A value of 255 is a special case: HW does NOT decrement this field to "0" and trigger activation is under direct control of TR_ACT: when TR_ACT is '1' the trigger is activated and when TR_ACT is '0' the trigger is deactivated. Default Value: 0

2.1.6 PERI_TR_CTL (continued)

11 : 8	TR_GROUP	Specifies the trigger group. Default Value: 0
6 : 0	TR_SEL	Specifies the activated trigger when TR_ACT is '1'. TR_OUT specifies whether the activated trigger is an input trigger or output trigger to the trigger multiplexer. During activation (TR_ACT is '1'), SW should not modify this register field. If the specified trigger is not present, the trigger activation has no effect. Default Value: 0

2.1.7 PERI_TR_GROUP0_TR_OUT_CTL0

Trigger control register

Address: 0x40012000

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None		RW					
HW Access	None		R					
Name	None [7:6]		SEL [5:0]					

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
5 : 0	SEL	Specifies input trigger. This field is typically set during the setup of a chip use case scenario. Changing this field while activated triggers are present on the input triggers may result in unpredictable behavior. Note that input trigger 0 (default value) is typically connected to a constant signal level of '0', and as a result will not cause HW activation of the output trigger. Default Value: 0

2.1.8 PERI_TR_GROUP1_TR_OUT_CTL0

Trigger control register

Address: 0x40012200

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None		RW					
HW Access	None		R					
Name	None [7:6]		SEL [5:0]					

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
5 : 0	SEL	Specifies input trigger. This field is typically set during the setup of a chip use case scenario. Changing this field while activated triggers are present on the input triggers may result in unpredictable behavior. Note that input trigger 0 (default value) is typically connected to a constant signal level of '0', and as a result will not cause HW activation of the output trigger. Default Value: 0

2.1.9 PERI_TR_GROUP2_TR_OUT_CTL

Trigger control register

Address: 0x40012400

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None		RW					
HW Access	None		R					
Name	None [7:6]		SEL [5:0]					

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
5 : 0	SEL	Specifies input trigger. This field is typically set during the setup of a chip use case scenario. Changing this field while activated triggers are present on the input triggers may result in unpredictable behavior. Note that input trigger 0 (default value) is typically connected to a constant signal level of '0', and as a result will not cause HW activation of the output trigger. Default Value: 0

2.1.10 PERI_TR_GROUP3_TR_OUT_CTL

Trigger control register

Address: 0x40012600

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None				RW			
HW Access	None				R			
Name	None [7:4]				SEL [3:0]			

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
3 : 0	SEL	Specifies input trigger. This field is typically set during the setup of a chip use case scenario. Changing this field while activated triggers are present on the input triggers may result in unpredictable behavior. Note that input trigger 0 (default value) is typically connected to a constant signal level of '0', and as a result will not cause HW activation of the output trigger. Default Value: 0

3 High Speed IO Matrix (HSIOM) Registers



This section discusses the HSIOM registers. It lists all the registers in mapping tables, in address order.

3.1 Register Details

Register	Address	Description
HSIOM_PORT_SEL0	0x40020000	Port selection register
HSIOM_PORT_SEL1	0x40020100	Port selection register. See HSIOM_PORT_SEL0 for the details of bit fields.
HSIOM_PORT_SEL2	0x40020200	Port selection register. See HSIOM_PORT_SEL0 for the details of bit fields.
HSIOM_PORT_SEL3	0x40020300	Port selection register. See HSIOM_PORT_SEL0 for the details of bit fields.
HSIOM_PORT_SEL4	0x40020400	Port selection register. See HSIOM_PORT_SEL0 for the details of bit fields.
HSIOM_PORT_SEL5	0x40020500	Port selection register. See HSIOM_PORT_SEL0 for the details of bit fields.
HSIOM_PORT_SEL6	0x40020600	Port selection register
HSIOM_PORT_SEL7	0x40020700	Port selection register
HSIOM_AMUX_SPLIT_CTL0	0x40022100	AMUX splitter cell control
HSIOM_AMUX_SPLIT_CTL1	0x40022104	AMUX splitter cell control. See HSIOM_AMUX_SPLIT_CTL0 for the details of bit fields.
HSIOM_AMUX_SPLIT_CTL2	0x40022108	AMUX splitter cell control. See HSIOM_AMUX_SPLIT_CTL0 for the details of bit fields.

3.1.1 HSIOM_PORT_SEL0

Port selection register

Address: 0x40020000

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW				RW			
HW Access	RW				RW			
Name	IO1_SEL [7:4]				IO0_SEL [3:0]			
Bits	15	14	13	12	11	10	9	8
SW Access	RW				RW			
HW Access	RW				RW			
Name	IO3_SEL [15:12]				IO2_SEL [11:8]			
Bits	23	22	21	20	19	18	17	16
SW Access	RW				RW			
HW Access	RW				RW			
Name	IO5_SEL [23:20]				IO4_SEL [19:16]			
Bits	31	30	29	28	27	26	25	24
SW Access	RW				RW			
HW Access	RW				RW			
Name	IO7_SEL [31:28]				IO6_SEL [27:24]			

Bits	Name	Description
31 : 28	IO7_SEL	Selects connection for IO pad 7 route. Default Value: 0
27 : 24	IO6_SEL	Selects connection for IO pad 6 route. Default Value: 0
23 : 20	IO5_SEL	Selects connection for IO pad 5 route. Default Value: 0
19 : 16	IO4_SEL	Selects connection for IO pad 4 route. Default Value: 0
15 : 12	IO3_SEL	Selects connection for IO pad 3 route. Default Value: 0
11 : 8	IO2_SEL	Selects connection for IO pad 2 route. Default Value: 0
7 : 4	IO1_SEL	Selects connection for IO pad 1 route. Default Value: 0
3 : 0	IO0_SEL	Selects connection for IO pad 0 route. Default Value: 0

3.1.1 HSIOM_PORT_SEL0 (continued)

0x0: GPIO :

SW controlled GPIO.

0x1: GPIO_DSI :

SW controlled "out", DSI controlled "oe_n".

0x2: DSI_DSI :

DSI controlled "out" and "oe_n".

0x3: DSI_GPIO :

DSI controlled "out", SW controlled "oe_n".

0x4: CSD_SENSE :

CSD sense connection (analog mode)

0x5: CSD_SHIELD :

CSD shield connection (analog mode)

0x6: AMUXA :

AMUXBUS A connection.

0x7: AMUXB :

AMUXBUS B connection. This mode is also used for CSD GPIO charging. When CSD GPIO charging is enabled in CSD_CONTROL, "oe_n" is connected to "!csd_charge" signal (and IO pad is also still connected to AMUXBUS B).

0x8: ACT_0 :

Chip specific Active source 0 connection.

0x9: ACT_1 :

Chip specific Active source 1 connection.

0xa: ACT_2 :

Chip specific Active source 2 connection.

0xb: ACT_3 :

Chip specific Active source 3 connection.

3.1.1 HSIOM_PORT_SEL0 (continued)

0xc: LCD_COM :

LCD common connection. This mode provides DeepSleep functionality (provided that the LCD block is enabled and properly configured).

0xd: LCD_SEG :

LCD segment connection. This mode provides DeepSleep functionality (provided that the LCD block is enabled and properly configured).

0xc: DS_0 :

Reserved

0xd: DS_1 :

Reserved

0xe: DS_2 :

Chip specific DeepSleep source 2 connection.

0xf: DS_3 :

Chip specific DeepSleep source 3 connection.

3.1.2 HSIOM_PORT_SEL6

Port selection register

Address: 0x40020600

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW				RW			
HW Access	RW				RW			
Name	IO1_SEL [7:4]				IO0_SEL [3:0]			

Bits	15	14	13	12	11	10	9	8
SW Access	RW				RW			
HW Access	RW				RW			
Name	IO3_SEL [15:12]				IO2_SEL [11:8]			

Bits	23	22	21	20	19	18	17	16
SW Access	RW				RW			
HW Access	RW				RW			
Name	IO5_SEL [23:20]				IO4_SEL [19:16]			

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
23 : 20	IO5_SEL	Selects connection for IO pad 5 route. Default Value: 0
19 : 16	IO4_SEL	Selects connection for IO pad 4 route. Default Value: 0
15 : 12	IO3_SEL	Selects connection for IO pad 3 route. Default Value: 0
11 : 8	IO2_SEL	Selects connection for IO pad 2 route. Default Value: 0
7 : 4	IO1_SEL	Selects connection for IO pad 1 route. Default Value: 0
3 : 0	IO0_SEL	Selects connection for IO pad 0 route. Default Value: 0
0x0: GPIO :		
SW controlled GPIO.		

3.1.2 HSIOM_PORT_SEL6 (continued)

0x1: GPIO_DSI :

SW controlled "out", DSI controlled "oe_n".

0x2: DSI_DSI :

DSI controlled "out" and "oe_n".

0x3: DSI_GPIO :

DSI controlled "out", SW controlled "oe_n".

0x4: CSD_SENSE :

CSD sense connection (analog mode)

0x5: CSD_SHIELD :

CSD shield connection (analog mode)

0x6: AMUXA :

AMUXBUS A connection.

0x7: AMUXB :

AMUXBUS B connection. This mode is also used for CSD GPIO charging. When CSD GPIO charging is enabled in CSD_CONTROL, "oe_n" is connected to "!csd_charge" signal (and IO pad is also still connected to AMUXBUS B).

0x8: ACT_0 :

Chip specific Active source 0 connection.

0x9: ACT_1 :

Chip specific Active source 1 connection.

0xa: ACT_2 :

Chip specific Active source 2 connection.

0xb: ACT_3 :

Chip specific Active source 3 connection.

0xc: LCD_COM :

LCD common connection. This mode provides DeepSleep functionality (provided that the LCD block is enabled and properly configured).

3.1.2 HSIOM_PORT_SEL6 (continued)

0xd: LCD_SEG :

LCD segment connection. This mode provides DeepSleep functionality (provided that the LCD block is enabled and properly configured).

0xc: DS_0 :

Reserved

0xd: DS_1 :

Reserved

0xe: DS_2 :

Chip specific DeepSleep source 2 connection.

0xf: DS_3 :

Chip specific DeepSleep source 3 connection.

3.1.3 HSIOM_PORT_SEL7

Port selection register

Address: 0x40020700

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW				RW			
HW Access	RW				RW			
Name	IO1_SEL [7:4]				IO0_SEL [3:0]			

Bits	15	14	13	12	11	10	9	8
SW Access	None				RW			
HW Access	None				RW			
Name	None [15:12]				IO2_SEL [11:8]			

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
11 : 8	IO2_SEL	Selects connection for IO pad 2 route. Default Value: 0
7 : 4	IO1_SEL	Selects connection for IO pad 1 route. Default Value: 0
3 : 0	IO0_SEL	Selects connection for IO pad 0 route. Default Value: 0
0x0: GPIO :		
SW controlled GPIO.		
0x1: GPIO_DSI :		
SW controlled "out", DSI controlled "oe_n".		
0x2: DSI_DSI :		
DSI controlled "out" and "oe_n".		

3.1.3 HSIOM_PORT_SEL7 (continued)

0x3: DSI_GPIO :

DSI controlled "out", SW controlled "oe_n".

0x4: CSD_SENSE :

CSD sense connection (analog mode)

0x5: CSD_SHIELD :

CSD shield connection (analog mode)

0x6: AMUXA :

AMUXBUS A connection.

0x7: AMUXB :

AMUXBUS B connection. This mode is also used for CSD GPIO charging. When CSD GPIO charging is enabled in CSD_CONTROL, "oe_n" is connected to "!csd_charge" signal (and IO pad is also still connected to AMUXBUS B).

0x8: ACT_0 :

Chip specific Active source 0 connection.

0x9: ACT_1 :

Chip specific Active source 1 connection.

0xa: ACT_2 :

Chip specific Active source 2 connection.

0xb: ACT_3 :

Chip specific Active source 3 connection.

0xc: LCD_COM :

LCD common connection. This mode provides DeepSleep functionality (provided that the LCD block is enabled and properly configured).

0xd: LCD_SEG :

LCD segment connection. This mode provides DeepSleep functionality (provided that the LCD block is enabled and properly configured).

3.1.3 HSIOM_PORT_SEL7 (continued)

0xc: DS_0 :

Reserved

0xd: DS_1 :

Reserved

0xe: DS_2 :

Chip specific DeepSleep source 2 connection.

0xf: DS_3 :

Chip specific DeepSleep source 3 connection.

3.1.4 HSIOM_AMUX_SPLIT_CTL0

AMUX splitter cell control

Address: 0x40022100

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None	RW	RW	RW	None	RW	RW	RW
HW Access	None	R	R	R	None	R	R	R
Name	None	SWITCH_BB_S0	SWITCH_BB_SR	SWITCH_BB_SL	None	SWITCH_AA_S0	SWITCH_AA_SR	SWITCH_AA_SL

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
6	SWITCH_BB_S0	T-switch control for AMUXBUSB vssa/ground switch. Default Value: 0
5	SWITCH_BB_SR	T-switch control for Right AMUXBUSB switch. Default Value: 0
4	SWITCH_BB_SL	T-switch control for Left AMUXBUSB switch. Default Value: 0
2	SWITCH_AA_S0	T-switch control for AMUXBUSA vssa/ground switch: '0': switch open. '1': switch closed. Default Value: 0
1	SWITCH_AA_SR	T-switch control for Right AMUXBUSA switch: '0': switch open. '1': switch closed. Default Value: 0

3.1.4 HSIOM_AMUX_SPLIT_CTL0 (continued)

0	SWITCH_AA_SL	T-switch control for Left AMUXBUS switch: '0': switch open. '1': switch closed. Default Value: 0
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4 System Resources Subsystem (SRSS) Registers



This section discusses the SRSS registers. It lists all the registers in mapping tables, in address order.

4.1 Register Details

Register	Address	Description
PWR_CONTROL	0x40030000	Power Mode Control
PWR_KEY_DELAY	0x40030004	Power System Key&Delay Register
PWR_DDFT_SELECT	0x4003000C	Power DDFT Mode Selection Register
TST_MODE	0x40030014	Test Mode Control Register
CLK_SELECT	0x40030028	Clock Select Register
CLK_ILO_CONFIG	0x4003002C	ILO Configuration
CLK_IMO_CONFIG	0x40030030	IMO Configuration
CLK_DFT_SELECT	0x40030034	Clock DFT Mode Selection Register
WDT_DISABLE_KEY	0x40030038	Watchdog Disable Key Register
WDT_COUNTER	0x4003003C	Watchdog Counter Register
WDT_MATCH	0x40030040	Watchdog Match Register
SRSS_INTR	0x40030044	SRSS Interrupt Register
SRSS_INTR_SET	0x40030048	SRSS Interrupt Set Register
SRSS_INTR_MASK	0x4003004C	SRSS Interrupt Mask Register
RES_CAUSE	0x40030054	Reset Cause Observation Register
PWR_BG_TRIM1	0x40030F00	Bandgap Trim Register
PWR_BG_TRIM2	0x40030F04	Bandgap Trim Register
CLK_IMO_SELECT	0x40030F08	IMO Frequency Select Register
CLK_IMO_TRIM1	0x40030F0C	IMO Trim Register
CLK_IMO_TRIM2	0x40030F10	IMO Trim Register
PWR_PWRSYS_TRIM1	0x40030F14	Power System Trim Register
CLK_IMO_TRIM3	0x40030F18	IMO Trim Register

4.1.1 PWR_CONTROL

Power Mode Control

Address: 0x40030000

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None		R	R	R			
HW Access	None		RW	RW	RW			
Name	None [7:6]		LP-M_READY	DEBUG_SESSION	POWER_MODE [3:0]			

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	RW	None			R		RW	RW
HW Access	A	None			RW		R	R
Name	EXT_VCCD	None [22:20]			SPARE [19:18]		OVER_TEMP_THRESH	OVER_TEMP_EN

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
23	EXT_VCCD	Always write 0 except as noted below. PSoC4-S0 and Streetfighter CapSense products may set this bit if Vccd is provided externally (on Vccd pin). Setting this bit turns off the active regulator and will lead to system reset (BOD) unless both Vddd and Vccd pins are supplied externally. This register bit only resets for XRES, POR, or a detected BOD. Default Value: 0
19 : 18	SPARE	Spare AHB readback bits that are hooked to PWR_PWRSYS_TRIM1.SPARE_TRIM[1:0] through spare logic equivalent to bitwise inversion. Engineering only. Default Value: 0
17	OVER_TEMP_THRESH	Over-temperature threshold. 0: TEMP_HIGH condition occurs between 120C and 125C. 1: TEMP_HIGH condition occurs between 60C and 75C (used for testing). Default Value: 0

4.1.1 PWR_CONTROL (continued)

16	OVER_TEMP_EN	Enables the die over temperature sensor. Must be enabled when using the TEMP_HIGH interrupt. Default Value: 0
5	LPM_READY	Indicates whether the low power mode regulator is ready to enter DEEPSLEEP mode. 0: If DEEPSLEEP mode is requested, device will enter SLEEP mode. When low power regulators are ready, device will automatically enter the originally requested mode. 1: Normal operation. DEEPSLEEP works as described. Default Value: 0
4	DEBUG_SESSION	Indicates whether a debug session is active (CDBGPWUPREQ signal is 1) Default Value: 0 0x0: NO_SESSION : No debug session active 0x1: SESSION_ACTIVE : Debug session is active
3 : 0	POWER_MODE	Current power mode of the device. Note that this field cannot be read in all power modes on actual silicon. Default Value: 0 0x0: RESET : RESET state 0x1: ACTIVE : ACTIVE state 0x2: SLEEP : SLEEP state 0x3: DEEP_SLEEP : DEEP_SLEEP state

4.1.2 PWR_KEY_DELAY

Power System Key Register

Address: 0x40030004

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access	R							
Name	WAKEUP_HOLDOFF [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	None						RW	
HW Access	None						R	
Name	None [15:10]						WAKEUP_HOLDOFF [9:8]	

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
9 : 0	WAKEUP_HOLDOFF	Delay to wait for references to settle on wakeup from deepsleep. BOD is ignored and system does not resume until this delay expires. Note that the same delay on POR is hard-coded. The default assumes the output of the predivider is 48MHz + 3%. Firmware may scale this setting according to the fastest actual clock frequency that can occur when waking from DEEPSLEEP. Default Value: 248

4.1.3 PWR_DDFT_SELECT

Power DDFT Mode Selection Register

Address: 0x4003000C

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW				RW			
HW Access	R				R			
Name	DDFT1_SEL [7:4]				DDFT0_SEL [3:0]			

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
7 : 4	DDFT1_SEL	Select signal for power DDFT output #1 Default Value: 0 0x0: WAKEUP : wakeup 0x1: AWAKE : awake 0x2: ACT_POWER_EN : act_power_en 0x3: ACT_POWER_UP : act_power_up

4.1.3 PWR_DDFT_SELECT (continued)

0x4: ACT_POWER_GOOD :

act_power_good

0x5: ACT_REF_VALID :

act_ref_valid

0x6: ACT_REG_VALID :

act_reg_valid

0x7: ACT_COMP_OUT :

act_comp_out

0x8: ACT_TEMP_HIGH :

act_temp_high

0x9: DPSLP_COMP_OUT :

dpslp_comp_out

0xa: DPSLP_POWER_UP :

dpslp_power_up

0xb: AWAKE_DELAYED :

awake_delayed

0xc: LPM_READY :

lpm_ready

0xd: SLEEPHOLDACK_N :

sleepholdack_n

0xe: GND :

1'b0

0xf: PWR :

1'b1

3 : 0	DDFT0_SEL	Select signal for power DDFT output #0 Default Value: 0
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4.1.3 PWR_DDFT_SELECT (continued)

0x0: WAKEUP :

wakeup

0x1: AWAKE :

awake

0x2: ACT_POWER_EN :

act_power_en

0x3: ACT_POWER_UP :

act_power_up

0x4: ACT_POWER_GOOD :

act_power_good

0x5: ACT_REF_EN :

srss_adft_control_act_ref_en

0x6: ACT_COMP_EN :

srss_adft_control_act_comp_en

0x7: DPSLP_REF_EN :

srss_adft_control_dpslp_ref_en

0x8: DPSLP_REG_EN :

srss_adft_control_dpslp_reg_en

0x9: DPSLP_COMP_EN :

srss_adft_control_dpslp_comp_en

0xa: OVER_TEMP_EN :

pwr_control_over_temp_en

0xb: SLEEPHOLDREQ_N :

sleepholdreq_n

4.1.3 PWR_DDFT_SELECT (continued)

0xc: ADFT_BUF_EN :

adft_buf_en

0xd: ATPG_OBSERVE :

ATPG observe point (no functional purpose)

0xe: GND :

1'b0

0xf: PWR :

1'b1

4.1.4 TST_MODE

Test Mode Control Register

Address: 0x40030014

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None					R	None	
HW Access	None					RW	None	
Name	None [7:3]					SWD_CONNECTED	None	

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	RW	R	None	RW	None			
HW Access	R	RW	None	A	None			
Name	TEST_MODE	TEST_KEY_DFT_EN	None	BLOCK_AL T_XRES	None [27:24]			

Bits	Name	Description
31	TEST_MODE	0: Normal operation mode 1: Test mode (any test mode) Setting this bit will prevent BootROM from yielding execution to Flash image. Default Value: 0
30	TEST_KEY_DFT_EN	This bit is set when a XRES test mode key is shifted in. It is the value of the test_key_dft_en signal. When this bit is set, the BootROM will not yield execution to the FLASH image (same function as setting TEST_MODE bit below). Default Value: 0
28	BLOCK_ALT_XRES	Relevant only for parts that have the alternate XRES mechanism of overloading a GPIO pin temporarily as alternate XRES during test. When set, this bit blocks the alternate XRES function, such that the pin can be used for normal I/O or for ddft/adft observation. See SAS Part-V and Part-IX for details. This register bit only resets for XRES, POR, or a detected BOD. Default Value: 0

4.1.4 TST_MODE (continued)

2	SWD_CONNECTED	0: SWD not active 1: SWD activated (Line Reset & Connect sequence passed) (Note: this bit replaces TST_CTRL.SWD_CONNECTED and is present in all M0S8 products except TSG4) Default Value: 0
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4.1.5 CLK_SELECT

Clock Select Register

Address: 0x40030028

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW		RW		RW		RW	
HW Access	R		R		R		R	
Name	SYSCLK_DIV [7:6]		PUMP_SEL [5:4]		HFCLK_DIV [3:2]		HFCLK_SEL [1:0]	

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
7 : 6	SYSCLK_DIV	Select clk_sys prescaler value. Default Value: 0 0x0: NO_DIV : clk_sys= clk_hf/1 0x1: DIV_BY_2 : clk_sys= clk_hf/2 0x2: DIV_BY_4 : clk_sys= clk_hf/4 0x3: DIV_BY_8 : clk_sys= clk_hf/8

4.1.5 CLK_SELECT (continued)

5 : 4	PUMP_SEL	Selects clock source for charge pump clock. This clock is not guaranteed to be glitch free when changing any of its sources or settings. Default Value: 0 0x0: GND : No clock, connect to gnd 0x1: IMO : Use main IMO output 0x2: HFCLK : Use clk_hf (using selected source after predivider but before prescaler)
3 : 2	HFCLK_DIV	Selects clk_hf predivider value. Default Value: 2 0x0: NO_DIV : Transparent mode, feed through selected clock source w/o dividing. 0x1: DIV_BY_2 : Divide selected clock source by 2 0x2: DIV_BY_4 : Divide selected clock source by 4 0x3: DIV_BY_8 : Divide selected clock source by 8
1 : 0	HFCLK_SEL	Selects a source for clk_hf and dsi_in[0]. Note that not all products support all clock sources. Selecting a clock source that is not supported will result in undefined behavior. Default Value: 0 0x0: IMO : IMO - Internal R/C Oscillator 0x1: EXTCLK : EXTCLK - External Clock Pin 0x2: ECO : ECO - External-Crystal Oscillator or PLL subsystem output

4.1.6 CLK_ILO_CONFIG

ILO Configuration

Address: 0x4003002C

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None							
HW Access	None							
Name	None [7:0]							
Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							
Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							
Bits	31	30	29	28	27	26	25	24
SW Access	RW	None						
HW Access	RW	None						
Name	ENABLE	None						

Bits	Name	Description
31	ENABLE	Master enable for ILO oscillator. This bit is hardware set whenever the WD_DISABLE_KEY is not set to the magic value. Default Value: 1

4.1.7 CLK_IMO_CONFIG

IMO Configuration

Address: 0x40030030

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None							
HW Access	None							
Name	None [7:0]							
Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							
Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							
Bits	31	30	29	28	27	26	25	24
SW Access	RW	None						
HW Access	R	None						
Name	ENABLE	None						

Bits	Name	Description
31	ENABLE	Master enable for IMO oscillator. Clearing this bit will disable the IMO. Don't do this if the system is running off it. Default Value: 1

4.1.8 CLK_DFT_SELECT

Clock DFT Mode Selection Register

Address: 0x40030034

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None	RW	RW		RW			
HW Access	None	R	R		R			
Name	None	DFT_EDGE 0	DFT_DIV0 [5:4]		DFT_SEL0 [3:0]			

Bits	15	14	13	12	11	10	9	8
SW Access	None	RW	RW		RW			
HW Access	None	R	R		R			
Name	None	DFT_EDGE 1	DFT_DIV1 [13:12]		DFT_SEL1 [11:8]			

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
14	DFT_EDGE1	Edge sensitivity for in-line divider on output #1 (only relevant when DIV1>0). Default Value: 0 0x0: POSEDGE : Use posedge for divider 0x1: NEGEDGE : Use negedge for divider
13 : 12	DFT_DIV1	DFT Output Divide Down. Default Value: 0 0x0: NO_DIV : Direct Output

4.1.8 CLK_DFT_SELECT (continued)

		0x1: DIV_BY_2 :
		Divide by 2
		0x2: DIV_BY_4 :
		Divide by 4
		0x3: DIV_BY_8 :
		Divide by 8
11 : 8	DFT_SEL1	Select signal for DFT output #1 Default Value: 0
		0x0: NC :
		Disabled - output is 0
		0x1: ILO :
		clk_ilo: ILO output
		0x2: IMO :
		clk_imo: IMO primary output
		0x3: ECO :
		clk_eco: ECO output
		0x4: EXTCLK :
		clk_ext: external clock input
		0x5: HFCLK :
		clk_hf: root of the high-speed clock tree
		0x6: LFCLK :
		clk_lf: root of the low-speed clock tree
		0x7: SYSCLK :
		clk_sys: root of the CPU/AHB clock tree (gated version of clk_hf)
		0x8: PUMPCLK :
		clk_pump: clock provided to charge pumps in FLASH and PA

4.1.8 CLK_DFT_SELECT (continued)

		0x9: SLPCTRLCLK : clk_slpctrl: clock provided to SleepController
6	DFT_EDGE0	Edge sensitivity for in-line divider on output #0 (only relevant when DIV0>0). Default Value: 0 0x0: POSEDGE : Use posedge for divider 0x1: NEGEDGE : Use negedge for divider
5 : 4	DFT_DIV0	DFT Output Divide Down. Default Value: 0 0x0: NO_DIV : Direct Output 0x1: DIV_BY_2 : Divide by 2 0x2: DIV_BY_4 : Divide by 4 0x3: DIV_BY_8 : Divide by 8
3 : 0	DFT_SEL0	Select signal for DFT output #0 Default Value: 0 0x0: NC : Disabled - output is 0 0x1: ILO : clk_ilo: ILO output 0x2: IMO : clk_imo: IMO primary output 0x3: ECO : clk_eco: ECO output

4.1.8 CLK_DFT_SELECT (continued)

0x4: EXTCLK :

clk_ext: external clock input

0x5: HFCLK :

clk_hf: root of the high-speed clock tree

0x6: LFCLK :

clk_lf: root of the low-speed clock tree

0x7: SYSCLK :

clk_sys: root of the CPU/AHB clock tree (gated version of clk_hf)

0x8: PUMPCLK :

clk_pump: clock provided to charge pumps in FLASH and PA

0x9: SLPCTRLCLK :

clk_slpctrl: clock provided to SleepController

4.1.9 WDT_DISABLE_KEY

Watchdog Disable Key Register

Address: 0x40030038

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access	R							
Name	KEY [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	RW							
HW Access	R							
Name	KEY [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	RW							
HW Access	R							
Name	KEY [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	RW							
HW Access	R							
Name	KEY [31:24]							

Bits	Name	Description
31 : 0	KEY	Disables WDT reset when equal to 0xACED8865. The WDT reset functions normally for any other setting. Default Value: 0

4.1.10 WDT_COUNTER

Watchdog Counter Register

Address: 0x4003003C

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access	RW							
Name	COUNTER [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access	RW							
Name	COUNTER [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
15 : 0	COUNTER	Current value of WDT Counter Default Value: 0

4.1.11 WDT_MATCH

Watchdog Match Register

Address: 0x40030040

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access	R							
Name	MATCH [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	RW							
HW Access	R							
Name	MATCH [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None				RW			
HW Access	None				R			
Name	None [23:20]				IGNORE_BITS [19:16]			

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
19 : 16	IGNORE_BITS	The number of MSB bits of the watchdog timer that are NOT checked against MATCH. This value provides control over the time-to-reset of the watchdog (which happens after 3 successive matches). Note that certain products may enforce a minimum value for this register through design time configuration. Default Value: 0
15 : 0	MATCH	Match value for Watchdog counter. Every time WDT_COUNTER reaches MATCH an interrupt is generated. Two unserved interrupts will lead to a system reset (i.e. at the third match). Default Value: 4096

4.1.12 SRSS_INTR

SRSS Interrupt Register

Address: 0x40030044

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None						RW1C	RW1C
HW Access	None						A	A
Name	None [7:2]						TEMP_HIGH	WDT_MATCH

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
1	TEMP_HIGH	Regulator over-temp interrupt. This interrupt can occur when a short circuit exists on the vccd pin or when extreme loads are applied on IO-cells causing the die to overheat. Firmware is encouraged to shutdown all IO cells and then go to DeepSleep mode when this interrupt occurs if protection against such conditions is desired. Default Value: 0
0	WDT_MATCH	WDT Interrupt Request. This bit is set each time WDT_COUNTER==WDT_MATCH. Clearing this bit also feeds the watch dog. Missing 2 interrupts in a row will generate brown-out reset. Due to internal synchronization, it takes 2 SYSCLK cycles to update after a W1C. Default Value: 0

4.1.13 SRSS_INTR_SET

SRSS Interrupt Set Register

Address: 0x40030048

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None						RW1S	None
HW Access	None						A	None
Name	None [7:2]						TEMP_HIGH	None

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
1	TEMP_HIGH	Writing 1 to this bit internally sets the overtemp interrupt. This can be observed by reading SRSS_INTR.TEMP_HIGH. This bit always reads back as zero. Default Value: 0

4.1.14 SRSS_INTR_MASK

SRSS Interrupt Mask Register

Address: 0x4003004C

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None						RW	RW
HW Access	None						R	R
Name	None [7:2]						TEMP_HIGH	WDT_MATCH

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
1	TEMP_HIGH	Masks REG_OVERTEMP interrupt Default Value: 0
0	WDT_MATCH	Clearing this bit will not forward the interrupt to the CPU. It will not, however, disable the WDT reset generation on 2 missed interrupts. Default Value: 0

4.1.15 RES_CAUSE

Reset Cause Observation Register

Address: 0x40030054

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None			RW1C	RW1C	None		RW1C
HW Access	None			A	A	None		A
Name	None [7:5]			RESET_- SOFT	RE- SET_PROT _FAULT	None [2:1]		RE- SET_WDT

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
4	RESET_SOFT	Cortex-M0 requested a system reset through it's SYSRESETREQ. This can be done via a debugger probe or in firmware. Default Value: 0
3	RESET_PROT_FAULT	A protection violation occurred that requires a RESET. This includes, but is not limited to, hitting a debug breakpoint while in Privileged Mode. Default Value: 0
0	RESET_WDT	A WatchDog Timer reset has occurred since last power cycle. Default Value: 0

4.1.16 PWR_BG_TRIM1

Bandgap Trim Register

Address: 0x40030F00

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None		RW					
HW Access	None		R					
Name	None [7:6]		REF_VTRIM [5:0]					

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
5 : 0	REF_VTRIM	Trims the bandgap reference voltage output. Used to trim the VBG to the voltage where its temperature curvature is minimal. Bit [5] is unused within the bandgap block. Default Value: 16

4.1.17 PWR_BG_TRIM2

Bandgap Trim Register

Address: 0x40030F04

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None		RW					
HW Access	None		R					
Name	None [7:6]		REF_ITRIM [5:0]					

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
5 : 0	REF_ITRIM	Trims the bandgap reference current output. Used to trim the IBG to the voltage where its temperature curvature is minimal. Default Value: 28

4.1.18 CLK_IMO_SELECT

IMO Frequency Select Register

Address: 0x40030F08

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None					RW		
HW Access	None					R		
Name	None [7:3]					FREQ [2:0]		

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
2 : 0	FREQ	Select operating frequency Default Value: 0 0x0: 24_MHZ : IMO runs at 24 MHz 0x1: 28_MHZ : IMO runs at 28 MHz 0x2: 32_MHZ : IMO runs at 32 MHz 0x3: 36_MHZ : IMO runs at 36 MHz

4.1.18 CLK_IMO_SELECT (continued)

0x4: 40_MHZ :

IMO runs at 40 MHz

0x5: 44_MHZ :

IMO runs at 44 MHz

0x6: 48_MHZ :

IMO runs at 48 MHz

4.1.19 CLK_IMO_TRIM1

IMO Trim Register

Address: 0x40030F0C

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access	RW							
Name	OFFSET [7:0]							
Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							
Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							
Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
7 : 0	OFFSET	Frequency trim bits. These bits are determined at manufacturing time for each FREQ setting (IMO_TRIM2) and stored in SFLASH. This field is hardware updated during USB osclock mode. This field is mapped to the most significant bits of the IMO trim imo_clk_trim[10:3]. The step size of 1 LSB on this field is approximately 120 kHz. Default Value: 128

4.1.20 CLK_IMO_TRIM2

IMO Trim Register

Address: 0x40030F10

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None					RW		
HW Access	None					RW		
Name	None [7:3]					FSOFFSET [2:0]		

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
2 : 0	FSOFFSET	Frequency trim bits. These bits are not trimmed during manufacturing and kept at 0 under normal operation. This field is hardware updated during USB osclock mode. This field is mapped to the least significant bits of the IMO trim imo_clk_trim[2:0]. The step size of 1 LSB on this field is approximately 15 kHz. Default Value: 0

4.1.21 PWR_PWRSYS_TRIM1

Power System Trim Register

Address: 0x40030F14

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW				RW			
HW Access	R				R			
Name	SPARE_TRIM [7:4]				DPSLP_REF_TRIM [3:0]			

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
7 : 4	SPARE_TRIM	Active-Reference temperature compensation trim (repurposed from spare bits). Bits [7:6] - trim the Active-Reference IREF temperature coefficient (TC). 00: TC = 0 (unchanged) 01: TC = +80ppm/C 10: TC = -80ppm/C 11: TC = -150ppm/C Bits [5:4] - trim the Active-Reference VREF temperature coefficient (TC). 00: TC = 0 (unchanged) 01: TC = -50ppm/C 10: TC = -80ppm/C 11: TC = +150ppm/C Default Value: 0
3 : 0	DPSLP_REF_TRIM	Trims the DeepSleep reference that is used by the DeepSleep regulator and DeepSleep power comparator. Default Value: 0

4.1.22 CLK_IMO_TRIM3

IMO Trim Register

Address: 0x40030F18

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None	RW		RW				
HW Access	None	R		R				
Name	None	TCTRIM [6:5]		STEPSIZE [4:0]				

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
6 : 5	TCTRIM	IMO temperature compensation trim. These bits are determined at manufacturing time to adjust for temperature dependence. These bits are dependent on frequency and need to be changed using the Cypress provided frequency change algorithm. Default Value: 2
4 : 0	STEPSIZE	IMO trim stepsize bits. These bits are determined at manufacturing time to adjust for process variation. They are used to tune the stepsize of the FSOFFSET and OFFSET trims. Default Value: 16

5 General Purpose IO (GPIO) Registers



This section discusses the GPIO registers. It lists all the registers in mapping tables, in address order.

5.1 Register Details

Register	Address	Description
GPIO_PRT0_DR	0x40040000	Port output data register
GPIO_PRT0_PS	0x40040004	Port IO pad state register
GPIO_PRT0_PC	0x40040008	Port configuration register
GPIO_PRT0_INTR_CFG	0x4004000C	Port interrupt configuration register
GPIO_PRT0_INTR	0x40040010	Port interrupt status register
GPIO_PRT0_PC2	0x40040018	Port configuration register 2
GPIO_PRT0_DR_SET	0x40040040	Port output data set register
GPIO_PRT0_DR_CLR	0x40040044	Port output data clear register
GPIO_PRT0_DR_INV	0x40040048	Port output data invert register
GPIO_PRT1_DR	0x40040100	Port output data register. See GPIO_PRT0_DR for the details of bit fields.
GPIO_PRT1_PS	0x40040104	Port IO pad state register. See GPIO_PRT0_PS for the details of bit fields.
GPIO_PRT1_PC	0x40040108	Port configuration register. See GPIO_PRT0_PC for the details of bit fields.
GPIO_PRT1_INTR_CFG	0x4004010C	Port interrupt configuration register. See GPIO_PRT0_INTR_CFG for the details of bit fields.
GPIO_PRT1_INTR	0x40040110	Port interrupt status register. See GPIO_PRT0_INTR for the details of bit fields.
GPIO_PRT1_PC2	0x40040118	Port configuration register 2. See GPIO_PRT0_PC2 for the details of bit fields.
GPIO_PRT1_DR_SET	0x40040140	Port output data set register. See GPIO_PRT0_DR_SET for the details of bit fields.
GPIO_PRT1_DR_CLR	0x40040144	Port output data clear register. See GPIO_PRT0_DR_CLR for the details of bit fields.
GPIO_PRT1_DR_INV	0x40040148	Port output data invert register. See GPIO_PRT0_DR_INV for the details of bit fields.
GPIO_PRT2_DR	0x40040200	Port output data register. See GPIO_PRT0_DR for the details of bit fields.
GPIO_PRT2_PS	0x40040204	Port IO pad state register. See GPIO_PRT0_PS for the details of bit fields.
GPIO_PRT2_PC	0x40040208	Port configuration register. See GPIO_PRT0_PC for the details of bit fields.
GPIO_PRT2_INTR_CFG	0x4004020C	Port interrupt configuration register. See GPIO_PRT0_INTR_CFG for the details of bit fields.
GPIO_PRT2_INTR	0x40040210	Port interrupt status register. See GPIO_PRT0_INTR for the details of bit fields.
GPIO_PRT2_PC2	0x40040218	Port configuration register 2. See GPIO_PRT0_PC2 for the details of bit fields.
GPIO_PRT2_DR_SET	0x40040240	Port output data set register. See GPIO_PRT0_DR_SET for the details of bit fields.
GPIO_PRT2_DR_CLR	0x40040244	Port output data clear register. See GPIO_PRT0_DR_CLR for the details of bit fields.
GPIO_PRT2_DR_INV	0x40040248	Port output data invert register. See GPIO_PRT0_DR_INV for the details of bit fields.

Register	Address	Description
GPIO_PRT3_DR	0x40040300	Port output data register. See GPIO_PRT0_DR for the details of bit fields.
GPIO_PRT3_PS	0x40040304	Port IO pad state register. See GPIO_PRT0_PS for the details of bit fields.
GPIO_PRT3_PC	0x40040308	Port configuration register. See GPIO_PRT0_PC for the details of bit fields.
GPIO_PRT3_INTR_CFG	0x4004030C	Port interrupt configuration register. See GPIO_PRT0_INTR_CFG for the details of bit fields.
GPIO_PRT3_INTR	0x40040310	Port interrupt status register. See GPIO_PRT0_INTR for the details of bit fields.
GPIO_PRT3_PC2	0x40040318	Port configuration register 2. See GPIO_PRT0_PC2 for the details of bit fields.
GPIO_PRT3_DR_SET	0x40040340	Port output data set register. See GPIO_PRT0_DR_SET for the details of bit fields.
GPIO_PRT3_DR_CLR	0x40040344	Port output data clear register. See GPIO_PRT0_DR_CLR for the details of bit fields.
GPIO_PRT3_DR_INV	0x40040348	Port output data invert register. See GPIO_PRT0_DR_INV for the details of bit fields.
GPIO_PRT4_DR	0x40040400	Port output data register. See GPIO_PRT0_DR for the details of bit fields.
GPIO_PRT4_PS	0x40040404	Port IO pad state register. See GPIO_PRT0_PS for the details of bit fields.
GPIO_PRT4_PC	0x40040408	Port configuration register. See GPIO_PRT0_PC for the details of bit fields.
GPIO_PRT4_INTR_CFG	0x4004040C	Port interrupt configuration register. See GPIO_PRT0_INTR_CFG for the details of bit fields.
GPIO_PRT4_INTR	0x40040410	Port interrupt status register. See GPIO_PRT0_INTR for the details of bit fields.
GPIO_PRT4_PC2	0x40040418	Port configuration register 2. See GPIO_PRT0_PC2 for the details of bit fields.
GPIO_PRT4_DR_SET	0x40040440	Port output data set register. See GPIO_PRT0_DR_SET for the details of bit fields.
GPIO_PRT4_DR_CLR	0x40040444	Port output data clear register. See GPIO_PRT0_DR_CLR for the details of bit fields.
GPIO_PRT4_DR_INV	0x40040448	Port output data invert register. See GPIO_PRT0_DR_INV for the details of bit fields.
GPIO_PRT5_DR	0x40040500	Port output data register. See GPIO_PRT0_DR for the details of bit fields.
GPIO_PRT5_PS	0x40040504	Port IO pad state register. See GPIO_PRT0_PS for the details of bit fields.
GPIO_PRT5_PC	0x40040508	Port configuration register. See GPIO_PRT0_PC for the details of bit fields.
GPIO_PRT5_INTR_CFG	0x4004050C	Port interrupt configuration register. See GPIO_PRT0_INTR_CFG for the details of bit fields.
GPIO_PRT5_INTR	0x40040510	Port interrupt status register. See GPIO_PRT0_INTR for the details of bit fields.
GPIO_PRT5_PC2	0x40040518	Port configuration register 2. See GPIO_PRT0_PC2 for the details of bit fields.
GPIO_PRT5_DR_SET	0x40040540	Port output data set register. See GPIO_PRT0_DR_SET for the details of bit fields.
GPIO_PRT5_DR_CLR	0x40040544	Port output data clear register. See GPIO_PRT0_DR_CLR for the details of bit fields.
GPIO_PRT5_DR_INV	0x40040548	Port output data invert register. See GPIO_PRT0_DR_INV for the details of bit fields.
GPIO_PRT6_DR	0x40040600	Port output data register
GPIO_PRT6_PS	0x40040604	Port IO pad state register
GPIO_PRT6_PC	0x40040608	Port configuration register
GPIO_PRT6_INTR_CFG	0x4004060C	Port interrupt configuration register
GPIO_PRT6_INTR	0x40040610	Port interrupt status register
GPIO_PRT6_PC2	0x40040618	Port configuration register 2
GPIO_PRT6_DR_SET	0x40040640	Port output data set register. See GPIO_PRT0_DR_SET for the details of bit fields.
GPIO_PRT6_DR_CLR	0x40040644	Port output data clear register. See GPIO_PRT0_DR_CLR for the details of bit fields.
GPIO_PRT6_DR_INV	0x40040648	Port output data invert register. See GPIO_PRT0_DR_INV for the details of bit fields.
GPIO_PRT7_DR	0x40040700	Port output data register
GPIO_PRT7_PS	0x40040704	Port IO pad state register
GPIO_PRT7_PC	0x40040708	Port configuration register
GPIO_PRT7_INTR_CFG	0x4004070C	Port interrupt configuration register
GPIO_PRT7_INTR	0x40040710	Port interrupt status register
GPIO_PRT7_PC2	0x40040718	Port configuration register 2

Register	Address	Description
GPIO_PRT7_DR_SET	0x40040740	Port output data set register. See GPIO_PRT0_DR_SET for the details of bit fields.
GPIO_PRT7_DR_CLR	0x40040744	Port output data clear register. See GPIO_PRT0_DR_CLR for the details of bit fields.
GPIO_PRT7_DR_INV	0x40040748	Port output data invert register. See GPIO_PRT0_DR_INV for the details of bit fields.
GPIO_INTR_CAUSE	0x40041000	Interrupt port cause register

5.1.1 GPIO_PRT0_DR

Port output data register

Address: 0x40040000

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW	RW	RW	RW	RW	RW	RW	RW
HW Access	RW	RW	RW	RW	RW	RW	RW	RW
Name	DATA7	DATA6	DATA5	DATA4	DATA3	DATA2	DATA1	DATA0

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
7	DATA7	IO pad 7 output data. Default Value: 0
6	DATA6	IO pad 6 output data. Default Value: 0
5	DATA5	IO pad 5 output data. Default Value: 0
4	DATA4	IO pad 4 output data. Default Value: 0
3	DATA3	IO pad 3 output data. Default Value: 0
2	DATA2	IO pad 2 output data. Default Value: 0
1	DATA1	IO pad 1 output data. Default Value: 0
0	DATA0	IO pad 0 output data. Default Value: 0

5.1.2 GPIO_PRT0_PS

Port IO pad state register

Address: 0x40040004

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R	R	R	R	R	R	R	R
HW Access	W	W	W	W	W	W	W	W
Name	DATA7	DATA6	DATA5	DATA4	DATA3	DATA2	DATA1	DATA0

Bits	15	14	13	12	11	10	9	8
SW Access	None							R
HW Access	None							W
Name	None [15:9]							FLT_DATA

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
8	FLT_DATA	Reads of this register return the logical state of the filtered pin. Default Value: 0
7	DATA7	IO pad 7 state. Default Value: 0
6	DATA6	IO pad 6 state. Default Value: 0
5	DATA5	IO pad 5 state. Default Value: 0
4	DATA4	IO pad 4 state. Default Value: 0
3	DATA3	IO pad 3 state. Default Value: 0
2	DATA2	IO pad 2 state. Default Value: 0
1	DATA1	IO pad 1 state. Default Value: 0

5.1.2 GPIO_PRT0_PS (continued)

0	DATA0	IO pad 0 state: 1: Logic high, if the pin voltage is above the input buffer threshold, logic high. 0: Logic low, if the pin voltage is below that threshold, logic low. If the drive mode for the pin is set to high Z Analog, the pin state will read 0 independent of the voltage on the pin. Default Value: 0
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5.1.3 GPIO_PRT0_PC

Port configuration register

Address: 0x40040008

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW		RW			RW		
HW Access	R		R			R		
Name	DM2 [7:6]		DM1 [5:3]			DM0 [2:0]		

Bits	15	14	13	12	11	10	9	8
SW Access	RW	RW			RW			RW
HW Access	R	R			R			R
Name	DM5	DM4 [14:12]			DM3 [11:9]			DM2

Bits	23	22	21	20	19	18	17	16
SW Access	RW			RW			RW	
HW Access	R			R			R	
Name	DM7 [23:21]			DM6 [20:18]			DM5 [17:16]	

Bits	31	30	29	28	27	26	25	24
SW Access	RW		None				RW	RW
HW Access	R		None				R	R
Name	PORT_IB_MODE_SEL [31:30]		None [29:26]				PORT_SLOW	PORT_VTRIP_SEL

Bits	Name	Description
31 : 30	PORT_IB_MODE_SEL	This field selects the input buffer reference. The size (1 or 2 bits) and functionality is dependent on the IO cell. For GPIOv2 IO cells, bit PORT_IB_MODE_SEL[1] is not used (GPIOv2 IO cell replaces GPIO IO cell): "0"/"2": CMOS input buffer (PORT_VTRIP_SEL is '0'), LVTTTL input buffer (PORT_VTRIP_SEL is '1') "1"/"3": vcchib. For GPIO_OVTV2 and SIOv2 IO cells: "0": CMOS input buffer (PORT_VTRIP_SEL is '0'), LVTTTL input buffer (PORT_VTRIP_SEL is '1') "1": vcchib. "2": OVT. "3": Reference (possibly from reference generator cell). For SIO IO cell, this field is present but not used as the SIO IO cell does not provide input buffer mode select functionality (SIOv2 IO cell will replace SIO IO cell, as soon as it is available). Default Value: 0
25	PORT_SLOW	This field controls the output edge rate of all pins on the port: '0': fast. '1': slow. Default Value: 0

5.1.3 GPIO_PRT0_PC (continued)

24	PORT_VTRIP_SEL	The GPIO cells include a VTRIP_SEL signal to alter the input buffer voltage. Note: this bit is ignored for SIO ports, the VTRIP_SEL settings in the SIO register are used instead (a separate VTRIP_SEL is provided for each pin pair). 0: input buffer functions as a CMOS input buffer. 1: input buffer functions as a LVTTTL input buffer. Default Value: 0
23 : 21	DM7	The GPIO drive mode for IO pad 7. Default Value: 0
20 : 18	DM6	The GPIO drive mode for IO pad 6. Default Value: 0
17 : 15	DM5	The GPIO drive mode for IO pad 5. Default Value: 0
14 : 12	DM4	The GPIO drive mode for IO pad 4. Default Value: 0
11 : 9	DM3	The GPIO drive mode for IO pad 3. Default Value: 0
8 : 6	DM2	The GPIO drive mode for IO pad 2. Default Value: 0
5 : 3	DM1	The GPIO drive mode for IO pad 1. Default Value: 0
2 : 0	DM0	The GPIO drive mode for IO pad 0. Note: when initializing IO's that are connected to a live bus (such as I2C), make sure the HSIOM is properly configured (HSIOM_PRT_SELx) before turning the IO on here to avoid producing glitches on the bus. Default Value: 0

0x0: OFF :

Mode 0 (analog mode): Output buffer off (high Z). Input buffer off.

0x1: INPUT :

Mode 1: Output buffer off (high Z). Input buffer on.

0x2: 0_PU :

Mode 2: Strong pull down ('0'), weak/resistive pull up (PU). Input buffer on.

0x3: PD_1 :

Mode 3: Weak/resistive pull down (PD), strong pull up ('1'). Input buffer on.

0x4: 0_Z :

Mode 4: Strong pull down ('0'), open drain (pull up off). Input buffer on.

0x5: Z_1 :

Mode 5: Open drain (pull down off), strong pull up ('1'). Input buffer on.

5.1.3 GPIO_PRT0_PC (continued)

0x6: 0_1 :

Mode 6: Strong pull down ('0'), strong pull up ('1'). Input buffer on.

0x7: PD_PU :

Mode 7: Weak/resistive pull down (PD), weak/resistive pull up (PU). Input buffer on.

5.1.4 GPIO_PRT0_INTR_CFG

Port interrupt configuration register

Address: 0x4004000C

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW		RW		RW		RW	
HW Access	R		R		R		R	
Name	EDGE3_SEL [7:6]		EDGE2_SEL [5:4]		EDGE1_SEL [3:2]		EDGE0_SEL [1:0]	

Bits	15	14	13	12	11	10	9	8
SW Access	RW		RW		RW		RW	
HW Access	R		R		R		R	
Name	EDGE7_SEL [15:14]		EDGE6_SEL [13:12]		EDGE5_SEL [11:10]		EDGE4_SEL [9:8]	

Bits	23	22	21	20	19	18	17	16
SW Access	None				RW		RW	
HW Access	None				R		R	
Name	None [23:21]				FLT_SEL [20:18]		FLT_EDGE_SEL [17:16]	

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
20 : 18	FLT_SEL	Selects which pin is routed through the 50ns glitch filter to provide a glitch-safe interrupt. Default Value: 0
17 : 16	FLT_EDGE_SEL	Same for the glitch filtered pin (selected by FLT_SELECT). Default Value: 0
		0x0: DISABLE :
		Disabled
		0x1: RISING :
		Rising edge
		0x2: FALLING :
		Falling edge

5.1.4 GPIO_PRT0_INTR_CFG (continued)

0x3: BOTH :

Both rising and falling edges

15 : 14	EDGE7_SEL	Sets which edge will trigger an IRQ for IO pad 7. Default Value: 0
13 : 12	EDGE6_SEL	Sets which edge will trigger an IRQ for IO pad 6. Default Value: 0
11 : 10	EDGE5_SEL	Sets which edge will trigger an IRQ for IO pad 5. Default Value: 0
9 : 8	EDGE4_SEL	Sets which edge will trigger an IRQ for IO pad 4. Default Value: 0
7 : 6	EDGE3_SEL	Sets which edge will trigger an IRQ for IO pad 3. Default Value: 0
5 : 4	EDGE2_SEL	Sets which edge will trigger an IRQ for IO pad 2. Default Value: 0
3 : 2	EDGE1_SEL	Sets which edge will trigger an IRQ for IO pad 1. Default Value: 0
1 : 0	EDGE0_SEL	Sets which edge will trigger an IRQ for IO pad 0. Default Value: 0

0x0: DISABLE :

Disabled

0x1: RISING :

Rising edge

0x2: FALLING :

Falling edge

0x3: BOTH :

Both rising and falling edges

5.1.5 GPIO_PRT0_INTR

Port interrupt status register

Address: 0x40040010

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW1C	RW1C	RW1C	RW1C	RW1C	RW1C	RW1C	RW1C
HW Access	A	A	A	A	A	A	A	A
Name	DATA7	DATA6	DATA5	DATA4	DATA3	DATA2	DATA1	DATA0

Bits	15	14	13	12	11	10	9	8
SW Access	None							RW1C
HW Access	None							A
Name	None [15:9]							FLT_DATA

Bits	23	22	21	20	19	18	17	16
SW Access	R	R	R	R	R	R	R	R
HW Access	W	W	W	W	W	W	W	W
Name	PS_DATA7	PS_DATA6	PS_DATA5	PS_DATA4	PS_DATA3	PS_DATA2	PS_DATA1	PS_DATA0

Bits	31	30	29	28	27	26	25	24
SW Access	None							R
HW Access	None							W
Name	None [31:25]							PS_FLT_- DATA

Bits	Name	Description
24	PS_FLT_DATA	This is a duplicate of the contents of the PS register, provided here to allow reading of both pin state and interrupt state of the port in a single read operation. Default Value: 0
23	PS_DATA7	Default Value: 0
22	PS_DATA6	Default Value: 0
21	PS_DATA5	Default Value: 0
20	PS_DATA4	Default Value: 0
19	PS_DATA3	Default Value: 0
18	PS_DATA2	Default Value: 0
17	PS_DATA1	Default Value: 0
16	PS_DATA0	Default Value: 0
8	FLT_DATA	Deglitched interrupt pending (selected by FLT_SELECT). Default Value: 0

5.1.5 GPIO_PRT0_INTR (continued)

7	DATA7	Interrupt pending on IO pad 7. Firmware writes 1 to clear the interrupt. Default Value: 0
6	DATA6	Interrupt pending on IO pad 6. Firmware writes 1 to clear the interrupt. Default Value: 0
5	DATA5	Interrupt pending on IO pad 5. Firmware writes 1 to clear the interrupt. Default Value: 0
4	DATA4	Interrupt pending on IO pad 4. Firmware writes 1 to clear the interrupt. Default Value: 0
3	DATA3	Interrupt pending on IO pad 3. Firmware writes 1 to clear the interrupt. Default Value: 0
2	DATA2	Interrupt pending on IO pad 2. Firmware writes 1 to clear the interrupt. Default Value: 0
1	DATA1	Interrupt pending on IO pad 1. Firmware writes 1 to clear the interrupt. Default Value: 0
0	DATA0	Interrupt pending on IO pad 0. Firmware writes 1 to clear the interrupt. Default Value: 0

5.1.6 GPIO_PRT0_PC2

Port configuration register 2

Address: 0x40040018

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW	RW	RW	RW	RW	RW	RW	RW
HW Access	R	R	R	R	R	R	R	R
Name	INP_DIS7	INP_DIS6	INP_DIS5	INP_DIS4	INP_DIS3	INP_DIS2	INP_DIS1	INP_DIS0

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
7	INP_DIS7	Disables the input buffer for IO pad 7. Default Value: 0
6	INP_DIS6	Disables the input buffer for IO pad 6. Default Value: 0
5	INP_DIS5	Disables the input buffer for IO pad 5. Default Value: 0
4	INP_DIS4	Disables the input buffer for IO pad 4. Default Value: 0
3	INP_DIS3	Disables the input buffer for IO pad 3. Default Value: 0
2	INP_DIS2	Disables the input buffer for IO pad 2. Default Value: 0
1	INP_DIS1	Disables the input buffer for IO pad 1. Default Value: 0

5.1.6 GPIO_PRT0_PC2 (continued)

0	INP_DIS0	Disables the input buffer for IO pad 0 independent of the port control drive mode (PC.DM). This bit should be set when analog signals are present on the pin and PC.DM != 0 is required to use the output driver. Default Value: 0
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5.1.7 GPIO_PRT0_DR_SET

Port output data set register

Address: 0x40040040

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access	A							
Name	DATA [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
7 : 0	DATA	IO pad i: '0': Output state DR.DATA[i] not affected. '1': Output state DR.DATA[i] set to '1'. Default Value: 0

5.1.8 GPIO_PRT0_DR_CLR

Port output data clear register

Address: 0x40040044

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access	A							
Name	DATA [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
7 : 0	DATA	IO pad i: '0': Output state DR.DATA[i] not affected. '1': Output state DR.DATA[i] set to '0'. Default Value: 0

5.1.9 GPIO_PRT0_DR_INV

Port output data invert register

Address: 0x40040048

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access	A							
Name	DATA [7:0]							
Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							
Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							
Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
7 : 0	DATA	IO pad i: '0': Output state DR.DATA[i] not affected. '1': Output state DR.DATA[i] inverted ('0' => '1', '1' => '0'). Default Value: 0

5.1.10 GPIO_PRT6_DR

Port output data register

Address: 0x40040600

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None		RW	RW	RW	RW	RW	RW
HW Access	None		RW	RW	RW	RW	RW	RW
Name	None [7:6]		DATA5	DATA4	DATA3	DATA2	DATA1	DATA0

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
5	DATA5	IO pad 5 output data. Default Value: 0
4	DATA4	IO pad 4 output data. Default Value: 0
3	DATA3	IO pad 3 output data. Default Value: 0
2	DATA2	IO pad 2 output data. Default Value: 0
1	DATA1	IO pad 1 output data. Default Value: 0
0	DATA0	IO pad 0 output data. Default Value: 0

5.1.11 GPIO_PRT6_PS

Port IO pad state register

Address: 0x40040604

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None		R	R	R	R	R	R
HW Access	None		W	W	W	W	W	W
Name	None [7:6]		DATA5	DATA4	DATA3	DATA2	DATA1	DATA0

Bits	15	14	13	12	11	10	9	8
SW Access	None							R
HW Access	None							W
Name	None [15:9]							FLT_DATA

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
8	FLT_DATA	Reads of this register return the logical state of the filtered pin. Default Value: 0
5	DATA5	IO pad 5 state. Default Value: 0
4	DATA4	IO pad 4 state. Default Value: 0
3	DATA3	IO pad 3 state. Default Value: 0
2	DATA2	IO pad 2 state. Default Value: 0
1	DATA1	IO pad 1 state. Default Value: 0

5.1.11 GPIO_PRT6_PS (continued)

0	DATA0	IO pad 0 state: 1: Logic high, if the pin voltage is above the input buffer threshold, logic high. 0: Logic low, if the pin voltage is below that threshold, logic low. If the drive mode for the pin is set to high Z Analog, the pin state will read 0 independent of the voltage on the pin. Default Value: 0
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5.1.12 GPIO_PRT6_PC

Port configuration register

Address: 0x40040608

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW		RW			RW		
HW Access	R		R			R		
Name	DM2 [7:6]		DM1 [5:3]			DM0 [2:0]		

Bits	15	14	13	12	11	10	9	8
SW Access	RW	RW			RW			RW
HW Access	R	R			R			R
Name	DM5	DM4 [14:12]			DM3 [11:9]			DM2

Bits	23	22	21	20	19	18	17	16
SW Access	None						RW	
HW Access	None						R	
Name	None [23:18]						DM5 [17:16]	

Bits	31	30	29	28	27	26	25	24
SW Access	RW		None				RW	RW
HW Access	R		None				R	R
Name	PORT_IB_MODE_SEL [31:30]		None [29:26]				PORT_SLOW	PORT_VTRIP_SEL

Bits	Name	Description
31 : 30	PORT_IB_MODE_SEL	This field selects the input buffer reference. The size (1 or 2 bits) and functionality is dependent on the IO cell. For GPIOv2 IO cells, bit PORT_IB_MODE_SEL[1] is not used (GPIOv2 IO cell replaces GPIO IO cell): "0"/"2": CMOS input buffer (PORT_VTRIP_SEL is '0'), LVTTTL input buffer (PORT_VTRIP_SEL is '1') "1"/"3": vcchib. For GPIO_OVTV2 and SIOv2 IO cells: "0": CMOS input buffer (PORT_VTRIP_SEL is '0'), LVTTTL input buffer (PORT_VTRIP_SEL is '1') "1": vcchib. "2": OVT. "3": Reference (possibly from reference generator cell). For SIO IO cell, this field is present but not used as the SIO IO cell does not provide input buffer mode select functionality (SIOv2 IO cell will replace SIO IO cell, as soon as it is available). Default Value: 0
25	PORT_SLOW	This field controls the output edge rate of all pins on the port: '0': fast. '1': slow. Default Value: 0

5.1.12 GPIO_PRT6_PC (continued)

24	PORT_VTRIP_SEL	The GPIO cells include a VTRIP_SEL signal to alter the input buffer voltage. Note: this bit is ignored for SIO ports, the VTRIP_SEL settings in the SIO register are used instead (a separate VTRIP_SEL is provided for each pin pair). 0: input buffer functions as a CMOS input buffer. 1: input buffer functions as a LVTTL input buffer. Default Value: 0
17 : 15	DM5	The GPIO drive mode for IO pad 5. Default Value: 0
14 : 12	DM4	The GPIO drive mode for IO pad 4. Default Value: 0
11 : 9	DM3	The GPIO drive mode for IO pad 3. Default Value: 0
8 : 6	DM2	The GPIO drive mode for IO pad 2. Default Value: 0
5 : 3	DM1	The GPIO drive mode for IO pad 1. Default Value: 0
2 : 0	DM0	The GPIO drive mode for IO pad 0. Note: when initializing IO's that are connected to a live bus (such as I2C), make sure the HSIOM is properly configured (HSIOM_PRT_SELx) before turning the IO on here to avoid producing glitches on the bus. Default Value: 0 0x0: OFF : Mode 0 (analog mode): Output buffer off (high Z). Input buffer off. 0x1: INPUT : Mode 1: Output buffer off (high Z). Input buffer on. 0x2: 0_PU : Mode 2: Strong pull down ('0'), weak/resistive pull up (PU). Input buffer on. 0x3: PD_1 : Mode 3: Weak/resistive pull down (PD), strong pull up ('1'). Input buffer on. 0x4: 0_Z : Mode 4: Strong pull down ('0'), open drain (pull up off). Input buffer on. 0x5: Z_1 : Mode 5: Open drain (pull down off), strong pull up ('1'). Input buffer on. 0x6: 0_1 : Mode 6: Strong pull down ('0'), strong pull up ('1'). Input buffer on.

5.1.12 GPIO_PRT6_PC (continued)

0x7: PD_PU :

Mode 7: Weak/resistive pull down (PD), weak/resistive pull up (PU). Input buffer on.

5.1.13 GPIO_PRT6_INTR_CFG

Port interrupt configuration register

Address: 0x4004060C

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW		RW		RW		RW	
HW Access	R		R		R		R	
Name	EDGE3_SEL [7:6]		EDGE2_SEL [5:4]		EDGE1_SEL [3:2]		EDGE0_SEL [1:0]	

Bits	15	14	13	12	11	10	9	8
SW Access	None				RW		RW	
HW Access	None				R		R	
Name	None [15:12]				EDGE5_SEL [11:10]		EDGE4_SEL [9:8]	

Bits	23	22	21	20	19	18	17	16
SW Access	None				RW		RW	
HW Access	None				R		R	
Name	None [23:21]				FLT_SEL [20:18]		FLT_EDGE_SEL [17:16]	

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
20 : 18	FLT_SEL	Selects which pin is routed through the 50ns glitch filter to provide a glitch-safe interrupt. Default Value: 0
17 : 16	FLT_EDGE_SEL	Same for the glitch filtered pin (selected by FLT_SELECT). Default Value: 0 0x0: DISABLE : Disabled 0x1: RISING : Rising edge 0x2: FALLING : Falling edge

5.1.13 GPIO_PRT6_INTR_CFG (continued)

0x3: BOTH :

Both rising and falling edges

11 : 10	EDGE5_SEL	Sets which edge will trigger an IRQ for IO pad 5. Default Value: 0
9 : 8	EDGE4_SEL	Sets which edge will trigger an IRQ for IO pad 4. Default Value: 0
7 : 6	EDGE3_SEL	Sets which edge will trigger an IRQ for IO pad 3. Default Value: 0
5 : 4	EDGE2_SEL	Sets which edge will trigger an IRQ for IO pad 2. Default Value: 0
3 : 2	EDGE1_SEL	Sets which edge will trigger an IRQ for IO pad 1. Default Value: 0
1 : 0	EDGE0_SEL	Sets which edge will trigger an IRQ for IO pad 0. Default Value: 0

0x0: DISABLE :

Disabled

0x1: RISING :

Rising edge

0x2: FALLING :

Falling edge

0x3: BOTH :

Both rising and falling edges

5.1.14 GPIO_PRT6_INTR

Port interrupt status register

Address: 0x40040610

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None		RW1C	RW1C	RW1C	RW1C	RW1C	RW1C
HW Access	None		A	A	A	A	A	A
Name	None [7:6]		DATA5	DATA4	DATA3	DATA2	DATA1	DATA0

Bits	15	14	13	12	11	10	9	8
SW Access	None							RW1C
HW Access	None							A
Name	None [15:9]							FLT_DATA

Bits	23	22	21	20	19	18	17	16
SW Access	None		R	R	R	R	R	R
HW Access	None		W	W	W	W	W	W
Name	None [23:22]		PS_DATA5	PS_DATA4	PS_DATA3	PS_DATA2	PS_DATA1	PS_DATA0

Bits	31	30	29	28	27	26	25	24
SW Access	None							R
HW Access	None							W
Name	None [31:25]							PS_FLT_- DATA

Bits	Name	Description
24	PS_FLT_DATA	This is a duplicate of the contents of the PS register, provided here to allow reading of both pin state and interrupt state of the port in a single read operation. Default Value: 0
21	PS_DATA5	Default Value: 0
20	PS_DATA4	Default Value: 0
19	PS_DATA3	Default Value: 0
18	PS_DATA2	Default Value: 0
17	PS_DATA1	Default Value: 0
16	PS_DATA0	` Default Value: 0
8	FLT_DATA	Deglitched interrupt pending (selected by FLT_SELECT). Default Value: 0
5	DATA5	Interrupt pending on IO pad 5. Firmware writes 1 to clear the interrupt. Default Value: 0

5.1.14 GPIO_PRT6_INTR (continued)

4	DATA4	Interrupt pending on IO pad 4. Firmware writes 1 to clear the interrupt. Default Value: 0
3	DATA3	Interrupt pending on IO pad 3. Firmware writes 1 to clear the interrupt. Default Value: 0
2	DATA2	Interrupt pending on IO pad 2. Firmware writes 1 to clear the interrupt. Default Value: 0
1	DATA1	Interrupt pending on IO pad 1. Firmware writes 1 to clear the interrupt. Default Value: 0
0	DATA0	Interrupt pending on IO pad 0. Firmware writes 1 to clear the interrupt. Default Value: 0

5.1.15 GPIO_PRT6_PC2

Port configuration register 2

Address: 0x40040618

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None		RW	RW	RW	RW	RW	RW
HW Access	None		R	R	R	R	R	R
Name	None [7:6]		INP_DIS5	INP_DIS4	INP_DIS3	INP_DIS2	INP_DIS1	INP_DIS0

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
5	INP_DIS5	Disables the input buffer for IO pad 5. Default Value: 0
4	INP_DIS4	Disables the input buffer for IO pad 4. Default Value: 0
3	INP_DIS3	Disables the input buffer for IO pad 3. Default Value: 0
2	INP_DIS2	Disables the input buffer for IO pad 2. Default Value: 0
1	INP_DIS1	Disables the input buffer for IO pad 1. Default Value: 0
0	INP_DIS0	Disables the input buffer for IO pad 0 independent of the port control drive mode (PC.DM). This bit should be set when analog signals are present on the pin and PC.DM != 0 is required to use the output driver. Default Value: 0

5.1.16 GPIO_PRT7_DR

Port output data register

Address: 0x40040700

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None					RW	RW	RW
HW Access	None					RW	RW	RW
Name	None [7:3]					DATA2	DATA1	DATA0

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
2	DATA2	IO pad 2 output data. Default Value: 0
1	DATA1	IO pad 1 output data. Default Value: 0
0	DATA0	IO pad 0 output data. Default Value: 0

5.1.17 GPIO_PRT7_PS

Port IO pad state register

Address: 0x40040704

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None					R	R	R
HW Access	None					W	W	W
Name	None [7:3]					DATA2	DATA1	DATA0

Bits	15	14	13	12	11	10	9	8
SW Access	None							R
HW Access	None							W
Name	None [15:9]							FLT_DATA

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
8	FLT_DATA	Reads of this register return the logical state of the filtered pin. Default Value: 0
2	DATA2	IO pad 2 state. Default Value: 0
1	DATA1	IO pad 1 state. Default Value: 0
0	DATA0	IO pad 0 state: 1: Logic high, if the pin voltage is above the input buffer threshold, logic high. 0: Logic low, if the pin voltage is below that threshold, logic low. If the drive mode for the pin is set to high Z Analog, the pin state will read 0 independent of the voltage on the pin. Default Value: 0

5.1.18 GPIO_PRT7_PC

Port configuration register

Address: 0x40040708

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW		RW			RW		
HW Access	R		R			R		
Name	DM2 [7:6]		DM1 [5:3]			DM0 [2:0]		

Bits	15	14	13	12	11	10	9	8
SW Access	None							RW
HW Access	None							R
Name	None [15:9]							DM2

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	RW		None				RW	RW
HW Access	R		None				R	R
Name	PORT_IB_MODE_SEL [31:30]		None [29:26]				PORT_SLOW	PORT_VTRIP_SEL

Bits	Name	Description
31 : 30	PORT_IB_MODE_SEL	This field selects the input buffer reference. The size (1 or 2 bits) and functionality is dependent on the IO cell. For GPIOv2 IO cells, bit PORT_IB_MODE_SEL[1] is not used (GPIOv2 IO cell replaces GPIO IO cell): "0"/"2": CMOS input buffer (PORT_VTRIP_SEL is '0'), LVTTTL input buffer (PORT_VTRIP_SEL is '1') "1"/"3": vcchib. For GPIO_OVTV2 and SIOv2 IO cells: "0": CMOS input buffer (PORT_VTRIP_SEL is '0'), LVTTTL input buffer (PORT_VTRIP_SEL is '1') "1": vcchib. "2": OVT. "3": Reference (possibly from reference generator cell). For SIO IO cell, this field is present but not used as the SIO IO cell does not provide input buffer mode select functionality (SIOv2 IO cell will replace SIO IO cell, as soon as it is available). Default Value: 0
25	PORT_SLOW	This field controls the output edge rate of all pins on the port: '0': fast. '1': slow. Default Value: 0

5.1.18 GPIO_PRT7_PC (continued)

24	PORT_VTRIP_SEL	The GPIO cells include a VTRIP_SEL signal to alter the input buffer voltage. Note: this bit is ignored for SIO ports, the VTRIP_SEL settings in the SIO register are used instead (a separate VTRIP_SEL is provided for each pin pair). 0: input buffer functions as a CMOS input buffer. 1: input buffer functions as a LVTTL input buffer. Default Value: 0
8 : 6	DM2	The GPIO drive mode for IO pad 2. Default Value: 0
5 : 3	DM1	The GPIO drive mode for IO pad 1. Default Value: 0
2 : 0	DM0	The GPIO drive mode for IO pad 0. Note: when initializing IO's that are connected to a live bus (such as I2C), make sure the HSIOM is properly configured (HSIOM_PRT_SELx) before turning the IO on here to avoid producing glitches on the bus. Default Value: 0 0x0: OFF : Mode 0 (analog mode): Output buffer off (high Z). Input buffer off. 0x1: INPUT : Mode 1: Output buffer off (high Z). Input buffer on. 0x2: 0_PU : Mode 2: Strong pull down ('0'), weak/resistive pull up (PU). Input buffer on. 0x3: PD_1 : Mode 3: Weak/resistive pull down (PD), strong pull up ('1'). Input buffer on. 0x4: 0_Z : Mode 4: Strong pull down ('0'), open drain (pull up off). Input buffer on. 0x5: Z_1 : Mode 5: Open drain (pull down off), strong pull up ('1'). Input buffer on. 0x6: 0_1 : Mode 6: Strong pull down ('0'), strong pull up ('1'). Input buffer on. 0x7: PD_PU : Mode 7: Weak/resistive pull down (PD), weak/resistive pull up (PU). Input buffer on.

5.1.19 GPIO_PRT7_INTR_CFG

Port interrupt configuration register

Address: 0x4004070C

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None		RW		RW		RW	
HW Access	None		R		R		R	
Name	None [7:6]		EDGE2_SEL [5:4]		EDGE1_SEL [3:2]		EDGE0_SEL [1:0]	

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None			RW			RW	
HW Access	None			R			R	
Name	None [23:21]			FLT_SEL [20:18]			FLT_EDGE_SEL [17:16]	

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
20 : 18	FLT_SEL	Selects which pin is routed through the 50ns glitch filter to provide a glitch-safe interrupt. Default Value: 0
17 : 16	FLT_EDGE_SEL	Same for the glitch filtered pin (selected by FLT_SELECT). Default Value: 0
		0x0: DISABLE :
		Disabled
		0x1: RISING :
		Rising edge
		0x2: FALLING :
		Falling edge

5.1.19 GPIO_PRT7_INTR_CFG (continued)

0x3: BOTH :

Both rising and falling edges

5 : 4	EDGE2_SEL	Sets which edge will trigger an IRQ for IO pad 2. Default Value: 0
3 : 2	EDGE1_SEL	Sets which edge will trigger an IRQ for IO pad 1. Default Value: 0
1 : 0	EDGE0_SEL	Sets which edge will trigger an IRQ for IO pad 0. Default Value: 0

0x0: DISABLE :

Disabled

0x1: RISING :

Rising edge

0x2: FALLING :

Falling edge

0x3: BOTH :

Both rising and falling edges

5.1.20 GPIO_PRT7_INTR

Port interrupt status register

Address: 0x40040710

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None					RW1C	RW1C	RW1C
HW Access	None					A	A	A
Name	None [7:3]					DATA2	DATA1	DATA0

Bits	15	14	13	12	11	10	9	8
SW Access	None							RW1C
HW Access	None							A
Name	None [15:9]							FLT_DATA

Bits	23	22	21	20	19	18	17	16
SW Access	None					R	R	R
HW Access	None					W	W	W
Name	None [23:19]					PS_DATA2	PS_DATA1	PS_DATA0

Bits	31	30	29	28	27	26	25	24
SW Access	None							R
HW Access	None							W
Name	None [31:25]							PS_FLT_- DATA

Bits	Name	Description
24	PS_FLT_DATA	This is a duplicate of the contents of the PS register, provided here to allow reading of both pin state and interrupt state of the port in a single read operation. Default Value: 0
18	PS_DATA2	Default Value: 0
17	PS_DATA1	Default Value: 0
16	PS_DATA0	` Default Value: 0
8	FLT_DATA	Deglitched interrupt pending (selected by FLT_SELECT). Default Value: 0
2	DATA2	Interrupt pending on IO pad 2. Firmware writes 1 to clear the interrupt. Default Value: 0
1	DATA1	Interrupt pending on IO pad 1. Firmware writes 1 to clear the interrupt. Default Value: 0
0	DATA0	Interrupt pending on IO pad 0. Firmware writes 1 to clear the interrupt. Default Value: 0

5.1.21 GPIO_PRT7_PC2

Port configuration register 2

Address: 0x40040718

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None					RW	RW	RW
HW Access	None					R	R	R
Name	None [7:3]					INP_DIS2	INP_DIS1	INP_DIS0

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
2	INP_DIS2	Disables the input buffer for IO pad 2. Default Value: 0
1	INP_DIS1	Disables the input buffer for IO pad 1. Default Value: 0
0	INP_DIS0	Disables the input buffer for IO pad 0 independent of the port control drive mode (PC.DM). This bit should be set when analog signals are present on the pin and PC.DM != 0 is required to use the output driver. Default Value: 0

5.1.22 GPIO_INTR_CAUSE

Interrupt port cause register

Address: 0x40041000

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access	W							
Name	PORT_INT [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
7 : 0	PORT_INT	Each IO port has an associated bit field in this register. The bit field reflects the IO port's interrupt line (bit field i reflects "gpio_interrupts[i]" for IO port i). The register is used when the system uses a shared/combined interrupt line "gpio_interrupt". The SW ISR reads the register to determine which IO port(s) is responsible for the shared/combined interrupt line "gpio_interrupt". Once, the IO port(s) is determined, the IO port's INTR register is read to determine the IO pad(s) in the IO port that caused the interrupt. Default Value: 0

6 Programmable IO (PRGIO) Registers



This section discusses the PRGIO registers. It lists all the registers in mapping tables, in address order.

6.1 Register Details

Register	Address	Description
PRGIO_PRT0_CTL	0x40050000	Control register
PRGIO_PRT0_SYNC_CTL	0x40050010	Synchronization control register
PRGIO_PRT0_LUT_SEL0	0x40050020	LUT component input selection
PRGIO_PRT0_LUT_SEL1	0x40050024	LUT component input selection. See PRGIO_PRT0_LUT_SEL0 for the details of bit fields.
PRGIO_PRT0_LUT_SEL2	0x40050028	LUT component input selection. See PRGIO_PRT0_LUT_SEL0 for the details of bit fields.
PRGIO_PRT0_LUT_SEL3	0x4005002C	LUT component input selection. See PRGIO_PRT0_LUT_SEL0 for the details of bit fields.
PRGIO_PRT0_LUT_SEL4	0x40050030	LUT component input selection. See PRGIO_PRT0_LUT_SEL0 for the details of bit fields.
PRGIO_PRT0_LUT_SEL5	0x40050034	LUT component input selection. See PRGIO_PRT0_LUT_SEL0 for the details of bit fields.
PRGIO_PRT0_LUT_SEL6	0x40050038	LUT component input selection. See PRGIO_PRT0_LUT_SEL0 for the details of bit fields.
PRGIO_PRT0_LUT_SEL7	0x4005003C	LUT component input selection. See PRGIO_PRT0_LUT_SEL0 for the details of bit fields.
PRGIO_PRT0_LUT_CTL0	0x40050040	LUT component control register
PRGIO_PRT0_LUT_CTL1	0x40050044	LUT component control register. See PRGIO_PRT0_LUT_CTL0 for the details of bit fields.
PRGIO_PRT0_LUT_CTL2	0x40050048	LUT component control register. See PRGIO_PRT0_LUT_CTL0 for the details of bit fields.
PRGIO_PRT0_LUT_CTL3	0x4005004C	LUT component control register. See PRGIO_PRT0_LUT_CTL0 for the details of bit fields.
PRGIO_PRT0_LUT_CTL4	0x40050050	LUT component control register. See PRGIO_PRT0_LUT_CTL0 for the details of bit fields.
PRGIO_PRT0_LUT_CTL5	0x40050054	LUT component control register. See PRGIO_PRT0_LUT_CTL0 for the details of bit fields.
PRGIO_PRT0_LUT_CTL6	0x40050058	LUT component control register. See PRGIO_PRT0_LUT_CTL0 for the details of bit fields.
PRGIO_PRT0_LUT_CTL7	0x4005005C	LUT component control register. See PRGIO_PRT0_LUT_CTL0 for the details of bit fields.
PRGIO_PRT0_DU_SEL	0x400500C0	Data unit component input selection
PRGIO_PRT0_DU_CTL	0x400500C4	Data unit component control register
PRGIO_PRT0_DATA	0x400500F0	Data register
PRGIO_PRT1_CTL	0x40050100	Control register. See PRGIO_PRT0_CTL for the details of bit fields.
PRGIO_PRT1_SYNC_CTL	0x40050110	Synchronization control register. See PRGIO_PRT0_SYNC_CTL for the details of bit fields.
PRGIO_PRT1_LUT_SEL0	0x40050120	LUT component input selection. See PRGIO_PRT0_LUT_SEL0 for the details of bit fields.
PRGIO_PRT1_LUT_SEL1	0x40050124	LUT component input selection. See PRGIO_PRT0_LUT_SEL0 for the details of bit fields.
PRGIO_PRT1_LUT_SEL2	0x40050128	LUT component input selection. See PRGIO_PRT0_LUT_SEL0 for the details of bit fields.
PRGIO_PRT1_LUT_SEL3	0x4005012C	LUT component input selection. See PRGIO_PRT0_LUT_SEL0 for the details of bit fields.

Register	Address	Description
PRGIO_PRT1_LUT_SEL4	0x40050130	LUT component input selection. See PRGIO_PRT0_LUT_SEL0 for the details of bit fields.
PRGIO_PRT1_LUT_SEL5	0x40050134	LUT component input selection. See PRGIO_PRT0_LUT_SEL0 for the details of bit fields.
PRGIO_PRT1_LUT_SEL6	0x40050138	LUT component input selection. See PRGIO_PRT0_LUT_SEL0 for the details of bit fields.
PRGIO_PRT1_LUT_SEL7	0x4005013C	LUT component input selection. See PRGIO_PRT0_LUT_SEL0 for the details of bit fields.
PRGIO_PRT1_LUT_CTL0	0x40050140	LUT component control register. See PRGIO_PRT0_LUT_CTL0 for the details of bit fields.
PRGIO_PRT1_LUT_CTL1	0x40050144	LUT component control register. See PRGIO_PRT0_LUT_CTL0 for the details of bit fields.
PRGIO_PRT1_LUT_CTL2	0x40050148	LUT component control register. See PRGIO_PRT0_LUT_CTL0 for the details of bit fields.
PRGIO_PRT1_LUT_CTL3	0x4005014C	LUT component control register. See PRGIO_PRT0_LUT_CTL0 for the details of bit fields.
PRGIO_PRT1_LUT_CTL4	0x40050150	LUT component control register. See PRGIO_PRT0_LUT_CTL0 for the details of bit fields.
PRGIO_PRT1_LUT_CTL5	0x40050154	LUT component control register. See PRGIO_PRT0_LUT_CTL0 for the details of bit fields.
PRGIO_PRT1_LUT_CTL6	0x40050158	LUT component control register. See PRGIO_PRT0_LUT_CTL0 for the details of bit fields.
PRGIO_PRT1_LUT_CTL7	0x4005015C	LUT component control register. See PRGIO_PRT0_LUT_CTL0 for the details of bit fields.
PRGIO_PRT1_DU_SEL	0x400501C0	Data unit component input selection. See PRGIO_PRT0_DU_SEL for the details of bit fields.
PRGIO_PRT1_DU_CTL	0x400501C4	Data unit component control register. See PRGIO_PRT0_DU_CTL for the details of bit fields.
PRGIO_PRT1_DATA	0x400501F0	Data register. See PRGIO_PRT0_DATA for the details of bit fields.

6.1.1 PRGIO_PRT0_CTL

Control register

Address: 0x40050000

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access	R							
Name	BYPASS [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	None			RW				
HW Access	None			R				
Name	None [15:13]			CLOCK_SRC [12:8]				

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	RW	None					RW	RW
HW Access	R	None					R	R
Name	ENABLED	None [30:26]					PIPE- LINE_EN	HLD_OVR

Bits	Name	Description
31	ENABLED	Enable for programmable IO. Should only be set to '1' when the programmable IO is completely configured: '0': Disabled (signals are bypassed; behavior as if BYPASS is 0xFF). When disabled, the fabric (data unit and LUTs) reset is activated. If the IP is disabled: - The PIPELINE_EN register field should be set to '1', to ensure low power consumption by preventing combinatorial loops. - The CLOCK_SRC register field should be set to "20"- "30" (clock is constant '0'), to ensure low power consumption. '1': Enabled. Once enabled, it takes 3 "clk_fabric" clock cycles till the fabric reset is de-activated and the fabric becomes fully functional. This ensures that the IO pins' input synchronizer states are flushed when the fabric is fully functional. Default Value: 0
25	PIPELINE_EN	Enable for pipeline register: '0': Disabled (register is bypassed). '1': Enabled. Default Value: 1

6.1.1 PRGIO_PRT0_CTL (continued)

24	HLD_OVR	IO cell hold override functionality. In DeepSleep and Hibernate power modes, the HSIOM holds the IO cell output and output enable signals if Active functionality is connected to the IO pads. This is undesirable if the PRGIO is supposed to deliver DeepSleep or Hibernate output functionality on these IO pads. This field is used to control the hold override functionality from the PRGIO: '0': The HSIOM controls the IO cell hold override functionality ("hsiom_hld_ovr"). '1': The PRGIO controls the IO cel hold override functionality: - In bypass mode (ENABLED is '0' or BYPASS[i] is '1'), the HSIOM control is used. - In NON bypass mode (ENABLED is '1' and BYPASS[i] is '0'), the PRGIO sets hold override to "pwr_hld_ovr_hib" to enable PRGIO functionality in DeepSleep and Hibernate power modes (but disables it in Stop power mode). Note that in Hibernate power mode, the PRGIO should not rely on the state of Active or DeepSleep functionality signals from the HSIOM: these signals are clamped to '0' in Hibernate" Default Value: Undefined
12 : 8	CLOCK_SRC	Clock ("clk_fabric") and reset ("rst_fabric_n") source selection: "0": io_data_in[0]/'1'. ... "7": io_data_in[7]/'1'. "8": chip_data[0]/'1'. ... "15": chip_data[7]/'1'. "16": clk_prgio/rst_sys_act_n. Used for both Active functionality synchronous logic on "clk_prgio". This selection is intended for synchronous operation on a PCLK specified clock frequency ("clock_prgio_en"). Note that the fabric's clocked elements are frequency aligned, but NOT phase aligned to "clk_sys". "17": clk_prgio/rst_sys_dpslp_n. Used for both DeepSleep functionality synchronous logic on "clk_prgio" (note that "clk_prgio" is NOT available in DeepSleep and Hibernate power modes). This selection is intended for synchronous operation on a PCLK specified clock frequency ("clock_prgio_en"). Note that the fabric's clocked elements are frequency aligned, but NOT phase aligned to "clk_sys". "18": clk_prgio/rst_sys_hib_n. Used for both Hibernate functionality synchronous logic on "clk_prgio" (note that "clk_prgio" is NOT available in DeepSleep and Hibernate power modes). This selection is intended for synchronous operation on a PCLK specified clock frequency ("clock_prgio_en"). Note that the fabric's clocked elements are frequency aligned, but NOT phase aligned to "clk_sys". "19": clk_lf/rst_lf_dpslp_n (note that "clk_lf" is only available in DeepSleep power mode). This selection is intended for synchronous operation on "clk_lf". Note that the fabric's clocked elements are frequency aligned, but NOT phase aligned to other "clk_lf" clocked elements. "20"-"30": Clock source is constant '0'. Any of these clock sources should be selected when the IP is disabled to ensure low power consumption. "31": clk_sys/'1'. This selection is NOT intended for "clk_sys" operation, but for asynchronous operation: three "clk_sys" cycles after enabling the IP, the IP is fully functional (reset is de-activated). To be used for asynchronous (clockless) fabric functionality. Default Value: 20
7 : 0	BYPASS	Bypass of the programmable IO, one bit for each IO pin: BYPASS[i] is for IO pin i. When ENABLED is '1', this field is used. When ENABLED is '0', this field is NOT used and PRGIO is always bypassed. '0': No bypass (programmable IO fabric is exposed). '1': Bypass (programmable IO fabric is hidden). Default Value: Undefined

6.1.2 PRGIO_PRT0_SYNC_CTL

Synchronization control register

Address: 0x40050010

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access	R							
Name	IO_SYNC_EN [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	RW							
HW Access	R							
Name	CHIP_SYNC_EN [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
15 : 8	CHIP_SYNC_EN	Synchronization of the chip input signals to "clk_fabric", one bit for each input: CHIP_SYNC_EN[i] is for input i. '0': No synchronization. '1': Synchronization. Default Value: Undefined
7 : 0	IO_SYNC_EN	Synchronization of the IO pin input signals to "clk_fabric", one bit for each IO pin: IO_SYNC_EN[i] is for IO pin i. '0': No synchronization. '1': Synchronization. Default Value: Undefined

6.1.3 PRGIO_PRT0_LUT_SEL0

LUT component input selection

Address: 0x40050020

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None				RW			
HW Access	None				R			
Name	None [7:4]				LUT_TR0_SEL [3:0]			
Bits	15	14	13	12	11	10	9	8
SW Access	None				RW			
HW Access	None				R			
Name	None [15:12]				LUT_TR1_SEL [11:8]			
Bits	23	22	21	20	19	18	17	16
SW Access	None				RW			
HW Access	None				R			
Name	None [23:20]				LUT_TR2_SEL [19:16]			
Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
19 : 16	LUT_TR2_SEL	LUT input signal "tr2_in" source selection. Encoding is the same as for LUT_TR1_SEL. Default Value: Undefined
11 : 8	LUT_TR1_SEL	LUT input signal "tr1_in" source selection: "0": LUT 0 output. "1": LUT 1 output. "2": LUT 2 output. "3": LUT 3 output. "4": LUT 4 output. "5": LUT 5 output. "6": LUT 6 output. "7": LUT 7 output. "8": chip_data[0] (for LUTs 0, 1, 2, 3); chip_data[4] (for LUTs 4, 5, 6, 7). "9": chip_data[1] (for LUTs 0, 1, 2, 3); chip_data[5] (for LUTs 4, 5, 6, 7). "10": chip_data[2] (for LUTs 0, 1, 2, 3); chip_data[6] (for LUTs 4, 5, 6, 7). "11": chip_data[3] (for LUTs 0, 1, 2, 3); chip_data[7] (for LUTs 4, 5, 6, 7). "12": io_data_in[0] (for LUTs 0, 1, 2, 3); io_data_in[4] (for LUTs 4, 5, 6, 7). "13": io_data_in[1] (for LUTs 0, 1, 2, 3); io_data_in[5] (for LUTs 4, 5, 6, 7). "14": io_data_in[2] (for LUTs 0, 1, 2, 3); io_data_in[6] (for LUTs 4, 5, 6, 7). "15": io_data_in[3] (for LUTs 0, 1, 2, 3); io_data_in[7] (for LUTs 4, 5, 6, 7). Default Value: Undefined

6.1.3 PRGIO_PRT0_LUT_SEL0 (continued)

3 : 0	LUT_TR0_SEL	LUT input signal "tr0_in" source selection: "0": Data unit output. "1": LUT 1 output. "2": LUT 2 output. "3": LUT 3 output. "4": LUT 4 output. "5": LUT 5 output. "6": LUT 6 output. "7": LUT 7 output. "8": chip_data[0] (for LUTs 0, 1, 2, 3); chip_data[4] (for LUTs 4, 5, 6, 7). "9": chip_data[1] (for LUTs 0, 1, 2, 3); chip_data[5] (for LUTs 4, 5, 6, 7). "10": chip_data[2] (for LUTs 0, 1, 2, 3); chip_data[6] (for LUTs 4, 5, 6, 7). "11": chip_data[3] (for LUTs 0, 1, 2, 3); chip_data[7] (for LUTs 4, 5, 6, 7). "12": io_data_in[0] (for LUTs 0, 1, 2, 3); io_data_in[4] (for LUTs 4, 5, 6, 7). "13": io_data_in[1] (for LUTs 0, 1, 2, 3); io_data_in[5] (for LUTs 4, 5, 6, 7). "14": io_data_in[2] (for LUTs 0, 1, 2, 3); io_data_in[6] (for LUTs 4, 5, 6, 7). "15": io_data_in[3] (for LUTs 0, 1, 2, 3); io_data_in[7] (for LUTs 4, 5, 6, 7). Default Value: Undefined
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6.1.4 PRGIO_PRT0_LUT_CTL0

LUT component control register

Address: 0x40050040

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access	R							
Name	LUT [7:0]							
Bits	15	14	13	12	11	10	9	8
SW Access	None						RW	
HW Access	None						R	
Name	None [15:10]						LUT_OPC [9:8]	
Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							
Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
9 : 8	LUT_OPC	LUT opcode specifies the LUT operation: "0": Combinatorial output, no feedback. $tr_out = LUT\{tr2_in, tr1_in, tr0_in\}$. "1": Combinatorial output, feedback. $tr_out = LUT\{lut_reg, tr1_in, tr0_in\}$. On clock: $lut_reg \leq tr_in2$. "2": Sequential output, no feedback. $temp = LUT\{tr2_in, tr1_in, tr0_in\}$. $tr_out = lut_reg$. On clock: $lut_reg \leq temp$. "3": Register with asynchronous set and reset. $tr_out = lut_reg$. $enable = (tr2_in \wedge LUT[4]) \vee LUT[5]$. $set = enable \wedge (tr1_in \wedge LUT[2]) \wedge LUT[3]$. $clr = enable \wedge (tr0_in \wedge LUT[0]) \wedge LUT[1]$. Asynchronously (no clock required): $lut_reg \leq if (clr) '0' else if (set) '1'$ Default Value: Undefined

6.1.4 PRGIO_PRT0_LUT_CTL0 (continued)

7 : 0	LUT	LUT configuration. Depending on the LUT opcode LUT_OPC, the internal state lut_reg (captured in a flip-flop) and the LUT input signals tr0_in, tr1_in, tr2_in, the LUT configuration is used to determine the LUT output signal and the next sequential state (lut_reg). Default Value: Undefined
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6.1.5 PRGIO_PRT0_DU_SEL

Data unit component input selection

Address: 0x400500C0

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None				RW			
HW Access	None				R			
Name	None [7:4]				DU_TR0_SEL [3:0]			

Bits	15	14	13	12	11	10	9	8
SW Access	None				RW			
HW Access	None				R			
Name	None [15:12]				DU_TR1_SEL [11:8]			

Bits	23	22	21	20	19	18	17	16
SW Access	None				RW			
HW Access	None				R			
Name	None [23:20]				DU_TR2_SEL [19:16]			

Bits	31	30	29	28	27	26	25	24
SW Access	None		RW		None		RW	
HW Access	None		R		None		R	
Name	None [31:30]		DU_DATA1_SEL [29:28]		None [27:26]		DU_DATA0_SEL [25:24]	

Bits	Name	Description
29 : 28	DU_DATA1_SEL	Data unit input data "data1_in" source selection. Encoding is the same as for DU_DATA0_SEL. Default Value: Undefined
25 : 24	DU_DATA0_SEL	Data unit input data "data0_in" source selection: "0": Constant "0". "1": chip_data[7:0]. "2": io_data_in[7:0]. "3": DATA.DATA MMIO register field. Default Value: Undefined
19 : 16	DU_TR2_SEL	Data unit input signal "tr2_in" source selection. Encoding is the same as for DU_TR0_SEL. Default Value: Undefined
11 : 8	DU_TR1_SEL	Data unit input signal "tr1_in" source selection. Encoding is the same as for DU_TR0_SEL. Default Value: Undefined

6.1.5 PRGIO_PRT0_DU_SEL (continued)

3 : 0	DU_TR0_SEL	Data unit input signal "tr0_in" source selection: "0": Constant '0'. "1": Constant '1'. "2": Data unit output. "10-3": LUT 7 - 0 outputs. Otherwise: Undefined. Default Value: Undefined
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6.1.6 PRGIO_PRT0_DU_CTL

Data unit component control register

Address: 0x400500C4

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None					RW		
HW Access	None					R		
Name	None [7:3]					DU_SIZE [2:0]		
Bits	15	14	13	12	11	10	9	8
SW Access	None					RW		
HW Access	None					R		
Name	None [15:12]					DU_OPC [11:8]		
Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							
Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
11 : 8	DU_OPC	Data unit opcode specifies the data unit operation: "1": INCR "2": DECR "3": INCR_WRAP "4": DECR_WRAP "5": INCR_DECR "6": INCR_DECR_WRAP "7": ROR "8": SHR "9": AND_OR "10": SHR_MAJ3 "11": SHR_EQL. Otherwise: Undefined. Default Value: Undefined
2 : 0	DU_SIZE	Size/width of the data unit data operands (in bits) is DU_SIZE+1. E.g., if DU_SIZE is 7, the width is 8 bits. Default Value: Undefined

6.1.7 PRGIO_PRT0_DATA

Data register

Address: 0x400500F0

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access	R							
Name	DATA [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
7 : 0	DATA	Data unit input data source. Default Value: Undefined

7 CPU Sub System (CPUSS) Registers



This section discusses the CPUSS registers. It lists all the registers in mapping tables, in address order.

7.1 Register Details

Register	Address	Description
CPUSS_SYSREQ	0x40100004	SYSCALL control register
CPUSS_SYSARG	0x40100008	SYSARG control register
CPUSS_PROTECTION	0x4010000C	Protection control register
CPUSS_PRIV_ROM	0x40100010	ROM privilege register
CPUSS_PRIV_RAM	0x40100014	RAM privilege register
CPUSS_PRIV_FLASH	0x40100018	Flash privilege register
CPUSS_WOUNDING	0x4010001C	Wounding register
CPUSS_FLASH_CTL	0x40100030	FLASH control register
CPUSS_ROM_CTL	0x40100034	ROM control register
CPUSS_RAM_CTL	0x40100038	RAM control register
CPUSS_DMAC_CTL	0x4010003C	DMA controller register
CPUSS_SL_CTL0	0x40100100	Slave control register
CPUSS_SL_CTL1	0x40100104	Slave control register. See CPUSS_SL_CTL0 for the details of bit fields.
CPUSS_SL_CTL2	0x40100108	Slave control register. See CPUSS_SL_CTL0 for the details of bit fields.
CPUSS_SL_CTL3	0x4010010C	Slave control register. See CPUSS_SL_CTL0 for the details of bit fields.

7.1.1 CPUSS_SYSREQ

SYSCALL control register

Address: 0x40100004

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access								
Name	SYSCALL_COMMAND [7:0]							
Bits	15	14	13	12	11	10	9	8
SW Access	RW							
HW Access								
Name	SYSCALL_COMMAND [15:8]							
Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							
Bits	31	30	29	28	27	26	25	24
SW Access	RW	R	R	RW	RW	None		
HW Access	R	A	RW	A	R	None		
Name	SY- SCALL_RE Q	HMAS- TER_0	ROM_AC- CESS_EN	PRIVI- LEGED	DIS_RE- SET_VECT _REL	None [26:24]		

Bits	Name	Description
31	SYSCALL_REQ	CPU/DAP writes a '1' to this field to request a SystemCall. The HMASTER_0 field indicates the source of the write access. Setting this field to '1' immediate results in a NMI. The SystemCall NMI interrupt handler sets this field to '0' after servicing the request. Default Value: 0
30	HMASTER_0	Indicates the source of the write access to the SYSREQ register. '0': CPU write access. '1': DAP write access. HW sets this field when the SYSREQ register is written to and SYSCALL_REQ is '0' (the last time it is set is when SW sets SYSCALL_REQ from '0' to '1'). Default Value: 0
29	ROM_ACCESS_EN	Indicates that executing from Boot ROM is enabled. HW sets this field to '1', on reset or when the SystemCall NMI vector is fetched from Boot ROM. HW sets this field to '0', when the CPU is NOT executing from either Boot or System ROM. This bit is used for debug purposes only. Default Value: 1

7.1.1 CPUSS_SYSREQ (continued)

28	PRIVILEGED	Indicates whether the system is in privileged ('1') or user mode ('0'). Only CPU SW executing from ROM can set this field to '1' when ROM_ACCESS_EN is '1' (the CPU is executing a SystemCall NMI interrupt handler). Any other write to this field sets is to '0'. This field is used as the AHB-Lite hprot[1] signal to implement Cypress proprietary user/privileged modes. These modes are used to enable/disable access to specific MMIO registers and memory regions. Default Value: 1
27	DIS_RESET_VECT_REL	Disable Reset Vector fetch relocation: '0': CPU accesses to locations 0x0000:0000 - 0x0000:0007 are redirected to ROM. '1': CPU accesses to locations 0x0000:0000 - 0x0000:0007 are made to flash. Note that this field defaults to '0' on reset, ensuring actual reset vector fetches are always made to ROM. Note that this field does not affect DAP accesses. Flash DfT routines may set this bit to '1' to enable uninhibited read-back of programmed data in the first flash page. Default Value: 0
15 : 0	SYSCALL_COMMAND	Opcode of the system call being requested. Default Value: 0

7.1.2 CPUSS_SYSARG

SYSARG control register

Address: 0x40100008

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access								
Name	SYSCALL_ARG [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	RW							
HW Access								
Name	SYSCALL_ARG [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	RW							
HW Access								
Name	SYSCALL_ARG [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	RW							
HW Access								
Name	SYSCALL_ARG [31:24]							

Bits	Name	Description
31 : 0	SYSCALL_ARG	Argument to System Call specified in SYSREQ. Semantics of argument depends on system call made. Typically a pointer to a parameter block. Default Value: 0

7.1.3 CPUSS_PROTECTION

Protection control register

Address: 0x4010000C

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None				RW			
HW Access	None				A			
Name	None [7:4]				PROTECTION_MODE [3:0]			

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	RW1S	RW	None					
HW Access	R	A	None					
Name	PROTECTION_LOCK	FLASH_LOCK	None [29:24]					

Bits	Name	Description
31	PROTECTION_LOCK	Setting this field will block (ignore) any further writes to the PROTECTION_MODE field in this register. Once '1', this field cannot be cleared. Default Value: 0
30	FLASH_LOCK	Setting this bit will force SPCIF.ADDRESS.AXA to be ignored, which prevents SM Flash from being erased or overwritten. It is used to indicate the DEAD protection mode. Writes to this field are ignored when PROTECTION_LOCK is '1' Default Value: 0
3 : 0	PROTECTION_MODE	Current protection mode; this field is available as a global signal everywhere in the system. Writes to this field are ignored when PROTECTION_LOCK is '1': 0b1xxx: BOOT 0b01xx: KILL 0b001x: PROTECTED 0b0001: OPEN 0b0000: VIRGIN (also used for DEAD mode, but then FLASH_LOCK is also set) Default Value: 15

7.1.4 CPOSS_PRIV_ROM

ROM privilege register

Address: 0x40100010

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access	R							
Name	BROM_PROT_LIMIT [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
7 : 0	BROM_PROT_LIMIT	Indicates the limit where the privileged area of the Boot ROM partition starts in increments of 256 Bytes. "0": Entire Boot ROM is Privileged. "1": First 256 Bytes are User accessible. ... BROM_PROT_LIMIT >= "Boot ROM partition capacity": Entire Boot ROM partition is user mode accessible. Default Value: 0

7.1.5 CPUSS_PRIV_RAM

RAM privilege register

Address: 0x40100014

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access	R							
Name	RAM_PROT_LIMIT [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	None							RW
HW Access	None							R
Name	None [15:9]							RAM_PROT_LIMIT

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
8 : 0	RAM_PROT_LIMIT	Indicates the limit where the privileged area of SRAM starts in increments of 256 Bytes. "0": Entire SRAM is Privileged. "1": First 256 Bytes are User accessible. Any number larger than the size of the SRAM indicates that the entire SRAM is user mode accessible. Default Value: 0

7.1.6 CPUSS_PRIV_FLASH

Flash privilege register

Address: 0x40100018

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access	R							
Name	FLASH_PROT_LIMIT [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	None				RW			
HW Access	None				R			
Name	None [15:12]				FLASH_PROT_LIMIT [11:8]			

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
11 : 0	FLASH_PROT_LIMIT	Indicates the limit where the privileged area of flash starts in increments of 256 Bytes. "0": Entire flash is Privileged. "1": First 256 Bytes are User accessible. Any number larger than the size of the flash indicates that the entire flash is user mode accessible. Note that SuperVisory rows are always User accessible. If FLASH_PROT_LIMIT defines a non-empty privileged area, the boot ROM will assume that a system call table exists at the beginning of the Flash privileged area and use it for all SystemCalls made using SYSREQ. Default Value: 0

7.1.7 CPUSS_WOUNDING

Wounding register

Address: 0x4010001C

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None							
HW Access	None							
Name	None [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None	RW1S			None	RW1S		
HW Access	None	R			None	R		
Name	None	FLASH_WOUND [22:20]			None	RAM_WOUND [18:16]		

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
22 : 20	FLASH_WOUND	Indicates the amount of accessible flash in this part. The value in this field is effectively write-once (it is only possible to set bits, not clear them). The remainder portion of flash is not accessible and will return an AHB-Lite bus error. "0": entire memory accessible "1": first 1/2 of the memory accessible "2": first 1/4 of the memory accessible "3": first 1/8 of the memory accessible "4": first 1/16 of the memory accessible "5": first 1/32 of the memory accessible "6": first 1/64 of the memory accessible "7": first 1/128 of the memory accessible (used for the DEAD protection mode) Default Value: 0

7.1.7 CPUSS_WOUNDING (continued)

18 : 16	RAM_WOUND	Indicates the amount of accessible RAM 0 memory capacity in this part. The value in this field is effectively write-once (it is only possible to set bits, not clear them). The remainder portion of SRAM is not accessible and will return an AHB-Lite bus error. "0": entire memory accessible "1": first 1/2 of the memory accessible "2": first 1/4 of the memory accessible "3": first 1/8 of the memory accessible "4": first 1/16 of the memory accessible "5": first 1/32 of the memory accessible "6": first 1/64 of the memory accessible "7": first 1/128 of the memory accessible Default Value: 0
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7.1.8 CPOSS_FLASH_CTL

FLASH control register

Address: 0x40100030

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None			RW	None		RW	
HW Access	None			R	None		R	
Name	None [7:5]			PREF_EN	None [3:2]		FLASH_WS [1:0]	

Bits	15	14	13	12	11	10	9	8
SW Access	None							RW
HW Access	None							RW1C
Name	None [15:9]							FLASH_IN- VALIDATE

Bits	23	22	21	20	19	18	17	16
SW Access	None						RW	
HW Access	None						R	
Name	None [23:18]						ARB [17:16]	

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
17 : 16	ARB	Arbitration policy: "0": CPU has priority "1": DW/DMA has priority "2": Roundrobin "3": Roundrobin - sticky Default Value: 0
8	FLASH_INVALIDATE	1': Invalidates the content of the flash controller's buffers. Default Value: 0
4	PREF_EN	Prefetch enable: '0': disabled. This is a desirable setting when FLASH_WS is "0" or when predictable execution behavior is required. '1': enabled. Default Value: 0

7.1.8 CPOSS_FLASH_CTL (continued)

1 : 0	FLASH_WS	Amount of ROM wait states: "0": 0 wait states (fast flash: [0, 24] MHz system frequency, slow flash: [0, 16] MHz system frequency) "1": 1 wait state (fast flash: [24, 48] MHz system frequency, slow flash: [16, 32] MHz system frequency) "2": 2 wait states (slow flash: [32, 48] MHz system frequency) "3": 3 wait states (can be used to give more time for flash access if 2 wait states are not sufficient) Default Value: 0
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7.1.9 CPUSS_ROM_CTL

ROM control register

Address: 0x40100034

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None							RW
HW Access	None							R
Name	None [7:1]							ROM_WS

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
0	ROM_WS	Amount of ROM wait states: '0': 0 wait states. Use this setting for newer, faster ROM design. Use this setting for older, slower ROM design and frequencies in the range [0, 24] MHz. '1': 1 wait state. Use this setting for older, slower ROM design and frequencies in the range <24, 48] MHz. CPUSSv2 supports two types of ROM memory: an older, slower design (operating at up to 24 MHz) and a newer, faster design (operating at up to 48 MHz). The older design requires 1 wait state for frequencies above 24 MHz. The newer design never requires wait states. All chips after Street Fighter will use the newer design. As a result, all chips after Street Fighter can always use 0 wait states. Default Value: 0

7.1.10 CPUSS_RAM_CTL

RAM control register

Address: 0x40100038

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None							
HW Access	None							
Name	None [7:0]							
Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							
Bits	23	22	21	20	19	18	17	16
SW Access	None						RW	
HW Access	None						R	
Name	None [23:18]						ARB [17:16]	
Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
17 : 16	ARB	Arbitration policy: "0": CPU has priority "1": DW/DMA has priority "2": Roundrobin "3": Roundrobin - sticky Default Value: 0

7.1.11 CPUSS_DMAC_CTL

DMA controller register

Address: 0x4010003C

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None							
HW Access	None							
Name	None [7:0]							
Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							
Bits	23	22	21	20	19	18	17	16
SW Access	None						RW	
HW Access	None						R	
Name	None [23:18]						ARB [17:16]	
Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
17 : 16	ARB	Arbitration policy: "0": CPU has priority "1": DW/DMA has priority "2": Roundrobin "3": Roundrobin - sticky Default Value: 0

7.1.12 CPUSS_SL_CTL0

Slave control register

Address: 0x40100100

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None							
HW Access	None							
Name	None [7:0]							
Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							
Bits	23	22	21	20	19	18	17	16
SW Access	None						RW	
HW Access	None						R	
Name	None [23:18]						ARB [17:16]	
Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
17 : 16	ARB	Arbitration policy: "0": CPU priority "1": DMA priority "2": Roundrobin "3": Roundrobin - sticky Default Value: 0

8 Direct Memory Access (DMA) Registers



This section discusses the DMA registers. It lists all the registers in mapping tables, in address order.

8.1 Register Details

Register	Address	Description
DMAC_CTL	0x40101000	Control register
DMAC_STATUS	0x40101010	Status register
DMAC_STATUS_SRC_ADDR	0x40101014	Source address status register
DMAC_STATUS_DST_ADDR	0x40101018	Destination address register
DMAC_STATUS_CH_ACT	0x4010101C	Channel activation status register
DMAC_CH_CTL0	0x40101080	Channel control register
DMAC_CH_CTL1	0x40101084	Channel control register. See DMAC_CH_CTL0 for the details of bit fields.
DMAC_CH_CTL2	0x40101088	Channel control register. See DMAC_CH_CTL0 for the details of bit fields.
DMAC_CH_CTL3	0x4010108C	Channel control register. See DMAC_CH_CTL0 for the details of bit fields.
DMAC_CH_CTL4	0x40101090	Channel control register. See DMAC_CH_CTL0 for the details of bit fields.
DMAC_CH_CTL5	0x40101094	Channel control register. See DMAC_CH_CTL0 for the details of bit fields.
DMAC_CH_CTL6	0x40101098	Channel control register. See DMAC_CH_CTL0 for the details of bit fields.
DMAC_CH_CTL7	0x4010109C	Channel control register. See DMAC_CH_CTL0 for the details of bit fields.
DMAC_INTR	0x401017F0	Interrupt register
DMAC_INTR_SET	0x401017F4	Interrupt set register
DMAC_INTR_MASK	0x401017F8	Interrupt mask register
DMAC_INTR_MASKED	0x401017FC	Interrupt masked register
DMAC_DESCR0_PING_SRC	0x40101800	Ping source address
DMAC_DESCR0_PING_DST	0x40101804	Ping destination address
DMAC_DESCR0_PING_CTL	0x40101808	Ping control word
DMAC_DESCR0_PING_STATUS	0x4010180C	Ping status word
DMAC_DESCR0_PONG_SRC	0x40101810	Pong source address
DMAC_DESCR0_PONG_DST	0x40101814	Pong destination address
DMAC_DESCR0_PONG_CTL	0x40101818	Pong control word
DMAC_DESCR0_PONG_STATUS	0x4010181C	Pong status word
DMAC_DESCR1_PING_SRC	0x40101820	Ping source address. See DMAC_DESCR0_PING_SRC for the details of bit fields.
DMAC_DESCR1_PING_DST	0x40101824	Ping destination address. See DMAC_DESCR0_PING_DST for the details of bit fields.

Register	Address	Description
DMAC_DESCR1_PING_CTL	0x40101828	Ping control word. See DMAC_DESCR0_PING_CTL for the details of bit fields.
DMAC_DESCR1_PING_STATUS	0x4010182C	Ping status word. See DMAC_DESCR0_PING_STATUS for the details of bit fields.
DMAC_DESCR1_PONG_SRC	0x40101830	Pong source address. See DMAC_DESCR0_PONG_SRC for the details of bit fields.
DMAC_DESCR1_PONG_DST	0x40101834	Pong destination address. See DMAC_DESCR0_PONG_DST for the details of bit fields.
DMAC_DESCR1_PONG_CTL	0x40101838	Pong control word. See DMAC_DESCR0_PONG_CTL for the details of bit fields.
DMAC_DESCR1_PONG_STATUS	0x4010183C	Pong status word. See DMAC_DESCR0_PONG_STATUS for the details of bit fields.
DMAC_DESCR2_PING_SRC	0x40101840	Ping source address. See DMAC_DESCR0_PING_SRC for the details of bit fields.
DMAC_DESCR2_PING_DST	0x40101844	Ping destination address. See DMAC_DESCR0_PING_DST for the details of bit fields.
DMAC_DESCR2_PING_CTL	0x40101848	Ping control word. See DMAC_DESCR0_PING_CTL for the details of bit fields.
DMAC_DESCR2_PING_STATUS	0x4010184C	Ping status word. See DMAC_DESCR0_PING_STATUS for the details of bit fields.
DMAC_DESCR2_PONG_SRC	0x40101850	Pong source address. See DMAC_DESCR0_PONG_SRC for the details of bit fields.
DMAC_DESCR2_PONG_DST	0x40101854	Pong destination address. See DMAC_DESCR0_PONG_DST for the details of bit fields.
DMAC_DESCR2_PONG_CTL	0x40101858	Pong control word. See DMAC_DESCR0_PONG_CTL for the details of bit fields.
DMAC_DESCR2_PONG_STATUS	0x4010185C	Pong status word. See DMAC_DESCR0_PONG_STATUS for the details of bit fields.
DMAC_DESCR3_PING_SRC	0x40101860	Ping source address. See DMAC_DESCR0_PING_SRC for the details of bit fields.
DMAC_DESCR3_PING_DST	0x40101864	Ping destination address. See DMAC_DESCR0_PING_DST for the details of bit fields.
DMAC_DESCR3_PING_CTL	0x40101868	Ping control word. See DMAC_DESCR0_PING_CTL for the details of bit fields.
DMAC_DESCR3_PING_STATUS	0x4010186C	Ping status word. See DMAC_DESCR0_PING_STATUS for the details of bit fields.
DMAC_DESCR3_PONG_SRC	0x40101870	Pong source address. See DMAC_DESCR0_PONG_SRC for the details of bit fields.
DMAC_DESCR3_PONG_DST	0x40101874	Pong destination address. See DMAC_DESCR0_PONG_DST for the details of bit fields.
DMAC_DESCR3_PONG_CTL	0x40101878	Pong control word. See DMAC_DESCR0_PONG_CTL for the details of bit fields.
DMAC_DESCR3_PONG_STATUS	0x4010187C	Pong status word. See DMAC_DESCR0_PONG_STATUS for the details of bit fields.
DMAC_DESCR4_PING_SRC	0x40101880	Ping source address. See DMAC_DESCR0_PING_SRC for the details of bit fields.
DMAC_DESCR4_PING_DST	0x40101884	Ping destination address. See DMAC_DESCR0_PING_DST for the details of bit fields.
DMAC_DESCR4_PING_CTL	0x40101888	Ping control word. See DMAC_DESCR0_PING_CTL for the details of bit fields.
DMAC_DESCR4_PING_STATUS	0x4010188C	Ping status word. See DMAC_DESCR0_PING_STATUS for the details of bit fields.
DMAC_DESCR4_PONG_SRC	0x40101890	Pong source address. See DMAC_DESCR0_PONG_SRC for the details of bit fields.
DMAC_DESCR4_PONG_DST	0x40101894	Pong destination address. See DMAC_DESCR0_PONG_DST for the details of bit fields.
DMAC_DESCR4_PONG_CTL	0x40101898	Pong control word. See DMAC_DESCR0_PONG_CTL for the details of bit fields.
DMAC_DESCR4_PONG_STATUS	0x4010189C	Pong status word. See DMAC_DESCR0_PONG_STATUS for the details of bit fields.
DMAC_DESCR5_PING_SRC	0x401018A0	Ping source address. See DMAC_DESCR0_PING_SRC for the details of bit fields.
DMAC_DESCR5_PING_DST	0x401018A4	Ping destination address. See DMAC_DESCR0_PING_DST for the details of bit fields.
DMAC_DESCR5_PING_CTL	0x401018A8	Ping control word. See DMAC_DESCR0_PING_CTL for the details of bit fields.
DMAC_DESCR5_PING_STATUS	0x401018AC	Ping status word. See DMAC_DESCR0_PING_STATUS for the details of bit fields.
DMAC_DESCR5_PONG_SRC	0x401018B0	Pong source address. See DMAC_DESCR0_PONG_SRC for the details of bit fields.
DMAC_DESCR5_PONG_DST	0x401018B4	Pong destination address. See DMAC_DESCR0_PONG_DST for the details of bit fields.
DMAC_DESCR5_PONG_CTL	0x401018B8	Pong control word. See DMAC_DESCR0_PONG_CTL for the details of bit fields.
DMAC_DESCR5_PONG_STATUS	0x401018BC	Pong status word. See DMAC_DESCR0_PONG_STATUS for the details of bit fields.
DMAC_DESCR6_PING_SRC	0x401018C0	Ping source address. See DMAC_DESCR0_PING_SRC for the details of bit fields.
DMAC_DESCR6_PING_DST	0x401018C4	Ping destination address. See DMAC_DESCR0_PING_DST for the details of bit fields.
DMAC_DESCR6_PING_CTL	0x401018C8	Ping control word. See DMAC_DESCR0_PING_CTL for the details of bit fields.
DMAC_DESCR6_PING_STATUS	0x401018CC	Ping status word. See DMAC_DESCR0_PING_STATUS for the details of bit fields.

Register	Address	Description
DMAC_DESCR6_PONG_SRC	0x401018D0	Pong source address. See DMAC_DESCR0_PONG_SRC for the details of bit fields.
DMAC_DESCR6_PONG_DST	0x401018D4	Pong destination address. See DMAC_DESCR0_PONG_DST for the details of bit fields.
DMAC_DESCR6_PONG_CTL	0x401018D8	Pong control word. See DMAC_DESCR0_PONG_CTL for the details of bit fields.
DMAC_DESCR6_PONG_STATUS	0x401018DC	Pong status word. See DMAC_DESCR0_PONG_STATUS for the details of bit fields.
DMAC_DESCR7_PING_SRC	0x401018E0	Ping source address. See DMAC_DESCR0_PING_SRC for the details of bit fields.
DMAC_DESCR7_PING_DST	0x401018E4	Ping destination address. See DMAC_DESCR0_PING_DST for the details of bit fields.
DMAC_DESCR7_PING_CTL	0x401018E8	Ping control word. See DMAC_DESCR0_PING_CTL for the details of bit fields.
DMAC_DESCR7_PING_STATUS	0x401018EC	Ping status word. See DMAC_DESCR0_PING_STATUS for the details of bit fields.
DMAC_DESCR7_PONG_SRC	0x401018F0	Pong source address. See DMAC_DESCR0_PONG_SRC for the details of bit fields.
DMAC_DESCR7_PONG_DST	0x401018F4	Pong destination address. See DMAC_DESCR0_PONG_DST for the details of bit fields.
DMAC_DESCR7_PONG_CTL	0x401018F8	Pong control word. See DMAC_DESCR0_PONG_CTL for the details of bit fields.
DMAC_DESCR7_PONG_STATUS	0x401018FC	Pong status word. See DMAC_DESCR0_PONG_STATUS for the details of bit fields.

8.1.1 DMAC_CTL

Control register

Address: 0x40101000

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None							
HW Access	None							
Name	None [7:0]							
Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							
Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							
Bits	31	30	29	28	27	26	25	24
SW Access	RW	None						
HW Access	R	None						
Name	ENABLED	None						

Bits	Name	Description
31	ENABLED	0': IP is disabled. Non-retainable MMIO registers and logic functionality are reset (retainable MMIO registers are NOT reset): - INTR register is set to "0". - DW/DMA functionality is aborted. - DW/DMA controller input/pending triggers are de-activated. - DW/DMA controller output triggers are de-activated. Disabling the IP has the same effect as an active "rst_sys_act_n" reset in DeepSleep power mode. To prevent a loss of active (pending) DW/DMA triggers when disabling the IP or when transitioning from Active to DeepSleep power mode, the STATUS.ACTIVE and STATUS.CH_ACTIVE.CH fields can be used. Note that most MMIO registers are retainable, and a transition from DeepSleep to Active/Sleep power modes makes the DW/DMA controller operational, and ready to react to DW/DMA input triggers that are activated after the transition. Triggers are Active/Sleep functionality. '1': IP is enabled. Default Value: 0

8.1.2 DMAC_STATUS

Status register

Address: 0x40101010

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access	W							
Name	DATA_NR [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access	W							
Name	DATA_NR [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None					R		
HW Access	None					W		
Name	None [23:19]					CH_ADDR [18:16]		

Bits	31	30	29	28	27	26	25	24
SW Access	R	R	R		None	R		
HW Access	W	W	W		None	W		
Name	ACTIVE	PING_PONG	PRIO [29:28]		None	STATE [26:24]		

Bits	Name	Description
31	ACTIVE	Specifies if there is a currently active (pending) channel in the data transfer engine: '0': no currently active channel. '1': currently active channel. Default Value: 0
30	PING_PONG	Specifies whether the PING descriptor ('0') or PONG descriptor ('1') of the channel is currently in use. Default Value: Undefined
29 : 28	PRIO	Specifies the priority of the currently active channel. Default Value: Undefined
26 : 24	STATE	State of the data transfer engine. "0": DEFAULT state. "1": Loading descriptor (SRC, DST, CONTROL and STATUS words). "2": Loading data element from source location. "3": Storing data element to destination location. "4": Storing descriptor (STATUS word). "5": Wait for trigger de-activation. "6": Storing descriptor with error response (STATUS word). Default Value: 0

8.1.2 DMAC_STATUS (continued)

18 : 16	CH_ADDR	Specifies the channel number of the currently active channel. E.g. if we have 32 channels, the channel number address with CH_ADDR_WIDTH is $\text{LOG}_2(32) = 5$, and this field is a 5-bit field. If channel 7 is active, STATUS.ACTIVE is '1' and STATUS.CH_ADDR is "7". Default Value: Undefined
15 : 0	DATA_NR	Specifies the index of the currently active data transfer. This value increases from "0" to CONTROL.DATA_NR. Default Value: Undefined

8.1.3 DMAC_STATUS_SRC_ADDR

Source address status register

Address: 0x40101014

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access	W							
Name	ADDR [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access	W							
Name	ADDR [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access	W							
Name	ADDR [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access	W							
Name	ADDR [31:24]							

Bits	Name	Description
31 : 0	ADDR	Base address or current address of source location of currently active channel. The specific address information is cycle dependent. This is field is provided for debug purposes. Functionally, no assumption should be made on whether the base or current address is provided. The specifics of the currently active channel are available through STATUS. Note while reading the STATUS, STATUS_SRC and STATUS_DST registers, the transfer engine may have moved from one active channel to another. Default Value: Undefined

8.1.4 DMAC_STATUS_DST_ADDR

Destination address register

Address: 0x40101018

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access	W							
Name	ADDR [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access	W							
Name	ADDR [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access	W							
Name	ADDR [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access	W							
Name	ADDR [31:24]							

Bits	Name	Description
31 : 0	ADDR	Base address or current address of destination location of currently active channel. The specific address information is cycle dependent. This is field is provided for debug purposes. Functionally, no assumption should be made on whether the base or current address is provided. The specifics of the currently active channel are available through STATUS. Note while reading the STATUS, STATUS_SRC and STATUS_DST registers, the transfer engine may have moved from one active channel to another. Default Value: Undefined

8.1.5 DMAC_STATUS_CH_ACT

Channel activation status register

Address: 0x4010101C

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access	W							
Name	CH [7:0]							
Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							
Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							
Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
7 : 0	CH	Channel activation status. Bit i is associated to channel i, with i = 0, ..., CH_NR-1. Software reads this field to get information on all actively pending channels (either in pending or in the data transfer engine). Default Value: 0

8.1.6 DMAC_CH_CTL0

Channel control register

Address: 0x40101080

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None							
HW Access	None							
Name	None [7:0]							
Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							
Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							
Bits	31	30	29	28	27	26	25	24
SW Access	RW	RW	RW		None			
HW Access	RW1C	RW	R		None			
Name	ENABLED	PING_PON G	PRIO [29:28]		None [27:24]			

Bits	Name	Description
31	ENABLED	'0': channel disabled. The channel's trigger is ignored and the channel cannot be activated. If the activated channel is disabled, the data transfer(s) are aborted. '1': channel enabled. Software sets this field to '1' to enable a specific channel. Hardware sets this field to '0' on erroneous channel behavior (the specific error is specified by STATUS.RESP in the channel's descriptor structure). Default Value: 0

8.1.6 DMAC_CH_CTL0 (continued)

30	PING_PONG	Each channel has two descriptor structures for double buffering purposes. As the controller operates on one structure, the main CPU can operate on the other structure. The descriptor structures are identified as PING ('0') and PONG ('1'). This field identifies the descriptor structure that is currently in use by the controller. Software sets this field to the desired descriptor structure. Hardware inverts the field value on the completion of a descriptor structure when CONTROL.FLIPPING of the current descriptor structure is set to '1'. Each channel has two descriptor structures for double buffering purposes. As the controller operates on one structure, the main CPU can operate on the other structure. The descriptor structures are identified as PING ('0') and PONG ('1'). This field identifies the descriptor structure that is currently in use by the controller. Default Value: 0
29 : 28	PRI0	Channel priority, with "0" representing the highest priority and "3" representing the lowest priority. Priority decoding uses the channel priority to determine the highest priority activated channel. If multiple activated channels have the same highest priority, the channel with the lowest index <i>i</i>, is considered the highest priority activated channel. Default Value: 0

8.1.7 DMAC_INTR

Interrupt register

Address: 0x401017F0

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW1C							
HW Access	RW1S							
Name	CH [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
7 : 0	CH	Set to '1', when event is detected. Write INTR field with '1', to clear bit. Write INTR_SET field with '1', to set bit. Default Value: 0

8.1.8 DMAC_INTR_SET

Interrupt set register

Address: 0x401017F4

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW1S							
HW Access	A							
Name	CH [7:0]							
Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							
Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							
Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
7 : 0	CH	Write INTR_SET field with '1' to set corresponding INTR field (a write of '0' has no effect). Default Value: 0

8.1.9 DMAC_INTR_MASK

Interrupt mask register

Address: 0x401017F8

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access	R							
Name	CH [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
7 : 0	CH	Mask for corresponding field in INTR register. Default Value: 0

8.1.10 DMAC_INTR_MASKED

Interrupt masked register

Address: 0x401017FC

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access	W							
Name	CH [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
7 : 0	CH	Logical and of corresponding request and mask fields. Default Value: 0

8.1.11 DMAC_DESCR0_PING_SRC

Ping source address

Address: 0x40101800

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access	RW							
Name	ADDR [7:0]							
Bits	15	14	13	12	11	10	9	8
SW Access	RW							
HW Access	RW							
Name	ADDR [15:8]							
Bits	23	22	21	20	19	18	17	16
SW Access	RW							
HW Access	RW							
Name	ADDR [23:16]							
Bits	31	30	29	28	27	26	25	24
SW Access	RW							
HW Access	RW							
Name	ADDR [31:24]							

Bits	Name	Description
31 : 0	ADDR	Base address of source location. The effective source location is calculated by adding on offset (derived from PING.STATUS.CURR_DATA_NR) to this base address. Default Value: Undefined

8.1.12 DMAC_DESCR0_PING_DST

Ping destination address

Address: 0x40101804

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access	RW							
Name	ADDR [7:0]							
Bits	15	14	13	12	11	10	9	8
SW Access	RW							
HW Access	RW							
Name	ADDR [15:8]							
Bits	23	22	21	20	19	18	17	16
SW Access	RW							
HW Access	RW							
Name	ADDR [23:16]							
Bits	31	30	29	28	27	26	25	24
SW Access	RW							
HW Access	RW							
Name	ADDR [31:24]							

Bits	Name	Description
31 : 0	ADDR	Base address of destination location. The effective destination location is calculated by adding on offset (derived from PING.STATUS.CURR_DATA_NR) to this base address. Default Value: Undefined

8.1.13 DMAC_DESCR0_PING_CTL

Ping control word

Address: 0x40101808

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access	RW							
Name	DATA_NR [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	RW							
HW Access	RW							
Name	DATA_NR [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	RW	RW	RW	RW	None		RW	
HW Access	RW	RW	RW	RW	None		RW	
Name	SRC_AD- DR_INCR	SRC_TRAN- SFER_SIZE	DST_AD- DR_INCR	DST_TRAN- SFER_SIZE	None [19:18]		DATA_SIZE [17:16]	

Bits	31	30	29	28	27	26	25	24
SW Access	RW		RW	RW	RW	RW	RW	
HW Access	RW		RW	RW	RW	RW	RW	
Name	OPCODE [31:30]		FLIPPING	PREEMPT- ABLE	SET_- CAUSE	INV_DE- SCR	WAIT_FOR_DEACT [25:24]	

Bits	Name	Description
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8.1.13 DMAC_DESCR0_PING_CTL (continued)

31 : 30	OPCODE	Specifies the specific data transfer (only when the VALID bit of the descriptor's STATUS word is '1'): "0": A single trigger initiates a single data element transfer (DW mode). This opcode specifies a transfer of a single data element. The current descriptor is completed when the amount of transferred single data elements equals the programmed buffer size (DATA_NR+1). "1": A single trigger initiates a single descriptor transfer (DMA mode). This opcode specifies a transfer of DATA_NR+1 data elements as specified by the current descriptor structure. The current descriptor is completed when its data transfer is completed. "2": A single trigger initiates a descriptor list transfer (DMA mode). This opcode specifies a transfer of DATA_NR+1 data elements as specified by the current descriptor structure and by successive valid descriptors. The current descriptor is completed when its data transfer is completed. This OPCODE relies on FLIPPING to be set to '1', such that the CHI_CTL.PING_PONG field is flipped/inverted and the successive descriptor is used. This continues for as long as the successive descriptor is valid. Note that as the HW is using the PING/PONG descriptor, the SW can prepare the alternate PONG/PING descriptor. The interrupt mechanism is used by HW to convey to the SW that the current descriptor is completed (and can be prepared for a successive transfer). After completion of the opcode (and waiting for de-activation, the channel's output trigger is activated). Default Value: Undefined
29	FLIPPING	'1': On completion of the current descriptor structure, the current descriptor identifier CHI_CTL.PING_PONG is flipped/inverted. In DMA mode, descriptor list transfer, flipping of the current descriptor identifier can be used to construct a linked list of descriptor structures. Default Value: Undefined
28	PREEMPTABLE	'1': Transfer is preemptable. In DMA mode (OPCODE is "1" or "2"), multi data element transfers are constructed out of multiple single data element load (from the source location) and store (to the destination location) sequences. This field allows higher priority activated channels to preempt the current transfer in between these atomic (load, store) sequences. Preemption will NOT deactivate the current channel. As a result, after completion of a higher priority activated channel, the current channel is rescheduled. Default Value: Undefined
27	SET_CAUSE	'1': On completion of the current descriptor structure, the interrupt cause field of the channel is set to '1' (INTR.CH[i]). Default Value: Undefined
26	INV_DESCR	'1': On completion of the current descriptor structure, the VALID bit of the descriptor's STATUS word is set to '0'. Default Value: Undefined

8.1.13 DMAC_DESCR0_PING_CTL (continued)

25 : 24	WAIT_FOR_DEACT	Specifies whether the data transfer engine should wait for the channel to be deactivated; i.e. the selected system trigger is not active. This field is used to synchronize the controller's data transfer(s) with the agent that generated the trigger. This field is ONLY used at the completion of an opcode. E.g., a FIFO indicates that it is empty and it needs a new data sample. The agent removes the trigger ONLY when the data sample has been written by the transfer engine AND received by the agent. Furthermore, the agent's trigger may be delayed by a few cycles before it reaches the DW/DMA controller. This field is used for level sensitive trigger, which reflect state (pulse sensitive triggers should have this field set to "0"). The wait cycles incurred by this field reduce DW/DMA controller performance. "0": Do not wait for de-activation (for pulse sensitive triggers). "1": Wait for up to 4 cycles. "2": Wait for up to 8 cycles. "3": Wait indefinitely. This option may result in DW/DMA controller lockup if the system trigger is not de-activated by the source agent. Default Value: Undefined
23	SRC_ADDR_INCR	Specifies whether the source location address is incremented by the SRC_TRANSFER_SIZE after each single data element transfer or not. '0': No increment, typically used for receive (RX) FIFO structures. '1': Increment, typically used for memory structures. Default Value: Undefined
22	SRC_TRANSFER_SIZE	Specifies the bus transfer size to the source location: '0': As specified by DATA_SIZE. '1': Word (32 bits). Distinguishing bus transfer size from data element size allows for source components with data elements that are smaller than their 32-bit bus interface width. E.g., an ADC source has a 32-bit bus transfer size, but only provides a 16-bit data element. Default Value: Undefined
21	DST_ADDR_INCR	Specifies whether the destination location address is incremented by the DST_TRANSFER_SIZE after each single data element transfer or not. '0' : No increment, typically used for transmit (TX) FIFO structures. '1': Increment, typically used for memory structures. Default Value: Undefined
20	DST_TRANSFER_SIZE	Specifies the bus transfer size to the destination location: '0': As specified by DATA_SIZE. '1': Word (32 bits). Distinguishing bus transfer size from data element size allows for destination components with data elements that are smaller than their 32-bit bus interface width. E.g., a DAC destination has a 32-bit bus transfer size, but only requires a 16-bit data element. Default Value: Undefined

8.1.13 DMAC_DESCR0_PING_CTL (continued)

17 : 16	DATA_SIZE	Specifies the data element size: "0": Byte (8 bits). "1": Halfword (16 bits). "2": Word (32 bits). DATA_SIZE, SRC_TRANSFER_SIZE and DST_TRANSFER_SIZE together determine how data elements are transferred. The following are the 9 legal settings: - DATA is 8 bit, SRC is 8 bit, DST is 8 bit - DATA is 8 bit, SRC is 32 bit (higher 24 bits are dropped), DST is 8 bit - DATA is 8 bit, SRC is 8 bit, DST is 32 bit (higher 24 bits are made "0") - DATA is 8 bit, SRC is 32 bit (higher 24 bits are dropped), DST is 32 bit (higher 24 bits are made "0") - DATA is 16 bit, SRC is 16 bit, DST is 16 bit - DATA is 16 bit, SRC is 32 bit (higher 16 bits are dropped), DST is 16 bit - DATA is 16 bit, SRC is 16 bit, DST is 32 bit (higher 16 bits are made "0") - DATA is 16 bit, SRC is 32 bit (higher 16 bits are dropped), DST is 32 bit (higher 16 bits are made "0") - DATA is 32 bit, SRC is 32 bit, DST is 32 bit Default Value: Undefined
15 : 0	DATA_NR	Number of data elements that are transferred by a single descriptor. In DW mode (OPCODE is "0") each trigger initiates the transfer of a single data element. This field specifies the source and/or destination buffer size in data elements: buffer size = DATA_NR+1. The buffer is typically associated to a memory structure. In DMA mode (OPCODE is "1" or "2") each trigger initiates the transfer of DATA_NR+1 data elements. Default Value: Undefined

8.1.14 DMAC_DESCR0_PING_STATUS

Ping status word

Address: 0x4010180C

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access	RW							
Name	CURR_DATA_NR [7:0]							
Bits	15	14	13	12	11	10	9	8
SW Access	RW							
HW Access	RW							
Name	CURR_DATA_NR [15:8]							
Bits	23	22	21	20	19	18	17	16
SW Access	None					RW		
HW Access	None					RW		
Name	None [23:19]					RESPONSE [18:16]		
Bits	31	30	29	28	27	26	25	24
SW Access	RW	None						
HW Access	RW	None						
Name	VALID	None [30:24]						

Bits	Name	Description
31	VALID	'0': Invalid, cannot be used for a data transfer. An attempt to use this descriptor for a data transfer will result in an INVALID_DESCR response code (and the interrupt cause bit is set to '1'). '1': Valid. Hardware set this field to '0' when a descriptor is done, but only if CONTROL.INV_DESCR is '1'. Software sets this field to '1' when a descriptor is initialized. Default Value: Undefined

8.1.14 DMAC_DESCR0_PING_STATUS (continued)

18 : 16	RESPONSE	Response code (the first two codes NO_ERROR and DONE are the result of normal behavior, the other codes are the result of erroneous behavior). "0"/NO_ERROR: No error. Setting this response does NOT set the interrupt cause bit to '1'. STATUS.VALID is NOT affected. CHi_CTL.ENABLED is NOT affected. CHi_CTL.PING_PONG is not updated. This response is used for an unused or not completed descriptor. Software should set the RESPONSE field to "0"/NO_ERROR during descriptor initialization. "1"/DONE: Descriptor is done (without errors). Setting this response sets the interrupt cause bit to '1' if CONTROL.SET_CAUSE is '1'. STATUS.VALID is set to '0' if CONTROL.INV_DESCR is '1'. CHi_CTL.ENABLED is NOT affected. CHi_CTL.PING_PONG is updated if CONTROL.FLIP_PING is '1'. "2"/SRC_BUS_ERROR: Bus error while loading data from the source location. Setting this response sets the interrupt cause bit to '1'. STATUS.VALID is set '0'. CHi_CTL.ENABLED is set to '0'. CHi_CTL.PING_PONG is not updated (it identifies the descriptor that caused the error). "3"/DST_BUS_ERROR: Bus error while storing data to the destination location. Setting this response sets the interrupt cause bit to '1'. STATUS.VALID is set '0'. CHi_CTL.ENABLED is set to '0'. CHi_CTL.PING_PONG is not updated (it identifies the descriptor that caused the error). "4"/SRC_MISAL: Misalignment of source address. This occurs on a 16-bit bus transfer that is not 2-byte aligned or on a 32-bit bus transfer that is not 4-byte aligned. Setting this response sets the interrupt cause bit to '1'. STATUS.VALID is set '0'. CHi_CTL.ENABLED is set to '0'. CHi_CTL.PING_PONG is not updated (it identifies the descriptor that caused the error). "5"/DST_MISAL: Misalignment of destination address. This occurs on a 16-bit bus transfer that is not 2-byte aligned or on a 32-bit bus transfer that is not 4-byte aligned. Setting this response sets the interrupt cause bit to '1'. STATUS.VALID is set '0'. CHi_CTL.ENABLED is set to '0'. CHi_CTL.PING_PONG is not updated (it identifies the descriptor that caused the error). "6"/INVALID_DESCR: Invalid descriptor (STATUS.VALID is '0'). This occurs when an activated channel has an invalid descriptor. CHi_CTL.ENABLED is set to '0'. CHi_CTL.PING_PONG is not updated (it identifies the descriptor that caused the error). Default Value: Undefined
15 : 0	CURR_DATA_NR	Specifies the index of the current data transfer. This value increases from 0 to CONTROL.DATA_NR. HW sets this field: - When a descriptor is done (RESPONSE is DONE), the field is set to "0" when PING_CTL.INV_DESCR is '0' and the field is set to PING_CTL.DATA_NR when PING_CTL.INV_DESCR is '1'. - When a descriptor is not done (RESPONSE is NO_ERROR), the field reflects the progress of a data transfer. - In case of erroneous behavior (RESPONSE is neither DONE or NO_ERROR), the field is not updated, but keeps its value to ease debugging. HW only modifies this field for an active descriptor (STATUS.VALID to be '1'). At descriptor initialization, SW should set this field to "0". This field allows software to read the progress of the data transfer. Note that SRC.ADDR and DST.ADDR represent base addresses and are not modified during data transfer. However, STATUS.CURR_DATA_NR is modified during data transfer and provides an offset wrt. the base addresses. Default Value: Undefined

8.1.15 DMAC_DESCR0_PONG_SRC

Pong source address

Address: 0x40101810

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access	RW							
Name	ADDR [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	RW							
HW Access	RW							
Name	ADDR [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	RW							
HW Access	RW							
Name	ADDR [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	RW							
HW Access	RW							
Name	ADDR [31:24]							

Bits	Name	Description
31 : 0	ADDR	See description of PING_SRC. Default Value: Undefined

8.1.16 DMAC_DESCR0_PONG_DST

Pong destination address

Address: 0x40101814

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access	RW							
Name	ADDR [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	RW							
HW Access	RW							
Name	ADDR [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	RW							
HW Access	RW							
Name	ADDR [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	RW							
HW Access	RW							
Name	ADDR [31:24]							

Bits	Name	Description
31 : 0	ADDR	See description of PING_DST. Default Value: Undefined

8.1.17 DMAC_DESCR0_PONG_CTL

Pong control word

Address: 0x40101818

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access	RW							
Name	DATA_NR [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	RW							
HW Access	RW							
Name	DATA_NR [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	RW	RW	RW	RW	None		RW	
HW Access	RW	RW	RW	RW	None		RW	
Name	SRC_ADDR_INCR	SRC_TRANSFER_SIZE	DST_ADDR_INCR	DST_TRANSFER_SIZE	None [19:18]		DATA_SIZE [17:16]	

Bits	31	30	29	28	27	26	25	24
SW Access	RW		RW	RW	RW	RW	RW	
HW Access	RW		RW	RW	RW	RW	RW	
Name	OPCODE [31:30]		FLIPPING	PREEMPTABLE	SET_CAUSE	INV_DESCRIPTOR	WAIT_FOR_DEACT [25:24]	

Bits	Name	Description
31 : 30	OPCODE	See description of PING_CTL. Default Value: Undefined
29	FLIPPING	See description of PING_CTL. Default Value: Undefined
28	PREEMPTABLE	See description of PING_CTL. Default Value: Undefined
27	SET_CAUSE	See description of PING_CTL. Default Value: Undefined
26	INV_DESCRIPTOR	See description of PING_CTL. Default Value: Undefined
25 : 24	WAIT_FOR_DEACT	See description of PING_CTL. Default Value: Undefined
23	SRC_ADDR_INCR	See description of PING_CTL. Default Value: Undefined

8.1.17 DMAC_DESCR0_PONG_CTL (continued)

22	SRC_TRANSFER_SIZE	See description of PING_CTL. Default Value: Undefined
21	DST_ADDR_INCR	See description of PING_CTL. Default Value: Undefined
20	DST_TRANSFER_SIZE	See description of PING_CTL. Default Value: Undefined
17 : 16	DATA_SIZE	See description of PING_CTL. Default Value: Undefined
15 : 0	DATA_NR	See description of PING_CTL. Default Value: Undefined

8.1.18 DMAC_DESCR0_PONG_STATUS

Pong status word

Address: 0x4010181C

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access	RW							
Name	CURR_DATA_NR [7:0]							
Bits	15	14	13	12	11	10	9	8
SW Access	RW							
HW Access	RW							
Name	CURR_DATA_NR [15:8]							
Bits	23	22	21	20	19	18	17	16
SW Access	None					RW		
HW Access	None					RW		
Name	None [23:19]					RESPONSE [18:16]		
Bits	31	30	29	28	27	26	25	24
SW Access	RW	None						
HW Access	RW	None						
Name	VALID	None [30:24]						

Bits	Name	Description
31	VALID	See description of PING_STATUS. Default Value: Undefined
18 : 16	RESPONSE	See description of PING_STATUS. Default Value: Undefined
15 : 0	CURR_DATA_NR	See description of PING_STATUS. Default Value: Undefined

9 SPC Interface (SPCIF) Registers



This section discusses the SPCIF registers. It lists all the registers in mapping tables, in address order.

9.1 Register Details

Register	Address	Description
SPCIF_GEOMETRY	0x40110000	Flash/NVL geometry information
SPCIF_INTR	0x401107F0	SPCIF interrupt request register
SPCIF_INTR_SET	0x401107F4	SPCIF interrupt set request register
SPCIF_INTR_MASK	0x401107F8	SPCIF interrupt mask register
SPCIF_INTR_MASKED	0x401107FC	SPCIF interrupt masked request register

9.1.1 SPCIF_GEOMETRY

Flash/NVL geometry information

Address: 0x40110000

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access	W							
Name	FLASH [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R		R					
HW Access	W		W					
Name	SFLASH [15:14]		FLASH [13:8]					

Bits	23	22	21	20	19	18	17	16
SW Access	R		R		R			
HW Access	W		W		W			
Name	FLASH_ROW [23:22]		NUM_FLASH [21:20]		SFLASH [19:16]			

Bits	31	30	29	28	27	26	25	24
SW Access	RW		None					
HW Access			None					
Name	DE_CPD_LP		None [30:24]					

Bits	Name	Description
31	DE_CPD_LP	0': SRAM busy wait loop has not been copied. '1': Busy wait loop has been written into SRAM. Default Value: 0
23 : 22	FLASH_ROW	Page size in 64 Byte multiples (chip dependent): "0": 64 byte "1": 128 byte "2": 192 byte "3": 256 byte The page size is used to determine the number of Bytes in a page for Flash page based operations (e.g. PGM_PAGE). Note: the field name FLASH_ROW is misleading, as this field specifies the number of Bytes in a page, rather than the number of Bytes in a row. In a single plane flash macro architecture, a page consists of a single row. However, in a multi plane flash macro architecture, a page consists of multiple rows from different planes. Default Value: Undefined

9.1.1 SPCIF_GEOMETRY (continued)

21 : 20	NUM_FLASH	Number of flash macros (chip dependent): "0": 1 flash macro "1": 2 flash macros "2": 3 flash macros "3": 4 flash macros Default Value: Undefined
19 : 14	SFLASH	Supervisory flash capacity in 256 Byte multiples (chip dependent). If multiple flash macros are present, this field provides the supervisory flash capacity of all flash macros together: "0": 256 Bytes. "1": 2*256 Bytes. ... "63": 64*256 Bytes. Default Value: Undefined
13 : 0	FLASH	Regular flash capacity in 256 Byte multiples (chip dependent). If multiple flash macros are present, this field provides the flash capacity of all flash macros together: "0": 256 Bytes. "1": 2*256 Bytes. ... "16383": 16384*256 Bytes. Default Value: Undefined

9.1.2 SPCIF_INTR

SPCIF interrupt request register

Address: 0x401107F0

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None							RW1C
HW Access	None							RW1S
Name	None [7:1]							TIMER

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
0	TIMER	Timer counter value reaches "0". Set to '1', when event is detected. Write INTR field with '1', to clear bit. Write INTR_SET field with '1', to set bit. Default Value: 0

9.1.3 SPCIF_INTR_SET

SPCIF interrupt set request register

Address: 0x401107F4

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None							RW1S
HW Access	None							A
Name	None [7:1]							TIMER

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
0	TIMER	Write INTR_SET field with '1' to set corresponding INTR field. Default Value: 0

9.1.4 SPCIF_INTR_MASK

SPCIF interrupt mask register

Address: 0x401107F8

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None							RW
HW Access	None							R
Name	None [7:1]							TIMER

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
0	TIMER	Mask for corresponding field in INTR register. Default Value: 0

9.1.5 SPCIF_INTR_MASKED

SPCIF interrupt masked request register

Address: 0x401107FC

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None							R
HW Access	None							W
Name	None [7:1]							TIMER

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
0	TIMER	Logical and of corresponding request and mask fields. Default Value: 0

10 Timer, Counter, PWM (TCPWM) Registers



This section discusses the TCPWM registers. It lists all the registers in mapping tables, in address order.

10.1 Register Details

Register	Address	Description
TCPWM_CTRL	0x40200000	TCPWM control register 0.
TCPWM_CMD	0x40200008	TCPWM command register.
TCPWM_INTR_CAUSE	0x4020000C	TCPWM Counter interrupt cause register.
TCPWM_CNT0_CTRL	0x40200100	Counter control register
TCPWM_CNT0_STATUS	0x40200104	Counter status register
TCPWM_CNT0_COUNTER	0x40200108	Counter count register
TCPWM_CNT0_CC	0x4020010C	Counter compare/capture register
TCPWM_CNT0_CC_BUFF	0x40200110	Counter buffered compare/capture register
TCPWM_CNT0_PERIOD	0x40200114	Counter period register
TCPWM_CNT0_PERIOD_BUFF	0x40200118	Counter buffered period register
TCPWM_CNT0_TR_CTRL0	0x40200120	Counter trigger control register 0
TCPWM_CNT0_TR_CTRL1	0x40200124	Counter trigger control register 1
TCPWM_CNT0_TR_CTRL2	0x40200128	Counter trigger control register 2
TCPWM_CNT0_INTR	0x40200130	Interrupt request register.
TCPWM_CNT0_INTR_SET	0x40200134	Interrupt set request register.
TCPWM_CNT0_INTR_MASK	0x40200138	Interrupt mask register.
TCPWM_CNT0_INTR_MASKED	0x4020013C	Interrupt masked request register
TCPWM_CNT1_CTRL	0x40200140	Counter control register. See TCPWM_CNT0_CTRL for the details of bit fields.
TCPWM_CNT1_STATUS	0x40200144	Counter status register. See TCPWM_CNT0_STATUS for the details of bit fields.
TCPWM_CNT1_COUNTER	0x40200148	Counter count register. See TCPWM_CNT0_COUNTER for the details of bit fields.
TCPWM_CNT1_CC	0x4020014C	Counter compare/capture register. See TCPWM_CNT0_CC for the details of bit fields.
TCPWM_CNT1_CC_BUFF	0x40200150	Counter buffered compare/capture register. See TCPWM_CNT0_CC_BUFF for the details of bit fields.
TCPWM_CNT1_PERIOD	0x40200154	Counter period register. See TCPWM_CNT0_PERIOD for the details of bit fields.
TCPWM_CNT1_PERIOD_BUFF	0x40200158	Counter buffered period register. See TCPWM_CNT0_PERIOD_BUFF for the details of bit fields.
TCPWM_CNT1_TR_CTRL0	0x40200160	Counter trigger control register 0. See TCPWM_CNT0_TR_CTRL0 for the details of bit fields.
TCPWM_CNT1_TR_CTRL1	0x40200164	Counter trigger control register 1. See TCPWM_CNT0_TR_CTRL1 for the details of bit fields.

Register	Address	Description
TCPWM_CNT1_TR_CTRL2	0x40200168	Counter trigger control register 2. See TCPWM_CNT0_TR_CTRL2 for the details of bit fields.
TCPWM_CNT1_INTR	0x40200170	Interrupt request register.. See TCPWM_CNT0_INTR for the details of bit fields.
TCPWM_CNT1_INTR_SET	0x40200174	Interrupt set request register.. See TCPWM_CNT0_INTR_SET for the details of bit fields.
TCPWM_CNT1_INTR_MASK	0x40200178	Interrupt mask register.. See TCPWM_CNT0_INTR_MASK for the details of bit fields.
TCPWM_CNT1_INTR_MASKED	0x4020017C	Interrupt masked request register. See TCPWM_CNT0_INTR_MASKED for the details of bit fields.
TCPWM_CNT2_CTRL	0x40200180	Counter control register. See TCPWM_CNT0_CTRL for the details of bit fields.
TCPWM_CNT2_STATUS	0x40200184	Counter status register. See TCPWM_CNT0_STATUS for the details of bit fields.
TCPWM_CNT2_COUNTER	0x40200188	Counter count register. See TCPWM_CNT0_COUNTER for the details of bit fields.
TCPWM_CNT2_CC	0x4020018C	Counter compare/capture register. See TCPWM_CNT0_CC for the details of bit fields.
TCPWM_CNT2_CC_BUFF	0x40200190	Counter buffered compare/capture register. See TCPWM_CNT0_CC_BUFF for the details of bit fields.
TCPWM_CNT2_PERIOD	0x40200194	Counter period register. See TCPWM_CNT0_PERIOD for the details of bit fields.
TCPWM_CNT2_PERIOD_BUFF	0x40200198	Counter buffered period register. See TCPWM_CNT0_PERIOD_BUFF for the details of bit fields.
TCPWM_CNT2_TR_CTRL0	0x402001A0	Counter trigger control register 0. See TCPWM_CNT0_TR_CTRL0 for the details of bit fields.
TCPWM_CNT2_TR_CTRL1	0x402001A4	Counter trigger control register 1. See TCPWM_CNT0_TR_CTRL1 for the details of bit fields.
TCPWM_CNT2_TR_CTRL2	0x402001A8	Counter trigger control register 2. See TCPWM_CNT0_TR_CTRL2 for the details of bit fields.
TCPWM_CNT2_INTR	0x402001B0	Interrupt request register.. See TCPWM_CNT0_INTR for the details of bit fields.
TCPWM_CNT2_INTR_SET	0x402001B4	Interrupt set request register.. See TCPWM_CNT0_INTR_SET for the details of bit fields.
TCPWM_CNT2_INTR_MASK	0x402001B8	Interrupt mask register.. See TCPWM_CNT0_INTR_MASK for the details of bit fields.
TCPWM_CNT2_INTR_MASKED	0x402001BC	Interrupt masked request register. See TCPWM_CNT0_INTR_MASKED for the details of bit fields.
TCPWM_CNT3_CTRL	0x402001C0	Counter control register. See TCPWM_CNT0_CTRL for the details of bit fields.
TCPWM_CNT3_STATUS	0x402001C4	Counter status register. See TCPWM_CNT0_STATUS for the details of bit fields.
TCPWM_CNT3_COUNTER	0x402001C8	Counter count register. See TCPWM_CNT0_COUNTER for the details of bit fields.
TCPWM_CNT3_CC	0x402001CC	Counter compare/capture register. See TCPWM_CNT0_CC for the details of bit fields.
TCPWM_CNT3_CC_BUFF	0x402001D0	Counter buffered compare/capture register. See TCPWM_CNT0_CC_BUFF for the details of bit fields.
TCPWM_CNT3_PERIOD	0x402001D4	Counter period register. See TCPWM_CNT0_PERIOD for the details of bit fields.
TCPWM_CNT3_PERIOD_BUFF	0x402001D8	Counter buffered period register. See TCPWM_CNT0_PERIOD_BUFF for the details of bit fields.
TCPWM_CNT3_TR_CTRL0	0x402001E0	Counter trigger control register 0. See TCPWM_CNT0_TR_CTRL0 for the details of bit fields.
TCPWM_CNT3_TR_CTRL1	0x402001E4	Counter trigger control register 1. See TCPWM_CNT0_TR_CTRL1 for the details of bit fields.
TCPWM_CNT3_TR_CTRL2	0x402001E8	Counter trigger control register 2. See TCPWM_CNT0_TR_CTRL2 for the details of bit fields.
TCPWM_CNT3_INTR	0x402001F0	Interrupt request register.. See TCPWM_CNT0_INTR for the details of bit fields.
TCPWM_CNT3_INTR_SET	0x402001F4	Interrupt set request register.. See TCPWM_CNT0_INTR_SET for the details of bit fields.
TCPWM_CNT3_INTR_MASK	0x402001F8	Interrupt mask register.. See TCPWM_CNT0_INTR_MASK for the details of bit fields.
TCPWM_CNT3_INTR_MASKED	0x402001FC	Interrupt masked request register. See TCPWM_CNT0_INTR_MASKED for the details of bit fields.
TCPWM_CNT4_CTRL	0x40200200	Counter control register. See TCPWM_CNT0_CTRL for the details of bit fields.
TCPWM_CNT4_STATUS	0x40200204	Counter status register. See TCPWM_CNT0_STATUS for the details of bit fields.
TCPWM_CNT4_COUNTER	0x40200208	Counter count register. See TCPWM_CNT0_COUNTER for the details of bit fields.
TCPWM_CNT4_CC	0x4020020C	Counter compare/capture register. See TCPWM_CNT0_CC for the details of bit fields.

Register	Address	Description
TCPWM_CNT4_CC_BUFF	0x40200210	Counter buffered compare/capture register. See TCPWM_CNT0_PERIOD_BUFF for the details of bit fields.
TCPWM_CNT4_PERIOD	0x40200214	Counter period register. See TCPWM_CNT0_PERIOD for the details of bit fields.
TCPWM_CNT4_PERIOD_BUFF	0x40200218	Counter buffered period register. See TCPWM_CNT0_PERIOD_BUFF for the details of bit fields.
TCPWM_CNT4_TR_CTRL0	0x40200220	Counter trigger control register 0. See TCPWM_CNT0_TR_CTRL0 for the details of bit fields.
TCPWM_CNT4_TR_CTRL1	0x40200224	Counter trigger control register 1. See TCPWM_CNT0_TR_CTRL1 for the details of bit fields.
TCPWM_CNT4_TR_CTRL2	0x40200228	Counter trigger control register 2. See TCPWM_CNT0_TR_CTRL2 for the details of bit fields.
TCPWM_CNT4_INTR	0x40200230	Interrupt request register.. See TCPWM_CNT0_INTR for the details of bit fields.
TCPWM_CNT4_INTR_SET	0x40200234	Interrupt set request register.. See TCPWM_CNT0_INTR_SET for the details of bit fields.
TCPWM_CNT4_INTR_MASK	0x40200238	Interrupt mask register.. See TCPWM_CNT0_INTR_MASK for the details of bit fields.
TCPWM_CNT4_INTR_MASKED	0x4020023C	Interrupt masked request register. See TCPWM_CNT0_INTR_MASKED for the details of bit fields.
TCPWM_CNT5_CTRL	0x40200240	Counter control register. See TCPWM_CNT0_CTRL for the details of bit fields.
TCPWM_CNT5_STATUS	0x40200244	Counter status register. See TCPWM_CNT0_STATUS for the details of bit fields.
TCPWM_CNT5_COUNTER	0x40200248	Counter count register. See TCPWM_CNT0_COUNTER for the details of bit fields.
TCPWM_CNT5_CC	0x4020024C	Counter compare/capture register. See TCPWM_CNT0_CC for the details of bit fields.
TCPWM_CNT5_CC_BUFF	0x40200250	Counter buffered compare/capture register. See TCPWM_CNT0_CC_BUFF for the details of bit fields.
TCPWM_CNT5_PERIOD	0x40200254	Counter period register. See TCPWM_CNT0_PERIOD for the details of bit fields.
TCPWM_CNT5_PERIOD_BUFF	0x40200258	Counter buffered period register. See TCPWM_CNT0_PERIOD_BUFF for the details of bit fields.
TCPWM_CNT5_TR_CTRL0	0x40200260	Counter trigger control register 0. See TCPWM_CNT0_TR_CTRL0 for the details of bit fields.
TCPWM_CNT5_TR_CTRL1	0x40200264	Counter trigger control register 1. See TCPWM_CNT0_TR_CTRL1 for the details of bit fields.
TCPWM_CNT5_TR_CTRL2	0x40200268	Counter trigger control register 2. See TCPWM_CNT0_TR_CTRL2 for the details of bit fields.
TCPWM_CNT5_INTR	0x40200270	Interrupt request register.. See TCPWM_CNT0_INTR for the details of bit fields.
TCPWM_CNT5_INTR_SET	0x40200274	Interrupt set request register.. See TCPWM_CNT0_INTR_SET for the details of bit fields.
TCPWM_CNT5_INTR_MASK	0x40200278	Interrupt mask register.. See TCPWM_CNT0_INTR_MASK for the details of bit fields.
TCPWM_CNT5_INTR_MASKED	0x4020027C	Interrupt masked request register. See TCPWM_CNT0_INTR_MASKED for the details of bit fields.
TCPWM_CNT6_CTRL	0x40200280	Counter control register. See TCPWM_CNT0_CTRL for the details of bit fields.
TCPWM_CNT6_STATUS	0x40200284	Counter status register. See TCPWM_CNT0_STATUS for the details of bit fields.
TCPWM_CNT6_COUNTER	0x40200288	Counter count register. See TCPWM_CNT0_COUNTER for the details of bit fields.
TCPWM_CNT6_CC	0x4020028C	Counter compare/capture register. See TCPWM_CNT0_CC for the details of bit fields.
TCPWM_CNT6_CC_BUFF	0x40200290	Counter buffered compare/capture register. See TCPWM_CNT0_CC_BUFF for the details of bit fields.
TCPWM_CNT6_PERIOD	0x40200294	Counter period register. See TCPWM_CNT0_PERIOD for the details of bit fields.
TCPWM_CNT6_PERIOD_BUFF	0x40200298	Counter buffered period register. See TCPWM_CNT0_PERIOD_BUFF for the details of bit fields.
TCPWM_CNT6_TR_CTRL0	0x402002A0	Counter trigger control register 0. See TCPWM_CNT0_TR_CTRL0 for the details of bit fields.
TCPWM_CNT6_TR_CTRL1	0x402002A4	Counter trigger control register 1. See TCPWM_CNT0_TR_CTRL1 for the details of bit fields.
TCPWM_CNT6_TR_CTRL2	0x402002A8	Counter trigger control register 2. See TCPWM_CNT0_TR_CTRL2 for the details of bit fields.
TCPWM_CNT6_INTR	0x402002B0	Interrupt request register.. See TCPWM_CNT0_INTR for the details of bit fields.

Register	Address	Description
TCPWM_CNT6_INTR_SET	0x402002B4	Interrupt set request register.. See TCPWM_CNT0_INTR_SET for the details of bit fields.
TCPWM_CNT6_INTR_MASK	0x402002B8	Interrupt mask register.. See TCPWM_CNT0_INTR_MASK for the details of bit fields.
TCPWM_CNT6_INTR_MASKED	0x402002BC	Interrupt masked request register. See TCPWM_CNT0_INTR_MASKED for the details of bit fields.
TCPWM_CNT7_CTRL	0x402002C0	Counter control register. See TCPWM_CNT0_CTRL for the details of bit fields.
TCPWM_CNT7_STATUS	0x402002C4	Counter status register. See TCPWM_CNT0_STATUS for the details of bit fields.
TCPWM_CNT7_COUNTER	0x402002C8	Counter count register. See TCPWM_CNT0_COUNTER for the details of bit fields.
TCPWM_CNT7_CC	0x402002CC	Counter compare/capture register. See TCPWM_CNT0_CC for the details of bit fields.
TCPWM_CNT7_CC_BUFF	0x402002D0	Counter buffered compare/capture register. See TCPWM_CNT0_PERIOD_BUFF for the details of bit fields.
TCPWM_CNT7_PERIOD	0x402002D4	Counter period register. See TCPWM_CNT0_PERIOD for the details of bit fields.
TCPWM_CNT7_PERIOD_BUFF	0x402002D8	Counter buffered period register. See TCPWM_CNT0_PERIOD_BUFF for the details of bit fields.
TCPWM_CNT7_TR_CTRL0	0x402002E0	Counter trigger control register 0. See TCPWM_CNT0_TR_CTRL0 for the details of bit fields.
TCPWM_CNT7_TR_CTRL1	0x402002E4	Counter trigger control register 1. See TCPWM_CNT0_TR_CTRL1 for the details of bit fields.
TCPWM_CNT7_TR_CTRL2	0x402002E8	Counter trigger control register 2. See TCPWM_CNT0_TR_CTRL2 for the details of bit fields.
TCPWM_CNT7_INTR	0x402002F0	Interrupt request register.. See TCPWM_CNT0_INTR for the details of bit fields.
TCPWM_CNT7_INTR_SET	0x402002F4	Interrupt set request register.. See TCPWM_CNT0_INTR_SET for the details of bit fields.
TCPWM_CNT7_INTR_MASK	0x402002F8	Interrupt mask register.. See TCPWM_CNT0_INTR_MASK for the details of bit fields.
TCPWM_CNT7_INTR_MASKED	0x402002FC	Interrupt masked request register. See TCPWM_CNT0_INTR_MASKED for the details of bit fields.

10.1.1 TCPWM_CTRL

TCPWM control register 0.

Address: 0x40200000

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access	R							
Name	COUNTER_ENABLED [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
7 : 0	COUNTER_ENABLED	Counter enables for counters 0 up to CNT_NR-1. '0': counter disabled. '1': counter enabled. Counter static configuration information (e.g. CTRL.MODE, all TR_CTRL0, TR_CTRL1, and TR_CTRL2 register fields) should only be modified when the counter is disabled. When a counter is disabled, command and status information associated to the counter is cleared by HW, this includes: - the associated counter triggers in the CMD register are set to '0'. - the counter's interrupt cause fields in counter's INTR register. - the counter's status fields in counter's STATUS register.. - the counter's trigger outputs ("tr_overflow", "tr_underflow" and "tr_compare_match"). - the counter's line outputs ("line_out" and "line_compl_out"). Default Value: 0

10.1.2 TCPWM_CMD

TCPWM command register.

Address: 0x40200008

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW1S							
HW Access	RW1C							
Name	COUNTER_CAPTURE [7:0]							
Bits	15	14	13	12	11	10	9	8
SW Access	RW1S							
HW Access	RW1C							
Name	COUNTER_RELOAD [15:8]							
Bits	23	22	21	20	19	18	17	16
SW Access	RW1S							
HW Access	RW1C							
Name	COUNTER_STOP [23:16]							
Bits	31	30	29	28	27	26	25	24
SW Access	RW1S							
HW Access	RW1C							
Name	COUNTER_START [31:24]							

Bits	Name	Description
31 : 24	COUNTER_START	Counters SW start trigger. For HW behavior, see COUNTER_CAPTURE field. Default Value: 0
23 : 16	COUNTER_STOP	Counters SW stop trigger. For HW behavior, see COUNTER_CAPTURE field. Default Value: 0
15 : 8	COUNTER_RELOAD	Counters SW reload trigger. For HW behavior, see COUNTER_CAPTURE field. Default Value: 0
7 : 0	COUNTER_CAPTURE	Counters SW capture trigger. When written with '1', a capture trigger is generated and the HW sets the field to '0' when the SW trigger has taken effect. It should be noted that the HW operates on the counter frequency. If the counter is disabled through CTRL.COUNTER_ENABLED, the field is immediately set to '0'. Default Value: 0

10.1.3 TCPWM_INTR_CAUSE

TCPWM Counter interrupt cause register.

Address: 0x4020000C

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access	W							
Name	COUNTER_INT [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
7 : 0	COUNTER_INT	Counters interrupt signal active. If the counter is disabled through CTRL.COUNTER_ENABLED, the associated interrupt field is immediately set to '0'. Default Value: 0

10.1.4 TCPWM_CNT0_CTRL

Counter control register

Address: 0x40200100

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None				RW	RW	RW	RW
HW Access	None				R	R	R	R
Name	None [7:4]				PW-M_STOP_ON_KILL	PWM_SYN-C_KILL	AUTO_RE-LOAD_PERIOD	AUTO_RE-LOAD_CC

Bits	15	14	13	12	11	10	9	8
SW Access	RW							
HW Access	R							
Name	GENERIC [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None		RW		None	RW	RW	
HW Access	None		R		None	R	R	
Name	None [23:22]		QUADRATURE_MODE [21:20]		None	ONE_SHOT	UP_DOWN_MODE [17:16]	

Bits	31	30	29	28	27	26	25	24
SW Access	None					RW		
HW Access	None					R		
Name	None [31:27]					MODE [26:24]		

Bits	Name	Description
26 : 24	MODE	Counter mode. Default Value: 0
0x0: TIMER :		
Timer mode		
0x2: CAPTURE :		
Capture mode		
0x3: QUAD :		
Quadrature encoding mode		

10.1.4 TCPWM_CNT0_CTRL (continued)

0x4: PWM :

Pulse width modulation (PWM) mode

0x5: PWM_DT :

PWM with deadtime insertion mode

0x6: PWM_PR :

Pseudo random pulse width modulation

21 : 20 QUADRATURE_MODE

In QUAD mode selects quadrature encoding mode (X1/X2/X4).
 In PWM, PWM_DT and PWM_PR modes, these two bits can be used to invert "dt_line_out" and "dt_line_compl_out". Inversion is the last step in generation of "dt_line_out" and "dt_line_compl_out"; i.e. a disabled output line "dt_line_out" has the value QUADRATURE_MODE[0] and a disabled output line "dt_line_compl_out" has the value QUADRATURE_MODE[1].
 Default Value: 0

0x0: X1 :

X1 encoding (QUAD mode)

0x1: X2 :

X2 encoding (QUAD mode)

0x2: X4 :

X4 encoding (QUAD mode)

0x1: INV_OUT :

When bit 0 is '1', QUADRATURE_MODE[0] inverts "dt_line_out" (PWM/PWM_DT modes)

0x2: INV_COMPL_OUT :

When bit 1 is '1', QUADRATURE_MODE[1] inverts "dt_line_compl_out" (PWM/PWM_DT modes)

18 ONE_SHOT

When '0', counter runs continuous. When '1', counter is turned off by hardware when a terminal count event is generated.
 Default Value: 0

17 : 16 UP_DOWN_MODE

Determines counter direction.
 Default Value: 0

0x0: COUNT_UP :

Count up (to PERIOD). An overflow event is generated when the counter reaches PERIOD. A terminal count event is generated when the counter reaches PERIOD.

10.1.4 TCPWM_CNT0_CTRL (continued)

0x1: COUNT_DOWN :

Count down (to "0"). An underflow event is generated when the counter reaches "0". A terminal count event is generated when the counter reaches "0".

0x2: COUNT_UPDN1 :

Count up (to PERIOD), then count down (to "0"). An overflow event is generated when the counter reaches PERIOD. An underflow event is generated when the counter reaches "0". A terminal count event is generated when the counter reaches "0".

0x3: COUNT_UPDN2 :

Count up (to PERIOD), then count down (to "0"). An overflow event is generated when the counter reaches PERIOD. An underflow event is generated when the counter reaches "0". A terminal count event is generated when the counter reaches "0" AND when the counter reaches PERIOD (this counter direction can be used for PWM functionality with asymmetrical updates).

15 : 8 GENERIC

Generic 8-bit control field. In PWM_DT mode, this field is used to determine the dead time: amount of dead time cycles in the counter clock domain. In all other modes, the lower 3 bits of this field determine pre-scaling of the selected counter clock.
Default Value: 0

0x0: DIVBY1 :

Divide by 1 (other-than-PWM_DT mode)

0x1: DIVBY2 :

Divide by 2 (other-than-PWM_DT mode)

0x2: DIVBY4 :

Divide by 4 (other-than-PWM_DT mode)

0x3: DIVBY8 :

Divide by 8 (other-than-PWM_DT mode)

0x4: DIVBY16 :

Divide by 16 (other-than-PWM_DT mode)

0x5: DIVBY32 :

Divide by 32 (other-than-PWM_DT mode)

0x6: DIVBY64 :

Divide by 64 (other-than-PWM_DT mode)

10.1.4 TCPWM_CNT0_CTRL (continued)

0x7: DIVBY128 :

Divide by 128 (other-than-PWM_DT mode)

3	PWM_STOP_ON_KILL	Specifies whether the counter stops on a kill events: '0': kill event does NOT stop counter. '1': kill event stops counter. This field has a function in PWM, PWM_DT and PWM_PR modes only. Default Value: 0
2	PWM_SYNC_KILL	Specifies asynchronous/synchronous kill behavior: '1': synchronous kill mode: the kill event disables the "dt_line_out" and "dt_line_compl_out" signals till the next terminal count event (synchronous kill). In synchronous kill mode, STOP_EDGE should be RISING_EDGE. '0': asynchronous kill mode: the kill event only disables the "dt_line_out" and "dt_line_compl_out" signals when present. In asynchronous kill mode, STOP_EDGE should be NO_EDGE_DET. This field has a function in PWM and PWM_DT modes only. This field is only used when PWM_STOP_ON_KILL is '0'. Default Value: 0
1	AUTO_RELOAD_PERIOD	Specifies switching of the PERIOD and buffered PERIOD values. This field has a function in PWM, PWM_DT and PWM_PR modes. '0': never switch. '1': switch on a terminal count event with and actively pending siwtch event. Default Value: 0
0	AUTO_RELOAD_CC	Specifies switching of the CC and buffered CC values. This field has a function in TIMER, PWM, PWM_DT and PWM_PR modes. Timer mode: '0': never switch. '1': switch on a compare match event. PWM, PWM_DT, PWM_PR modes: '0': never switch. '1': switch on a terminal count event with an actively pending switch event. Default Value: 0

10.1.5 TCPWM_CNT0_STATUS

Counter status register

Address: 0x40200104

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None							R
HW Access	None							RW
Name	None [7:1]							DOWN

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access	RW							
Name	GENERIC [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R	None						
HW Access	RW	None						
Name	RUNNING	None [30:24]						

Bits	Name	Description
31	RUNNING	When '0', the counter is NOT running. When '1', the counter is running. Default Value: 0
15 : 8	GENERIC	Generic 8-bit counter field. In PWM_DT mode, this counter is used for dead time insertion. In all other modes, this counter is used for pre-scaling the selected counter clock. PWM_DT mode can NOT use prescaled clock functionality. Default Value: 0
0	DOWN	When '0', counter is counting up. When '1', counter is counting down. In QUAD mode, this field indicates the direction of the latest counter change: '0' when last incremented and '1' when last decremented. Default Value: 0

10.1.6 TCPWM_CNT0_COUNTER

Counter count register

Address: 0x40200108

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access	RW							
Name	COUNTER [7:0]							
Bits	15	14	13	12	11	10	9	8
SW Access	RW							
HW Access	RW							
Name	COUNTER [15:8]							
Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							
Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
15 : 0	COUNTER	16-bit counter value. It is advised to not write to this field when the counter is running. Default Value: 0

10.1.7 TCPWM_CNT0_CC

Counter compare/capture register

Address: 0x4020010C

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access	RW							
Name	CC [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	RW							
HW Access	RW							
Name	CC [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
15 : 0	CC	In CAPTURE mode, captures the counter value. In other modes, compared to counter value. Default Value: 65535

10.1.8 TCPWM_CNT0_CC_BUFF

Counter buffered compare/capture register

Address: 0x40200110

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access	RW							
Name	CC [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	RW							
HW Access	RW							
Name	CC [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
15 : 0	CC	Additional buffer for counter CC register. Default Value: 65535

10.1.9 TCPWM_CNT0_PERIOD

Counter period register

Address: 0x40200114

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access	RW							
Name	PERIOD [7:0]							
Bits	15	14	13	12	11	10	9	8
SW Access	RW							
HW Access	RW							
Name	PERIOD [15:8]							
Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							
Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
15 : 0	PERIOD	Period value: upper value of the counter. When the counter should count for n cycles, this field should be set to n-1. Default Value: 65535

10.1.10 TCPWM_CNT0_PERIOD_BUFF

Counter buffered period register

Address: 0x40200118

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access	RW							
Name	PERIOD [7:0]							
Bits	15	14	13	12	11	10	9	8
SW Access	RW							
HW Access	RW							
Name	PERIOD [15:8]							
Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							
Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
15 : 0	PERIOD	Additional buffer for counter PERIOD register. Default Value: 65535

10.1.11 TCPWM_CNT0_TR_CTRL0

Counter trigger control register 0

Address: 0x40200120

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW				RW			
HW Access	R				R			
Name	COUNT_SEL [7:4]				CAPTURE_SEL [3:0]			

Bits	15	14	13	12	11	10	9	8
SW Access	RW				RW			
HW Access	R				R			
Name	STOP_SEL [15:12]				RELOAD_SEL [11:8]			

Bits	23	22	21	20	19	18	17	16
SW Access	None				RW			
HW Access	None				R			
Name	None [23:20]				START_SEL [19:16]			

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
19 : 16	START_SEL	Selects one of the 16 input triggers as a start trigger. In QUAD mode, this is the second phase (phi B). Default Value: 0
15 : 12	STOP_SEL	Selects one of the 16 input triggers as a stop trigger. In PWM, PWM_DT and PWM_PR modes, this is the kill trigger. In these modes, the kill trigger is used to either temporarily block the PWM outputs (PWM_STOP_ON_KILL is '0') or stop the functionality (PWM_STOP_ON_KILL is '1'). For the PWM and PWM_DT modes, the blocking of the output signals can be asynchronous (STOP_EDGE should be NO_EDGE_DET) in which case the blocking is as long as the trigger is '1' or synchronous (STOP_EDGE should be RISING_EDGE) in which case it extends till the next terminal count event. Default Value: 0
11 : 8	RELOAD_SEL	Selects one of the 16 input triggers as a reload trigger. In QUAD mode, this is the index or revolution pulse. In this mode, it will update the counter with the value in the TCPWM_CNTn_PERIOD register. Default Value: 0
7 : 4	COUNT_SEL	Selects one of the 16 input triggers as a count trigger. In QUAD mode, this is the first phase (phi A). Default setting selects input trigger 1, which is always '1'. Default Value: 1

10.1.11 TCPWM_CNT0_TR_CTRL0 (continued)

3 : 0	CAPTURE_SEL	Selects one of the 16 input triggers as a capture trigger. Input trigger 0 is always '0' and input trigger is always '1'. In the PWM, PWM_DT and PWM_PR modes this trigger is used to switch the values if the compare and period registers with their buffer counterparts. Default Value: 0
-------	-------------	---

10.1.12 TCPWM_CNT0_TR_CTRL1

Counter trigger control register 1

Address: 0x40200124

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW		RW		RW		RW	
HW Access	R		R		R		R	
Name	STOP_EDGE [7:6]		RELOAD_EDGE [5:4]		COUNT_EDGE [3:2]		CAPTURE_EDGE [1:0]	

Bits	15	14	13	12	11	10	9	8
SW Access	None						RW	
HW Access	None						R	
Name	None [15:10]						START_EDGE [9:8]	

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
9 : 8	START_EDGE	A start event will start the counter; i.e. the counter will become running. Starting does NOT enable the counter. A start event will not initialize the counter whereas the reload event does. Default Value: 3 0x0: RISING_EDGE : Rising edge. Any rising edge generates an event. 0x1: FALLING_EDGE : Falling edge. Any falling edge generates an event. 0x2: BOTH_EDGES : Rising AND falling edge. Any odd amount of edges generates an event.

10.1.12 TCPWM_CNT0_TR_CTRL1 (continued)

		0x3: NO_EDGE_DET : No edge detection, use trigger as is.
7 : 6	STOP_EDGE	A stop event, will stop the counter; i.e. it will no longer be running. Stopping will NOT disable the counter. Default Value: 3 0x0: RISING_EDGE : Rising edge. Any rising edge generates an event. 0x1: FALLING_EDGE : Falling edge. Any falling edge generates an event. 0x2: BOTH_EDGES : Rising AND falling edge. Any odd amount of edges generates an event. 0x3: NO_EDGE_DET : No edge detection, use trigger as is.
5 : 4	RELOAD_EDGE	A reload event will initialize the counter. When counting up, the counter is initialized to "0". When counting down, the counter is initialized with PERIOD. Default Value: 3 0x0: RISING_EDGE : Rising edge. Any rising edge generates an event. 0x1: FALLING_EDGE : Falling edge. Any falling edge generates an event. 0x2: BOTH_EDGES : Rising AND falling edge. Any odd amount of edges generates an event. 0x3: NO_EDGE_DET : No edge detection, use trigger as is.
3 : 2	COUNT_EDGE	A counter event will increase or decrease the counter by '1'. Default Value: 3 0x0: RISING_EDGE : Rising edge. Any rising edge generates an event.

10.1.12 TCPWM_CNT0_TR_CTRL1 (continued)

0x1: FALLING_EDGE :

Falling edge. Any falling edge generates an event.

0x2: BOTH_EDGES :

Rising AND falling edge. Any odd amount of edges generates an event.

0x3: NO_EDGE_DET :

No edge detection, use trigger as is.

1 : 0 CAPTURE_EDGE

A capture event will copy the counter value into the CC register.
 Default Value: 3

0x0: RISING_EDGE :

Rising edge. Any rising edge generates an event.

0x1: FALLING_EDGE :

Falling edge. Any falling edge generates an event.

0x2: BOTH_EDGES :

Rising AND falling edge. Any odd amount of edges generates an event.

0x3: NO_EDGE_DET :

No edge detection, use trigger as is.

10.1.13 TCPWM_CNT0_TR_CTRL2

Counter trigger control register 2

Address: 0x40200128

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None		RW		RW		RW	
HW Access	None		R		R		R	
Name	None [7:6]		UNDERFLOW_MODE [5:4]		OVERFLOW_MODE [3:2]		CC_MATCH_MODE [1:0]	

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
5 : 4	UNDERFLOW_MODE	Determines the effect of a counter underflow event (COUNTER reaches "0") on the "line_out" output signals. Default Value: 3 0x0: SET : Set to '1' 0x1: CLEAR : Set to '0' 0x2: INVERT : Invert

10.1.13 TCPWM_CNT0_TR_CTRL2 (continued)

		0x3: NO_CHANGE : No Change
3 : 2	OVERFLOW_MODE	Determines the effect of a counter overflow event (COUNTER reaches PERIOD) on the "line_out" output signals. Default Value: 3 0x0: SET : Set to '1' 0x1: CLEAR : Set to '0' 0x2: INVERT : Invert 0x3: NO_CHANGE : No Change
1 : 0	CC_MATCH_MODE	Determines the effect of a compare match event (COUNTER equals CC register) on the "line_out" output signals. Note that INVERT is especially useful for center aligned pulse width modulation. To generate a duty cycle of 0%, the counter CC register should be set to "0". For a 100% duty cycle, the counter CC register should be set to larger than the counter PERIOD register. Default Value: 3 0x0: SET : Set to '1' 0x1: CLEAR : Set to '0' 0x2: INVERT : Invert 0x3: NO_CHANGE : No Change

10.1.14 TCPWM_CNT0_INTR

Interrupt request register.

Address: 0x40200130

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None						RW1C	RW1C
HW Access	None						RW1S	RW1S
Name	None [7:2]						CC_MATCH	TC

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
1	CC_MATCH	Counter matches CC register event. Set to '1', when event is detected. Write with '1' to clear bit. Default Value: 0
0	TC	Terminal count event. Set to '1', when event is detected. Write with '1' to clear bit. Default Value: 0

10.1.15 TCPWM_CNT0_INTR_SET

Interrupt set request register.

Address: 0x40200134

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None						RW1S	RW1S
HW Access	None						A	A
Name	None [7:2]						CC_MATCH	TC

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
1	CC_MATCH	Write with '1' to set corresponding bit in interrupt request register. Default Value: 0
0	TC	Write with '1' to set corresponding bit in interrupt request register. Default Value: 0

10.1.16 TCPWM_CNT0_INTR_MASK

Interrupt mask register.

Address: 0x40200138

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None						RW	RW
HW Access	None						R	R
Name	None [7:2]						CC_MATCH	TC

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
1	CC_MATCH	Mask bit for corresponding bit in interrupt request register. Default Value: 0
0	TC	Mask bit for corresponding bit in interrupt request register. Default Value: 0

10.1.17 TCPWM_CNT0_INTR_MASKED

Interrupt masked request register

Address: 0x4020013C

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None						R	R
HW Access	None						W	W
Name	None [7:2]						CC_MATCH	TC

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
1	CC_MATCH	Logical and of corresponding request and mask bits. Default Value: 0
0	TC	Logical and of corresponding request and mask bits. Default Value: 0

11 Watch Crystal Oscillator (WCO) Registers



This section discusses the WCO registers. It lists all the registers in mapping tables, in address order.

11.1 Register Details

Register	Address	Description
WCO_CONFIG	0x40220000	WCO Configuration Register
WCO_STATUS	0x40220004	WCO Status Register
WCO_DPLL	0x40220008	WCO DPLL Register
WCO_WDT_CTRLLOW	0x40220200	Watchdog Counters 0/1
WCO_WDT_CTRHIGH	0x40220204	Watchdog Counter 2
WCO_WDT_MATCH	0x40220208	Watchdog counter match values
WCO_WDT_CONFIG	0x4022020C	Watchdog Counters Configuration
WCO_WDT_CONTROL	0x40220210	Watchdog Counters Control
WCO_WDT_CLKEN	0x40220214	Watchdog Counters Clock Enable
WCO_TRIM	0x40220F00	WCO Trim Register

11.1.1 WCO_CONFIG

WCO Configuration Register

Address: 0x40220000

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None					RW	RW	RW
HW Access	None					R	R	R
Name	None [7:3]					EXT_IN- PUT_EN	LPM_AUTO	LPM_EN

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	RW							
HW Access	R							
Name	ENBUS [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	RW	RW	None					
HW Access	R	R	None					
Name	IP_ENABLE	DPLL_EN- ABLE	None [29:24]					

Bits	Name	Description
31	IP_ENABLE	Master enable for IP - disables both WCO and DPLL Default Value: 0
30	DPLL_ENABLE	Enable DPLL operation. The Oscillator is specified to be stable after 500 ms thus the DPLL should be asserted no sooner than that after IP_ENABLE is set. Default Value: 0
23 : 16	ENBUS	Test Mode Control bits enbus[7] - N/A enbus[6] - 1=enable both primary Beta Multipliers enbus[5] - N/A enbus[4] - N/A enbus[3] - Load Resistor Control enbus[2] - Load Resistor Control enbus[1] - Load Resistor Control enbus[0] - Load Resistor Control Default Value: 71
2	EXT_INPUT_EN	Disables the load resistor and allows external clock input for pad_xin Default Value: 0

11.1.1 WCO_CONFIG (continued)

1	LPM_AUTO	Automatically control low power mode (only relevant when LPM_EN=0): 0: Do not enter low power mode (LPM) in DeepSleep 1: Enter low power mode (LPM) in DeepSleep. The logic monitors !act_power_en to determine the device has entered DeepSleep. Default Value: 1
0	LPM_EN	Force block into Low Power Mode: 0: Do not force low power mode (LPM) on 1: Force low power mode (LPM) on Default Value: 0

11.1.2 WCO_STATUS

WCO Status Register

Address: 0x40220004

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None							R
HW Access	None							RW
Name	None [7:1]							OUT- _BLNK_A

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
0	OUT_BLNK_A	Indicates that output has transitioned - This bit is intended for Test Mode Only and is not a reliable indicator. Default Value: 0

11.1.3 WCO_DPLL

WCO DPLL Register

Address: 0x40220008

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access	R							
Name	DPLL_MULT [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	None					RW		
HW Access	None					R		
Name	None [15:11]					DPLL_MULT [10:8]		

Bits	23	22	21	20	19	18	17	16
SW Access	RW		RW			RW		
HW Access	R		R			R		
Name	DPLL_LF_LIMIT [23:22]		DPLL_LF_PGAIN [21:19]			DPLL_LF_IGAIN [18:16]		

Bits	31	30	29	28	27	26	25	24
SW Access	None		RW					
HW Access	None		R					
Name	None [31:30]		DPLL_LF_LIMIT [29:24]					

Bits	Name	Description
29 : 22	DPLL_LF_LIMIT	Maximum IMO offset allowed (used to prevent DPLL dynamics from selecting an IMO frequency that the logic cannot support) Default Value: 255
21 : 19	DPLL_LF_PGAIN	DPLL Loop Filter Proportional Gain Setting 0x0 - 0.0625 0x1 - 0.125 0x2 - 0.25 0x3 - 0.5 0x4 - 1.0 0x5 - 2.0 0x6 - 4.0 0x7 - 8.0 Default Value: 0

11.1.3 WCO_DPLL (continued)

18 : 16	DPLL_LF_IGAIN	DPLL Loop Filter Integral Gain Setting 0x0 - 0.0625 0x1 - 0.125 0x2 - 0.25 0x3 - 0.5 0x4 - 1.0 0x5 - 2.0 0x6 - 4.0 0x7 - 8.0 Default Value: 0
10 : 0	DPLL_MULT	Multiplier to determine IMO frequency in multiples of the WCO frequency $F_{imo} = (DPLL_MULT + 1) * F_{wco}$ Default Value: 0

11.1.4 WCO_WDT_CTRL0W

Watchdog Counters 0/1

Address: 0x40220200

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access	RW							
Name	WDT_CTRL0 [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access	RW							
Name	WDT_CTRL0 [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access	RW							
Name	WDT_CTRL1 [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access	RW							
Name	WDT_CTRL1 [31:24]							

Bits	Name	Description
31 : 16	WDT_CTRL1	Current value of WDT Counter 1 Default Value: 0
15 : 0	WDT_CTRL0	Current value of WDT Counter 0 Default Value: 0

11.1.5 WCO_WDT_CTRHIGH

Watchdog Counter 2

Address: 0x40220204

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access	RW							
Name	WDT_CTR2 [7:0]							
Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access	RW							
Name	WDT_CTR2 [15:8]							
Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access	RW							
Name	WDT_CTR2 [23:16]							
Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access	RW							
Name	WDT_CTR2 [31:24]							

Bits	Name	Description
31 : 0	WDT_CTR2	Current value of WDT Counter 2 Default Value: 0

11.1.6 WCO_WDT_MATCH

Watchdog counter match values

Address: 0x40220208

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access	R							
Name	WDT_MATCH0 [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	RW							
HW Access	R							
Name	WDT_MATCH0 [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	RW							
HW Access	R							
Name	WDT_MATCH1 [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	RW							
HW Access	R							
Name	WDT_MATCH1 [31:24]							

Bits	Name	Description
31 : 16	WDT_MATCH1	Match value for Watchdog Counter 1 Default Value: 0
15 : 0	WDT_MATCH0	Match value for Watchdog Counter 0 Default Value: 0

11.1.7 WCO_WDT_CONFIG

Watchdog Counters Configuration

Address: 0x4022020C

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None				RW	RW	RW	
HW Access	None				R	R	R	
Name	None [7:4]				WDT_CAS- CADE0_1	WDT_- CLEAR0	WDT_MODE0 [1:0]	

Bits	15	14	13	12	11	10	9	8
SW Access	None				RW	RW	RW	
HW Access	None				R	R	R	
Name	None [15:12]				WDT_CAS- CADE1_2	WDT_- CLEAR1	WDT_MODE1 [9:8]	

Bits	23	22	21	20	19	18	17	16
SW Access	None							RW
HW Access	None							R
Name	None [23:17]							WDT_- MODE2

Bits	31	30	29	28	27	26	25	24
SW Access	RW		None	RW				
HW Access	R		None	R				
Name	LFCLK_SEL [31:30]		None	WDT_BITS2 [28:24]				

Bits	Name	Description
31 : 30	LFCLK_SEL	Select source for LFCLK: 0: ILO - Internal R/C Oscillator 1: WCO - Internal Crystal Oscillator 2-3: Reserved - do not use Note that not all products support all clock sources. Selecting a clock source that is not supported will result in undefined behavior. To safely change LFCLK_SEL wait for WDT_CTLLOW/WDT_CTLHIGH to change then change the setting immediately. Default Value: 0
28 : 24	WDT_BITS2	Bit to observe for WDT_INT2: 0: Assert when bit0 of WDT_CTL2 toggles (one int every tick) ... 31: Assert when bit31 of WDT_CTL2 toggles (one int every 2^31 ticks) Default Value: 0
16	WDT_MODE2	Watchdog Counter 2 Mode. Default Value: 0

11.1.7 WCO_WDT_CONFIG (continued)

0x0: NOTHING :

Free running counter with no interrupt requests

0x1: INT :

Free running counter with interrupt request when a specified bit in CTR2 toggles (see WDT_BITS2)

11	WDT_CASCADE1_2	Cascade Watchdog Counters 1,2. Counter 2 increments the cycle after WDT_CTR1=WDT_MATCH1. It is allowed to cascade all three WDT counters. 0: Independent counters 1: Cascaded counters. When cascading all three counters, WDT_CLEAR1 must be 1 Default Value: 0
----	----------------	---

10	WDT_CLEAR1	Clear Watchdog Counter when WDT_CTR1=WDT_MATCH1. In other words WDT_CTR1 divides LFCLK by (WDT_MATCH1+1). 0: Free running counter 1: Clear on match Default Value: 0
----	------------	---

9 : 8	WDT_MODE1	Watchdog Counter Action on Match (WDT_CTR1=WDT_MATCH1). Default Value: 0
-------	-----------	---

0x0: NOTHING :

Do nothing

0x1: INT :

Assert WDT_INTx

0x2: RESET :

Assert WDT Reset - Not Supported - here for backwards compatibility

0x3: INT_THEN_RESET :

Assert WDT_INTx, assert WDT Reset after 3rd unhandled interrupt - Not supported - here for backwards compatibility.

3	WDT_CASCADE0_1	Cascade Watchdog Counters 0,1. Counter 1 increments the cycle after WDT_CTR0=WDT_MATCH0. 0: Independent counters 1: Cascaded counters Default Value: 0
---	----------------	---

2	WDT_CLEAR0	Clear Watchdog Counter when WDT_CTR0=WDT_MATCH0. In other words WDT_CTR0 divides LFCLK by (WDT_MATCH0+1). 0: Free running counter 1: Clear on match Default Value: 0
---	------------	---

1 : 0	WDT_MODE0	Watchdog Counter Action on Match (WDT_CTR0=WDT_MATCH0). Default Value: 0
-------	-----------	---

11.1.7 WCO_WDT_CONFIG (continued)

0x0: NOTHING :

Do nothing

0x1: INT :

Assert WDT_INTx

0x2: RESET :

Assert WDT Reset - Not Supported - here for backwards compatibility

0x3: INT_THEN_RESET :

Assert WDT_INTx, assert WDT Reset after 3rd unhandled interrupt. Not supported - here for Backwards compatibility.

11.1.8 WCO_WDT_CONTROL

Watchdog Counters Control

Address: 0x40220210

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None				RW1S	RW1C	R	RW
HW Access	None				RW0C	A	RW	R
Name	None [7:4]				WDT_RE-SET0	WDT_INT0	WDT_ENABLED0	WDT_ENABLED0

Bits	15	14	13	12	11	10	9	8
SW Access	None				RW1S	RW1C	R	RW
HW Access	None				RW0C	A	RW	R
Name	None [15:12]				WDT_RE-SET1	WDT_INT1	WDT_ENABLED1	WDT_ENABLED1

Bits	23	22	21	20	19	18	17	16
SW Access	None				RW1S	RW1C	R	RW
HW Access	None				RW0C	A	RW	R
Name	None [23:20]				WDT_RE-SET2	WDT_INT2	WDT_ENABLED2	WDT_ENABLED2

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
19	WDT_RESET2	Resets counter 2 back to 0000_0000. Hardware will reset this bit after counter was reset. This will take several LFCLK cycles to take effect. Wait until the reset completes before enabling the WDT. Default Value: 0
18	WDT_INT2	WDT Interrupt Request. This bit is set by hardware as configured by this registers. This bit must be cleared by firmware. After W1C, WDT_CONTROL must be read for the hardware to internally remove the clear flag. Failure to do this may result in missing the next interrupt. Default Value: 0
17	WDT_ENABLED2	Indicates actual state of counter. May lag WDT_ENABLE2 by up to 3 LFCLK cycles. After changing WDT_ENABLE2, do not enter DEEPSLEEP mode until this field acknowledges the change. Default Value: 0

11.1.8 WCO_WDT_CONTROL (continued)

16	WDT_ENABLE2	Enable Counter 2 0: Counter is disabled (not clocked) 1: Counter is enabled (counting up) Note: This field takes considerable time (up to 3 LFCLK cycles) to take effect. It must not be changed more than once in that period. Default Value: 0
11	WDT_RESET1	Resets counter 1 back to 0000. Hardware will reset this bit after counter was reset. This will take several LFCLK cycles to take effect. Wait until the reset completes before enabling the WDT. Default Value: 0
10	WDT_INT1	WDT Interrupt Request. This bit is set by hardware as configured by this registers. This bit must be cleared by firmware. After W1C, WDT_CONTROL must be read for the hardware to internally remove the clear flag. Failure to do this may result in missing the next interrupt. Default Value: 0
9	WDT_ENABLED1	Indicates actual state of counter. May lag WDT_ENABLE1 by up to 3 LFCLK cycles. After changing WDT_ENABLE1, do not enter DEEPSLEEP mode until this field acknowledges the change. Default Value: 0
8	WDT_ENABLE1	Enable Counter 1 0: Counter is disabled (not clocked) 1: Counter is enabled (counting up) Note: This field takes considerable time (up to 3 LFCLK cycles) to take effect. It must not be changed more than once in that period. Default Value: 0
3	WDT_RESET0	Resets counter 0 back to 0000. Hardware will reset this bit after counter was reset. This will take several LFCLK cycles to take effect. Wait until the reset completes before enabling the WDT. Default Value: 0
2	WDT_INT0	WDT Interrupt Request. This bit is set by hardware as configured by this registers. This bit must be cleared by firmware. Clearing this bit also prevents Reset from happening when WDT_MODEx=3. After W1C, WDT_CONTROL must be read for the hardware to internally remove the clear flag. Failure to do this may result in missing the next interrupt. Default Value: 0
1	WDT_ENABLED0	Indicates actual state of counter. May lag WDT_ENABLE0 by up to 3 LFCLK cycles. After changing WDT_ENABLE0, do not enter DEEPSLEEP mode until this field acknowledges the change. Default Value: 0
0	WDT_ENABLE0	Enable Counter 0 0: Counter is disabled (not clocked) 1: Counter is enabled (counting up) Note: This field takes considerable time (up to 3 LFCLK cycles) to take effect. It must not be changed more than once in that period. Default Value: 0

11.1.9 WCO_WDT_CLKEN

Watchdog Counters Clock Enable

Address: 0x40220214

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None						RW	RW
HW Access	None						R	R
Name	None [7:2]						CLK_ILO_EN_FOR_WDT	CLK_WCO_EN_FOR_WDT

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
1	CLK_ILO_EN_FOR_WDT	Enables the ILO clock for use by the WDT logic. Wait at least 4 ILO clock cycles for a change to take effect. Must be 0 when switching WDT_CONFIG.LFCLK_SEL. Should be 0 if CLK_WCO_EN_FOR_WDT=1. Default Value: 0
0	CLK_WCO_EN_FOR_WDT	Enables the WCO clock for use by the WDT logic. Wait at least 4 WCO clock cycles for a change to take effect. Must be 0 when switching WDT_CONFIG.LFCLK_SEL. Should be 0 if CLK_ILO_EN_FOR_WDT=1 Default Value: 0

11.1.10 WCO_TRIM

WCO Trim Register

Address: 0x40220F00

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None		RW		None	RW		
HW Access	None		R		None	R		
Name	None [7:6]		LPM_GM [5:4]		None	XGM [2:0]		

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
5 : 4	LPM_GM	GM setting for LPM (bandwidth = DC/ms) - Used when WCO.LPM_AUTO=0 or when LPM_AUTO=1 and not in DeepSleep mode. Default Value: 1
2 : 0	XGM	Amplifier GM setting - Used when WCO.LPM_AUTO=0 or when LPM_AUTO=1 and not in DeepSleep mode. 0x0 - 3370 nA 0x1 - 2620 nA 0x2 - 2250 nA 0x3 - 1500 nA 0x4 - 1870 nA 0x5 - 1120 nA 0x6 - 750 nA 0x7 - 0 nA Default Value: 1

12 Serial Communication Block (SCB) Registers



This section discusses the SCB registers. It lists all the registers in mapping tables, in address order.

12.1 Register Details

Register	Address	Description
SCB0_CTRL	0x40240000	Generic control register.
SCB0_STATUS	0x40240004	Generic status register.
SCB0_SPI_CTRL	0x40240020	SPI control register.
SCB0_SPI_STATUS	0x40240024	SPI status register.
SCB0_UART_CTRL	0x40240040	UART control register.
SCB0_UART_TX_CTRL	0x40240044	UART transmitter control register.
SCB0_UART_RX_CTRL	0x40240048	UART receiver control register.
SCB0_UART_RX_STATUS	0x4024004C	UART receiver status register.
SCB0_UART_FLOW_CTRL	0x40240050	UART flow control register
SCB0_I2C_CTRL	0x40240060	I2C control register.
SCB0_I2C_STATUS	0x40240064	I2C status register.
SCB0_I2C_M_CMD	0x40240068	I2C master command register.
SCB0_I2C_S_CMD	0x4024006C	I2C slave command register.
SCB0_I2C_CFG	0x40240070	I2C configuration register.
SCB0_TX_CTRL	0x40240200	Transmitter control register.
SCB0_TX_FIFO_CTRL	0x40240204	Transmitter FIFO control register.
SCB0_TX_FIFO_STATUS	0x40240208	Transmitter FIFO status register.
SCB0_TX_FIFO_WR	0x40240240	Transmitter FIFO write register.
SCB0_RX_CTRL	0x40240300	Receiver control register.
SCB0_RX_FIFO_CTRL	0x40240304	Receiver FIFO control register.
SCB0_RX_FIFO_STATUS	0x40240308	Receiver FIFO status register.
SCB0_RX_MATCH	0x40240310	Slave address and mask register.
SCB0_RX_FIFO_RD	0x40240340	Receiver FIFO read register.
SCB0_RX_FIFO_RD_SILENT	0x40240344	Receiver FIFO read register.
SCB0_EZ_DATA0	0x40240400	Memory buffer registers.
SCB0_EZ_DATA1	0x40240404	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB0_EZ_DATA2	0x40240408	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.

Register	Address	Description
SCB0_EZ_DATA3	0x4024040C	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB0_EZ_DATA4	0x40240410	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB0_EZ_DATA5	0x40240414	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB0_EZ_DATA6	0x40240418	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB0_EZ_DATA7	0x4024041C	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB0_EZ_DATA8	0x40240420	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB0_EZ_DATA9	0x40240424	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB0_EZ_DATA10	0x40240428	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB0_EZ_DATA11	0x4024042C	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB0_EZ_DATA12	0x40240430	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB0_EZ_DATA13	0x40240434	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB0_EZ_DATA14	0x40240438	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB0_EZ_DATA15	0x4024043C	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB0_EZ_DATA16	0x40240440	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB0_EZ_DATA17	0x40240444	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB0_EZ_DATA18	0x40240448	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB0_EZ_DATA19	0x4024044C	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB0_EZ_DATA20	0x40240450	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB0_EZ_DATA21	0x40240454	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB0_EZ_DATA22	0x40240458	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB0_EZ_DATA23	0x4024045C	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB0_EZ_DATA24	0x40240460	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB0_EZ_DATA25	0x40240464	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB0_EZ_DATA26	0x40240468	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB0_EZ_DATA27	0x4024046C	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB0_EZ_DATA28	0x40240470	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB0_EZ_DATA29	0x40240474	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB0_EZ_DATA30	0x40240478	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB0_EZ_DATA31	0x4024047C	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB0_INTR_CAUSE	0x40240E00	Active clocked interrupt signal register
SCB0_INTR_I2C_EC	0x40240E80	Externally clocked I2C interrupt request register
SCB0_INTR_I2C_EC_MASK	0x40240E88	Externally clocked I2C interrupt mask register
SCB0_INTR_I2C_EC_MASKED	0x40240E8C	Externally clocked I2C interrupt masked register
SCB0_INTR_SPI_EC	0x40240EC0	Externally clocked SPI interrupt request register
SCB0_INTR_SPI_EC_MASK	0x40240EC8	Externally clocked SPI interrupt mask register
SCB0_INTR_SPI_EC_MASKED	0x40240ECC	Externally clocked SPI interrupt masked register
SCB0_INTR_M	0x40240F00	Master interrupt request register.
SCB0_INTR_M_SET	0x40240F04	Master interrupt set request register
SCB0_INTR_M_MASK	0x40240F08	Master interrupt mask register.
SCB0_INTR_M_MASKED	0x40240F0C	Master interrupt masked request register
SCB0_INTR_S	0x40240F40	Slave interrupt request register.
SCB0_INTR_S_SET	0x40240F44	Slave interrupt set request register.

Register	Address	Description
SCB0_INTR_S_MASK	0x40240F48	Slave interrupt mask register.
SCB0_INTR_S_MASKED	0x40240F4C	Slave interrupt masked request register
SCB0_INTR_TX	0x40240F80	Transmitter interrupt request register.
SCB0_INTR_TX_SET	0x40240F84	Transmitter interrupt set request register
SCB0_INTR_TX_MASK	0x40240F88	Transmitter interrupt mask register.
SCB0_INTR_TX_MASKED	0x40240F8C	Transmitter interrupt masked request register
SCB0_INTR_RX	0x40240FC0	Receiver interrupt request register.
SCB0_INTR_RX_SET	0x40240FC4	Receiver interrupt set request register.
SCB0_INTR_RX_MASK	0x40240FC8	Receiver interrupt mask register.
SCB0_INTR_RX_MASKED	0x40240FCC	Receiver interrupt masked request register
SCB1_CTRL	0x40250000	Generic control register.. See SCB0_CTRL for the details of bit fields.
SCB1_STATUS	0x40250004	Generic status register.. See SCB0_STATUS for the details of bit fields.
SCB1_SPI_CTRL	0x40250020	SPI control register.. See SCB0_SPI_CTRL for the details of bit fields.
SCB1_SPI_STATUS	0x40250024	SPI status register.. See SCB0_SPI_STATUS for the details of bit fields.
SCB1_UART_CTRL	0x40250040	UART control register.. See SCB0_UART_CTRL for the details of bit fields.
SCB1_UART_TX_CTRL	0x40250044	UART transmitter control register.. See SCB0_UART_TX_CTRL for the details of bit fields.
SCB1_UART_RX_CTRL	0x40250048	UART receiver control register.. See SCB0_UART_RX_CTRL for the details of bit fields.
SCB1_UART_RX_STATUS	0x4025004C	UART receiver status register.. See SCB0_UART_RX_STATUS for the details of bit fields.
SCB1_UART_FLOW_CTRL	0x40250050	UART flow control register. See SCB0_UART_FLOW_CTRL for the details of bit fields.
SCB1_I2C_CTRL	0x40250060	I2C control register.. See SCB0_I2C_CTRL for the details of bit fields.
SCB1_I2C_STATUS	0x40250064	I2C status register.. See SCB0_I2C_STATUS for the details of bit fields.
SCB1_I2C_M_CMD	0x40250068	I2C master command register.. See SCB0_I2C_M_CMD for the details of bit fields.
SCB1_I2C_S_CMD	0x4025006C	I2C slave command register.. See SCB0_I2C_S_CMD for the details of bit fields.
SCB1_I2C_CFG	0x40250070	I2C configuration register.. See SCB0_I2C_CFG for the details of bit fields.
SCB1_TX_CTRL	0x40250200	Transmitter control register.. See SCB0_TX_CTRL for the details of bit fields.
SCB1_TX_FIFO_CTRL	0x40250204	Transmitter FIFO control register.. See SCB0_TX_FIFO_CTRL for the details of bit fields.
SCB1_TX_FIFO_STATUS	0x40250208	Transmitter FIFO status register.. See SCB0_TX_FIFO_STATUS for the details of bit fields.
SCB1_TX_FIFO_WR	0x40250240	Transmitter FIFO write register.. See SCB0_TX_FIFO_WR for the details of bit fields.
SCB1_RX_CTRL	0x40250300	Receiver control register.. See SCB0_RX_CTRL for the details of bit fields.
SCB1_RX_FIFO_CTRL	0x40250304	Receiver FIFO control register.. See SCB0_RX_FIFO_CTRL for the details of bit fields.
SCB1_RX_FIFO_STATUS	0x40250308	Receiver FIFO status register.. See SCB0_RX_FIFO_STATUS for the details of bit fields.
SCB1_RX_MATCH	0x40250310	Slave address and mask register.. See SCB0_RX_MATCH for the details of bit fields.
SCB1_RX_FIFO_RD	0x40250340	Receiver FIFO read register.. See SCB0_RX_FIFO_RD for the details of bit fields.
SCB1_RX_FIFO_RD_SILENT	0x40250344	Receiver FIFO read register.. See SCB0_RX_FIFO_RD_SILENT for the details of bit fields.
SCB1_EZ_DATA0	0x40250400	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB1_EZ_DATA1	0x40250404	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB1_EZ_DATA2	0x40250408	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB1_EZ_DATA3	0x4025040C	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB1_EZ_DATA4	0x40250410	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB1_EZ_DATA5	0x40250414	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB1_EZ_DATA6	0x40250418	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB1_EZ_DATA7	0x4025041C	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.

Register	Address	Description
SCB1_EZ_DATA8	0x40250420	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB1_EZ_DATA9	0x40250424	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB1_EZ_DATA10	0x40250428	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB1_EZ_DATA11	0x4025042C	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB1_EZ_DATA12	0x40250430	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB1_EZ_DATA13	0x40250434	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB1_EZ_DATA14	0x40250438	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB1_EZ_DATA15	0x4025043C	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB1_EZ_DATA16	0x40250440	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB1_EZ_DATA17	0x40250444	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB1_EZ_DATA18	0x40250448	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB1_EZ_DATA19	0x4025044C	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB1_EZ_DATA20	0x40250450	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB1_EZ_DATA21	0x40250454	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB1_EZ_DATA22	0x40250458	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB1_EZ_DATA23	0x4025045C	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB1_EZ_DATA24	0x40250460	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB1_EZ_DATA25	0x40250464	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB1_EZ_DATA26	0x40250468	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB1_EZ_DATA27	0x4025046C	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB1_EZ_DATA28	0x40250470	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB1_EZ_DATA29	0x40250474	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB1_EZ_DATA30	0x40250478	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB1_EZ_DATA31	0x4025047C	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB1_INTR_CAUSE	0x40250E00	Active clocked interrupt signal register. See SCB0_INTR_CAUSE for the details of bit fields.
SCB1_INTR_I2C_EC	0x40250E80	Externally clocked I2C interrupt request register. See SCB0_INTR_I2C_EC for the details of bit fields.
SCB1_INTR_I2C_EC_MASK	0x40250E88	Externally clocked I2C interrupt mask register. See SCB0_INTR_I2C_EC_MASK for the details of bit fields.
SCB1_INTR_I2C_EC_MASKED	0x40250E8C	Externally clocked I2C interrupt masked register. See SCB0_INTR_I2C_EC_MASKED for the details of bit fields.
SCB1_INTR_SPI_EC	0x40250EC0	Externally clocked SPI interrupt request register. See SCB0_INTR_SPI_EC for the details of bit fields.
SCB1_INTR_SPI_EC_MASK	0x40250EC8	Externally clocked SPI interrupt mask register. See SCB0_INTR_SPI_EC_MASK for the details of bit fields.
SCB1_INTR_SPI_EC_MASKED	0x40250ECC	Externally clocked SPI interrupt masked register. See SCB0_INTR_SPI_EC_MASKED for the details of bit fields.
SCB1_INTR_M	0x40250F00	Master interrupt request register.. See SCB0_INTR_M for the details of bit fields.
SCB1_INTR_M_SET	0x40250F04	Master interrupt set request register. See SCB0_INTR_M_SET for the details of bit fields.
SCB1_INTR_M_MASK	0x40250F08	Master interrupt mask register.. See SCB0_INTR_M_MASK for the details of bit fields.
SCB1_INTR_M_MASKED	0x40250F0C	Master interrupt masked request register. See SCB0_INTR_M_MASKED for the details of bit fields.
SCB1_INTR_S	0x40250F40	Slave interrupt request register.. See SCB0_INTR_S for the details of bit fields.
SCB1_INTR_S_SET	0x40250F44	Slave interrupt set request register.. See SCB0_INTR_S_SET for the details of bit fields.
SCB1_INTR_S_MASK	0x40250F48	Slave interrupt mask register.. See SCB0_INTR_S_MASK for the details of bit fields.
SCB1_INTR_S_MASKED	0x40250F4C	Slave interrupt masked request register. See SCB0_INTR_S_MASKED for the details of bit fields.

Register	Address	Description
SCB1_INTR_TX	0x40250F80	Transmitter interrupt request register.. See SCB0_INTR_TX for the details of bit fields.
SCB1_INTR_TX_SET	0x40250F84	Transmitter interrupt set request register. See SCB0_INTR_TX_SET for the details of bit fields.
SCB1_INTR_TX_MASK	0x40250F88	Transmitter interrupt mask register.. See SCB0_INTR_TX_MASK for the details of bit fields.
SCB1_INTR_TX_MASKED	0x40250F8C	Transmitter interrupt masked request register. See SCB0_INTR_TX_MASKED for the details of bit fields.
SCB1_INTR_RX	0x40250FC0	Receiver interrupt request register.. See SCB0_INTR_RX for the details of bit fields.
SCB1_INTR_RX_SET	0x40250FC4	Receiver interrupt set request register.. See SCB0_INTR_RX_SET for the details of bit fields.
SCB1_INTR_RX_MASK	0x40250FC8	Receiver interrupt mask register.. See SCB0_INTR_RX_MASK for the details of bit fields.
SCB1_INTR_RX_MASKED	0x40250FCC	Receiver interrupt masked request register. See SCB0_INTR_RX_MASKED for the details of bit fields.
SCB2_CTRL	0x40260000	Generic control register.. See SCB0_CTRL for the details of bit fields.
SCB2_STATUS	0x40260004	Generic status register.. See SCB0_STATUS for the details of bit fields.
SCB2_SPI_CTRL	0x40260020	SPI control register.. See SCB0_SPI_CTRL for the details of bit fields.
SCB2_SPI_STATUS	0x40260024	SPI status register.. See SCB0_SPI_STATUS for the details of bit fields.
SCB2_UART_CTRL	0x40260040	UART control register.. See SCB0_UART_CTRL for the details of bit fields.
SCB2_UART_TX_CTRL	0x40260044	UART transmitter control register.. See SCB0_UART_TX_CTRL for the details of bit fields.
SCB2_UART_RX_CTRL	0x40260048	UART receiver control register.. See SCB0_UART_RX_CTRL for the details of bit fields.
SCB2_UART_RX_STATUS	0x4026004C	UART receiver status register.. See SCB0_UART_RX_STATUS for the details of bit fields.
SCB2_UART_FLOW_CTRL	0x40260050	UART flow control register. See SCB0_UART_FLOW_CTRL for the details of bit fields.
SCB2_I2C_CTRL	0x40260060	I2C control register.. See SCB0_I2C_CTRL for the details of bit fields.
SCB2_I2C_STATUS	0x40260064	I2C status register.. See SCB0_I2C_STATUS for the details of bit fields.
SCB2_I2C_M_CMD	0x40260068	I2C master command register.. See SCB0_I2C_M_CMD for the details of bit fields.
SCB2_I2C_S_CMD	0x4026006C	I2C slave command register.. See SCB0_I2C_S_CMD for the details of bit fields.
SCB2_I2C_CFG	0x40260070	I2C configuration register.. See SCB0_I2C_CFG for the details of bit fields.
SCB2_TX_CTRL	0x40260200	Transmitter control register.. See SCB0_TX_CTRL for the details of bit fields.
SCB2_TX_FIFO_CTRL	0x40260204	Transmitter FIFO control register.. See SCB0_TX_FIFO_CTRL for the details of bit fields.
SCB2_TX_FIFO_STATUS	0x40260208	Transmitter FIFO status register.. See SCB0_TX_FIFO_STATUS for the details of bit fields.
SCB2_TX_FIFO_WR	0x40260240	Transmitter FIFO write register.. See SCB0_TX_FIFO_WR for the details of bit fields.
SCB2_RX_CTRL	0x40260300	Receiver control register.. See SCB0_RX_CTRL for the details of bit fields.
SCB2_RX_FIFO_CTRL	0x40260304	Receiver FIFO control register.. See SCB0_RX_FIFO_CTRL for the details of bit fields.
SCB2_RX_FIFO_STATUS	0x40260308	Receiver FIFO status register.. See SCB0_RX_FIFO_STATUS for the details of bit fields.
SCB2_RX_MATCH	0x40260310	Slave address and mask register.. See SCB0_RX_MATCH for the details of bit fields.
SCB2_RX_FIFO_RD	0x40260340	Receiver FIFO read register.. See SCB0_RX_FIFO_RD for the details of bit fields.
SCB2_RX_FIFO_RD_SILENT	0x40260344	Receiver FIFO read register.. See SCB0_RX_FIFO_RD_SILENT for the details of bit fields.
SCB2_EZ_DATA0	0x40260400	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB2_EZ_DATA1	0x40260404	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB2_EZ_DATA2	0x40260408	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB2_EZ_DATA3	0x4026040C	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB2_EZ_DATA4	0x40260410	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB2_EZ_DATA5	0x40260414	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB2_EZ_DATA6	0x40260418	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB2_EZ_DATA7	0x4026041C	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB2_EZ_DATA8	0x40260420	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.

Register	Address	Description
SCB2_EZ_DATA9	0x40260424	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB2_EZ_DATA10	0x40260428	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB2_EZ_DATA11	0x4026042C	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB2_EZ_DATA12	0x40260430	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB2_EZ_DATA13	0x40260434	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB2_EZ_DATA14	0x40260438	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB2_EZ_DATA15	0x4026043C	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB2_EZ_DATA16	0x40260440	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB2_EZ_DATA17	0x40260444	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB2_EZ_DATA18	0x40260448	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB2_EZ_DATA19	0x4026044C	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB2_EZ_DATA20	0x40260450	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB2_EZ_DATA21	0x40260454	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB2_EZ_DATA22	0x40260458	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB2_EZ_DATA23	0x4026045C	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB2_EZ_DATA24	0x40260460	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB2_EZ_DATA25	0x40260464	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB2_EZ_DATA26	0x40260468	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB2_EZ_DATA27	0x4026046C	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB2_EZ_DATA28	0x40260470	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB2_EZ_DATA29	0x40260474	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB2_EZ_DATA30	0x40260478	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB2_EZ_DATA31	0x4026047C	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB2_INTR_CAUSE	0x40260E00	Active clocked interrupt signal register. See SCB0_INTR_CAUSE for the details of bit fields.
SCB2_INTR_I2C_EC	0x40260E80	Externally clocked I2C interrupt request register. See SCB0_INTR_I2C_EC for the details of bit fields.
SCB2_INTR_I2C_EC_MASK	0x40260E88	Externally clocked I2C interrupt mask register. See SCB0_INTR_I2C_EC_MASK for the details of bit fields.
SCB2_INTR_I2C_EC_MASKED	0x40260E8C	Externally clocked I2C interrupt masked register. See SCB0_INTR_I2C_EC_MASKED for the details of bit fields.
SCB2_INTR_SPI_EC	0x40260EC0	Externally clocked SPI interrupt request register. See SCB0_INTR_SPI_EC for the details of bit fields.
SCB2_INTR_SPI_EC_MASK	0x40260EC8	Externally clocked SPI interrupt mask register. See SCB0_INTR_SPI_EC_MASK for the details of bit fields.
SCB2_INTR_SPI_EC_MASKED	0x40260ECC	Externally clocked SPI interrupt masked register. See SCB0_INTR_SPI_EC_MASKED for the details of bit fields.
SCB2_INTR_M	0x40260F00	Master interrupt request register.. See SCB0_INTR_M for the details of bit fields.
SCB2_INTR_M_SET	0x40260F04	Master interrupt set request register. See SCB0_INTR_M_SET for the details of bit fields.
SCB2_INTR_M_MASK	0x40260F08	Master interrupt mask register.. See SCB0_INTR_M_MASK for the details of bit fields.
SCB2_INTR_M_MASKED	0x40260F0C	Master interrupt masked request register. See SCB0_INTR_M_MASKED for the details of bit fields.
SCB2_INTR_S	0x40260F40	Slave interrupt request register.. See SCB0_INTR_S for the details of bit fields.
SCB2_INTR_S_SET	0x40260F44	Slave interrupt set request register.. See SCB0_INTR_S_SET for the details of bit fields.
SCB2_INTR_S_MASK	0x40260F48	Slave interrupt mask register.. See SCB0_INTR_S_MASK for the details of bit fields.
SCB2_INTR_S_MASKED	0x40260F4C	Slave interrupt masked request register. See SCB0_INTR_S_MASKED for the details of bit fields.
SCB2_INTR_TX	0x40260F80	Transmitter interrupt request register.. See SCB0_INTR_TX for the details of bit fields.

Register	Address	Description
SCB2_INTR_TX_SET	0x40260F84	Transmitter interrupt set request register. See SCB0_INTR_TX_SET for the details of bit fields.
SCB2_INTR_TX_MASK	0x40260F88	Transmitter interrupt mask register.. See SCB0_INTR_TX_MASK for the details of bit fields.
SCB2_INTR_TX_MASKED	0x40260F8C	Transmitter interrupt masked request register. See SCB0_INTR_TX_MASKED for the details of bit fields.
SCB2_INTR_RX	0x40260FC0	Receiver interrupt request register.. See SCB0_INTR_RX for the details of bit fields.
SCB2_INTR_RX_SET	0x40260FC4	Receiver interrupt set request register.. See SCB0_INTR_RX_SET for the details of bit fields.
SCB2_INTR_RX_MASK	0x40260FC8	Receiver interrupt mask register.. See SCB0_INTR_RX_MASK for the details of bit fields.
SCB2_INTR_RX_MASKED	0x40260FCC	Receiver interrupt masked request register. See SCB0_INTR_RX_MASKED for the details of bit fields.
SCB3_CTRL	0x40270000	Generic control register.. See SCB0_CTRL for the details of bit fields.
SCB3_STATUS	0x40270004	Generic status register.. See SCB0_STATUS for the details of bit fields.
SCB3_SPI_CTRL	0x40270020	SPI control register.. See SCB0_SPI_CTRL for the details of bit fields.
SCB3_SPI_STATUS	0x40270024	SPI status register.. See SCB0_SPI_STATUS for the details of bit fields.
SCB3_UART_CTRL	0x40270040	UART control register.. See SCB0_UART_CTRL for the details of bit fields.
SCB3_UART_TX_CTRL	0x40270044	UART transmitter control register.. See SCB0_UART_TX_CTRL for the details of bit fields.
SCB3_UART_RX_CTRL	0x40270048	UART receiver control register.. See SCB0_UART_RX_CTRL for the details of bit fields.
SCB3_UART_RX_STATUS	0x4027004C	UART receiver status register.. See SCB0_UART_RX_STATUS for the details of bit fields.
SCB3_UART_FLOW_CTRL	0x40270050	UART flow control register. See SCB0_UART_FLOW_CTRL for the details of bit fields.
SCB3_I2C_CTRL	0x40270060	I2C control register.. See SCB0_I2C_CTRL for the details of bit fields.
SCB3_I2C_STATUS	0x40270064	I2C status register.. See SCB0_I2C_STATUS for the details of bit fields.
SCB3_I2C_M_CMD	0x40270068	I2C master command register.. See SCB0_I2C_M_CMD for the details of bit fields.
SCB3_I2C_S_CMD	0x4027006C	I2C slave command register.. See SCB0_I2C_S_CMD for the details of bit fields.
SCB3_I2C_CFG	0x40270070	I2C configuration register.. See SCB0_I2C_CFG for the details of bit fields.
SCB3_TX_CTRL	0x40270200	Transmitter control register.. See SCB0_TX_CTRL for the details of bit fields.
SCB3_TX_FIFO_CTRL	0x40270204	Transmitter FIFO control register.. See SCB0_TX_FIFO_CTRL for the details of bit fields.
SCB3_TX_FIFO_STATUS	0x40270208	Transmitter FIFO status register.. See SCB0_TX_FIFO_STATUS for the details of bit fields.
SCB3_TX_FIFO_WR	0x40270240	Transmitter FIFO write register.. See SCB0_TX_FIFO_WR for the details of bit fields.
SCB3_RX_CTRL	0x40270300	Receiver control register.. See SCB0_RX_CTRL for the details of bit fields.
SCB3_RX_FIFO_CTRL	0x40270304	Receiver FIFO control register.. See SCB0_RX_FIFO_CTRL for the details of bit fields.
SCB3_RX_FIFO_STATUS	0x40270308	Receiver FIFO status register.. See SCB0_RX_FIFO_STATUS for the details of bit fields.
SCB3_RX_MATCH	0x40270310	Slave address and mask register.. See SCB0_RX_MATCH for the details of bit fields.
SCB3_RX_FIFO_RD	0x40270340	Receiver FIFO read register.. See SCB0_RX_FIFO_RD for the details of bit fields.
SCB3_RX_FIFO_RD_SILENT	0x40270344	Receiver FIFO read register.. See SCB0_RX_FIFO_RD_SILENT for the details of bit fields.
SCB3_EZ_DATA0	0x40270400	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB3_EZ_DATA1	0x40270404	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB3_EZ_DATA2	0x40270408	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB3_EZ_DATA3	0x4027040C	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB3_EZ_DATA4	0x40270410	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB3_EZ_DATA5	0x40270414	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB3_EZ_DATA6	0x40270418	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB3_EZ_DATA7	0x4027041C	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB3_EZ_DATA8	0x40270420	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB3_EZ_DATA9	0x40270424	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.

Register	Address	Description
SCB3_EZ_DATA10	0x40270428	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB3_EZ_DATA11	0x4027042C	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB3_EZ_DATA12	0x40270430	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB3_EZ_DATA13	0x40270434	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB3_EZ_DATA14	0x40270438	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB3_EZ_DATA15	0x4027043C	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB3_EZ_DATA16	0x40270440	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB3_EZ_DATA17	0x40270444	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB3_EZ_DATA18	0x40270448	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB3_EZ_DATA19	0x4027044C	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB3_EZ_DATA20	0x40270450	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB3_EZ_DATA21	0x40270454	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB3_EZ_DATA22	0x40270458	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB3_EZ_DATA23	0x4027045C	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB3_EZ_DATA24	0x40270460	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB3_EZ_DATA25	0x40270464	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB3_EZ_DATA26	0x40270468	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB3_EZ_DATA27	0x4027046C	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB3_EZ_DATA28	0x40270470	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB3_EZ_DATA29	0x40270474	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB3_EZ_DATA30	0x40270478	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB3_EZ_DATA31	0x4027047C	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB3_INTR_CAUSE	0x40270E00	Active clocked interrupt signal register. See SCB0_INTR_CAUSE for the details of bit fields.
SCB3_INTR_I2C_EC	0x40270E80	Externally clocked I2C interrupt request register. See SCB0_INTR_I2C_EC for the details of bit fields.
SCB3_INTR_I2C_EC_MASK	0x40270E88	Externally clocked I2C interrupt mask register. See SCB0_INTR_I2C_EC_MASK for the details of bit fields.
SCB3_INTR_I2C_EC_MASKED	0x40270E8C	Externally clocked I2C interrupt masked register. See SCB0_INTR_I2C_EC_MASKED for the details of bit fields.
SCB3_INTR_SPI_EC	0x40270EC0	Externally clocked SPI interrupt request register. See SCB0_INTR_SPI_EC for the details of bit fields.
SCB3_INTR_SPI_EC_MASK	0x40270EC8	Externally clocked SPI interrupt mask register. See SCB0_INTR_SPI_EC_MASK for the details of bit fields.
SCB3_INTR_SPI_EC_MASKED	0x40270ECC	Externally clocked SPI interrupt masked register. See SCB0_INTR_SPI_EC_MASKED for the details of bit fields.
SCB3_INTR_M	0x40270F00	Master interrupt request register.. See SCB0_INTR_M for the details of bit fields.
SCB3_INTR_M_SET	0x40270F04	Master interrupt set request register. See SCB0_INTR_M_SET for the details of bit fields.
SCB3_INTR_M_MASK	0x40270F08	Master interrupt mask register.. See SCB0_INTR_M_MASK for the details of bit fields.
SCB3_INTR_M_MASKED	0x40270F0C	Master interrupt masked request register. See SCB0_INTR_M_MASKED for the details of bit fields.
SCB3_INTR_S	0x40270F40	Slave interrupt request register.. See SCB0_INTR_S for the details of bit fields.
SCB3_INTR_S_SET	0x40270F44	Slave interrupt set request register.. See SCB0_INTR_S_SET for the details of bit fields.
SCB3_INTR_S_MASK	0x40270F48	Slave interrupt mask register.. See SCB0_INTR_S_MASK for the details of bit fields.
SCB3_INTR_S_MASKED	0x40270F4C	Slave interrupt masked request register. See SCB0_INTR_S_MASKED for the details of bit fields.
SCB3_INTR_TX	0x40270F80	Transmitter interrupt request register.. See SCB0_INTR_TX for the details of bit fields.
SCB3_INTR_TX_SET	0x40270F84	Transmitter interrupt set request register. See SCB0_INTR_TX_SET for the details of bit fields.

Register	Address	Description
SCB3_INTR_TX_MASK	0x40270F88	Transmitter interrupt mask register.. See SCB0_INTR_TX_MASK for the details of bit fields.
SCB3_INTR_TX_MASKED	0x40270F8C	Transmitter interrupt masked request register. See SCB0_INTR_TX_MASKED for the details of bit fields.
SCB3_INTR_RX	0x40270FC0	Receiver interrupt request register.. See SCB0_INTR_RX for the details of bit fields.
SCB3_INTR_RX_SET	0x40270FC4	Receiver interrupt set request register.. See SCB0_INTR_RX_SET for the details of bit fields.
SCB3_INTR_RX_MASK	0x40270FC8	Receiver interrupt mask register.. See SCB0_INTR_RX_MASK for the details of bit fields.
SCB3_INTR_RX_MASKED	0x40270FCC	Receiver interrupt masked request register. See SCB0_INTR_RX_MASKED for the details of bit fields.
SCB4_CTRL	0x40280000	Generic control register.. See SCB0_CTRL for the details of bit fields.
SCB4_STATUS	0x40280004	Generic status register.. See SCB0_STATUS for the details of bit fields.
SCB4_SPI_CTRL	0x40280020	SPI control register.. See SCB0_SPI_CTRL for the details of bit fields.
SCB4_SPI_STATUS	0x40280024	SPI status register.. See SCB0_SPI_STATUS for the details of bit fields.
SCB4_UART_CTRL	0x40280040	UART control register.. See SCB0_UART_CTRL for the details of bit fields.
SCB4_UART_TX_CTRL	0x40280044	UART transmitter control register.. See SCB0_UART_TX_CTRL for the details of bit fields.
SCB4_UART_RX_CTRL	0x40280048	UART receiver control register.. See SCB0_UART_RX_CTRL for the details of bit fields.
SCB4_UART_RX_STATUS	0x4028004C	UART receiver status register.. See SCB0_UART_RX_STATUS for the details of bit fields.
SCB4_UART_FLOW_CTRL	0x40280050	UART flow control register. See SCB0_UART_FLOW_CTRL for the details of bit fields.
SCB4_I2C_CTRL	0x40280060	I2C control register.. See SCB0_I2C_CTRL for the details of bit fields.
SCB4_I2C_STATUS	0x40280064	I2C status register.. See SCB0_I2C_STATUS for the details of bit fields.
SCB4_I2C_M_CMD	0x40280068	I2C master command register.. See SCB0_I2C_M_CMD for the details of bit fields.
SCB4_I2C_S_CMD	0x4028006C	I2C slave command register.. See SCB0_I2C_S_CMD for the details of bit fields.
SCB4_I2C_CFG	0x40280070	I2C configuration register.. See SCB0_I2C_CFG for the details of bit fields.
SCB4_TX_CTRL	0x40280200	Transmitter control register.. See SCB0_TX_CTRL for the details of bit fields.
SCB4_TX_FIFO_CTRL	0x40280204	Transmitter FIFO control register.. See SCB0_TX_FIFO_CTRL for the details of bit fields.
SCB4_TX_FIFO_STATUS	0x40280208	Transmitter FIFO status register.. See SCB0_TX_FIFO_STATUS for the details of bit fields.
SCB4_TX_FIFO_WR	0x40280240	Transmitter FIFO write register.. See SCB0_TX_FIFO_WR for the details of bit fields.
SCB4_RX_CTRL	0x40280300	Receiver control register.. See SCB0_RX_CTRL for the details of bit fields.
SCB4_RX_FIFO_CTRL	0x40280304	Receiver FIFO control register.. See SCB0_RX_FIFO_CTRL for the details of bit fields.
SCB4_RX_FIFO_STATUS	0x40280308	Receiver FIFO status register.. See SCB0_RX_FIFO_STATUS for the details of bit fields.
SCB4_RX_MATCH	0x40280310	Slave address and mask register.. See SCB0_RX_MATCH for the details of bit fields.
SCB4_RX_FIFO_RD	0x40280340	Receiver FIFO read register.. See SCB0_RX_FIFO_RD for the details of bit fields.
SCB4_RX_FIFO_RD_SILENT	0x40280344	Receiver FIFO read register.. See SCB0_RX_FIFO_RD_SILENT for the details of bit fields.
SCB4_EZ_DATA0	0x40280400	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB4_EZ_DATA1	0x40280404	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB4_EZ_DATA2	0x40280408	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB4_EZ_DATA3	0x4028040C	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB4_EZ_DATA4	0x40280410	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB4_EZ_DATA5	0x40280414	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB4_EZ_DATA6	0x40280418	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB4_EZ_DATA7	0x4028041C	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB4_EZ_DATA8	0x40280420	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB4_EZ_DATA9	0x40280424	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB4_EZ_DATA10	0x40280428	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.

Register	Address	Description
SCB4_EZ_DATA11	0x4028042C	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB4_EZ_DATA12	0x40280430	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB4_EZ_DATA13	0x40280434	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB4_EZ_DATA14	0x40280438	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB4_EZ_DATA15	0x4028043C	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB4_EZ_DATA16	0x40280440	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB4_EZ_DATA17	0x40280444	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB4_EZ_DATA18	0x40280448	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB4_EZ_DATA19	0x4028044C	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB4_EZ_DATA20	0x40280450	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB4_EZ_DATA21	0x40280454	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB4_EZ_DATA22	0x40280458	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB4_EZ_DATA23	0x4028045C	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB4_EZ_DATA24	0x40280460	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB4_EZ_DATA25	0x40280464	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB4_EZ_DATA26	0x40280468	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB4_EZ_DATA27	0x4028046C	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB4_EZ_DATA28	0x40280470	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB4_EZ_DATA29	0x40280474	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB4_EZ_DATA30	0x40280478	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB4_EZ_DATA31	0x4028047C	Memory buffer registers.. See SCB0_EZ_DATA0 for the details of bit fields.
SCB4_INTR_CAUSE	0x40280E00	Active clocked interrupt signal register. See SCB0_INTR_CAUSE for the details of bit fields.
SCB4_INTR_I2C_EC	0x40280E80	Externally clocked I2C interrupt request register. See SCB0_INTR_I2C_EC for the details of bit fields.
SCB4_INTR_I2C_EC_MASK	0x40280E88	Externally clocked I2C interrupt mask register. See SCB0_INTR_I2C_EC_MASK for the details of bit fields.
SCB4_INTR_I2C_EC_MASKED	0x40280E8C	Externally clocked I2C interrupt masked register. See SCB0_INTR_I2C_EC_MASKED for the details of bit fields.
SCB4_INTR_SPI_EC	0x40280EC0	Externally clocked SPI interrupt request register. See SCB0_INTR_SPI_EC for the details of bit fields.
SCB4_INTR_SPI_EC_MASK	0x40280EC8	Externally clocked SPI interrupt mask register. See SCB0_INTR_SPI_EC_MASK for the details of bit fields.
SCB4_INTR_SPI_EC_MASKED	0x40280ECC	Externally clocked SPI interrupt masked register. See SCB0_INTR_SPI_EC_MASKED for the details of bit fields.
SCB4_INTR_M	0x40280F00	Master interrupt request register.. See SCB0_INTR_M for the details of bit fields.
SCB4_INTR_M_SET	0x40280F04	Master interrupt set request register. See SCB0_INTR_M_SET for the details of bit fields.
SCB4_INTR_M_MASK	0x40280F08	Master interrupt mask register.. See SCB0_INTR_M_MASK for the details of bit fields.
SCB4_INTR_M_MASKED	0x40280F0C	Master interrupt masked request register. See SCB0_INTR_M_MASKED for the details of bit fields.
SCB4_INTR_S	0x40280F40	Slave interrupt request register.. See SCB0_INTR_S for the details of bit fields.
SCB4_INTR_S_SET	0x40280F44	Slave interrupt set request register.. See SCB0_INTR_S_SET for the details of bit fields.
SCB4_INTR_S_MASK	0x40280F48	Slave interrupt mask register.. See SCB0_INTR_S_MASK for the details of bit fields.
SCB4_INTR_S_MASKED	0x40280F4C	Slave interrupt masked request register. See SCB0_INTR_S_MASKED for the details of bit fields.
SCB4_INTR_TX	0x40280F80	Transmitter interrupt request register.. See SCB0_INTR_TX for the details of bit fields.
SCB4_INTR_TX_SET	0x40280F84	Transmitter interrupt set request register. See SCB0_INTR_TX_SET for the details of bit fields.
SCB4_INTR_TX_MASK	0x40280F88	Transmitter interrupt mask register.. See SCB0_INTR_TX_MASK for the details of bit fields.

Register	Address	Description
SCB4_INTR_TX_MASKED	0x40280F8C	Transmitter interrupt masked request register. See SCB0_INTR_TX_MASKED for the details of bit fields.
SCB4_INTR_RX	0x40280FC0	Receiver interrupt request register.. See SCB0_INTR_RX for the details of bit fields.
SCB4_INTR_RX_SET	0x40280FC4	Receiver interrupt set request register.. See SCB0_INTR_RX_SET for the details of bit fields.
SCB4_INTR_RX_MASK	0x40280FC8	Receiver interrupt mask register.. See SCB0_INTR_RX_MASK for the details of bit fields.
SCB4_INTR_RX_MASKED	0x40280FCC	Receiver interrupt masked request register. See SCB0_INTR_RX_MASKED for the details of bit fields.

12.1.1 SCB0_CTRL

Generic control register.

Address: 0x40240000

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None				RW			
HW Access	None				R			
Name	None [7:4]				OVS [3:0]			

Bits	15	14	13	12	11	10	9	8
SW Access	None				RW	RW	RW	RW
HW Access	None				R	R	R	R
Name	None [15:12]				BYTE_- MODE	EZ_MODE	EC_OP_- MODE	EC_AM_- MODE

Bits	23	22	21	20	19	18	17	16
SW Access	None						RW	RW
HW Access	None						R	R
Name	None [23:18]						BLOCK	ADDR_AC- CEPT

Bits	31	30	29	28	27	26	25	24
SW Access	RW	None					RW	
HW Access	R	None					R	
Name	ENABLED	None [30:26]					MODE [25:24]	

Bits	Name	Description
31	ENABLED	SCB block is enabled ('1') or not ('0'). The proper order in which to initialize SCB is as follows: - Program protocol specific information using SPI_CTRL, UART_CTRL (and UART_TX_CTRL and UART_RX_CTRL) or I2C_CTRL registers. This includes selection of a submode, master/slave functionality and transmitter/receiver functionality when applicable. - Program generic transmitter (TX_CTRL) and receiver (RX_CTRL) information. This includes enabling of the transmitter and receiver functionality. - Program transmitter FIFO (TX_FIFO_CTRL) and receiver FIFO (RX_FIFO_CTRL) information. - Program CTRL register to enable SCB, select the specific operation mode and oversampling factor. When this block is enabled, no control information should be changed. Changes should be made AFTER disabling this block, e.g. to modify the operation mode (from I2C to SPI) or to go from externally to internally clocked. The change takes effect after the block is re-enabled. Note that disabling the block will cause re-initialization of the design and associated state is lost (e.g. FIFO content). Default Value: 0
25 : 24	MODE	Mode of operation (3: Reserved) Default Value: 3

12.1.1 SCB0_CTRL (continued)

0x0: I2C :

Inter-Integrated Circuits (I2C) mode.

0x1: SPI :

Serial Peripheral Interface (SPI) mode.

0x2: UART :

Universal Asynchronous Receiver/Transmitter (UART) mode.

17	BLOCK	Only used in externally clocked mode. If the externally clocked logic and the internal CPU accesses to EZ memory coincide/collide, this bit determines whether the CPU access should block and result in bus wait states ('BLOCK is 1') or not ('BLOCK is 0'). If BLOCK is '0' and the accesses collide, CPU read operations return 0xffff:ffff and CPU write operations are ignored. Colliding accesses are registered as interrupt causes: INTR_TX.BLOCKED and INTR_RX.BLOCKED. Default Value: 0
16	ADDR_ACCEPT	Determines whether a received matching address is accepted in the RX FIFO ('1') or not ('0'). In I2C mode, this field is used to allow the slave to put the received slave address or general call address in the RX FIFO. Note that a received matching address is put in the RX FIFO when this bit is '1' for both I2C read and write transfers. In multi-processor UART receiver mode, this field is used to allow the receiver to put the received address in the RX FIFO. Note: non-matching addresses are never put in the RX FIFO. Default Value: 0
11	BYTE_MODE	Determines the number of bits per FIFO data element: '0': 16-bit FIFO data elements. '1': 8-bit FIFO data elements. This mode doubles the amount of FIFO entries, but TX_CTRL.DATA_WIDTH and RX_CTRL.DATA_WIDTH are restricted to [0, 7]. Default Value: 0
10	EZ_MODE	Non EZ mode ('0') or EZ mode ('1'). In EZ mode, a meta protocol is applied to the serial interface protocol. This meta protocol adds meaning to the data frames transferred by the serial interface protocol: a data frame can represent a memory address, a write memory data element or a read memory data element. EZ mode is only used for synchronous serial interface protocols: SPI and I2C. In SPI mode, only Motorola submode (all Motorola modes: 0, 1, 2, 3) is supported and the transmitter should use continuous data frames; i.e. data frames not separated by slave deselection. This mode is only applicable to slave functionality. In EZ mode, the slave can read from and write to an addressable memory structure of 32 bytes. In EZ mode, data frames should 8-bit in size and should be transmitted and received with the Most Significant Bit (MSB) first. In UART mode this field should be '0'. Default Value: 0

12.1.1 SCB0_CTRL (continued)

9	EC_OP_MODE	This field specifies the clocking for the SCB block '0': Internally clocked mode '1': externally clocked mode In internally clocked mode, the serial interface protocols run off the SCB clock. In externally clocked mode, the serial interface protocols run off the clock as provided by the serial interface. Externally clocked operation mode is only used for synchronous serial interface protocols (SPI and I2C) in slave mode AND EZ mode. In SPI mode, only Motorola submode (all Motorola modes: 0, 1, 2, 3) is supported. The maximum SPI slave, EZ mode bitrate is 48 Mbps (transmission and IO delays outside the IP will degrade the effective bitrate). In UART mode this field should be '0'. Default Value: 0
8	EC_AM_MODE	This field specifies the clocking for the address matching (I2C) or slave selection detection logic (SPI) '0': Internally clocked mode '1': Externally clocked mode In internally clocked mode, the serial interface protocols run off the SCB clock. In externally clocked mode, the serial interface protocols run off the clock as provided by the serial interface. The clocking for the rest of the logic is determined by CTRL.EC_OP_MODE. Externally clocked mode is only used for synchronous serial interface protocols (SPI and I2C) in slave mode. In SPI mode, only Motorola submode (all Motorola modes: 0, 1, 2, 3) is supported. In UART mode this field should be '0'. Default Value: 0

12.1.1 SCB0_CTRL (continued)

3 : 0 OVS

Serial interface bit period oversampling factor expressed in SCB clock cycles. Used for SPI and UART functionality. $OVS + 1$ SCB clock cycles constitute a single serial interface clock/bit cycle. This field is NOT used in externally clocked mode. If OVS is odd, the oversampling factor is even and the low and high phase of the interface clock period are the same. If OVS is even, the oversampling factor is odd and the low and high phase can differ by 1 SCB clock period.

In SPI master mode, the valid range is [3, 15]. At an SCB frequency of 48 MHz, the maximum SPI bit rate is 12 Mbps, assuming ideal (0 ns) IO cell and routing (chip and board) delay. The calculated maximum bit rate decreases, when realistic chip and routing delays are taken into account. The effective system bit rate is dependent on the external SPI slave that we communicate with. If the SPI output clock to SPI MISO input round trip delay is significant (multiple SPI output clock cycles), it may be necessary to increase OVS and/or to set SPI_CTRL.LATE_MISO_SAMPLE to '1' to achieve the maximum possible system bit rate.

In SPI slave mode, the OVS field is NOT used. However, there is a frequency requirement for the SCB clock wrt. the SPI input clock (IF) on the interface to guarantee functional correct behavior. This requirement is expressed as a ratio: SCB clock/IF clock. The ratio is dependent on the setting of RX_CTRL.MEDIAN and the external SPI master's capability to support "late MISO sample" functionality (similar to our SPI master functionality represented by SPI_CTRL.LATE_MISO_SAMPLE):

- MEDIAN is '0' and external SPI master has NO "late MISO sample functionality": SCB clock/IF clock ≥ 6 . At a SCB frequency of 48 MHz, the maximum bit rate is 8 Mbps.
- MEDIAN is '0' and external SPI master has "late MISO sample functionality": SCB clock/IF clock ≥ 3 . At a SCB frequency of 48 MHz, the maximum bit rate is 16 Mbps.
- MEDIAN is '1' and external SPI master has NO "late MISO sample functionality": SCB clock/IF clock ≥ 8 . At a SCB frequency of 48 MHz, the maximum bit rate is 6 Mbps.
- MEDIAN is '1' and external SPI master has "late MISO sample functionality": SCB clock/IF clock ≥ 4 . At a SCB frequency of 48 MHz, the maximum bit rate is 12 Mbps.

As discussed earlier, the calculated maximum bit rate decreases, when realistic chip and routing delays are taken into account.

In UART standard submode (including LIN), the valid range is [7, 15]. In UART SmartCard submode, the valid range is [7, 15].

In UART TX IrDA submode this field indirectly specifies the oversampling. The oversampling determines the interface clock/bit cycle and the width of the pulse. Only normal transmission mode is supported, the pulse is roughly 3/16 of the bit period (for all bit rates). There is only one valid OVS value:

- 0: 16 times oversampling.
 SCB clock frequency of 16×115.2 KHz for 115.2 Kbps.
 SCB clock frequency of 16×57.6 KHz for 57.6 Kbps.
 SCB clock frequency of 16×38.4 KHz for 38.4 Kbps.
 SCB clock frequency of 16×19.2 KHz for 19.2 Kbps.
 SCB clock frequency of 16×9.6 KHz for 9.6 Kbps.
 SCB clock frequency of 16×2.4 KHz for 2.4 Kbps.
 SCB clock frequency of 16×1.2 KHz for 1.2 Kbps.
 - all other values are not used in normal mode.

In UART RX IrDA submode (1.2, 2.4, 9.6, 19.2, 38.4, 57.6 and 115.2 Kbps) this field indirectly specifies the oversampling. The oversampling determines the interface clock/bit cycle and the width of the pulse. In normal transmission mode, this pulse is roughly 3/16 of the bit period (for all bit rates). In low power transmission mode, this pulse is potentially smaller (down to 1.62 μ s typical and 1.41 μ s minimal) than 3/16 of the bit period (for < 115.2 Kbps bitrates). Pulse widths greater or equal than two SCB clock cycles are guaranteed to be detected by the receiver. Pulse widths less than two SCB clock cycles and greater or equal than one SCB clock cycle may be detected by the receiver. Pulse widths less than one SCB clock cycle will not be detected by the receiver. RX_CTRL.MEDIAN should be set to '1' for IrDA receiver functionality. The SCB clock (as provided by the programmable clock block) and the oversampling together determine the IrDA bitrate. Normal mode, OVS field values (with the required SCB clock frequency):

- 0: 16 times oversampling.
 SCB clock frequency of 16×115.2 KHz for 115.2 Kbps.
 SCB clock frequency of 16×57.6 KHz for 57.6 Kbps.
 SCB clock frequency of 16×38.4 KHz for 38.4 Kbps.
 SCB clock frequency of 16×19.2 KHz for 19.2 Kbps.
 SCB clock frequency of 16×9.6 KHz for 9.6 Kbps.
 SCB clock frequency of 16×2.4 KHz for 2.4 Kbps.
 SCB clock frequency of 16×1.2 KHz for 1.2 Kbps.
 - all other values are not used in normal mode.

Low power mode, OVS field values (with the required SCB clock frequency):

- 0: 16 times oversampling.
 SCB clock frequency of 16×115.2 KHz for 115.2 Kbps.
- 1: 32 times oversampling.
 SCB clock frequency of 32×57.6 KHz for 57.6 Kbps.
- 2: 48 times oversampling.
 SCB clock frequency of 48×38.4 KHz for 38.4 Kbps.
- 3: 96 times oversampling.
 SCB clock frequency of 96×19.2 KHz for 19.2 Kbps.
- 4: 192 times oversampling.
 SCB clock frequency of 192×9.6 KHz for 9.6 Kbps.
- 5: 768 times oversampling.
 SCB clock frequency of 768×2.4 KHz for 2.4 Kbps.
- 6: 1536 times oversampling.
 SCB clock frequency of 1536×1.2 KHz for 1.2 Kbps.
- all other values are not used in low power mode.

Default Value: 15

12.1.2 SCB0_STATUS

Generic status register.

Address: 0x40240004

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None							R
HW Access	None							W
Name	None [7:1]							EC_BUSY

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
0	EC_BUSY	Indicates whether the externally clocked logic is potentially accessing the EZ memory (this is only possible in EZ mode). This bit can be used by SW to determine whether it is safe to issue a SW access to the EZ memory (without bus wait states (a blocked SW access) or bus errors being generated). Note that the INTR_TX.BLOCKED and INTR_RX.BLOCKED interrupt causes are used to indicate whether a SW access was actually blocked by externally clocked logic. Default Value: Undefined

12.1.3 SCB0_SPI_CTRL

SPI control register.

Address: 0x40240020

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None		RW	RW	RW	RW	RW	RW
HW Access	None		R	R	R	R	R	R
Name	None [7:6]		SCLK_- CONTINU- OUS	LATE_MISO_ SAMPLE	CPOL	CPHA	SE- LECT_PRE- CEDE	CONTINU- OUS

Bits	15	14	13	12	11	10	9	8
SW Access	None				RW	RW	RW	RW
HW Access	None				R	R	R	R
Name	None [15:12]				SSEL_PO- LARITY3	SSEL_PO- LARITY2	SSEL_PO- LARITY1	SSEL_PO- LARITY0

Bits	23	22	21	20	19	18	17	16
SW Access	None							RW
HW Access	None							R
Name	None [23:17]							LOOPBACK

Bits	31	30	29	28	27	26	25	24
SW Access	RW	None			RW		RW	
HW Access	R	None			R		R	
Name	MASTER_- MODE	None [30:28]			SLAVE_SELECT [27:26]		MODE [25:24]	

Bits	Name	Description
31	MASTER_MODE	Master ('1') or slave ('0') mode. In master mode, transmission will commence on availability of data frames in the TX FIFO. In slave mode, when selected and there is no data frame in the TX FIFO, the slave will transmit all '1's. In both master and slave modes, received data frames will be lost if the RX FIFO is full. Default Value: 0
27 : 26	SLAVE_SELECT	Selects one of the four outgoing SPI slave select signals: - 0: Slave 0, SPI_SELECT[0]. - 1: Slave 1, SPI_SELECT[1]. - 2: Slave 2, SPI_SELECT[2]. - 3: Slave 3, SPI_SELECT[3]. Only used in master mode. SCB block should be disabled when changes are made to this field. Default Value: 0
25 : 24	MODE	Submode of SPI operation (3: Reserved). Default Value: 3

12.1.3 SCB0_SPI_CTRL (continued)

0x0: SPI_MOTOROLA :

SPI Motorola submode. In master mode, when not transmitting data (SELECT is inactive), SCLK is stable at CPOL. In slave mode, when not selected, SCLK is ignored; i.e. it can be either stable or clocking. In master mode, when there is no data to transmit (TX FIFO is empty), SELECT is inactive.

0x1: SPI_TI :

SPI Texas Instruments submode. In master mode, when not transmitting data, SCLK is stable at '0'. In slave mode, when not selected, SCLK is ignored; i.e. it can be either stable or clocking. In master mode, when there is no data to transmit (TX FIFO is empty), SELECT is inactive; i.e. no pulse is generated.

0x2: SPI_NS :

SPI National Semiconducturs submode. In master mode, when not transmitting data, SCLK is stable at '0'. In slave mode, when not selected, SCLK is ignored; i.e. it can be either stable or clocking. In master mode, when there is no data to transmit (TX FIFO is empty), SELECT is inactive.

16	LOOPBACK	Local loopback control (does NOT affect the information on the pins). Only used in master mode. Not used in National Semiconductors submode. '0': No local loopback '1': the SPI master MISO line is connected to the SPI master MOSI line. In other words, in loop-back mode the SPI master receives on MISO what it transmits on MOSI. Default Value: 0
11	SSEL_POLARITY3	Slave select polarity. SSEL_POLARITY3 applies to the outgoing SPI slave select signal 3 (master mode). Default Value: 0
10	SSEL_POLARITY2	Slave select polarity. SSEL_POLARITY2 applies to the outgoing SPI slave select signal 2 (master mode). Default Value: 0
9	SSEL_POLARITY1	Slave select polarity. SSEL_POLARITY1 applies to the outgoing SPI slave select signal 1 (master mode). Default Value: 0
8	SSEL_POLARITY0	Slave select polarity. SSEL_POLARITY0 applies to the outgoing SPI slave select signal 0 (master mode) and to the incoming SPI slave select signal (slave mode). only SPI_SELECT[0] is used in slave mode. For Motorola and National Semiconductors submodes: '0': slave select is low/'0' active. '1': slave select is high/'1' active. For Texas Instruments submode: '0': high/'1' active precede/coincide pulse. '1': low/'0' active precede/coincide pulse. Default Value: 0
5	SCLK_CONTINUOUS	Only applicable in master mode. '0': SCLK is generated, when the SPI master is enabled and data is transmitted. '1': SCLK is generated, when the SPI master is enabled. This mode is useful for slave devices that use SCLK for functional operation other than just SPI functionality. Default Value: 0

12.1.3 SCB0_SPI_CTRL (continued)

4	LATE_MISO_SAMPLE	Changes the SCLK edge on which MISO is captured. Only used in master mode. When '0', the default applies (for Motorola as determined by CPOL and CPHA, for Texas Instruments on the falling edge of SCLK and for National Semiconductors on the rising edge of SCLK). When '1', the alternate clock edge is used (which comes half a SPI SCLK period later). Late sampling addresses the round trip delay associated with transmitting SCLK from the master to the slave and transmitting MISO from the slave to the master. Default Value: 0
3	CPOL	Indicates the clock polarity. Only used in SPI Motorola submode. This field, together with the CPHA field, indicates when MOSI data is driven and MISO data is captured: - CPOL is '0': SCLK is '0' when not transmitting data. - CPOL is '1': SCLK is '1' when not transmitting data. Default Value: 0
2	CPHA	Only applicable in SPI Motorola submode. Indicates the clock phase. This field, together with the CPOL field, indicates when MOSI data is driven and MISO data is captured: - Motorola mode 0. CPOL is '0', CPHA is '0': MOSI is driven on a falling edge of SCLK. MISO is captured on a rising edge of SCLK. - Motorola mode 1. CPOL is '0', CPHA is '1': MOSI is driven on a rising edge of SCLK. MISO is captured on a falling edge of SCLK. - Motorola mode 2. CPOL is '1', CPHA is '0': MOSI is driven on a rising edge of SCLK. MISO is captured on a falling edge of SCLK. - Motorola mode 3. CPOL is '1', CPHA is '1': MOSI is driven on a falling edge of SCLK. MISO is captured on a rising edge of SCLK. Default Value: 0
1	SELECT_PRECEDE	Only used in SPI Texas Instruments' submode. When '1', the data frame start indication is a pulse on the Slave SELECT line that precedes the transfer of the first data frame bit. When '0', the data frame start indication is a pulse on the Slave SELECT line that coincides with the transfer of the first data frame bit. Default Value: 0
0	CONTINUOUS	Continuous SPI data transfers enabled ('1') or not ('0'). This field is used in master mode. In slave mode, both continuous and non-continuous SPI data transfers are supported independent of this field. When continuous transfers are enabled individual data frame transfers are not necessarily separated by slave deselection (as indicated by the level or pulse on the SELECT line): if the TX FIFO has multiple data frames, data frames are sent out without slave deselection. When continuous transfers are not enabled individual data frame transfers are always separated by slave deselection: independent of the availability of TX FIFO data frames, data frames are sent out with slave deselection. Default Value: 0

12.1.4 SCB0_SPI_STATUS

SPI status register.

Address: 0x40240024

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None						R	R
HW Access	None						W	W
Name	None [7:2]						SPI_EC_BUSY	BUS_BUSY

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access	W							
Name	CURR_EZ_ADDR [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access	W							
Name	BASE_EZ_ADDR [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
23 : 16	BASE_EZ_ADDR	SPI base EZ address. Address as provided by a SPI write transfer. This field is only reliable in internally clocked mode. In externally clocked mode the field may be unreliable, as clock domain synchronization is not performed in the design. Default Value: Undefined
15 : 8	CURR_EZ_ADDR	SPI current EZ address. Current address pointer. This field is only reliable in internally clocked mode. In externally clocked mode the field may be unreliable (during an ongoing transfer when SPI_EC_BUSY is '1'), as clock domain synchronization is not performed in the design. Default Value: Undefined
1	SPI_EC_BUSY	Indicates whether the externally clocked logic is potentially accessing the EZ memory and/or updating BASE_ADDR or CURR_ADDR (this is only possible in EZ mode). This bit can be used by SW to determine whether BASE_ADDR and CURR_ADDR are reliable. Default Value: Undefined
0	BUS_BUSY	SPI bus is busy. The bus is considered busy ('1') during an ongoing transaction. For Motorola and National submodes, the busy bit is '1', when the slave selection (low active) is activated. For TI submode, the busy bit is '1' from the time the preceding/coinciding slave select (high active) is activated for the first transmitted data frame, till the last MOSI/MISO bit of the last data frame is transmitted. Default Value: Undefined

12.1.5 SCB0_UART_CTRL

UART control register.

Address: 0x40240040

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None							
HW Access	None							
Name	None [7:0]							
Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							
Bits	23	22	21	20	19	18	17	16
SW Access	None							RW
HW Access	None							R
Name	None [23:17]							LOOPBACK
Bits	31	30	29	28	27	26	25	24
SW Access	None						RW	
HW Access	None						R	
Name	None [31:26]						MODE [25:24]	

Bits	Name	Description
25 : 24	MODE	Submode of UART operation (3: Reserved) Default Value: 3 0x0: UART_STD : Standard UART submode. 0x1: UART_SMARTCARD : SmartCard (ISO7816) submode. Support for negative acknowledgement (NACK) on the receiver side and retransmission on the transmitter side. 0x2: UART_IRDA : Infrared Data Association (IrDA) submode. Return to Zero modulation scheme. In this mode, the oversampling factor should be 16, that is OVS is 15.

12.1.5 SCB0_UART_CTRL (continued)

16	LOOPBACK	Local loopback control (does NOT affect the information on the pins). 0: Loopback is not enabled 1: UART_TX is connected to UART_RX. UART_RTS is connected to UART_CTS. This allows a SCB UART transmitter to communicate with its receiver counterpart. Default Value: 0
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12.1.6 SCB0_UART_TX_CTRL

UART transmitter control register.

Address: 0x40240044

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None		RW	RW	None	RW		
HW Access	None		R	R	None	R		
Name	None [7:6]		PARITY_ENABLED	PARITY	None	STOP_BITS [2:0]		

Bits	15	14	13	12	11	10	9	8
SW Access	None							RW
HW Access	None							R
Name	None [15:9]							RETRY_ON_NACK

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
8	RETRY_ON_NACK	When '1', a data frame is retransmitted when a negative acknowledgement is received. Only applicable to the SmartCard submode. Default Value: 0
5	PARITY_ENABLED	Parity generation enabled ('1') or not ('0'). Only applicable in standard UART submodes. In SmartCard submode, parity generation is always enabled through hardware. In IrDA submode, parity generation is always disabled through hardware Default Value: 0
4	PARITY	Parity bit. When '0', the transmitter generates an even parity. When '1', the transmitter generates an odd parity. Only applicable in standard UART and SmartCard submodes. Default Value: 0
2 : 0	STOP_BITS	Stop bits. STOP_BITS + 1 is the duration of the stop period in terms of halve bit periods. Valid range is [1, 7]; i.e. a stop period should last at least one bit period. Default Value: 2

12.1.7 SCB0_UART_RX_CTRL

UART receiver control register.

Address: 0x40240048

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None	RW	RW	RW	None	RW		
HW Access	None	R	R	R	None	R		
Name	None	POLARITY	PARITY_ENABLED	PARITY	None	STOP_BITS [2:0]		

Bits	15	14	13	12	11	10	9	8
SW Access	None		RW	RW	None	RW	RW	RW
HW Access	None		R	R	None	R	R	R
Name	None [15:14]		SKIP_START	LIN_MODE	None	MP_MODE	DROP_ON_FRAME_ERROR	DROP_ON_PARITY_ERROR

Bits	23	22	21	20	19	18	17	16
SW Access	None				RW			
HW Access	None				R			
Name	None [23:20]				BREAK_WIDTH [19:16]			

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
19 : 16	BREAK_WIDTH	Break width. BREAK_WIDTH + 1 is the minimum width in bit periods of a break. During a break the transmitted/received line value is '0'. This feature is useful for standard UART submode and LIN submode ("break field" detection). Once, the break is detected, the INTR_RX.BREAK_DETECT bit is set to '1'. Note that break detection precedes baud rate detection, which is used to synchronize/refine the receiver clock to the transmitter clock. As a result, break detection operates with an unsynchronized/unrefined receiver clock. Therefore, the receiver's definition of a bit period is imprecise and the setting of this field should take this imprecision into account. The LIN standard also accounts for this imprecision: a LIN start bit followed by 8 data bits allows for up to 9 consecutive '0' bit periods during regular transmission, whereas the LIN break detection should be at least 13 consecutive '0' bit periods. This provides for a margin of 4 bit periods. Therefore, the default value of this field is set to 10, representing a minimal break field with of 10+1 = 11 bit periods; a value in between the 9 consecutive bit periods of a regular transmission and the 13 consecutive bit periods of a break field. This provides for slight imprecisions of the receiver clock wrt. the transmitter clock. There should not be a need to program this field to any value other than its default value. Default Value: 10

12.1.7 SCB0_UART_RX_CTRL (continued)

13	SKIP_START	Only applicable in standard UART submode. When '1', the receiver skips start bit detection for the first received data frame. Instead, it synchronizes on the first received data frame bit, which should be a '1'. This functionality is intended for wake up from DeepSleep when receiving a data frame. The transition from idle ('1') to START ('0') on the RX line is used to wake up the CPU. The transition detection (and the associated wake up functionality) is performed by the GPIO2 IP. The woken up CPU will enable the SCB's UART receiver functionality. Once enabled, it is assumed that the START bit is ongoing (the CPU wakeup and SCB enable time should be less than the START bit period). The SCB will synchronize to a '0' to '1' transition, which indicates the first data frame bit is received (first data frame bit should be '1'). After synchronization to the first data frame bit, the SCB will resume normal UART functionality: subsequent data frames will be synchronized on the receipt of a START bit. Default Value: 0
12	LIN_MODE	Only applicable in standard UART submode. When '1', the receiver performs break detection and baud rate detection on the incoming data. First, break detection counts the amount of bit periods that have a line value of '0'. BREAK_WIDTH specifies the minum required amount of bit periods. Successful break detection sets the INTR_RX.BREAK_DETECT interrupt cause to '1'. Second, baud rate detection counts the amount of peripheral clock periods that are use to receive the synchronization byte (0x55; least significant bit first). The count is available through UART_RX_STATUS.BR_COUNTER. Successful baud rate detection sets the INTR_RX.BAUD_DETECT interrupt cause to '1' (BR_COUNTER is reliable). This functionality is used to synchronize/refine the receiver clock to the transmitter clock. The receiver software can use the BR_COUNTER value to set the right IP clock (from the programmable clock IP) to guarantee successful receipt of the first LIN data frame (Protected Identifier Field) after the synchronization byte. Default Value: 0
10	MP_MODE	Multi-processor mode. When '1', multi-processor mode is enabled. In this mode, RX_CTRL.DATA_WIDTH should indicate a 9-bit data frame. In multi-processor mode, the 9th received bit of a data frame separates addresses (bit is '1') from data (bit is '0'). A received address is matched with RX_MATCH.DATA and RX_MATCH.MASK. In the case of a match, subsequent received data are sent to the RX FIFO. In the case of NO match, subsequent received data are dropped. Default Value: 0
9	DROP_ON_FRAME_ERROR	Behaviour when an error is detected in a start or stop period. When '0', received data is sent to the RX FIFO. When '1', received data is dropped and lost. Default Value: 0
8	DROP_ON_PARITY_ERROR	Behaviour when a parity check fails. When '0', received data is sent to the RX FIFO. When '1', received data is dropped and lost. Only applicable in standard UART and SmartCard submodes (negatively acknowledged SmartCard data frames may be dropped with this field). Default Value: 0
6	POLARITY	Inverts incoming RX line signal. Inversion is after local loopback. This functionality is intended for IrDA receiver functionality. Default Value: 0
5	PARITY_ENABLED	Parity checking enabled ('1') or not ('0'). Only applicable in standard UART submode. In SmartCard submode, parity checking is always enabled through hardware. In IrDA submode, parity checking is always disabled through hardware. Default Value: 0
4	PARITY	Parity bit. When '0', the receiver expects an even parity. When '1', the receiver expects an odd parity. Only applicable in standard UART and SmartCard submodes. Default Value: 0

12.1.7 SCB0_UART_RX_CTRL (continued)

2 : 0	STOP_BITS	Stop bits. STOP_BITS + 1 is the duration of the stop period in terms of halve bit periods. Valid range is [1, 7]; i.e. a stop period should last at least one bit period. If STOP_BITS is '1', stop bits error detection is NOT performed. If STOP_BITS is in [2, 7], stop bits error detection is performed and the associated interrupt cause INTR_RX.FRAME_ERROR is set to '1' if an error is detected. In other words, the receiver supports data frames with a 1 bit period stop bit sequence, but requires at least 1.5 bit period stop bit sequences to detect errors. This limitation is due to possible transmitter and receiver clock skew that prevents the design from doing reliable stop bit detection for short (1 bit bit period) stop bit sequences. Note that in case of a stop bits error, the successive data frames may get lost as the receiver needs to resynchronize its start bit detection. The amount of lost data frames depends on both the amount of stop bits, the idle ('1') time between data frames and the data frame value. Default Value: 2
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12.1.8 SCB0_UART_RX_STATUS

UART receiver status register.

Address: 0x4024004C

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access	W							
Name	BR_COUNTER [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	None				R			
HW Access	None				W			
Name	None [15:12]				BR_COUNTER [11:8]			

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
11 : 0	BR_COUNTER	Amount of SCB clock periods that constitute the transmission of a 0x55 data frame (sent least significant bit first) as determined by the receiver. BR_COUNTER / 8 is the amount of SCB clock periods that constitute a bit period. This field has valid data when INTR_RX.BAUD_DETECT is set to '1'. Default Value: Undefined

12.1.9 SCB0_UART_FLOW_CTRL

UART flow control register

Address: 0x40240050

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None				RW			
HW Access	None				R			
Name	None [7:4]				TRIGGER_LEVEL [3:0]			
Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							
Bits	23	22	21	20	19	18	17	16
SW Access	None							RW
HW Access	None							R
Name	None [23:17]							RTS_POLARITY
Bits	31	30	29	28	27	26	25	24
SW Access	None						RW	RW
HW Access	None						R	R
Name	None [31:26]						CTS_ENABLED	CTS_POLARITY

Bits	Name	Description
25	CTS_ENABLED	Enable use of CTS input signal by the UART transmitter: '0': Disabled. The UART transmitter ignores the CTS input signal and transmits when a data frame is available for transmission in the TX FIFO or the TX shift register. '1': Enabled. The UART transmitter uses CTS input signal to qualify the transmission of data. It transmits when CTS input signal is active and a data frame is available for transmission in the TX FIFO or the TX shift register. If UART_CTRL.LOOPBACK is '1', the CTS input signal is driven by the RTS output signal locally in SCB (both signals are subjected to signal polarity changes as indicated by RTS_POLARITY and CTS_POLARITY). Default Value: 0
24	CTS_POLARITY	Polarity of the CTS input signal '0': CTS is active low ; '1': CTS is active high; Default Value: 0

12.1.9 SCB0_UART_FLOW_CTRL (continued)

16	RTS_POLARITY	Polarity of the RTS output signal: '0': RTS is active low; '1': RTS is active high; During SCB reset (Hibernate system power mode), RTS output signal is '1'. This represents an inactive state assuming an active low polarity. Default Value: 0
3 : 0	TRIGGER_LEVEL	Trigger level. When the receiver FIFO has less entries than the amount of this field, a Ready To Send (RTS) output signal is activated. By setting this field to "0", flow control is effectively disabled (may be useful for debug purposes). Default Value: 0

12.1.10 SCB0_I2C_CTRL

I2C control register.

Address: 0x40240060

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW				RW			
HW Access	R				R			
Name	LOW_PHASE_OVS [7:4]				HIGH_PHASE_OVS [3:0]			

Bits	15	14	13	12	11	10	9	8
SW Access	RW	RW	RW	RW	RW	None	RW	RW
HW Access	R	R	R	R	R	None	R	R
Name	S_NOT_READY_DATA_NACK	S_NOT_READY_ADDR_NACK	S_READY_DATA_ACK	S_READY_ADDR_ACK	S_GENERAL_IGNORE	None	M_NOT_READY_DATA_NACK	M_READY_DATA_ACK

Bits	23	22	21	20	19	18	17	16
SW Access	None							RW
HW Access	None							R
Name	None [23:17]							LOOPBACK

Bits	31	30	29	28	27	26	25	24
SW Access	RW	RW	None					
HW Access	R	R	None					
Name	MASTER_MODE	SLAVE_MODE	None [29:24]					

Bits	Name	Description
31	MASTER_MODE	Master mode enabled ('1') or not ('0'). Note that both master and slave modes can be enabled at the same time. This allows the IP to address itself. Default Value: 0
30	SLAVE_MODE	Slave mode enabled ('1') or not ('0'). Default Value: 0
16	LOOPBACK	Local loopback control (does NOT affect the information on the pins). Only applicable in master/slave mode. When '0', no loopback When '1', loopback is enabled internally in the peripheral, and as a result unaffected by other I2C devices. This allows a SCB I2C peripheral to address itself. Default Value: 0

12.1.10 SCB0_I2C_CTRL (continued)

15	S_NOT_READY_DATA_NACK	Only used when: - non EZ mode Functionality is as follows: - 1: a received data element by the slave is immediately NACK'd when the receiver FIFO is full. - 0: clock stretching is performed (till the receiver FIFO is no longer full). Default Value: 1
14	S_NOT_READY_ADDR_NACK	This field is used during an address match or general call address in internally clocked mode Only used when: - EC_AM_MODE is '0', EC_OP_MODE is '0', S_GENERAL_IGNORE is '0' and non EZ mode. Functionality is as follows: - 1: a received (matching) slave address is immediately NACK'd when the receiver FIFO is full. - 0: clock stretching is performed (till the receiver FIFO is no longer full). For externally clocked logic (EC_AM is '1') on an address match or general call address (and S_GENERAL_IGNORE is '0'). Only used when (NOT used when EC_AM is '1' and EC_OP is '1' and address match and EZ mode): - EC_AM is '1' and EC_OP is '0'. - EC_AM is '1' and general call address match. - EC_AM is '1' and non EZ mode. Functionality is as follows: - 1: a received (matching or general) slave address is always immediately NACK'd. There are two possibilities: 1). the SCB clock is available (in Active system power mode) and it handles the rest of the current transfer. In this case the I2C master will not observe the NACK. 2). SCB clock is not present (in DeepSleep system power mode). In this case the I2C master will observe the NACK and may retry the transfer in the future (which gives the internally clocked logic the time to wake up from DeepSleep system power mode). - 0: clock stretching is performed (till the SCB clock is available). The logic will handle the ongoing transfer as soon as the clock is enabled. Default Value: 1
13	S_READY_DATA_ACK	When '1', a received data element by the slave is immediately ACK'd when the receiver FIFO is not full. In EZ mode, this field should be set to '1'. Default Value: 1
12	S_READY_ADDR_ACK	When '1', a received (matching) slave address is immediately ACK'd when the receiver FIFO is not full. In EZ mode, this field should be set to '1'. Default Value: 1
11	S_GENERAL_IGNORE	When '1', a received general call slave address is immediately NACK'd (no ACK or clock stretching) and treated as a non matching slave address. This is useful for slaves that do not need any data supplied within the general call structure. Default Value: 1
9	M_NOT_READY_DATA_NACK	When '1', a received data element byte the master is immediately NACK'd when the receiver FIFO is full. When '0', clock stretching is used instead (till the receiver FIFO is no longer full). Default Value: 1
8	M_READY_DATA_ACK	When '1', a received data element by the master is immediately ACK'd when the receiver FIFO is not full. Default Value: 1

12.1.10 SCB0_I2C_CTRL (continued)

7 : 4	LOW_PHASE_OVS	Serial I2C interface low phase oversampling factor. LOW_PHASE_OVS + 1 SCB clock periods constitute the low phase of a bit period. The valid range is [7, 15] with input signal median filtering and [6, 15] without input signal median filtering. The field is only used in master mode. In slave mode, the field is NOT used. However, there is a frequency requirement for the SCB clock wrt. the regular (no stretching) interface (IF) low time to guarantee functionally correct behavior. With input signal median filtering, the IF low time should be ≥ 8 SCB clock cycles and ≤ 16 IP clock cycles. Without input signal median filtering, the IF low time should be ≥ 7 SCB clock cycles and ≤ 16 SCB clock cycles. Default Value: 8
3 : 0	HIGH_PHASE_OVS	Serial I2C interface high phase oversampling factor. HIGH_PHASE_OVS + 1 SCB clock periods constitute the high phase of a bit period. The valid range is [5, 15] with input signal median filtering and [4, 15] without input signal median filtering. The field is only used in master mode. In slave mode, the field is NOT used. However, there is a frequency requirement for the SCB clock wrt. the regular interface (IF) high time to guarantee functional correct behavior. With input signal median filtering, the IF high time should be ≥ 6 SCB clock cycles and ≤ 16 SCB clock cycles. Without input signal median filtering, the IF high time should be ≥ 5 SCB clock cycles and ≤ 16 SCB clock cycles. Default Value: 8

12.1.11 SCB0_I2C_STATUS

I2C status register.

Address: 0x40240064

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None		R	R	None		R	R
HW Access	None		W	W	None		W	W
Name	None [7:6]		M_READ	S_READ	None [3:2]		I2C_EC_BSY	BUS_BUSY

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access	W							
Name	CURR_EZ_ADDR [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access	W							
Name	BASE_EZ_ADDR [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
23 : 16	BASE_EZ_ADDR	I2C slave base EZ address. Address as provided by an I2C write transfer. This field is only reliable in internally clocked mode. In externally clocked mode the field may be unreliable, as clock domain synchronization is not performed in the design. Default Value: Undefined
15 : 8	CURR_EZ_ADDR	I2C slave current EZ address. Current address pointer. This field is only reliable in internally clocked mode. In externally clocked mode the field may be unreliable (during an ongoing transfer when I2C_EC_BSY is '1'), as clock domain synchronization is not performed in the design. Default Value: Undefined
5	M_READ	I2C master read transfer ('1') or I2C master write transfer ('0'). When the I2C master is inactive/ idle or transmitting START, REPEATED START, STOP or an address, this field is '0'. Default Value: 0
4	S_READ	I2C slave read transfer ('1') or I2C slave write transfer ('0'). When the I2C slave is inactive/idle or receiving START, REPEATED START, STOP or an address, this field is '0'. Default Value: 0

12.1.11 SCB0_I2C_STATUS (continued)

1	I2C_EC_BUSY	Indicates whether the externally clocked logic is potentially accessing the EZ memory and/or updating BASE_EZ_ADDR or CURR_EZ_ADDR (this is only possible in EZ mode). This bit can be used by SW to determine whether BASE_EZ_ADDR and CURR_EZ_ADDR are reliable. Default Value: Undefined
0	BUS_BUSY	I2C bus is busy. The bus is considered busy ('1'), from the time a START is detected or from the time the SCL line is '0'. The bus is considered idle ('0'), from the time a STOP is detected. If SCB block is disabled, BUS_BUSY is '0'. After enabling the block, it takes time for the BUS_BUSY to detect a busy bus. This time is the maximum high time of the SCL line. For a 100 kHz interface frequency, this maximum high time may last roughly 5 us (half a bit period). For single master systems, BUS_BUSY does not have to be used to detect an idle bus before a master starts a transfer using I2C_M_CMD.M_START (no bus collisions). For multi-master systems, BUS_BUSY can be used to detect an idle bus before a master starts a transfer using I2C_M_CMD.M_START_ON_IDLE (to prevent bus collisions). Default Value: 0

12.1.12 SCB0_I2C_M_CMD

I2C master command register.

Address: 0x40240068

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None			RW	RW	RW	RW	RW
HW Access	None			RW1C	RW1C	RW1C	RW1C	RW1C
Name	None [7:5]			M_STOP	M_NACK	M_ACK	M_START_ON_IDLE	M_START

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
4	M_STOP	When '1', attempt to transmit a STOP. When this action is performed, the hardware sets this field to '0'. I2C_M_CMD.M_START has a higher priority than this command: in situations where both a STOP and a REPEATED START could be transmitted, M_START takes precedence over M_STOP. Default Value: 0
3	M_NACK	When '1', attempt to transmit a negative acknowledgement (NACK). When this action is performed, the hardware sets this field to '0'. Default Value: 0
2	M_ACK	When '1', attempt to transmit an acknowledgement (ACK). When this action is performed, the hardware sets this field to '0'. Default Value: 0

12.1.12 SCB0_I2C_M_CMD (continued)

1	M_START_ON_IDLE	When '1', transmit a START as soon as the bus is idle (I2C_STATUS.BUS_BUSY is '0', note that BUSY has a default value of '0'). For bus idle detection the hardware relies on STOP detection. As a result, bus idle detection is only functional after at least one I2C bus transfer has been detected on the bus (default/reset value of BUSY is '0'). A START is only transmitted when the master state machine is in the default state. When this action is performed, the hardware sets this field to '0'. Default Value: 0
0	M_START	When '1', transmit a START or REPEATED START. Whether a START or REPEATED START is transmitted depends on the state of the master state machine. A START is only transmitted when the master state machine is in the default state. A REPEATED START is transmitted when the master state machine is not in the default state, but is working on an ongoing transaction. The REPEATED START can only be transmitted after a NACK or ACK has been received for a transmitted data element or after a NACK has been transmitted for a received data element. When this action is performed, the hardware sets this field to '0'. Default Value: 0

12.1.13 SCB0_I2C_S_CMD

I2C slave command register.

Address: 0x4024006C

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None						RW	RW
HW Access	None						RW1C	RW1C
Name	None [7:2]						S_NACK	S_ACK

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
1	S_NACK	When '1', attempt to transmit a negative acknowledgement (NACK). When this action is performed, the hardware sets this field to '0'. In EZ mode, this field should be set to '0' (it is only to be used in non EZ mode). This command has a higher priority than I2C_S_CMD.S_ACK, I2C_CTRL.S_READY_ADDR_ACK or I2C_CTRL.S_READY_DATA_ACK. Default Value: 0
0	S_ACK	When '1', attempt to transmit an acknowledgement (ACK). When this action is performed, the hardware sets this field to '0'. In EZ mode, this field should be set to '0' (it is only to be used in non EZ mode). Default Value: 0

12.1.14 SCB0_I2C_CFG

I2C configuration register.

Address: 0x40240070

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None			RW	None		RW	
HW Access	None			R	None		R	
Name	None [7:5]			SDA_IN_- FILT_SEL	None [3:2]		SDA_IN_FILT_TRIM [1:0]	

Bits	15	14	13	12	11	10	9	8
SW Access	None			RW	None		RW	
HW Access	None			R	None		R	
Name	None [15:13]			SCL_IN_- FILT_SEL	None [11:10]		SCL_IN_FILT_TRIM [9:8]	

Bits	23	22	21	20	19	18	17	16
SW Access	None		RW		RW		RW	
HW Access	None		R		R		R	
Name	None [23:22]		SDA_OUT_FILT2_TRIM [21:20]		SDA_OUT_FILT1_TRIM [19:18]		SDA_OUT_FILT0_TRIM [17:16]	

Bits	31	30	29	28	27	26	25	24
SW Access	None		RW		None			
HW Access	None		R		None			
Name	None [31:30]		SDA_OUT_FILT_SEL [29:28]		None [27:24]			

Bits	Name	Description
29 : 28	SDA_OUT_FILT_SEL	Selection of cumulative filter delay on SDA output to meet tHD_DAT parameter "0": 0 ns. "1": 50 ns (filter 0 enabled). "2": 100 ns (filters 0 and 1 enabled). "3": 150 ns (filters 0, 1 and 2 enabled). Default Value: 0
21 : 20	SDA_OUT_FILT2_TRIM	Trim settings for the 50ns delay filter on SDA output used to guarantee tHD_DAT I2C parameter. Default setting meets the I2C spec. Programmability available if required Default Value: 2
19 : 18	SDA_OUT_FILT1_TRIM	Trim settings for the 50ns delay filter on SDA output used to guarantee tHD_DAT I2C parameter. Default setting meets the I2C spec. Programmability available if required Default Value: 2
17 : 16	SDA_OUT_FILT0_TRIM	Trim settings for the 50ns delay filter on SDA output used to guarantee tHD_DAT I2C parameter. Default setting meets the I2C spec. Programmability available if required Default Value: 2

12.1.14 SCB0_I2C_CFG (continued)

12	SCL_IN_FILT_SEL	Enable for 50ns glitch filter on SCL input '0': 0 ns. '1': 50 ns (filter enabled). Default Value: 1
9 : 8	SCL_IN_FILT_TRIM	Trim settings for the 50ns glitch filter on the SDA input. Default setting meets the I2C glitch rejections specs. Programmability available if required Default Value: 0
4	SDA_IN_FILT_SEL	Enable for 50ns glitch filter on SDA input '0': 0 ns. '1': 50 ns (filter enabled). Default Value: 1
1 : 0	SDA_IN_FILT_TRIM	Trim settings for the 50ns glitch filter on the SDA input. Default setting meets the I2C glitch rejections specs. Programmability available if required Default Value: 3

12.1.15 SCB0_TX_CTRL

Transmitter control register.

Address: 0x40240200

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None				RW			
HW Access	None				R			
Name	None [7:4]				DATA_WIDTH [3:0]			

Bits	15	14	13	12	11	10	9	8
SW Access	None							RW
HW Access	None							R
Name	None [15:9]							MSB_ - FIRST

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
8	MSB_FIRST	Least significant bit first ('0') or most significant bit first ('1'). For I2C, this field should be '1'. Default Value: 1
3 : 0	DATA_WIDTH	Dataframe width. DATA_WIDTH + 1 is the amount of bits in a transmitted data frame. This number does not include start, parity and stop bits. For UART mode, the valid range is [3, 8]. For SPI, the valid range is [3, 15]. For I2C the only valid value is 7. Default Value: 7

12.1.16 SCB0_TX_FIFO_CTRL

Transmitter FIFO control register.

Address: 0x40240204

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None				RW			
HW Access	None				R			
Name	None [7:4]				TRIGGER_LEVEL [3:0]			

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None						RW	RW
HW Access	None						R	R
Name	None [23:18]						FREEZE	CLEAR

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
17	FREEZE	When '1', hardware reads from the transmitter FIFO do not remove FIFO entries. Freeze will not advance the TX FIFO read pointer. Default Value: 0
16	CLEAR	When '1', the transmitter FIFO and transmitter shift register are cleared/invalidated. Invalidation will last for as long as this field is '1'. If a quick clear/invalidation is required, the field should be set to '1' and be followed by a set to '0'. If a clear/invalidation is required for an extended time period, the field should be set to '1' during the complete time period. Default Value: 0
3 : 0	TRIGGER_LEVEL	Trigger level. When the transmitter FIFO has less entries than the number of this field, a transmitter trigger event INTR_TX.TRIGGER is generated. Default Value: 0

12.1.17 SCB0_TX_FIFO_STATUS

Transmitter FIFO status register.

Address: 0x40240208

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None			R				
HW Access	None			W				
Name	None [7:5]			USED [4:0]				

Bits	15	14	13	12	11	10	9	8
SW Access	R	None						
HW Access	W	None						
Name	SR_VALID	None [14:8]						

Bits	23	22	21	20	19	18	17	16
SW Access	None				R			
HW Access	None				W			
Name	None [23:20]				RD_PTR [19:16]			

Bits	31	30	29	28	27	26	25	24
SW Access	None				R			
HW Access	None				W			
Name	None [31:28]				WR_PTR [27:24]			

Bits	Name	Description
27 : 24	WR_PTR	FIFO write pointer: FIFO location at which a new data frame is written. Default Value: 0
19 : 16	RD_PTR	FIFO read pointer: FIFO location from which a data frame is read by the hardware. Default Value: 0
15	SR_VALID	Indicates whether the TX shift registers holds a valid data frame ('1') or not ('0'). The shift register can be considered the top of the TX FIFO (the data frame is not included in the USED field of the TX FIFO). The shift register is a working register and holds the data frame that is currently transmitted (when the protocol state machine is transmitting a data frame) or the data frame that is transmitted next (when the protocol state machine is not transmitting a data frame). Default Value: 0
4 : 0	USED	Amount of entries in the transmitter FIFO. The value of this field ranges from 0 to FF_DATA_NR (EZ_DATA_NR/2). Default Value: 0

12.1.18 SCB0_TX_FIFO_WR

Transmitter FIFO write register.

Address: 0x40240240

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	W							
HW Access	R							
Name	DATA [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	W							
HW Access	R							
Name	DATA [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
15 : 0	DATA	Data frame written into the transmitter FIFO. Behavior is similar to that of a PUSH operation. Note that when CTRL.BYTE_MODE is '1', only DATA[7:0] are used. A write to a full TX FIFO sets INTR_TX.OVERFLOW to '1'. Default Value: 0

12.1.19 SCB0_RX_CTRL

Receiver control register.

Address: 0x40240300

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None				RW			
HW Access	None				R			
Name	None [7:4]				DATA_WIDTH [3:0]			

Bits	15	14	13	12	11	10	9	8
SW Access	None						RW	RW
HW Access	None						R	R
Name	None [15:10]						MEDIAN	MSB_ - FIRST

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
9	MEDIAN	Median filter. When '1', a digital 3 taps median filter is performed on input interface lines. This filter should reduce the susceptibility to errors. However, its requires higher oversampling values. For UART IrDA submode, this field should always be '1'. Default Value: 0
8	MSB_FIRST	Least significant bit first ('0') or most significant bit first ('1'). For I2C, this field should be '1'. Default Value: 1
3 : 0	DATA_WIDTH	Dataframe width. DATA_WIDTH + 1 is the expected amount of bits in received data frame. This number does not include start, parity and stop bits. For UART mode, the valid range is [3, 8]. For SPI, the valid range is [3, 15]. For I2C the only valid value is 7. In EZ mode (for both SPI and I2C), the only valid value is 7. Default Value: 7

12.1.20 SCB0_RX_FIFO_CTRL

Receiver FIFO control register.

Address: 0x40240304

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None				RW			
HW Access	None				R			
Name	None [7:4]				TRIGGER_LEVEL [3:0]			

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None						RW	RW
HW Access	None						R	R
Name	None [23:18]						FREEZE	CLEAR

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
17	FREEZE	When '1', hardware writes to the receiver FIFO have no effect. Freeze will not advance the RX FIFO write pointer. Default Value: 0
16	CLEAR	When '1', the receiver FIFO and receiver shift register are cleared/invalidated. Invalidation will last for as long as this field is '1'. If a quick clear/invalidation is required, the field should be set to '1' and be followed by a set to '0'. If a clear/invalidation is required for an extended time period, the field should be set to '1' during the complete time period. Default Value: 0
3 : 0	TRIGGER_LEVEL	Trigger level. When the receiver FIFO has more entries than the number of this field, a receiver trigger event INTR_RX.TRIGGER is generated. Default Value: 0

12.1.21 SCB0_RX_FIFO_STATUS

Receiver FIFO status register.

Address: 0x40240308

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None			R				
HW Access	None			W				
Name	None [7:5]			USED [4:0]				

Bits	15	14	13	12	11	10	9	8
SW Access	R	None						
HW Access	W	None						
Name	SR_VALID	None [14:8]						

Bits	23	22	21	20	19	18	17	16
SW Access	None				R			
HW Access	None				W			
Name	None [23:20]				RD_PTR [19:16]			

Bits	31	30	29	28	27	26	25	24
SW Access	None				R			
HW Access	None				W			
Name	None [31:28]				WR_PTR [27:24]			

Bits	Name	Description
27 : 24	WR_PTR	FIFO write pointer: FIFO location at which a new data frame is written by the hardware. Default Value: 0
19 : 16	RD_PTR	FIFO read pointer: FIFO location from which a data frame is read. Default Value: 0
15	SR_VALID	Indicates whether the RX shift registers holds a (partial) valid data frame ('1') or not ('0'). The shift register can be considered the bottom of the RX FIFO (the data frame is not included in the USED field of the RX FIFO). The shift register is a working register and holds the data frame that is currently being received (when the protocol state machine is receiving a data frame). Default Value: 0
4 : 0	USED	Amount of entries in the receiver FIFO. The value of this field ranges from 0 to FF_DATA_NR (EZ_DATA_NR/2). Default Value: 0

12.1.22 SCB0_RX_MATCH

Slave address and mask register.

Address: 0x40240310

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access	R							
Name	ADDR [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	RW							
HW Access	R							
Name	MASK [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
23 : 16	MASK	Slave device address mask. This field is a mask that specifies which of the slave address bits take part in the matching. $MATCH = ((ADDR \& MASK) == ("slave\ address" \& MASK))$. Default Value: 0
7 : 0	ADDR	Slave device address. In UART multi-processor mode, all 8 bits are used. In I2C slave mode, only bits 7 down to 1 are used. This reflects the organization of the first transmitted byte in a I2C transfer: the first 7 bits represent the address of the addressed slave, and the last 1 bit is a read/write indicator ('0': write, '1': read). Default Value: 0

12.1.23 SCB0_RX_FIFO_RD

Receiver FIFO read register.

Address: 0x40240340

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access	W							
Name	DATA [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access	W							
Name	DATA [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
15 : 0	DATA	Data read from the receiver FIFO. Reading a data frame will remove the data frame from the FIFO; i.e. behavior is similar to that of a POP operation. Note that when CTRL.BYTE_MODE is '1', only DATA[7:0] are used. A read from an empty RX FIFO sets INTR_RX.UNDERFLOW to '1'. When this register is read through the debugger, the data frame will not be removed from the FIFO. Similar in operation to RX_FIFO_RD_SILENT Default Value: Undefined

12.1.24 SCB0_RX_FIFO_RD_SILENT

Receiver FIFO read register.

Address: 0x40240344

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access	W							
Name	DATA [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access	W							
Name	DATA [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
15 : 0	DATA	Data read from the receiver FIFO. Reading a data frame will NOT remove the data frame from the FIFO; i.e. behavior is similar to that of a PEEK operation. Note that when CTRL.BYTE_MODE is '1', only DATA[7:0] are used. A read from an empty RX FIFO sets INTR_RX.UNDERFLOW to '1'. Default Value: Undefined

12.1.25 SCB0_EZ_DATA0

Memory buffer registers.

Address: 0x40240400

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access	RW							
Name	EZ_DATA [7:0]							
Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							
Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							
Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
7 : 0	EZ_DATA	Data in buffer memory location. In case of a blocked discarded access, a read access returns 0xffff:ffff and a write access is dropped. Note that the 0xffff:ffff value is unique (not a legal EZ_DATA byte value) and can be detected by SW. Note that a discarded write access can be detected by reading back the written value. Default Value: Undefined

12.1.26 SCB0_INTR_CAUSE

Active clocked interrupt signal register

Address: 0x40240E00

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None		R	R	R	R	R	R
HW Access	None		W	W	W	W	W	W
Name	None [7:6]		SPI_EC	I2C_EC	RX	TX	S	M

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
5	SPI_EC	Externally clocked SPI interrupt active ("interrupt_spi_ec"): INTR_SPI_EC_MASKED != 0. Default Value: 0
4	I2C_EC	Externally clock I2C interrupt active ("interrupt_i2c_ec"): INTR_I2C_EC_MASKED != 0. Default Value: 0
3	RX	Receiver interrupt active ("interrupt_rx"): INTR_RX_MASKED != 0. Default Value: 0
2	TX	Transmitter interrupt active ("interrupt_tx"): INTR_TX_MASKED != 0. Default Value: 0
1	S	Slave interrupt active ("interrupt_slave"): INTR_S_MASKED != 0. Default Value: 0
0	M	Master interrupt active ("interrupt_master"): INTR_M_MASKED != 0. Default Value: 0

12.1.27 SCB0_INTR_I2C_EC

Externally clocked I2C interrupt request register

Address: 0x40240E80

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None				RW1C	RW1C	RW1C	RW1C
HW Access	None				A	A	A	A
Name	None [7:4]				EZ_READ_STOP	EZ_WRITE_STOP	EZ_STOP	WAKE_UP

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
3	EZ_READ_STOP	STOP detection after a read transfer occurred. Activated on the end of a read transfer (I2C STOP). This event is an indication that a buffer memory location has been read from. Only available for a slave request with an address match, in EZ and CMD_RESP modes, when CTRL.EC_OP_MODE is '1'. Default Value: 0
2	EZ_WRITE_STOP	STOP detection after a write transfer occurred. Activated on the end of a write transfer (I2C STOP). This event is an indication that a buffer memory location has been written to. For EZ mode: a transfer that only writes the base address does NOT activate this event. Only available for a slave request with an address match, in EZ and CMD_RESP modes, when CTRL.EC_OP_MODE is '1'. Default Value: 0
1	EZ_STOP	STOP detection. Activated on the end of a every transfer (I2C STOP). Only available for a slave request with an address match, in EZ and CMD_RESP modes, when CTRL.EC_OP_MODE is '1'. Default Value: 0

12.1.27 SCB0_INTR_I2C_EC (continued)

0	WAKE_UP	Wake up request. Active on incoming slave request (with address match). Only used when CTRL.EC_AM_MODE is '1'. Default Value: 0
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12.1.28 SCB0_INTR_I2C_EC_MASK

Externally clocked I2C interrupt mask register

Address: 0x40240E88

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None				RW	RW	RW	RW
HW Access	None				R	R	R	R
Name	None [7:4]				EZ_READ_STOP	EZ_WRITE_STOP	EZ_STOP	WAKE_UP

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
3	EZ_READ_STOP	Mask bit for corresponding bit in interrupt request register. Default Value: 0
2	EZ_WRITE_STOP	Mask bit for corresponding bit in interrupt request register. Default Value: 0
1	EZ_STOP	Mask bit for corresponding bit in interrupt request register. Default Value: 0
0	WAKE_UP	Mask bit for corresponding bit in interrupt request register. Default Value: 0

12.1.29 SCB0_INTR_I2C_EC_MASKED

Externally clocked I2C interrupt masked register

Address: 0x40240E8C

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None				R	R	R	R
HW Access	None				W	W	W	W
Name	None [7:4]				EZ_READ_STOP	EZ_WRITE_STOP	EZ_STOP	WAKE_UP

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
3	EZ_READ_STOP	Logical and of corresponding request and mask bits. Default Value: 0
2	EZ_WRITE_STOP	Logical and of corresponding request and mask bits. Default Value: 0
1	EZ_STOP	Logical and of corresponding request and mask bits. Default Value: 0
0	WAKE_UP	Logical and of corresponding request and mask bits. Default Value: 0

12.1.30 SCB0_INTR_SPI_EC

Externally clocked SPI interrupt request register

Address: 0x40240EC0

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None				RW1C	RW1C	RW1C	RW1C
HW Access	None				A	A	A	A
Name	None [7:4]				EZ_READ_STOP	EZ_WRITE_STOP	EZ_STOP	WAKE_UP

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
3	EZ_READ_STOP	STOP detection after a read transfer occurred. Activated on the end of a read transfer (SPI deselection). This event is an indication that a buffer memory location has been read from. Only used in EZ and CMD_RESP modes and when CTRL.EC_OP_MODE is '1'. Default Value: 0
2	EZ_WRITE_STOP	STOP detection after a write transfer occurred. Activated on the end of a write transfer (SPI deselection). This event is an indication that a buffer memory location has been written to. For EZ mode: a transfer that only writes the base address does NOT activate this event. Only used in EZ and CMD_RESP modes and when CTRL.EC_OP_MODE is '1'. Default Value: 0
1	EZ_STOP	STOP detection. Activated on the end of a every transfer (SPI deselection). Only available in EZ and CMD_RESP mode and when CTRL.EC_OP_MODE is '1'. Default Value: 0

12.1.30 SCB0_INTR_SPI_EC (continued)

0	WAKE_UP	Wake up request. Active on incoming slave request when externally clocked selection is '1'. Only used when CTRL.EC_AM_MODE is '1'. Default Value: 0
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12.1.31 SCB0_INTR_SPI_EC_MASK

Externally clocked SPI interrupt mask register

Address: 0x40240EC8

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None				RW	RW	RW	RW
HW Access	None				R	R	R	R
Name	None [7:4]				EZ_READ_STOP	EZ_WRITE_STOP	EZ_STOP	WAKE_UP

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
3	EZ_READ_STOP	Mask bit for corresponding bit in interrupt request register. Default Value: 0
2	EZ_WRITE_STOP	Mask bit for corresponding bit in interrupt request register. Default Value: 0
1	EZ_STOP	Mask bit for corresponding bit in interrupt request register. Default Value: 0
0	WAKE_UP	Mask bit for corresponding bit in interrupt request register. Default Value: 0

12.1.32 SCB0_INTR_SPI_EC_MASKED

Externally clocked SPI interrupt masked register

Address: 0x40240ECC

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None				R	R	R	R
HW Access	None				W	W	W	W
Name	None [7:4]				EZ_READ_STOP	EZ_WRITE_STOP	EZ_STOP	WAKE_UP

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
3	EZ_READ_STOP	Logical and of corresponding request and mask bits. Default Value: 0
2	EZ_WRITE_STOP	Logical and of corresponding request and mask bits. Default Value: 0
1	EZ_STOP	Logical and of corresponding request and mask bits. Default Value: 0
0	WAKE_UP	Logical and of corresponding request and mask bits. Default Value: 0

12.1.33 SCB0_INTR_M

Master interrupt request register.

Address: 0x40240F00

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None			RW1C	None	RW1C	RW1C	RW1C
HW Access	None			RW1S	None	RW1S	RW1S	RW1S
Name	None [7:5]			I2C_STOP	None	I2C_ACK	I2C_NACK	I2C_ARB_LOST

Bits	15	14	13	12	11	10	9	8
SW Access	None						RW1C	RW1C
HW Access	None						RW1S	RW1S
Name	None [15:10]						SPI_DONE	I2C_BUS_ERROR

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
9	SPI_DONE	SPI master transfer done event: all data frames in the transmit FIFO are sent and the transmit FIFO and shift register are empty. Default Value: 0
8	I2C_BUS_ERROR	I2C master bus error (unexpected detection of START or STOP condition). Default Value: 0
4	I2C_STOP	I2C master STOP. Set to '1', when the master has transmitted a STOP. Default Value: 0
2	I2C_ACK	I2C master acknowledgement. Set to '1', when the master receives a ACK (typically after the master transmitted the slave address or TX data). Default Value: 0
1	I2C_NACK	I2C master negative acknowledgement. Set to '1', when the master receives a NACK (typically after the master transmitted the slave address or TX data). Default Value: 0
0	I2C_ARB_LOST	I2C master lost arbitration: the value driven by the master on the SDA line is not the same as the value observed on the SDA line. Default Value: 0

12.1.34 SCB0_INTR_M_SET

Master interrupt set request register

Address: 0x40240F04

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None			RW1S	None	RW1S	RW1S	RW1S
HW Access	None			A	None	A	A	A
Name	None [7:5]			I2C_STOP	None	I2C_ACK	I2C_NACK	I2C_ARB_LOST

Bits	15	14	13	12	11	10	9	8
SW Access	None						RW1S	RW1S
HW Access	None						A	A
Name	None [15:10]						SPI_DONE	I2C_BUS_ERROR

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
9	SPI_DONE	Write with '1' to set corresponding bit in interrupt request register. Default Value: 0
8	I2C_BUS_ERROR	Write with '1' to set corresponding bit in interrupt request register. Default Value: 0
4	I2C_STOP	Write with '1' to set corresponding bit in interrupt request register. Default Value: 0
2	I2C_ACK	Write with '1' to set corresponding bit in interrupt request register. Default Value: 0
1	I2C_NACK	Write with '1' to set corresponding bit in interrupt request register. Default Value: 0
0	I2C_ARB_LOST	Write with '1' to set corresponding bit in interrupt request register. Default Value: 0

12.1.35 SCB0_INTR_M_MASK

Master interrupt mask register.

Address: 0x40240F08

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None			RW	None	RW	RW	RW
HW Access	None			R	None	R	R	R
Name	None [7:5]			I2C_STOP	None	I2C_ACK	I2C_NACK	I2C_ARB_LOST

Bits	15	14	13	12	11	10	9	8
SW Access	None						RW	RW
HW Access	None						R	R
Name	None [15:10]						SPI_DONE	I2C_BUS_ERROR

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
9	SPI_DONE	Mask bit for corresponding bit in interrupt request register. Default Value: 0
8	I2C_BUS_ERROR	Mask bit for corresponding bit in interrupt request register. Default Value: 0
4	I2C_STOP	Mask bit for corresponding bit in interrupt request register. Default Value: 0
2	I2C_ACK	Mask bit for corresponding bit in interrupt request register. Default Value: 0
1	I2C_NACK	Mask bit for corresponding bit in interrupt request register. Default Value: 0
0	I2C_ARB_LOST	Mask bit for corresponding bit in interrupt request register. Default Value: 0

12.1.36 SCB0_INTR_M_MASKED

Master interrupt masked request register

Address: 0x40240F0C

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None			R	None	R	R	R
HW Access	None			W	None	W	W	W
Name	None [7:5]			I2C_STOP	None	I2C_ACK	I2C_NACK	I2C_ARB_LOST

Bits	15	14	13	12	11	10	9	8
SW Access	None						R	R
HW Access	None						W	W
Name	None [15:10]						SPI_DONE	I2C_BUS_ERROR

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
9	SPI_DONE	Logical and of corresponding request and mask bits. Default Value: 0
8	I2C_BUS_ERROR	Logical and of corresponding request and mask bits. Default Value: 0
4	I2C_STOP	Logical and of corresponding request and mask bits. Default Value: 0
2	I2C_ACK	Logical and of corresponding request and mask bits. Default Value: 0
1	I2C_NACK	Logical and of corresponding request and mask bits. Default Value: 0
0	I2C_ARB_LOST	Logical and of corresponding request and mask bits. Default Value: 0

12.1.37 SCB0_INTR_S

Slave interrupt request register.

Address: 0x40240F40

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW1C	RW1C	RW1C	RW1C	RW1C	RW1C	RW1C	RW1C
HW Access	RW1S	RW1S	RW1S	RW1S	RW1S	RW1S	RW1S	RW1S
Name	I2C_GEN- ERAL	I2C_AD- DR_MATCH	I2C_START	I2C_STOP	I2C_WRITE _STOP	I2C_ACK	I2C_NACK	I2C_AR- B_LOST

Bits	15	14	13	12	11	10	9	8
SW Access	None				RW1C	RW1C	RW1C	RW1C
HW Access	None				RW1S	RW1S	RW1S	RW1S
Name	None [15:12]				SPI_BUS_E RROR	SPI_EZ_ST OP	SPI_EZ_W RITE_STOP	I2C_BUS_E RROR

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
11	SPI_BUS_ERROR	SPI slave deselected at an unexpected time in the SPI transfer. The Firmware may decide to clear the TX and RX FIFOs in case of this error. Default Value: 0
10	SPI_EZ_STOP	SPI slave deselected after any EZ SPI transfer occurred. Default Value: 0
9	SPI_EZ_WRITE_STOP	SPI slave deselected after a write EZ SPI transfer occurred. Default Value: 0
8	I2C_BUS_ERROR	I2C slave bus error (unexpected detection of START or STOP condition). This should not occur, it represents erroneous I2C bus behavior. In case of a bus error, the I2C slave state machine abort the ongoing transfer. The Firmware may decide to clear the TX and RX FIFOs in case of this error. Default Value: 0

12.1.37 SCB0_INTR_S (continued)

7	I2C_GENERAL	I2C slave general call address received. If CTRL.ADDR_ACCEPT, the received address 0x00 (including the R/W bit) is available in the RX FIFO. In the case of externally clocked address matching (CTRL.EC_AM_MODE is '1') and internally clocked operation (CTRL.EC_OP_MODE is '0'), this field is set when the event is detected. Default Value: 0
6	I2C_ADDR_MATCH	I2C slave matching address received. If CTRL.ADDR_ACCEPT, the received address (including the R/W bit) is available in the RX FIFO. In the case of externally clocked address matching (CTRL.EC_AM_MODE is '1') and internally clocked operation (CTRL.EC_OP_MODE is '0'), this field is set when the event is detected. Default Value: 0
5	I2C_START	I2C slave START received. Set to '1', when START or REPEATED START event is detected. In the case of externally clocked address matching (CTRL.EC_AM_MODE is '1') AND clock stretching is performed (I2C_CTRL.S_NOT_READY_ADDR_NACK is '0'), this field is NOT set. The Firmware should use INTR_S_EC.WAKE_UP, INTR_S.I2C_ADDR_MATCH and INTR_S.I2C_GENERAL. Default Value: 0
4	I2C_STOP	I2C STOP event for I2C (read or write) transfer intended for this slave (address matching is performed). Set to '1', when STOP or REPEATED START event is detected. The REPEATED START event is included in this interrupt cause such that the I2C transfers separated by a REPEATED START can be distinguished and potentially treated separately by the Firmware. Note that the second I2C transfer (after a REPEATED START) may be to a different slave address. The event is detected on any I2C transfer intended for this slave. Note that a I2C address intended for the slave (address is matching) will result in a I2C_STOP event independent of whether the I2C address is ACK'd or NACK'd. Default Value: 0
3	I2C_WRITE_STOP	I2C STOP event for I2C write transfer intended for this slave (address matching is performed). Set to '1', when STOP or REPEATED START event is detected. The REPEATED START event is included in this interrupt cause such that the I2C transfers separated by a REPEATED START can be distinguished and potentially treated separately by the Firmware. Note that the second I2C transfer (after a REPEATED START) may be to a different slave address. In non EZ mode, the event is detected on any I2C write transfer intended for this slave. Note that a I2C write address intended for the slave (address is matching and it is a write transfer) will result in a I2C_WRITE_STOP event independent of whether the I2C address is ACK'd or NACK'd. In EZ mode, the event is detected only on I2C write transfers that have EZ data written to the memory structure (an I2C write transfer that only communicates an I2C address and EZ address, will not result in this event being detected). Default Value: 0
2	I2C_ACK	I2C slave acknowledgement received. Set to '1', when the slave receives a ACK (typically after the slave transmitted TX data). Default Value: 0
1	I2C_NACK	I2C slave negative acknowledgement received. Set to '1', when the slave receives a NACK (typically after the slave transmitted TX data). Default Value: 0
0	I2C_ARB_LOST	I2C slave lost arbitration: the value driven on the SDA line is not the same as the value observed on the SDA line (while the SCL line is '1'). This should not occur, it represents erroneous I2C bus behavior. In case of lost arbitration, the I2C slave state machine aborts the ongoing transfer. The Firmware may decide to clear the TX and RX FIFOs in case of this error. Default Value: 0

12.1.38 SCB0_INTR_S_SET

Slave interrupt set request register.

Address: 0x40240F44

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW1S	RW1S	RW1S	RW1S	RW1S	RW1S	RW1S	RW1S
HW Access	A	A	A	A	A	A	A	A
Name	I2C_GEN- ERAL	I2C_AD- DR_MATCH	I2C_START	I2C_STOP	I2C_WRITE _STOP	I2C_ACK	I2C_NACK	I2C_AR- B_LOST

Bits	15	14	13	12	11	10	9	8
SW Access	None				RW1S	RW1S	RW1S	RW1S
HW Access	None				A	A	A	A
Name	None [15:12]				SPI_BUS_E RROR	SPI_EZ_ST OP	SPI_EZ_W RITE_STOP	I2C_BUS_E RROR

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
11	SPI_BUS_ERROR	Write with '1' to set corresponding bit in interrupt request register. Default Value: 0
10	SPI_EZ_STOP	Write with '1' to set corresponding bit in interrupt request register. Default Value: 0
9	SPI_EZ_WRITE_STOP	Write with '1' to set corresponding bit in interrupt request register. Default Value: 0
8	I2C_BUS_ERROR	Write with '1' to set corresponding bit in interrupt request register. Default Value: 0
7	I2C_GENERAL	Write with '1' to set corresponding bit in interrupt request register. Default Value: 0
6	I2C_ADDR_MATCH	Write with '1' to set corresponding bit in interrupt request register. Default Value: 0
5	I2C_START	Write with '1' to set corresponding bit in interrupt request register. Default Value: 0

12.1.38 SCB0_INTR_S_SET (continued)

4	I2C_STOP	Write with '1' to set corresponding bit in interrupt request register. Default Value: 0
3	I2C_WRITE_STOP	Write with '1' to set corresponding bit in interrupt request register. Default Value: 0
2	I2C_ACK	Write with '1' to set corresponding bit in interrupt request register. Default Value: 0
1	I2C_NACK	Write with '1' to set corresponding bit in interrupt request register. Default Value: 0
0	I2C_ARB_LOST	Write with '1' to set corresponding bit in interrupt request register. Default Value: 0

12.1.39 SCB0_INTR_S_MASK

Slave interrupt mask register.

Address: 0x40240F48

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW	RW	RW	RW	RW	RW	RW	RW
HW Access	R	R	R	R	R	R	R	R
Name	I2C_GENERAL	I2C_ADDR_MATCH	I2C_START	I2C_STOP	I2C_WRITE_STOP	I2C_ACK	I2C_NACK	I2C_ARB_LOST

Bits	15	14	13	12	11	10	9	8
SW Access	None				RW	RW	RW	RW
HW Access	None				R	R	R	R
Name	None [15:12]				SPI_BUS_ERROR	SPI_EZ_STOP	SPI_EZ_WRITE_STOP	I2C_BUS_ERROR

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
11	SPI_BUS_ERROR	Mask bit for corresponding bit in interrupt request register. Default Value: 0
10	SPI_EZ_STOP	Mask bit for corresponding bit in interrupt request register. Default Value: 0
9	SPI_EZ_WRITE_STOP	Mask bit for corresponding bit in interrupt request register. Default Value: 0
8	I2C_BUS_ERROR	Mask bit for corresponding bit in interrupt request register. Default Value: 0
7	I2C_GENERAL	Mask bit for corresponding bit in interrupt request register. Default Value: 0
6	I2C_ADDR_MATCH	Mask bit for corresponding bit in interrupt request register. Default Value: 0
5	I2C_START	Mask bit for corresponding bit in interrupt request register. Default Value: 0

12.1.39 SCB0_INTR_S_MASK (continued)

4	I2C_STOP	Mask bit for corresponding bit in interrupt request register. Default Value: 0
3	I2C_WRITE_STOP	Mask bit for corresponding bit in interrupt request register. Default Value: 0
2	I2C_ACK	Mask bit for corresponding bit in interrupt request register. Default Value: 0
1	I2C_NACK	Mask bit for corresponding bit in interrupt request register. Default Value: 0
0	I2C_ARB_LOST	Mask bit for corresponding bit in interrupt request register. Default Value: 0

12.1.40 SCB0_INTR_S_MASKED

Slave interrupt masked request register

Address: 0x40240F4C

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R	R	R	R	R	R	R	R
HW Access	W	W	W	W	W	W	W	W
Name	I2C_GENERAL	I2C_ADDR_MATCH	I2C_START	I2C_STOP	I2C_WRITE_STOP	I2C_ACK	I2C_NACK	I2C_ARB_LOST

Bits	15	14	13	12	11	10	9	8
SW Access	None				R	R	R	R
HW Access	None				W	W	W	W
Name	None [15:12]				SPI_BUS_ERROR	SPI_EZ_STOP	SPI_EZ_WRITE_STOP	I2C_BUS_ERROR

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
11	SPI_BUS_ERROR	Logical and of corresponding request and mask bits. Default Value: 0
10	SPI_EZ_STOP	Logical and of corresponding request and mask bits. Default Value: 0
9	SPI_EZ_WRITE_STOP	Logical and of corresponding request and mask bits. Default Value: 0
8	I2C_BUS_ERROR	Logical and of corresponding request and mask bits. Default Value: 0
7	I2C_GENERAL	Logical and of corresponding request and mask bits. Default Value: 0
6	I2C_ADDR_MATCH	Logical and of corresponding request and mask bits. Default Value: 0
5	I2C_START	Logical and of corresponding request and mask bits. Default Value: 0

12.1.40 SCB0_INTR_S_MASKED (continued)

4	I2C_STOP	Logical and of corresponding request and mask bits. Default Value: 0
3	I2C_WRITE_STOP	Logical and of corresponding request and mask bits. Default Value: 0
2	I2C_ACK	Logical and of corresponding request and mask bits. Default Value: 0
1	I2C_NACK	Logical and of corresponding request and mask bits. Default Value: 0
0	I2C_ARB_LOST	Logical and of corresponding request and mask bits. Default Value: 0

12.1.41 SCB0_INTR_TX

Transmitter interrupt request register.

Address: 0x40240F80

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW1C	RW1C	RW1C	RW1C	None		RW1C	RW1C
HW Access	RW1S	RW1S	RW1S	RW1S	None		RW1S	RW1S
Name	BLOCKED	UNDER-FLOW	OVER-FLOW	EMPTY	None [3:2]		NOT_FULL	TRIGGER

Bits	15	14	13	12	11	10	9	8
SW Access	None					RW1C	RW1C	RW1C
HW Access	None					RW1S	RW1S	RW1S
Name	None [15:11]					UART_ARB_LOST	UART_DONE	UART_NACK

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
10	UART_ARB_LOST	UART lost arbitration: the value driven on the TX line is not the same as the value observed on the RX line. This condition event is usefull when transmitter and receiver share a TX/RX line. This is the case in LIN or SmartCard modes. Set to '1', when event is detected. Write with '1' to clear bit. Default Value: 0
9	UART_DONE	UART transmitter done event. This happens when the IP is done transferring all data in the TX FIFO; i.e. EMPTY is '1'. Set to '1', when event is detected. Write with '1' to clear bit. Default Value: 0
8	UART_NACK	UART transmitter received a negative acknowledgement in SmartCard mode. Set to '1', when event is detected. Write with '1' to clear bit. Default Value: 0
7	BLOCKED	SW cannot get access to the EZ memory (EZ data access), due to an externally clocked EZ access. This may happen when STATUS.EC_BUSY is '1'. Default Value: 0

12.1.41 SCB0_INTR_TX (continued)

6	UNDERFLOW	Attempt to read from an empty TX FIFO. This happens when SCB is ready to transfer data and EMPTY is '1'. Only used in FIFO mode. Default Value: 0
5	OVERFLOW	Attempt to write to a full TX FIFO. Only used in FIFO mode. Default Value: 0
4	EMPTY	TX FIFO is empty; i.e. it has 0 entries. Only used in FIFO mode. Default Value: 0
1	NOT_FULL	TX FIFO is not full. Dependent on CTRL.BYTE_MODE: (FF_DATA_NR = EZ_DATA_NR/2) BYTE_MODE is '0': # entries != FF_DATA_NR/2. BYTE_MODE is '1': # entries != FF_DATA_NR. Only used in FIFO mode. Default Value: 0
0	TRIGGER	Less entries in the TX FIFO than the value specified by TX_FIFO_CTRL. Only used in FIFO mode. Default Value: 0

12.1.42 SCB0_INTR_TX_SET

Transmitter interrupt set request register

Address: 0x40240F84

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW1S	RW1S	RW1S	RW1S	None		RW1S	RW1S
HW Access	A	A	A	A	None		A	A
Name	BLOCKED	UNDER-FLOW	OVER-FLOW	EMPTY	None [3:2]		NOT_FULL	TRIGGER

Bits	15	14	13	12	11	10	9	8
SW Access	None					RW1S	RW1S	RW1S
HW Access	None					A	A	A
Name	None [15:11]					UART_ARB_LOST	UART_DONE	UART_NACK

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
10	UART_ARB_LOST	Write with '1' to set corresponding bit in interrupt request register. Default Value: 0
9	UART_DONE	Write with '1' to set corresponding bit in interrupt request register. Default Value: 0
8	UART_NACK	Write with '1' to set corresponding bit in interrupt request register. Default Value: 0
7	BLOCKED	Write with '1' to set corresponding bit in interrupt request register. Default Value: 0
6	UNDERFLOW	Write with '1' to set corresponding bit in interrupt request register. Default Value: 0
5	OVERFLOW	Write with '1' to set corresponding bit in interrupt request register. Default Value: 0
4	EMPTY	Write with '1' to set corresponding bit in interrupt request register. Default Value: 0

12.1.42 SCB0_INTR_TX_SET (continued)

1	NOT_FULL	Write with '1' to set corresponding bit in interrupt request register. Default Value: 0
0	TRIGGER	Write with '1' to set corresponding bit in interrupt request register. Default Value: 0

12.1.43 SCB0_INTR_TX_MASK

Transmitter interrupt mask register.

Address: 0x40240F88

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW	RW	RW	RW	None		RW	RW
HW Access	R	R	R	R	None		R	R
Name	BLOCKED	UNDER-FLOW	OVER-FLOW	EMPTY	None [3:2]		NOT_FULL	TRIGGER

Bits	15	14	13	12	11	10	9	8
SW Access	None					RW	RW	RW
HW Access	None					R	R	R
Name	None [15:11]					UART_ARB_LOST	UART_DONE	UART_NACK

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
10	UART_ARB_LOST	Mask bit for corresponding bit in interrupt request register. Default Value: 0
9	UART_DONE	Mask bit for corresponding bit in interrupt request register. Default Value: 0
8	UART_NACK	Mask bit for corresponding bit in interrupt request register. Default Value: 0
7	BLOCKED	Mask bit for corresponding bit in interrupt request register. Default Value: 0
6	UNDERFLOW	Mask bit for corresponding bit in interrupt request register. Default Value: 0
5	OVERFLOW	Mask bit for corresponding bit in interrupt request register. Default Value: 0
4	EMPTY	Mask bit for corresponding bit in interrupt request register. Default Value: 0

12.1.43 SCB0_INTR_TX_MASK (continued)

1	NOT_FULL	Mask bit for corresponding bit in interrupt request register. Default Value: 0
0	TRIGGER	Mask bit for corresponding bit in interrupt request register. Default Value: 0

12.1.44 SCB0_INTR_TX_MASKED

Transmitter interrupt masked request register

Address: 0x40240F8C

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R	R	R	R	None		R	R
HW Access	W	W	W	W	None		W	W
Name	BLOCKED	UNDER-FLOW	OVER-FLOW	EMPTY	None [3:2]		NOT_FULL	TRIGGER

Bits	15	14	13	12	11	10	9	8
SW Access	None					R	R	R
HW Access	None					W	W	W
Name	None [15:11]					UART_ARB_LOST	UART_DONE	UART_NACK

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
10	UART_ARB_LOST	Logical and of corresponding request and mask bits. Default Value: 0
9	UART_DONE	Logical and of corresponding request and mask bits. Default Value: 0
8	UART_NACK	Logical and of corresponding request and mask bits. Default Value: 0
7	BLOCKED	Logical and of corresponding request and mask bits. Default Value: 0
6	UNDERFLOW	Logical and of corresponding request and mask bits. Default Value: 0
5	OVERFLOW	Logical and of corresponding request and mask bits. Default Value: 0
4	EMPTY	Logical and of corresponding request and mask bits. Default Value: 0

12.1.44 SCB0_INTR_TX_MASKED (continued)

1	NOT_FULL	Logical and of corresponding request and mask bits. Default Value: 0
0	TRIGGER	Logical and of corresponding request and mask bits. Default Value: 0

12.1.45 SCB0_INTR_RX

Receiver interrupt request register.

Address: 0x40240FC0

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW1C	RW1C	RW1C	None	RW1C	RW1C	None	RW1C
HW Access	RW1S	RW1S	RW1S	None	RW1S	RW1S	None	RW1S
Name	BLOCKED	UNDER-FLOW	OVER-FLOW	None	FULL	NOT_EMP-TY	None	TRIGGER

Bits	15	14	13	12	11	10	9	8
SW Access	None				RW1C	RW1C	RW1C	RW1C
HW Access	None				RW1S	RW1S	RW1S	RW1S
Name	None [15:12]				BREAK - DETECT	BAUD_DE-TECT	PARI-TY_ERROR	FRAME_ER-ROR

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
11	BREAK_DETECT	Break detection is successful: the line is '0' for UART_RX_CTRL.BREAK_WIDTH + 1 bit period. Can occur at any time to address unanticipated break fields; i.e. "break-in-data" is supported. This feature is supported for the UART standard and LIN submodes. For the UART standard submodes, ongoing receipt of data frames is NOT affected; i.e. Firmware is expected to take the proper action. For the LIN submode, possible ongoing receipt of a data frame is stopped and the (partially) received data frame is dropped and baud rate detection is started. Set to '1', when event is detected. Write with '1' to clear bit. Default Value: 0
10	BAUD_DETECT	LIN baudrate detection is completed. The receiver software uses the UART_RX_STATUS.BR_COUNTER value to set the right IP clock (from the programmable clock IP) to guarantee successful receipt of the first LIN data frame (Protected Identifier Field) after the synchronization byte. Set to '1', when event is detected. Write with '1' to clear bit. Default Value: 0

12.1.45 SCB0_INTR_RX (continued)

9	PARITY_ERROR	Parity error in received data frame. Set to '1', when event is detected. Write with '1' to clear bit. If UART_RX_CTL.DROP_ON_PARITY_ERROR is '1', the received frame is dropped. If UART_RX_CTL.DROP_ON_PARITY_ERROR is '0', the received frame is send to the RX FIFO. In SmartCard submode, negatively acknowledged data frames generate a parity error. Note that Firmware can only identify the erroneous data frame in the RX FIFO if it is fast enough to read the data frame before the hardware writes a next data frame into the RX FIFO. Default Value: 0
8	FRAME_ERROR	Frame error in received data frame. Set to '1', when event is detected. Write with '1' to clear bit. This can be either a start or stop bit(s) error: Start bit error: after the detection of the beginning of a start bit period (RX line changes from '1' to '0'), the middle of the start bit period is sampled erroneously (RX line is '1'). Note: a start bit error is detected BEFORE a data frame is received. Stop bit error: the RX line is sampled as '0', but a '1' was expected. Note: a stop bit error may result in failure to receive successive data frame(s). Note: a stop bit error is detected AFTER a data frame is received. A stop bit error is detected after a data frame is received, and the UART_RX_CTL.DROP_ON_FRAME_ERROR field specifies whether the received frame is dropped or send to the RX FIFO. If UART_RX_CTL.DROP_ON_FRAME_ERROR is '1', the received data frame is dropped. If UART_RX_CTL.DROP_ON_FRAME_ERROR is '0', the received data frame is send to the RX FIFO. Note that Firmware can only identify the erroneous data frame in the RX FIFO if it is fast enough to read the data frame before the hardware writes a next data frame into the RX FIFO; i.e. the RX FIFO does not have error flags to tag erroneous data frames. Default Value: 0
7	BLOCKED	SW cannot get access to the EZ memory (EZ_DATA accesses), due to an externally clocked EZ access. This may happen when STATUS.EC_BUSY is '1'. Default Value: 0
6	UNDERFLOW	Attempt to read from an empty RX FIFO. Only used in FIFO mode. Default Value: 0
5	OVERFLOW	Attempt to write to a full RX FIFO. Note: in I2C mode, the OVERFLOW is set when a data frame is received and the RX FIFO is full, independent of whether it is ACK'd or NACK'd. Only used in FIFO mode. Default Value: 0
3	FULL	RX FIFO is full. Note that received data frames are lost when the RX FIFO is full. Dependent on CTRL.BYTE_MODE: (FF_DATA_NR = EZ_DATA_NR/2) BYTE_MODE is '0': # entries == FF_DATA_NR/2. BYTE_MODE is '1': # entries == FF_DATA_NR. Only used in FIFO mode. Default Value: 0
2	NOT_EMPTY	RX FIFO is not empty. Only used in FIFO mode. Default Value: 0
0	TRIGGER	More entries in the RX FIFO than the value specified by TRIGGER_LEVEL in SCB_RX_FIFO_CTL. Only used in FIFO mode. Default Value: 0

12.1.46 SCB0_INTR_RX_SET

Receiver interrupt set request register.

Address: 0x40240FC4

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW1S	RW1S	RW1S	None	RW1S	RW1S	None	RW1S
HW Access	A	A	A	None	A	A	None	A
Name	BLOCKED	UNDER-FLOW	OVER-FLOW	None	FULL	NOT_EMP-TY	None	TRIGGER

Bits	15	14	13	12	11	10	9	8
SW Access	None				RW1S	RW1S	RW1S	RW1S
HW Access	None				A	A	A	A
Name	None [15:12]				BREAK - DETECT	BAUD_DE-TECT	PARI-TY_ERROR	FRAME_ER-ROR

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
11	BREAK_DETECT	Write with '1' to set corresponding bit in interrupt status register. Default Value: 0
10	BAUD_DETECT	Write with '1' to set corresponding bit in interrupt status register. Default Value: 0
9	PARITY_ERROR	Write with '1' to set corresponding bit in interrupt status register. Default Value: 0
8	FRAME_ERROR	Write with '1' to set corresponding bit in interrupt status register. Default Value: 0
7	BLOCKED	Write with '1' to set corresponding bit in interrupt status register. Default Value: 0
6	UNDERFLOW	Write with '1' to set corresponding bit in interrupt status register. Default Value: 0
5	OVERFLOW	Write with '1' to set corresponding bit in interrupt status register. Default Value: 0

12.1.46 SCB0_INTR_RX_SET (continued)

3	FULL	Write with '1' to set corresponding bit in interrupt status register. Default Value: 0
2	NOT_EMPTY	Write with '1' to set corresponding bit in interrupt status register. Default Value: 0
0	TRIGGER	Write with '1' to set corresponding bit in interrupt request register. Default Value: 0

12.1.47 SCB0_INTR_RX_MASK

Receiver interrupt mask register.

Address: 0x40240FC8

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW	RW	RW	None	RW	RW	None	RW
HW Access	R	R	R	None	R	R	None	R
Name	BLOCKED	UNDER-FLOW	OVER-FLOW	None	FULL	NOT_EMP-TY	None	TRIGGER

Bits	15	14	13	12	11	10	9	8
SW Access	None				RW	RW	RW	RW
HW Access	None				R	R	R	R
Name	None [15:12]				BREAK - DETECT	BAUD_DE-TECT	PARI-TY_ERROR	FRAME_ER-ROR

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
11	BREAK_DETECT	Mask bit for corresponding bit in interrupt request register. Default Value: 0
10	BAUD_DETECT	Mask bit for corresponding bit in interrupt request register. Default Value: 0
9	PARITY_ERROR	Mask bit for corresponding bit in interrupt request register. Default Value: 0
8	FRAME_ERROR	Mask bit for corresponding bit in interrupt request register. Default Value: 0
7	BLOCKED	Mask bit for corresponding bit in interrupt request register. Default Value: 0
6	UNDERFLOW	Mask bit for corresponding bit in interrupt request register. Default Value: 0
5	OVERFLOW	Mask bit for corresponding bit in interrupt request register. Default Value: 0

12.1.47 SCB0_INTR_RX_MASK (continued)

3	FULL	Mask bit for corresponding bit in interrupt request register. Default Value: 0
2	NOT_EMPTY	Mask bit for corresponding bit in interrupt request register. Default Value: 0
0	TRIGGER	Mask bit for corresponding bit in interrupt request register. Default Value: 0

12.1.48 SCB0_INTR_RX_MASKED

Receiver interrupt masked request register

Address: 0x40240FCC

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R	R	R	None	R	R	None	R
HW Access	W	W	W	None	W	W	None	W
Name	BLOCKED	UNDER-FLOW	OVER-FLOW	None	FULL	NOT_EMP-TY	None	TRIGGER

Bits	15	14	13	12	11	10	9	8
SW Access	None				R	R	R	R
HW Access	None				W	W	W	W
Name	None [15:12]				BREAK - DETECT	BAUD_DE-TECT	PARI-TY_ERROR	FRAME_ER-ROR

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
11	BREAK_DETECT	Logical and of corresponding request and mask bits. Default Value: 0
10	BAUD_DETECT	Logical and of corresponding request and mask bits. Default Value: 0
9	PARITY_ERROR	Logical and of corresponding request and mask bits. Default Value: 0
8	FRAME_ERROR	Logical and of corresponding request and mask bits. Default Value: 0
7	BLOCKED	Logical and of corresponding request and mask bits. Default Value: 0
6	UNDERFLOW	Logical and of corresponding request and mask bits. Default Value: 0
5	OVERFLOW	Logical and of corresponding request and mask bits. Default Value: 0

12.1.48 SCB0_INTR_RX_MASKED (continued)

3	FULL	Logical and of corresponding request and mask bits. Default Value: 0
2	NOT_EMPTY	Logical and of corresponding request and mask bits. Default Value: 0
0	TRIGGER	Logical and of corresponding request and mask bits. Default Value: 0

13 LCD Direct Registers



This section discusses the LCD Direct registers. It lists all the registers in mapping tables, in address order.

13.1 Register Details

Register	Address	Description
LCD_ID	0x402A0000	ID & Revision
LCD_DIVIDER	0x402A0004	LCD Divider Register
LCD_CONTROL	0x402A0008	LCD Configuration Register
LCD_DATA00	0x402A0100	LCD Pin Data Registers
LCD_DATA01	0x402A0104	LCD Pin Data Registers. See LCD_DATA00 for the details of bit fields.
LCD_DATA02	0x402A0108	LCD Pin Data Registers. See LCD_DATA00 for the details of bit fields.
LCD_DATA03	0x402A010C	LCD Pin Data Registers. See LCD_DATA00 for the details of bit fields.
LCD_DATA04	0x402A0110	LCD Pin Data Registers. See LCD_DATA00 for the details of bit fields.
LCD_DATA05	0x402A0114	LCD Pin Data Registers. See LCD_DATA00 for the details of bit fields.
LCD_DATA06	0x402A0118	LCD Pin Data Registers. See LCD_DATA00 for the details of bit fields.
LCD_DATA07	0x402A011C	LCD Pin Data Registers. See LCD_DATA00 for the details of bit fields.
LCD_DATA10	0x402A0200	LCD Pin Data Registers
LCD_DATA11	0x402A0204	LCD Pin Data Registers. See LCD_DATA10 for the details of bit fields.
LCD_DATA12	0x402A0208	LCD Pin Data Registers. See LCD_DATA10 for the details of bit fields.
LCD_DATA13	0x402A020C	LCD Pin Data Registers. See LCD_DATA10 for the details of bit fields.
LCD_DATA14	0x402A0210	LCD Pin Data Registers. See LCD_DATA10 for the details of bit fields.
LCD_DATA15	0x402A0214	LCD Pin Data Registers. See LCD_DATA10 for the details of bit fields.
LCD_DATA16	0x402A0218	LCD Pin Data Registers. See LCD_DATA10 for the details of bit fields.
LCD_DATA17	0x402A021C	LCD Pin Data Registers. See LCD_DATA10 for the details of bit fields.

13.1.1 LCD_ID

ID & Revision

Address: 0x402A0000

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access								
Name	ID [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access								
Name	ID [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access								
Name	REVISION [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access								
Name	REVISION [31:24]							

Bits	Name	Description
31 : 16	REVISION	the version number is 0x0001 Default Value: 1
15 : 0	ID	the ID of LCD controller peripheral is 0xF0F0 Default Value: 61680

13.1.2 LCD_DIVIDER

LCD Divider Register

Address: 0x402A0004

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access	R							
Name	SUBFR_DIV [7:0]							
Bits	15	14	13	12	11	10	9	8
SW Access	RW							
HW Access	R							
Name	SUBFR_DIV [15:8]							
Bits	23	22	21	20	19	18	17	16
SW Access	RW							
HW Access	R							
Name	DEAD_DIV [23:16]							
Bits	31	30	29	28	27	26	25	24
SW Access	RW							
HW Access	R							
Name	DEAD_DIV [31:24]							

Bits	Name	Description
31 : 16	DEAD_DIV	Length of the dead time period in cycles. When set to zero, no dead time period exists. Default Value: 0
15 : 0	SUBFR_DIV	Input clock frequency divide value, to generate the 1/4 sub-frame period. The sub-frame period is 4*(SUBFR_DIV+1) cycles long. Default Value: 0

13.1.3 LCD_CONTROL

LCD Configuration Register

Address: 0x402A0008

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None	RW		RW	RW	RW	RW	RW
HW Access	None	R		R	R	R	R	R
Name	None	BIAS [6:5]		OP_MODE	TYPE	LCD_- MODE	HS_EN	LS_EN

Bits	15	14	13	12	11	10	9	8
SW Access	None				RW			
HW Access	None				R			
Name	None [15:12]				COM_NUM [11:8]			

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R	None						
HW Access	W	None						
Name	LS_EN_STAT	None [30:24]						

Bits	Name	Description
31	LS_EN_STAT	Reserved
11 : 8	COM_NUM	The number of COM connections minus 2. So: 0: 2 COM's 1: 3 COM's ... 13: 15 COM's 14: 16 COM's 15: undefined Default Value: 0
6 : 5	BIAS	PWM bias selection Default Value: 0 0x0: HALF : 1/2 Bias

13.1.3 LCD_CONTROL (continued)

		0x1: THIRD : 1/3 Bias
		0x2: FOURTH : 1/4 Bias
		0x3: FIFTH : 1/5 Bias
4	OP_MODE	Driving mode configuration Default Value: 0 0x0: PWM : PWM Mode
3	TYPE	LCD driving waveform type configuration. Default Value: 0 0x0: TYPE_A : Type A - Each frame addresses each COM pin only once with a balanced (DC=0) waveform. 0x1: TYPE_B : Type B - Each frame addresses each COM pin twice in sequence with a positive and negative waveform that together are balanced (DC=0).
2	LCD_MODE	HS/LS Mode selection Default Value: 0 0x0: LS : Reserved 0x1: HS : Select High Speed (system clock) Generator (Works in Active and Sleep power modes only).
1	HS_EN	High speed (HS) generator enable 1: enable 0: disable Default Value: 0
0	LS_EN	Reserved

13.1.4 LCD_DATA00

LCD Pin Data Registers

Address: 0x402A0100

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access	R							
Name	DATA [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	RW							
HW Access	R							
Name	DATA [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	RW							
HW Access	R							
Name	DATA [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	RW							
HW Access	R							
Name	DATA [31:24]							

Bits	Name	Description
31 : 0	DATA	Bits [4i+3:4i] represent the pin data for pin [i] for COMS 1-4 (COM1 is lsb). Default Value: 0

13.1.5 LCD_DATA10

LCD Pin Data Registers

Address: 0x402A0200

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access	R							
Name	DATA [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	RW							
HW Access	R							
Name	DATA [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	RW							
HW Access	R							
Name	DATA [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	RW							
HW Access	R							
Name	DATA [31:24]							

Bits	Name	Description
31 : 0	DATA	Bits [4i+3:4i] represent the pin data for pin [i] for COMS 5-8 (COM5 is lsb). Default Value: 0

14 Low-Power Comparator (LPCOMP) Registers



This section discusses the LPCOMP registers. It lists all the registers in mapping tables, in address order.

14.1 Register Details

Register	Address	Description
LPCOMP_ID	0x402B0000	ID & Revision
LPCOMP_CONFIG	0x402B0004	LPCOMP Configuration Register
LPCOMP_INTR	0x402B0010	LPCOMP Interrupt request register
LPCOMP_INTR_SET	0x402B0014	LPCOMP Interrupt set register
LPCOMP_INTR_MASK	0x402B0018	LPCOMP Interrupt request mask
LPCOMP_INTR_MASKED	0x402B001C	LPCOMP Interrupt request masked
LPCOMP_TRIM1	0x402BFF00	LPCOMP Trim Register
LPCOMP_TRIM2	0x402BFF04	LPCOMP Trim Register
LPCOMP_TRIM3	0x402BFF08	LPCOMP Trim Register
LPCOMP_TRIM4	0x402BFF0C	LPCOMP Trim Register

14.1.1 LPCOMP_ID

ID & Revision

Address: 0x402B0000

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access								
Name	ID [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access								
Name	ID [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access								
Name	REVISION [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access								
Name	REVISION [31:24]							

Bits	Name	Description
31 : 16	REVISION	the version number is 0x0001 Default Value: 1
15 : 0	ID	the ID of LPCOMP peripheral is 0xE0E0 Default Value: 57568

14.1.2 LPCOMP_CONFIG

LPCOMP Configuration Register

Address: 0x402B0004

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW	R	RW		RW	RW	RW	
HW Access	R	RW	R		R	R	R	
Name	ENABLE1	OUT1	INTTYPE1 [5:4]		FILTER1	HYST1	MODE1 [1:0]	

Bits	15	14	13	12	11	10	9	8
SW Access	RW	R	RW		RW	RW	RW	
HW Access	R	RW	R		R	R	R	
Name	ENABLE2	OUT2	INTTYPE2 [13:12]		FILTER2	HYST2	MODE2 [9:8]	

Bits	23	22	21	20	19	18	17	16
SW Access	None		RW	RW	None		RW	RW
HW Access	None		R	R	None		R	R
Name	None [23:22]		DSI_LEV-EL2	DSI_BY-PASS2	None [19:18]		DSI_LEV-EL1	DSI_BY-PASS1

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
21	DSI_LEVEL2	Opamp2 comparator DSI (trigger) out level : 0=pulse, 1=level Default Value: 0
20	DSI_BYPASS2	Opamp2 bypass comparator output synchronization for DSI output: 0=synchronize (level or pulse), 1=bypass (output async) Note that in DeepSleep mode this bit needs to be set to observe the DSI output on the dedicated pin. Default Value: 0
17	DSI_LEVEL1	Opamp1 comparator DSI (trigger) out level : 0=pulse, 1=level Default Value: 0
16	DSI_BYPASS1	Opamp1 bypass comparator output synchronization for DSI output: 0=synchronize (level or pulse), 1=bypass (output async) Note that in DeepSleep mode this bit needs to be set to observe the DSI output on the dedicated pin. Default Value: 0
15	ENABLE2	Enable comparator #2 Default Value: 0

14.1.2 LPCOMP_CONFIG (continued)

14	OUT2	Current output value of the comparator. Default Value: 0
13 : 12	INTTYPE2	Sets which edge will trigger an IRQ Default Value: 0 0x0: DISABLE : Disabled, no interrupts will be detected 0x1: RISING : Rising edge 0x2: FALLING : Falling edge 0x3: BOTH : Both rising and falling edges
11	FILTER2	Deprecated, Reserved, must be written 0 Default Value: 0
10	HYST2	Add 10mV hysteresis to the comparator - 0: Enable Hysteresis - 1: Disable Hysteresis Default Value: 0
9 : 8	MODE2	Operating mode for the comparator Default Value: 0 0x0: SLOW : Slow operating mode (uses less power, <50uA) 0x1: FAST : Fast operating mode (uses more power, <400uA) 0x2: ULP : Ultra low power operating mode (uses ~2-4uA)
7	ENABLE1	Enable comparator #1 Default Value: 0
6	OUT1	Current output value of the comparator. Default Value: 0
5 : 4	INTTYPE1	Sets which edge will trigger an IRQ Default Value: 0

14.1.2 LPCOMP_CONFIG (continued)

0x0: DISABLE :

Disabled, no interrupts will be detected

0x1: RISING :

Rising edge

0x2: FALLING :

Falling edge

0x3: BOTH :

Both rising and falling edges

3	FILTER1	Deprecated, Reserved, must be written 0 Default Value: 0
---	---------	---

2	HYST1	Add 10mV hysteresis to the comparator - 0: Enable Hysteresis - 1: Disable Hysteresis Default Value: 0
---	-------	--

1 : 0	MODE1	Operating mode for the comparator Default Value: 0
-------	-------	---

0x0: SLOW :

Slow operating mode (uses less power, <50uA)

0x1: FAST :

Fast operating mode (uses more power, <400uA)

0x2: ULP :

Ultra low power operating mode (uses ~2-4uA)

14.1.3 LPCOMP_INTR

LPCOMP Interrupt request register

Address: 0x402B0010

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None						RW1C	RW1C
HW Access	None						RW1S	RW1S
Name	None [7:2]						COMP2	COMP1

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
1	COMP2	Comparator 2 Interrupt: hardware sets this interrupt when comparator 0 triggers. Write with '1' to clear bit. Default Value: 0
0	COMP1	Comparator 1 Interrupt: hardware sets this interrupt when comparator 0 triggers. Write with '1' to clear bit. Default Value: 0

14.1.4 LPCOMP_INTR_SET

LPCOMP Interrupt set register

Address: 0x402B0014

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None						RW1S	RW1S
HW Access	None						A	A
Name	None [7:2]						COMP2	COMP1

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
1	COMP2	Write with '1' to set corresponding bit in interrupt request register. Default Value: 0
0	COMP1	Write with '1' to set corresponding bit in interrupt request register. Default Value: 0

14.1.5 LPCOMP_INTR_MASK

LPCOMP Interrupt request mask

Address: 0x402B0018

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None						RW	RW
HW Access	None						R	R
Name	None [7:2]						COMP2_M ASK	COMP1_M ASK

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
1	COMP2_MASK	Mask bit for corresponding bit in interrupt request register. Default Value: 0
0	COMP1_MASK	Mask bit for corresponding bit in interrupt request register. Default Value: 0

14.1.6 LPCOMP_INTR_MASKED

LPCOMP Interrupt request masked

Address: 0x402B001C

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None						R	R
HW Access	None						W	W
Name	None [7:2]						COMP2_M ASKED	COMP1_M ASKED

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
1	COMP2_MASKED	Logical and of corresponding request and mask bits. Default Value: 0
0	COMP1_MASKED	Logical and of corresponding request and mask bits. Default Value: 0

14.1.7 LPCOMP_TRIM1

LPCOMP Trim Register

Address: 0x402BFF00

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None			RW				
HW Access	None			R				
Name	None [7:5]			COMP1_TRIMA [4:0]				

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
4 : 0	COMP1_TRIMA	Trim A for Comparator #1 Default Value: 0

14.1.8 LPCOMP_TRIM2

LPCOMP Trim Register

Address: 0x402BFF04

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None			RW				
HW Access	None			R				
Name	None [7:5]			COMP1_TRIMB [4:0]				

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
4 : 0	COMP1_TRIMB	Trim B for Comparator #1 Default Value: 0

14.1.9 LPCOMP_TRIM3

LPCOMP Trim Register

Address: 0x402BFF08

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None			RW				
HW Access	None			R				
Name	None [7:5]			COMP2_TRIMA [4:0]				

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
4 : 0	COMP2_TRIMA	Trim A for Comparator #2 Default Value: 0

14.1.10 LPCOMP_TRIM4

LPCOMP Trim Register

Address: 0x402BFF0C

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None			RW				
HW Access	None			R				
Name	None [7:5]			COMP2_TRIMB [4:0]				

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
4 : 0	COMP2_TRIMB	Trim B for Comparator #2 Default Value: 0

15 External Clock (EXCO) Registers



This section discusses the EXCO registers. It lists all the registers in mapping tables, in address order.

15.1 Register Details

Register	Address	Description
EXCO_CLK_SELECT	0x402F0000	Clock Select Register
EXCO_ECO_CONFIG	0x402F0008	ECO Configuration Register
EXCO_ECO_STATUS	0x402F000C	ECO Status Register
EXCO_PLL_CONFIG	0x402F0014	PLL Configuration Register
EXCO_PLL_STATUS	0x402F0018	PLL Status Register
EXCO_PLL_TEST	0x402F001C	PLL Test Register
EXCO_EXCO_PGM_CLK	0x402F0020	EXCO Program Clock
EXCO_REF_CTL	0x402F0030	Clock Supervision Reference Control
EXCO_REF_LIMIT	0x402F0034	Clock Supervision Reference Limits
EXCO_MON_CTL	0x402F0038	Clock Supervision Monitor Control
EXCO_INTR	0x402F0040	Interrupt Request Register
EXCO_INTR_SET	0x402F0044	Interrupt Set Register
EXCO_INTR_MASK	0x402F0048	Interrupt Mask Register
EXCO_INTR_MASKED	0x402F004C	Interrupt Masked Register
EXCO_RSTDLY_CTL	0x402F0050	Programmable Delay Counter Control
EXCO_RSTDLY	0x402F0054	Programmable Delay Counter Initial Amount
EXCO_RSTDLY_COUNT_VAL	0x402F0058	Programmable Delay Counter Value
EXCO_ECO_TRIM0	0x402FFF00	ECO Trim0 Register
EXCO_ECO_TRIM1	0x402FFF04	ECO Trim1 Register
EXCO_ECO_TRIM2	0x402FFF08	ECO Trim2 Register
EXCO_PLL_TRIM	0x402FFF0C	PLL Trim Register

15.1.1 EXCO_CLK_SELECT

Clock Select Register

Address: 0x402F0000

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None					RW		RW
HW Access	None					R		R
Name	None [7:3]					REF_SEL [2:1]		CLK_SELECT

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
2 : 1	REF_SEL	Select source for PLL reference 0: from ECO 1: from external reference 2: from clk_imo 3: from clk_imo Default Value: 0 0x0: SEL_ECO : From ECO 0x1: SEL_EXT_REF : From external reference 0x2: SEL_CLK_IMO : From CLK_IMO

15.1.1 EXCO_CLK_SELECT (continued)

0	CLK_SELECT	Select a clock source for clk_eco: If PLL_CONFIG.ENABLE=0, then clk_eco=clk_osc. When PLL_CONFIG.ENABLE=1, and PLL_CONFIG.BYPASS=2'b10: 0: clk_osc 1: clk_pll Default Value: 0
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15.1.2 EXCO_ECO_CONFIG

ECO Configuration Register

Address: 0x402F0008

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None						RW	RW
HW Access	None						R	R
Name	None [7:2]						AGC_EN	CLK_EN

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	RW	None						
HW Access	R	None						
Name	ENABLE	None [30:24]						

Bits	Name	Description
31	ENABLE	Master enable for ECO oscillator. Refer to CLK_EN for sequencing. Default Value: 0
1	AGC_EN	Automatic Gain Control (AGC) enable. When set, the oscillation amplitude is controlled to the level selected by ECO_TRIM0.ATRIM. When low, the amplitude is not explicitly controlled and will grow until it saturates to the supply rail (1.8V nom). WARNING: use care when disabling AGC because driving a crystal beyond its rated limit can permanently damage the crystal. Default Value: 1
0	CLK_EN	Clock Enable. When enabling the clock, first write ENABLE=1, wait at least 10us, and then write CLK_EN=1. When disabling, clearing both CLK_EN=0 and ENABLE=0 can be done in the same AHB write Default Value: 0

15.1.3 EXCO_ECO_STATUS

ECO Status Register

Address: 0x402F000C

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None							R
HW Access	None							W
Name	None [7:1]							WATCH- DOG_ER- ROR

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
0	WATCHDOG_ERROR	This bit is set to 1 if the oscillator is stuck. The ECO clock is gated off during a watchdog error condition. Due to internal synchronization, the clock is stopped two cycles after an error condition is observed and ungated two cycles after the error condition is resolved. Note: This bit setting reflected in INT.WD_ERR if Interrupts are enabled. Default Value: 0

15.1.4 EXCO_PLL_CONFIG

PLL Configuration Register

Address: 0x402F0014

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access	R							
Name	FEEDBACK_DIV [7:0]							
Bits	15	14	13	12	11	10	9	8
SW Access	RW		RW					
HW Access	R		R					
Name	OUTPUT_DIV [15:14]		REFERENCE_DIV [13:8]					
Bits	23	22	21	20	19	18	17	16
SW Access	None		RW		None	RW		
HW Access	None		R		None	R		
Name	None [23:22]		BYPASS_SEL [21:20]		None	ICP_SEL [18:16]		
Bits	31	30	29	28	27	26	25	24
SW Access	RW	RW	None					
HW Access	R	R	None					
Name	ENABLE	ISOLATE_N	None [29:24]					

Bits	Name	Description
31	ENABLE	Master enable for PLL power gate. Refer to ISOLATE_N field for required sequencing. 0: Block is powered off, also forces clk_eco = clk_osc - See CLK_SELECT.CLK_SELECT description. 1: Block is powered on Default Value: 0
30	ISOLATE_N	Isolation control of PLL outputs. This also internally resets the PLL. De-assert >= 5us after ENABLE=1. Assertion can happen in same write as ENABLE=0. 0: Isolate outputs 1: Do not isolate outputs Default Value: 0
21 : 20	BYPASS_SEL	Selects the source of the system PLL0 clock. See also CLK_SELECT.CLK_SELECT for effect on clk_eco selection. Default Value: 0 0x0: AUTO : Automatic using lock indicator. When unlocked, automatically selects PLL reference input (bypass mode). When locked, automatically selects PLL output.

15.1.4 EXCO_PLL_CONFIG (continued)

0x1: AUTO1 :

Same as AUTO

0x2: PLL_REF :

Select PLL reference input (bypass mode). Ignores lock indicator

0x3: PLL_OUT :

Select PLL output. Ignores lock indicator.

18 : 16 ICP_SEL

Programmable charge pump current between 0uA and 7uA. For functional operation, the value must be set according to the PLL output frequency Fout (measured before the output divider):
 0,1: Illegal
 2: 2uA. Use when Fout <= 67MHz
 3: 3uA. Use when Fout > 67MHz
 4-7: Illegal
 Default Value: 2

15 : 14 OUTPUT_DIV

Control bits for Output divider:

$F_{OUT} = FVCO / \{1,2,4,8\}$
 Default Value: 0

0x0: PASS :

Pass Through

0x1: DIV2 :

Divide by 2

0x2: DIV4 :

Divide by 4

0x3: DIV8 :

Divide by 8

13 : 8 REFERENCE_DIV

Control bits for reference divider:

5x00 = Divide by 1
 5x01 = Divide by 2
 ...
 5x1F = Divide by 64

Must be set so that $1 \text{ MHz} \leq F_{PPD} \leq 3 \text{ MHz}$, where
 $F_{PPD} = F_{REF} / (\text{REFERENCE_DIV} + 1)$

Do not change while PLL output is selected.
 Default Value: 0

15.1.4 EXCO_PLL_CONFIG (continued)

7 : 0	FEEDBACK_DIV	Control bits for feedback divider: 8x08 = Divide by 8 (minimum legal value) 8x09 = Divide by 9 ... 8xFF = Divide by 255 Must be set so that $22.5\text{MHz} \leq \text{FVCO} \leq 104\text{MHz}$, where $\text{FVCO} = \text{FREF} * \text{FEEDBACK_DIV} / (\text{REFERENCE_DIV} + 1)$ Do not change while PLL output is selected. Additional limitations on PLL output frequency apply if PLL is providing the system clock. Default Value: 0
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15.1.5 EXCO_PLL_STATUS

PLL Status Register

Address: 0x402F0018

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None							R
HW Access	None							W
Name	None [7:1]							LOCKED

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
0	LOCKED	PLL Lock Indicator Default Value: 0

15.1.6 EXCO_PLL_TEST

PLL Test Register

Address: 0x402F001C

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None			RW1C	RW	RW		
HW Access	None			A	R	R		
Name	None [7:5]			UN- LOCK_OC- CURRED	FAST_LOC K_EN	TEST_MODE [2:0]		

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
4	UNLOCK_OCCURRED	This bit is not retained. Default Value: 0
3	FAST_LOCK_EN	Reserved. Leave at default value. Default Value: 1
2 : 0	TEST_MODE	Reserved. Leave at default value. Default Value: 0
0x0: NORMAL :		
Normal Operation		
0x1: TEST_VC_LKG :		
Vcontrol Leakage Test Mode Measure frequency drift over time to indirectly measure leakage on Vcontrol		

15.1.6 EXCO_PLL_TEST (continued)

0x2: TEST_CP_DN :

Charge Pump Down Current Test Mode

With ICPSEL>0, directly measure charge pump up current on Vcontrol

With ICPSEL=0, directly measure leakage on Vcontrol

With ICPSEL=0, FAST_LOCK_EN=0 and ISOLATE_N=0, directly measure discharge current on Vcontrol

With ICPSEL=0, FAST_LOCK_EN=1 and ISOLATE_N=0, directly measure precharge current on Vcontrol

0x3: TEST_CP_UP :

Charge Pump Up Current Test Mode

With ICPSEL>0, directly measure charge pump up current on Vcontrol

With ICPSEL=0, directly measure leakage on Vcontrol

With ICPSEL=0, FAST_LOCK_EN=0 and ISOLATE_N=0, directly measure discharge current on Vcontrol

With ICPSEL=0, FAST_LOCK_EN=1 and ISOLATE_N=0, directly measure precharge current on Vcontrol

0x4: USER_EXT_FL :

User Mode with Extended Fast Lock Precharge

0x5: TEST_CTR_PQ :

Reference and Feedback Counter Test Mode

0x6: TEST_LD_DLY :

Lock Detector Delay Line Test Mode

0x7: TEST_CTR_ALT :

Lock Detector Wait and Extended Fast Lock Counter Test Mode

With ICPSEL=0 and Reference Clock stopped directly measure precharge current on Vcontrol

With ICPSEL=0 and Reference Clock running directly measure leakage on Vcontrol

15.1.7 EXCO_EXCO_PGM_CLK

EXCO Program Clock

Address: 0x402F0020

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None			RW	RW	R	RW	None
HW Access	None			R	R	RW	R	None
Name	None [7:5]			EN_ CLK_PLL0	CLK_PLL0_ OUT	CLK_PLL0_ IN	CLK_ECO	None

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	RW	None						
HW Access	R	None						
Name	ENABLE	None [30:24]						

Bits	Name	Description
31	ENABLE	Enable bit-banging test capability in this register. Default Value: 0
4	EN_CLK_PLL0	Bit bang en_clk_pll0 Default Value: 0
3	CLK_PLL0_OUT	Bit bang clk_pll0_out Default Value: 0
2	CLK_PLL0_IN	Observation point for clk_pll0_in, not retained. Default Value: 0
1	CLK_ECO	Bit bang clk_eco Default Value: 0

15.1.8 EXCO_REF_CTL

Clock Supervision Reference Control

Address: 0x402F0030

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access	R							
Name	STARTUP [7:0]							
Bits	15	14	13	12	11	10	9	8
SW Access	RW							
HW Access	R							
Name	STARTUP [15:8]							
Bits	23	22	21	20	19	18	17	16
SW Access	None					RW	RW	RW
HW Access	None					R	R	R
Name	None [23:19]					CSV_CLK_SW_EN	CSV_TRIG_EN	CSV_INT_EN
Bits	31	30	29	28	27	26	25	24
SW Access	RW	None						
HW Access	R	None						
Name	CSV_EN	None [30:24]						

Bits	Name	Description
31	CSV_EN	Enables clock supervision, both frequency and loss. CSV in Active domain: Clock supervision is reset during DeepSleep and Hibernate modes. When enabled it begins operating automatically after a DeepSleep wakeup, but it must be reconfigured after Hibernate wakeup. When CSV error detection occurs - the CSV_CLK_SW, CSV_INT_EN and CVS_TRIG_EN control where the error is reported. Default Value: 0
18	CSV_CLK_SW_EN	Enable CSV to cause Clock Switch to IMO when set. Default Value: 1
17	CSV_TRIG_EN	Enable CSV to cause trigger if a clock switch occurs to IMO. Default Value: 0
16	CSV_INT_EN	Enable CSV setting INT.CSV_CLK_SW if a clock switch occurs to IMO. Default Value: 0

15.1.8 EXCO_REF_CTL (continued)

15 : 0	STARTUP	Startup delay time -1 (in reference clock cycles), after enable or DeepSleep wakeup, from reference clock start to monitored clock start. At a minimum (both clocks running): $STARTUP \geq (PERIOD + 3) * FREQ_RATIO - UPPER$, with $FREQ_RATIO = (Reference\ frequency / Monitored\ frequency)$ On top of that the actual clock startup delay and the margin for accuracy of both clocks must be added. Default Value: 0
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15.1.9 EXCO_REF_LIMIT

Clock Supervision Reference Limits

Address: 0x402F0034

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access	R							
Name	LOWER [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	RW							
HW Access	R							
Name	LOWER [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	RW							
HW Access	R							
Name	UPPER [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	RW							
HW Access	R							
Name	UPPER [31:24]							

Bits	Name	Description
31 : 16	UPPER	Cycle time upper limit. Set the upper limit -1, in reference clock cycles, before (or same time) the next monitored clock event must happen. If a monitored clock event does not happen before this limit is reached, or does not happen at all (clock loss), a CSV error is detected. Default Value: 0
15 : 0	LOWER	Cycle time lower limit. Set the lower limit -1, in reference clock cycles, before the next monitored clock event is allowed to happen. If a monitored clock event happens before this limit is reached a CSV error is detected. LOWER must be at least 1 less than UPPER. In case the clocks are asynchronous LOWER must be at least 3 less than UPPER. Default Value: 0

15.1.10 EXCO_MON_CTL

Clock Supervision Monitor Control

Address: 0x402F0038

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access	R							
Name	PERIOD [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	RW							
HW Access	R							
Name	PERIOD [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
15 : 0	PERIOD	Period time. Set the Period -1, in monitored clock cycles, before the next monitored clock event happens. $PERIOD \leq (UPPER+1) / FREQ_RATIO - 1$, with $FREQ_RATIO = (Reference\ frequency / Monitored\ frequency)$ In case the clocks are asynchronous: $PERIOD \leq UPPER / FREQ_RATIO - 1$ Additionally margin must be added for accuracy of both clocks. Default Value: 0

15.1.11 EXCO_INTR

Interrupt Request Register

Address: 0x402F0040

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None					RW1C	RW1C	RW1C
HW Access	None					RW1S	RW1S	RW1S
Name	None [7:3]					CSV_- CLK_SW	WD_ERR	PLL_LOCK

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
2	CSV_CLK_SW	Clock Supervisor Switched Clock Source to IMO Default Value: 0
1	WD_ERR	EXCO Watch Dog Error detected - Oscillator stopped oscillating Default Value: 0
0	PLL_LOCK	HW will set this bit when PLL loses lock (PLL "locked" output goes low). Default Value: 0

15.1.12 EXCO_INTR_SET

Interrupt Set Register

Address: 0x402F0044

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None					RW1S	RW1S	RW1S
HW Access	None					A	A	A
Name	None [7:3]					CSV_- CLK_SW	WD_ERR	PLL_LOCK

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
2	CSV_CLK_SW	Write with '1' to set corresponding bit in interrupt request register. Default Value: 0
1	WD_ERR	Write with '1' to set corresponding bit in interrupt request register. Default Value: 0
0	PLL_LOCK	Write with '1' to set corresponding bit in interrupt request register. Default Value: 0

15.1.13 EXCO_INTR_MASK

Interrupt Mask Register

Address: 0x402F0048

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None					RW	RW	RW
HW Access	None					R	R	R
Name	None [7:3]					CSV_- CLK_SW	WD_ERR	PLL_LOCK

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
2	CSV_CLK_SW	Mask bit for corresponding bit in interrupt request register. Default Value: 0
1	WD_ERR	Mask bit for corresponding bit in interrupt request register. Default Value: 0
0	PLL_LOCK	Mask bit for corresponding bit in interrupt request register. Default Value: 0

15.1.14 EXCO_INTR_MASKED

Interrupt Masked Register

Address: 0x402F004C

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None					R	R	R
HW Access	None					W	W	W
Name	None [7:3]					CSV_ CLK_SW	WD_ERR	PLL_LOCK

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
2	CSV_CLK_SW	Logical and of corresponding request and mask bits. Default Value: 0
1	WD_ERR	Logical and of corresponding request and mask bits. Default Value: 0
0	PLL_LOCK	Logical and of corresponding request and mask bits. Default Value: 0

15.1.15 EXCO_RSTDLY_CTL

Programmable Delay Counter Control

Address: 0x402F0050

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None							RW1S
HW Access	None							RW1C
Name	None [7:1]							LOAD

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	RW	None						
HW Access	R	None						
Name	EN	None [30:24]						

Bits	Name	Description
31	EN	Programmable Delay Counter Enable "0": Disable "1": Enable Default Value: 0
0	LOAD	Programmable Delay Counter Load - Reloads the DLYCOUNT into the COUNT_VAL register 0: No Action 1: Generate a 1 clock pulse that loads the Initial value into the COUNT_VAL register. Default Value: 0

15.1.16 EXCO_RSTDLY

Programmable Delay Counter Initial Amount

Address: 0x402F0054

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access	R							
Name	DLYCOUNT [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	RW							
HW Access	R							
Name	DLYCOUNT [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
15 : 0	DLYCOUNT	Delay Count Value Default Value: 256

15.1.17 EXCO_RSTDLY_COUNT_VAL

Programmable Delay Counter Value

Address: 0x402F0058

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access	RW							
Name	COUNT_VAL [7:0]							
Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access	RW							
Name	COUNT_VAL [15:8]							
Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							
Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
15 : 0	COUNT_VAL	Current Programmable Delay Counter value Default Value: 256

15.1.18 EXCO_ECO_TRIM0

ECO Trim0 Register

Address: 0x402FFF00

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None			RW			RW	
HW Access	None			R			R	
Name	None [7:5]			ATRIM [4:2]			WDTRIM [1:0]	

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
4 : 2	ATRIM	Amplitude trim to set the crystal drive level when ECO_CONFIG.AGC_EN=1. WARNING: use care when setting this field because driving a crystal beyond its rated limit can permanently damage the crystal. 0x0 - 0.3Vpp 0x1 - 0.4Vpp 0x2 - 0.5Vpp 0x3 - 0.6Vpp 0x4 - 0.7Vpp 0x5 - 0.8Vpp 0x6 - 0.9Vpp 0x7 - 1.0Vpp Default Value: 0
1 : 0	WDTRIM	Watch Dog Trim - Delta voltage below stead state level 0x0 - 0.05V 0x1 - 0.1V 0x2 - 0.15V 0x3 - 0.2V Default Value: 0

15.1.19 EXCO_ECO_TRIM1

ECO Trim1 Register

Address: 0x402FFF04

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None		RW		RW		RW	
HW Access	None		R		R		R	
Name	None [7:6]		GTRIM [5:4]		RTRIM [3:2]		FTRIM [1:0]	

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
5 : 4	GTRIM	Gain Trim - Startup time Default Value: 1
3 : 2	RTRIM	Feedback resistor Trim Default Value: 1
1 : 0	FTRIM	Filter Trim - 3rd harmonic oscillation Default Value: 1

15.1.20 EXCO_ECO_TRIM2

ECO Trim2 Register

Address: 0x402FFF08

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None		RW					
HW Access	None		R					
Name	None [7:6]		ITRIM [5:0]					

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
5 : 0	ITRIM	Current Trim Default Value: 32

15.1.21 EXCO_PLL_TRIM

PLL Trim Register

Address: 0x402FFF0C

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None		RW		RW		RW	
HW Access	None		R		R		R	
Name	None [7:6]		LOCK_DELAY [5:4]		LOCK_WINDOW [3:2]		VCO_GAIN [1:0]	

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
5 : 4	LOCK_DELAY	Selects the number of PLL phase frequency detector cycles that the phase error must be in range before declaring lock. (PFD clock cycle = Clock Reference Period/REFERENCE_DIV) Default Value: 1 0x0: PFD_CLK_16 : 16 PFD clock cycles 0x1: PFD_CLK_32 : 32 PFD clock cycles 0x2: PFD_CLK_48 : 48 PFD clock cycles

15.1.21 EXCO_PLL_TRIM (continued)

		0x3: PFD_CLK_64 : 64 PFD clock cycles
3 : 2	LOCK_WINDOW	This register determines how much phase error is tolerated before declaring the PLL unlocked (lock_out goes low). Phase error is a measure of the phase difference between the PLL feedforward path (CLK_REF PERIOD/REFERENCE_DIV[1:0]) and feedback path (CLK_VCO/FEEDBACK_DIV[1:0]). The average phase error when locked is zero, but accumulated jitter - particularly on CLK_REF - leads to the need for a relatively large allowed phase error window. Default Value: 0 0x0: DELAY_25NS : Delay 25 ns 0x1: DELAY_50NS : Delay 50 ns 0x2: DELAY_75NS : Delay 75 ns 0x3: DELAY_100NS : Delay 100 ns
1 : 0	VCO_GAIN	Programmable VCO frequency characteristic at high freq - set to <10> Default Value: 2

16 Continuous Time Block Mini (CTBM) Registers



This section discusses the CTBM registers. It lists all the registers in mapping tables, in address order.

16.1 Register Details

Register	Address	Description
CTBM0_CTB_CTRL	0x40300000	global CTB and power control
CTBM0_OA_RES0_CTRL	0x40300004	Opamp0 and resistor0 control
CTBM0_OA_RES1_CTRL	0x40300008	Opamp1 and resistor1 control
CTBM0_COMP_STAT	0x4030000C	Comparator status
CTBM0_INTR	0x40300020	Interrupt request register
CTBM0_INTR_SET	0x40300024	Interrupt request set register
CTBM0_INTR_MASK	0x40300028	Interrupt request mask
CTBM0_INTR_MASKED	0x4030002C	Interrupt request masked
CTBM0_OA0_SW	0x40300080	Opamp0 switch control
CTBM0_OA0_SW_CLEAR	0x40300084	Opamp0 switch control clear
CTBM0_OA1_SW	0x40300088	Opamp1 switch control
CTBM0_OA1_SW_CLEAR	0x4030008C	Opamp1 switch control clear
CTBM0_OA0_OFFSET_TRIM	0x40300F00	Opamp0 trim control
CTBM0_OA0_SLOPE_OFFSET_TRIM	0x40300F04	Opamp0 trim control
CTBM0_OA0_COMP_TRIM	0x40300F08	Opamp0 trim control
CTBM0_OA1_OFFSET_TRIM	0x40300F0C	Opamp1 trim control
CTBM0_OA1_SLOPE_OFFSET_TRIM	0x40300F10	Opamp1 trim control
CTBM0_OA1_COMP_TRIM	0x40300F14	Opamp1 trim control

16.1.1 CTBM0_CTLB_CTRL

global CTB and power control

Address: 0x40300000

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None							
HW Access	None							
Name	None [7:0]							
Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							
Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							
Bits	31	30	29	28	27	26	25	24
SW Access	RW	RW	None					
HW Access	R	R	None					
Name	ENABLED	DEEPSLEEP_ON	None					

Bits	Name	Description
31	ENABLED	- 0: CTBm disabled (put analog in power down, open all switches) - 1: CTBm enabled Default Value: 0
30	DEEPSLEEP_ON	- 0: CTBm disabled off during DeepSleep power mode - 1: CTBm remains enabled during DeepSleep power mode (if ENABLED=1) Default Value: 0

16.1.2 CTBM0_OA_RES0_CTRL

Opamp0 and resistor0 control

Address: 0x40300004

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW	RW	RW	RW	None	RW	RW	
HW Access	R	R	R	R	None	R	R	
Name	OA0_DSI_L EVEL	OA0_BY- PASS_DSI_ SYNC	OA0_HYST _EN	OA0_COM- P_EN	None	OA0_DRIV E_STR_- SEL	OA0_PWR_MODE [1:0]	

Bits	15	14	13	12	11	10	9	8
SW Access	None				RW	None	RW	
HW Access	None				R	None	R	
Name	None [15:12]				OA0_PUMP _EN	None	OA0_COMPINT [9:8]	

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
11	OA0_PUMP_EN	Opamp0 pump enable Default Value: 0
9 : 8	OA0_COMPINT	Opamp0 comparator edge detect for interrupt and pulse mode trigger Default Value: 0
0x0: DISABLE :		
Disabled, no interrupts will be detected		
0x1: RISING :		
Rising edge		

16.1.2 CTBM0_OA_RES0_CTRL (continued)

0x2: FALLING :

Falling edge

0x3: BOTH :

Both rising and falling edges

7	OA0_DSI_LEVEL	Opamp0 comparator trigger out level : 0=pulse, each time an edge is detected (see OA0_COMPINT) a pulse is sent out on trigger 1=level, trigger output is a synchronized version of the comparator output Default Value: 0
6	OA0_BY-PASS_DSI_SYNC	Opamp0 bypass comparator output synchronization for trigger output: 0=synchronize (level or pulse), 1=bypass (output async) Default Value: 0
5	OA0_HYST_EN	Opamp hysteresis enable. See the device Datasheet for hysteresis specifications. Default Value: 0
4	OA0_COMP_EN	Opamp0 comparator enable Default Value: 0
2	OA0_DRIVE_STR_SEL	Opamp output drive strength: 0=1x, 1=10x. See the device Datasheet for exact current ranges and related specifications. This setting sets specific requirements for OA0_BOOST_EN and OA0_COMP_TRIM. Default Value: 0
1 : 0	OA0_PWR_MODE	Opamp power level. Reduced power levels also reduce gain-bandwidth (GBW). See the "Opamp Specifications" table in the device Datasheet for more details. Default Value: 0

0x0: OFF :

Off

0x1: LOW :

Power setting = Low (low current consumption and GBW).

0x2: MEDIUM :

Power setting = Medium (moderate current consumption and GBW).

0x3: HIGH :

Power setting = High (high current consumption and GBW).

16.1.3 CTBM0_OA_RES1_CTRL

Opamp1 and resistor1 control

Address: 0x40300008

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW	RW	RW	RW	None	RW	RW	
HW Access	R	R	R	R	None	R	R	
Name	OA1_DSI_L EVEL	OA1_BY- PASS_DSI_ SYNC	OA1_HYST _EN	OA1_COM- P_EN	None	OA1_DRIV E_STR_- SEL	OA1_PWR_MODE [1:0]	

Bits	15	14	13	12	11	10	9	8
SW Access	None				RW	None	RW	
HW Access	None				R	None	R	
Name	None [15:12]				OA1_PUMP _EN	None	OA1_COMPINT [9:8]	

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
11	OA1_PUMP_EN	Opamp1 pump enable Default Value: 0
9 : 8	OA1_COMPINT	Opamp1 comparator edge detect for interrupt and pulse mode of trigger Default Value: 0 0x0: DISABLE : Disabled, no interrupts will be detected 0x1: RISING : Rising edge

16.1.3 CTBM0_OA_RES1_CTRL (continued)

0x2: FALLING :

Falling edge

0x3: BOTH :

Both rising and falling edges

7	OA1_DSI_LEVEL	Opamp1 comparator trigger out level : 0=pulse, each time an edge is detected (see OA1_COMPINT) a pulse is sent out on trigger 1=level, trigger output is a synchronized version of the comparator output Default Value: 0
6	OA1_BY-PASS_DSI_SYNC	Opamp bypass comparator output synchronization for trigger output: 0=synchronize (level or pulse), 1=bypass (asynchronous output). Default Value: 0
5	OA1_HYST_EN	Opamp hysteresis enable. See the device Datasheet for hysteresis specifications. Default Value: 0
4	OA1_COMP_EN	Opamp1 comparator enable Default Value: 0
2	OA1_DRIVE_STR_SEL	Opamp output drive strength: 0=1x, 1=10x. See the device Datasheet for exact current ranges and related specifications. This setting sets specific requirements for OA1_BOOST_EN and OA1_COMP_TRIM Default Value: 0
1 : 0	OA1_PWR_MODE	Opamp power level. Reduced power levels also reduce gain-bandwidth (GBW). See the "Opamp Specifications" table in the device Datasheet for more details. Default Value: 0

16.1.4 CTBM0_COMP_STAT

Comparator status

Address: 0x4030000C

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None							R
HW Access	None							W
Name	None [7:1]							OA0_- COMP

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							R
HW Access	None							W
Name	None [23:17]							OA1_- COMP

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
16	OA1_COMP	Opamp1 current comparator status Default Value: 0
0	OA0_COMP	Opamp0 current comparator status Default Value: 0

16.1.5 CTBM0_INTR

Interrupt request register

Address: 0x40300020

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None						RW1C	RW1C
HW Access	None						RW1S	RW1S
Name	None [7:2]						COMP1	COMP0

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
1	COMP1	Comparator 1 Interrupt: hardware sets this interrupt when comparator 1 triggers. Write with '1' to clear bit. Default Value: 0
0	COMP0	Comparator 0 Interrupt: hardware sets this interrupt when comparator 0 triggers. Write with '1' to clear bit. Default Value: 0

16.1.6 CTBM0_INTR_SET

Interrupt request set register

Address: 0x40300024

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None						RW1S	RW1S
HW Access	None						A	A
Name	None [7:2]						COMP1_SET	COMP0_SET

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
1	COMP1_SET	Write with '1' to set corresponding bit in interrupt request register. Default Value: 0
0	COMP0_SET	Write with '1' to set corresponding bit in interrupt request register. Default Value: 0

16.1.7 CTBM0_INTR_MASK

Interrupt request mask

Address: 0x40300028

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None						RW	RW
HW Access	None						R	R
Name	None [7:2]						COMP1_M ASK	COMP0_M ASK

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
1	COMP1_MASK	Mask bit for corresponding bit in interrupt request register. Default Value: 0
0	COMP0_MASK	Mask bit for corresponding bit in interrupt request register. Default Value: 0

16.1.8 CTBM0_INTR_MASKED

Interrupt request masked

Address: 0x4030002C

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None						R	R
HW Access	None						W	W
Name	None [7:2]						COMP1_M ASKED	COMP0_M ASKED

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
1	COMP1_MASKED	Logical and of corresponding request and mask bits. Default Value: 0
0	COMP0_MASKED	Logical and of corresponding request and mask bits. Default Value: 0

16.1.9 CTBM0_OA0_SW

Opamp0 switch control

Address: 0x40300080

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None				RW1S	RW1S	None	RW1S
HW Access	None				RW1C	RW1C	None	RW1C
Name	None [7:4]				OA0P_A30	OA0P_A20	None	OA0P_A00

Bits	15	14	13	12	11	10	9	8
SW Access	None	RW1S	None					RW1S
HW Access	None	RW1C	None					RW1C
Name	None	OA0M_A81	None [13:9]					OA0M_A11

Bits	23	22	21	20	19	18	17	16
SW Access	None		RW1S	None		RW1S	None	
HW Access	None		RW1C	None		RW1C	None	
Name	None [23:22]		OA0O_D81	None [20:19]		OA0O_D51	None [17:16]	

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
21	OA0O_D81	Switch that shorts Opamp's 1x and 10x outputs. Default Value: 0
18	OA0O_D51	Switch that connects Opamp's output to SARBUS 0. Default Value: 0
14	OA0M_A81	Switch that connects Opamp's inverting terminal to Opamp's output for follower mode. Default Value: 0
8	OA0M_A11	Switch that connects Opamp's inverting terminal to a pin of CTBm port . Default Value: 0
3	OA0P_A30	Switch that connects Opamp's non-inverting terminal to a secondary pin of the CTBm port. Default Value: 0
2	OA0P_A20	Switch that connects Opamp's non-inverting terminal to a pin of CTBm port. See the device Data-sheet for the location of CTBm port and pins. Default Value: 0
0	OA0P_A00	Switch that connects Opamp's non-inverting terminal to AMUXBUS A. Default Value: 0

16.1.10 CTBM0_OA0_SW_CLEAR

Opamp0 switch control clear

Address: 0x40300084

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None				RW1C	RW1C	None	RW1C
HW Access	None				A	A	None	A
Name	None [7:4]				OA0P_A30	OA0P_A20	None	OA0P_A00

Bits	15	14	13	12	11	10	9	8
SW Access	None	RW1C	None					RW1C
HW Access	None	A	None					A
Name	None	OA0M_A81	None [13:9]					OA0M_A11

Bits	23	22	21	20	19	18	17	16
SW Access	None		RW1C	None		RW1C	None	
HW Access	None		A	None		A	None	
Name	None [23:22]		OA0O_D81	None [20:19]		OA0O_D51	None [17:16]	

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
21	OA0O_D81	see corresponding bit in OA0_SW Default Value: 0
18	OA0O_D51	see corresponding bit in OA0_SW Default Value: 0
14	OA0M_A81	see corresponding bit in OA0_SW Default Value: 0
8	OA0M_A11	see corresponding bit in OA0_SW Default Value: 0
3	OA0P_A30	see corresponding bit in OA0_SW Default Value: 0
2	OA0P_A20	see corresponding bit in OA0_SW Default Value: 0
0	OA0P_A00	see corresponding bit in OA0_SW Default Value: 0

16.1.11 CTBM0_OA1_SW

Opamp1 switch control

Address: 0x40300088

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None			RW1S	None		RW1S	RW1S
HW Access	None			RW1C	None		RW1C	RW1C
Name	None [7:5]			OA1P_A43	None [3:2]		OA1P_A13	OA1P_A03

Bits	15	14	13	12	11	10	9	8
SW Access	None	RW1S	None					RW1S
HW Access	None	RW1C	None					RW1C
Name	None	OA1M_A82	None [13:9]					OA1M_A22

Bits	23	22	21	20	19	18	17	16
SW Access	None		RW1S	None	RW1S	RW1S	None	
HW Access	None		RW1C	None	RW1C	RW1C	None	
Name	None [23:22]		OA1O_D82	None	OA1O_D62	OA1O_D52	None [17:16]	

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
21	OA1O_D82	Switch that shorts Opamp's 1x and 10x outputs. Default Value: 0
19	OA1O_D62	Switch that connects Opamp's output to SARBUS 1. Default Value: 0
18	OA1O_D52	Switch that connects Opamp's output to SARBUS 0. Default Value: 0
14	OA1M_A82	Switch that connects Opamp's inverting terminal to Opamp's output for follower mode. Default Value: 0
8	OA1M_A22	Switch that connects Opamp's inverting terminal to a pin of CTBm port. Default Value: 0
4	OA1P_A43	Switch that connects Opamp's non-inverting terminal to a secondary pin of CTBm port. Default Value: 0
1	OA1P_A13	Switch that connects Opamp's non-inverting terminal to a pin of CTBm port. See the device Data-sheet for the location of CTBm port and pins. Default Value: 0

16.1.11 CTBM0_OA1_SW (continued)

0	OA1P_A03	Switch that connects Opamp's non-inverting terminal to AMUXBUS B. Default Value: 0
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16.1.12 CTBM0_OA1_SW_CLEAR

Opamp1 switch control clear

Address: 0x4030008C

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None			RW1C	None		RW1C	RW1C
HW Access	None			A	None		A	A
Name	None [7:5]			OA1P_A43	None [3:2]		OA1P_A13	OA1P_A03

Bits	15	14	13	12	11	10	9	8
SW Access	None	RW1C	None					RW1C
HW Access	None	A	None					A
Name	None	OA1M_A82	None [13:9]					OA1M_A22

Bits	23	22	21	20	19	18	17	16
SW Access	None		RW1C	None	RW1C	RW1C	None	
HW Access	None		A	None	A	A	None	
Name	None [23:22]		OA1O_D82	None	OA1O_D62	OA1O_D52	None [17:16]	

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
21	OA1O_D82	see corresponding bit in OA1_SW Default Value: 0
19	OA1O_D62	see corresponding bit in OA1_SW Default Value: 0
18	OA1O_D52	see corresponding bit in OA1_SW Default Value: 0
14	OA1M_A82	see corresponding bit in OA1_SW Default Value: 0
8	OA1M_A22	see corresponding bit in OA1_SW Default Value: 0
4	OA1P_A43	see corresponding bit in OA1_SW Default Value: 0
1	OA1P_A13	see corresponding bit in OA1_SW Default Value: 0
0	OA1P_A03	see corresponding bit in OA1_SW Default Value: 0

16.1.13 CTBM0_OA0_OFFSET_TRIM

Opamp0 trim control

Address: 0x40300F00

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None		RW					
HW Access	None		R					
Name	None [7:6]		OA0_OFFSET_TRIM [5:0]					

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
5 : 0	OA0_OFFSET_TRIM	Opamp0 offset trim Default Value: 0

16.1.14 CTBM0_OA0_SLOPE_OFFSET_TRIM

Opamp0 trim control

Address: 0x40300F04

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None		RW					
HW Access	None		R					
Name	None [7:6]		OA0_SLOPE_OFFSET_TRIM [5:0]					

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
5 : 0	OA0_SLOPE_OFFSET_TRIM	Opamp0 slope offset drift trim Default Value: 0

16.1.15 CTBM0_OA0_COMP_TRIM

Opamp0 trim control

Address: 0x40300F08

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None						RW	
HW Access	None						R	
Name	None [7:2]						OA0_COMP_TRIM [1:0]	

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
1 : 0	OA0_COMP_TRIM	Opamp0 Compensation Capacitor Trim Default Value: 0

16.1.16 CTBM0_OA1_OFFSET_TRIM

Opamp1 trim control

Address: 0x40300F0C

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None		RW					
HW Access	None		R					
Name	None [7:6]		OA1_OFFSET_TRIM [5:0]					

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
5 : 0	OA1_OFFSET_TRIM	Opamp1 offset trim Default Value: 0

16.1.17 CTBM0_OA1_SLOPE_OFFSET_TRIM

Opamp1 trim control

Address: 0x40300F10

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None		RW					
HW Access	None		R					
Name	None [7:6]		OA1_SLOPE_OFFSET_TRIM [5:0]					

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
5 : 0	OA1_SLOPE_OFFSET_TRIM	Opamp1 slope offset drift trim Default Value: 0

16.1.18 CTBM0_OA1_COMP_TRIM

Opamp1 trim control

Address: 0x40300F14

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None						RW	
HW Access	None						R	
Name	None [7:2]						OA1_COMP_TRIM [1:0]	

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
1 : 0	OA1_COMP_TRIM	Opamp1 Compensation Capacitor Trim Default Value: 0

17 SAR ADC Registers



This section discusses the SAR ADC registers. It lists all the registers in mapping tables, in address order.

17.1 Register Details

Register	Address	Description
SAR0_CTRL	0x403A0000	Analog control register.
SAR0_SAMPLE_CTRL	0x403A0004	Sample control register.
SAR0_SAMPLE_TIME01	0x403A0010	Sample time specification ST0 and ST1
SAR0_SAMPLE_TIME23	0x403A0014	Sample time specification ST2 and ST3
SAR0_RANGE_THRES	0x403A0018	Global range detect threshold register.
SAR0_RANGE_COND	0x403A001C	Global range detect mode register.
SAR0_CHAN_EN	0x403A0020	Enable bits for the channels
SAR0_START_CTRL	0x403A0024	Start control register (firmware trigger).
SAR0_CHAN_CONFIG0	0x403A0080	Channel configuration register.
SAR0_CHAN_CONFIG1	0x403A0084	Channel configuration register.. See SAR0_CHAN_CONFIG0 for the details of bit fields.
SAR0_CHAN_CONFIG2	0x403A0088	Channel configuration register.. See SAR0_CHAN_CONFIG0 for the details of bit fields.
SAR0_CHAN_CONFIG3	0x403A008C	Channel configuration register.. See SAR0_CHAN_CONFIG0 for the details of bit fields.
SAR0_CHAN_CONFIG4	0x403A0090	Channel configuration register.. See SAR0_CHAN_CONFIG0 for the details of bit fields.
SAR0_CHAN_CONFIG5	0x403A0094	Channel configuration register.. See SAR0_CHAN_CONFIG0 for the details of bit fields.
SAR0_CHAN_CONFIG6	0x403A0098	Channel configuration register.. See SAR0_CHAN_CONFIG0 for the details of bit fields.
SAR0_CHAN_CONFIG7	0x403A009C	Channel configuration register.. See SAR0_CHAN_CONFIG0 for the details of bit fields.
SAR0_CHAN_CONFIG8	0x403A00A0	Channel configuration register.. See SAR0_CHAN_CONFIG0 for the details of bit fields.
SAR0_CHAN_CONFIG9	0x403A00A4	Channel configuration register.. See SAR0_CHAN_CONFIG0 for the details of bit fields.
SAR0_CHAN_CONFIG10	0x403A00A8	Channel configuration register.. See SAR0_CHAN_CONFIG0 for the details of bit fields.
SAR0_CHAN_CONFIG11	0x403A00AC	Channel configuration register.. See SAR0_CHAN_CONFIG0 for the details of bit fields.
SAR0_CHAN_CONFIG12	0x403A00B0	Channel configuration register.. See SAR0_CHAN_CONFIG0 for the details of bit fields.
SAR0_CHAN_CONFIG13	0x403A00B4	Channel configuration register.. See SAR0_CHAN_CONFIG0 for the details of bit fields.
SAR0_CHAN_CONFIG14	0x403A00B8	Channel configuration register.. See SAR0_CHAN_CONFIG0 for the details of bit fields.
SAR0_CHAN_CONFIG15	0x403A00BC	Channel configuration register.. See SAR0_CHAN_CONFIG0 for the details of bit fields.
SAR0_CHAN_WORK0	0x403A0100	Channel working data register
SAR0_CHAN_WORK1	0x403A0104	Channel working data register. See SAR0_CHAN_WORK0 for the details of bit fields.
SAR0_CHAN_WORK2	0x403A0108	Channel working data register. See SAR0_CHAN_WORK0 for the details of bit fields.
SAR0_CHAN_WORK3	0x403A010C	Channel working data register. See SAR0_CHAN_WORK0 for the details of bit fields.

Register	Address	Description
SAR0_CHAN_WORK4	0x403A0110	Channel working data register. See SAR0_CHAN_WORK0 for the details of bit fields.
SAR0_CHAN_WORK5	0x403A0114	Channel working data register. See SAR0_CHAN_WORK0 for the details of bit fields.
SAR0_CHAN_WORK6	0x403A0118	Channel working data register. See SAR0_CHAN_WORK0 for the details of bit fields.
SAR0_CHAN_WORK7	0x403A011C	Channel working data register. See SAR0_CHAN_WORK0 for the details of bit fields.
SAR0_CHAN_WORK8	0x403A0120	Channel working data register. See SAR0_CHAN_WORK0 for the details of bit fields.
SAR0_CHAN_WORK9	0x403A0124	Channel working data register. See SAR0_CHAN_WORK0 for the details of bit fields.
SAR0_CHAN_WORK10	0x403A0128	Channel working data register. See SAR0_CHAN_WORK0 for the details of bit fields.
SAR0_CHAN_WORK11	0x403A012C	Channel working data register. See SAR0_CHAN_WORK0 for the details of bit fields.
SAR0_CHAN_WORK12	0x403A0130	Channel working data register. See SAR0_CHAN_WORK0 for the details of bit fields.
SAR0_CHAN_WORK13	0x403A0134	Channel working data register. See SAR0_CHAN_WORK0 for the details of bit fields.
SAR0_CHAN_WORK14	0x403A0138	Channel working data register. See SAR0_CHAN_WORK0 for the details of bit fields.
SAR0_CHAN_WORK15	0x403A013C	Channel working data register. See SAR0_CHAN_WORK0 for the details of bit fields.
SAR0_CHAN_RESULT0	0x403A0180	Channel result data register
SAR0_CHAN_RESULT1	0x403A0184	Channel result data register. See SAR0_CHAN_RESULT0 for the details of bit fields.
SAR0_CHAN_RESULT2	0x403A0188	Channel result data register. See SAR0_CHAN_RESULT0 for the details of bit fields.
SAR0_CHAN_RESULT3	0x403A018C	Channel result data register. See SAR0_CHAN_RESULT0 for the details of bit fields.
SAR0_CHAN_RESULT4	0x403A0190	Channel result data register. See SAR0_CHAN_RESULT0 for the details of bit fields.
SAR0_CHAN_RESULT5	0x403A0194	Channel result data register. See SAR0_CHAN_RESULT0 for the details of bit fields.
SAR0_CHAN_RESULT6	0x403A0198	Channel result data register. See SAR0_CHAN_RESULT0 for the details of bit fields.
SAR0_CHAN_RESULT7	0x403A019C	Channel result data register. See SAR0_CHAN_RESULT0 for the details of bit fields.
SAR0_CHAN_RESULT8	0x403A01A0	Channel result data register. See SAR0_CHAN_RESULT0 for the details of bit fields.
SAR0_CHAN_RESULT9	0x403A01A4	Channel result data register. See SAR0_CHAN_RESULT0 for the details of bit fields.
SAR0_CHAN_RESULT10	0x403A01A8	Channel result data register. See SAR0_CHAN_RESULT0 for the details of bit fields.
SAR0_CHAN_RESULT11	0x403A01AC	Channel result data register. See SAR0_CHAN_RESULT0 for the details of bit fields.
SAR0_CHAN_RESULT12	0x403A01B0	Channel result data register. See SAR0_CHAN_RESULT0 for the details of bit fields.
SAR0_CHAN_RESULT13	0x403A01B4	Channel result data register. See SAR0_CHAN_RESULT0 for the details of bit fields.
SAR0_CHAN_RESULT14	0x403A01B8	Channel result data register. See SAR0_CHAN_RESULT0 for the details of bit fields.
SAR0_CHAN_RESULT15	0x403A01BC	Channel result data register. See SAR0_CHAN_RESULT0 for the details of bit fields.
SAR0_CHAN_WORK_VALID	0x403A0200	Channel working data register valid bits
SAR0_CHAN_RESULT_VALID	0x403A0204	Channel result data register valid bits
SAR0_STATUS	0x403A0208	Current status of internal SAR registers (mostly for debug)
SAR0_AVG_STAT	0x403A020C	Current averaging status (for debug)
SAR0_INTR	0x403A0210	Interrupt request register.
SAR0_INTR_SET	0x403A0214	Interrupt set request register
SAR0_INTR_MASK	0x403A0218	Interrupt mask register.
SAR0_INTR_MASKED	0x403A021C	Interrupt masked request register
SAR0_SATURATE_INTR	0x403A0220	Saturate interrupt request register.
SAR0_SATURATE_INTR_SET	0x403A0224	Saturate interrupt set request register
SAR0_SATURATE_INTR_MASK	0x403A0228	Saturate interrupt mask register.
SAR0_SATURATE_INTR_MASKED	0x403A022C	Saturate interrupt masked request register
SAR0_RANGE_INTR	0x403A0230	Range detect interrupt request register.
SAR0_RANGE_INTR_SET	0x403A0234	Range detect interrupt set request register
SAR0_RANGE_INTR_MASK	0x403A0238	Range detect interrupt mask register.
SAR0_RANGE_INTR_MASKED	0x403A023C	Range interrupt masked request register

Register	Address	Description
SAR0_INTR_CAUSE	0x403A0240	Interrupt cause register
SAR0_INJ_CHAN_CONFIG	0x403A0280	Injection channel configuration register.
SAR0_INJ_RESULT	0x403A0290	Injection channel result register
SAR0_MUX_SWITCH0	0x403A0300	SARMUX Firmware switch controls
SAR0_MUX_SWITCH_CLEAR0	0x403A0304	SARMUX Firmware switch control clear
SAR0_MUX_SWITCH_HW_CTRL	0x403A0340	SARMUX switch hardware control
SAR0_MUX_SWITCH_STATUS	0x403A0348	SARMUX switch status
SAR0_PUMP_CTRL	0x403A0380	Switch pump control

17.1.1 SAR0_CTRL

Analog control register.

Address: 0x403A0000

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW	RW			None			
HW Access	R	R			None			
Name	VREF_BY- P_CAP_EN	VREF_SEL [6:4]			None [3:0]			

Bits	15	14	13	12	11	10	9	8
SW Access	RW		RW	None	RW			None
HW Access	R		R	None	R			None
Name	PWR_CTRL_VREF [15:14]		SAR_HW_CTRL_NEG VREF	None	NEG_SEL [11:9]			None

Bits	23	22	21	20	19	18	17	16
SW Access	None			RW	RW			
HW Access	None			R	R			
Name	None [23:21]			BOOST- PUMP_EN	SPARE [19:16]			

Bits	31	30	29	28	27	26	25	24
SW Access	RW	RW	RW	RW	RW	None	RW	
HW Access	R	R	R	R	R	None	R	
Name	ENABLED	SWITCH_- DISABLE	DSI_MODE	DSI_SYNC- C_CONFIG	DEEPS- LEEP_ON	None	ICONT_LV [25:24]	

Bits	Name	Description
31	ENABLED	- 0: SAR disabled (put analog in power down and stop clocks), also can clear FW_TRIGGER on write. - 1: SAR enabled. Default Value: 0
30	SWITCH_DISABLE	Disable SAR sequencer from enabling routing switches - 0: Normal mode, SAR sequencer changes switches according to pin address in channel configurations - 1: Switches disabled, SAR sequencer does not enable any switches, Other methods such as firmware control can be used to set the switches to route the signal to be converted through the SARMUX Default Value: 0
29	DSI_MODE	Reserved
28	DSI_SYNC_CONFIG	Reserved

17.1.1 SAR0_CTRL (continued)

27	DEEPSLEEP_ON	- 0: SARMUX IP disabled off during DeepSleep power mode - 1: SARMUX IP remains enabled during DeepSleep power mode (if ENABLED=1) Default Value: 0
25 : 24	ICONT_LV	SARADC low power mode. Default Value: 0 0x0: NORMAL_PWR : normal power (default), max clk_sar is 18MHz. 0x1: HALF_PWR : 1/2 power mode, max clk_sar is 9MHz. 0x2: MORE_PWR : 1.333 power mode, max clk_sar is 18MHz. 0x3: QUARTER_PWR : 1/4 power mode, max clk_sar is 4.5MHz.
20	BOOSTPUMP_EN	SARADC internal pump: 0=disabled: pump output is VDDA, 1=enabled: pump output is boosted. Default Value: 0
19 : 16	SPARE	Reserved
15 : 14	PWR_CTRL_VREF	VREF buffer low power mode. Default Value: 0 0x0: NORMAL_PWR : normal power (default), bypass cap, max clk_sar is 18MHz. 0x1: HALF_PWR : Reserved 0x2: THIRD_PWR : Invalid for PSoC4A, otherwise 2X power, no bypass cap, max clk_sar is 1.8MHz 0x3: QUARTER_PWR : Reserved
13	SAR_HW_CTRL_NEGVREF	Hardware control: 0=only firmware control, 1=hardware control masked by firmware setting for VREF to NEG switch. Default Value: 0
11 : 9	NEG_SEL	SARADC internal NEG selection for Single ended conversion Default Value: 0

17.1.1 SAR0_CTRL (continued)

0x0: VSSA_KELVIN :

NEG input of SARADC is connected to "vssa_kelvin", gives more precision around zero. Note this opens both SARADC internal switches, therefore use this value to insert a break-before-make cycle on those switches when SWITCH_DISABLE is high.

0x1: ART_VSSA :

NEG input of SARADC is connected to VSSA in AROUTE close to the SARADC

0x2: P1 :

NEG input of SARADC is connected to P1 pin of SARMUX

0x3: P3 :

NEG input of SARADC is connected to P3 pin of SARMUX

0x4: P5 :

NEG input of SARADC is connected to P5 pin of SARMUX

0x5: P7 :

NEG input of SARADC is connected to P7 pin of SARMUX

0x6: ACORE :

NEG input of SARADC is connected to an ACORE in AROUTE

0x7: VREF :

NEG input of SARADC is shorted with VREF input of SARADC.

7	VREF_BYP_CAP_EN	VREF bypass cap enable for when VREF buffer is on Default Value: 0
---	-----------------	---

6 : 4	VREF_SEL	SARADC VREF selection. Default Value: 0
-------	----------	--

0x0: VREF0 :

Reserved

0x1: VREF1 :

Reserved

0x2: VREF2 :

Reserved

17.1.1 SAR0_CTRL (continued)

0x3: VREF_AROUTE :

Reserved

0x4: VBGR :

Internal reference (VREF buffer on)

0x5: VREF_EXT :

External precision Vref direct from a pin (low impedance path).

0x6: VDDA_DIV_2 :

Vdda/2 (VREF buffer on)

0x7: VDDA :

Vdda.

17.1.2 SAR0_SAMPLE_CTRL

Sample control register.

Address: 0x403A0004

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW	RW			RW	RW	RW	RW
HW Access	R	R			R	R	R	R
Name	AVG_SHIFT	AVG_CNT [6:4]			DIFFERENTIAL_SIGNED	SINGLE_ENDED_SIGNAL	LEFT_ALIGN	SUB_RESOLUTION

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None				RW	RW	RW	RW
HW Access	None				R	R	R	R
Name	None [23:20]				DSI_SYNC_TRIGGER	DSI_TRIGGER_LEVEL	DSI_TRIGGER_EN	CONTINUOUS

Bits	31	30	29	28	27	26	25	24
SW Access	RW	None						
HW Access	R	None						
Name	EOS_DSI_OUT_EN	None [30:24]						

Bits	Name	Description
31	EOS_DSI_OUT_EN	Enable to output EOS_INTR. When enabled each time EOS_INTR is set by the hardware also a trigger plus is send on the tr_sar_out signal. Default Value: 0
19	DSI_SYNC_TRIGGER	- 0: bypass clock domain synchronisation of the trigger signal. - 1: synchronize the trigger signal to the SAR clock domain, if needed an edge detect is done in the peripheral clock domain. Default Value: 1
18	DSI_TRIGGER_LEVEL	- 0: trigger signal is a pulse input, a positive edge detected on the trigger signal triggers a new scan. - 1: trigger signal is a level input, as long as the trigger signal remains high the SAR will do continuous scans. Default Value: 0

17.1.2 SAR0_SAMPLE_CTRL (continued)

17	DSI_TRIGGER_EN	- 0: firmware trigger only: disable hardware trigger. - 1: enable hardware trigger (e.g. from TCPWM, GPIO or UDB). Default Value: 0
16	CONTINUOUS	- 0: Wait for next FW_TRIGGER (one shot) or hardware trigger (e.g. from TPWM for periodic triggering) before scanning enabled channels. - 1: Continuously scan enabled channels, ignore triggers. Default Value: 0
7	AVG_SHIFT	Averaging shifting: after averaging the result is shifted right to fit in the sample resolution. For averaging the sample resolution is the highest resolution allowed by rounding. Default Value: 0
6 : 4	AVG_CNT	Averaging Count for channels that have over sampling enabled (AVG_EN). A channel will be sampled back to back $(1 \leq (AVG_CNT+1)) = [2..256]$ times before the result is stored and the next enabled channel is sampled (1st order accumulate and dump filter). If shifting is not enabled (AVG_SHIFT=0) then the result is forced to shift right so that it fits in 16 bits, so right shift is done by $\max(0, AVG_CNT-3)$. Default Value: 0
3	DIFFERENTIAL_SIGNED	Output data from a differential conversion as a signed value Default Value: 1 0x0: UNSIGNED : result data is unsigned (zero extended if needed) 0x1: SIGNED : Default: result data is signed (sign extended if needed)
2	SINGLE_ENDED_SIGNED	Output data from a single ended conversion as a signed value Default Value: 0 0x0: UNSIGNED : Default: result data is unsigned (zero extended if needed) 0x1: SIGNED : result data is signed (sign extended if needed)
1	LEFT_ALIGN	Left align data in data[15:0], default data is right aligned in data[11:0], with sign extension to 16 bits if the channel is differential. Default Value: 0
0	SUB_RESOLUTION	Conversion resolution for channels that have sub-resolution enabled (RESOLUTION=1) (otherwise resolution is 12-bit). Default Value: 0 0x0: 8B : 8-bit. 0x1: 10B : 10-bit.

17.1.3 SAR0_SAMPLE_TIME01

Sample time specification ST0 and ST1

Address: 0x403A0010

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access	R							
Name	SAMPLE_TIME0 [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	None						RW	
HW Access	None						R	
Name	None [15:10]						SAMPLE_TIME0 [9:8]	

Bits	23	22	21	20	19	18	17	16
SW Access	RW							
HW Access	R							
Name	SAMPLE_TIME1 [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None						RW	
HW Access	None						R	
Name	None [31:26]						SAMPLE_TIME1 [25:24]	

Bits	Name	Description
25 : 16	SAMPLE_TIME1	Sample time1 Default Value: 4
9 : 0	SAMPLE_TIME0	Sample time0 (aperture) in ADC clock cycles. Note that actual sample time is half a clock less than specified here. The minimum sample time is 194ns, which is 3.5 cycles (4 in this field) with an 18MHz clock. Minimum legal value in this register is 2. Default Value: 4

17.1.4 SAR0_SAMPLE_TIME23

Sample time specification ST2 and ST3

Address: 0x403A0014

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access	R							
Name	SAMPLE_TIME2 [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	None						RW	
HW Access	None						R	
Name	None [15:10]						SAMPLE_TIME2 [9:8]	

Bits	23	22	21	20	19	18	17	16
SW Access	RW							
HW Access	R							
Name	SAMPLE_TIME3 [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None						RW	
HW Access	None						R	
Name	None [31:26]						SAMPLE_TIME3 [25:24]	

Bits	Name	Description
25 : 16	SAMPLE_TIME3	Sample time3 Default Value: 4
9 : 0	SAMPLE_TIME2	Sample time2 Default Value: 4

17.1.5 SAR0_RANGE_THRES

Global range detect threshold register.

Address: 0x403A0018

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access	R							
Name	RANGE_LOW [7:0]							
Bits	15	14	13	12	11	10	9	8
SW Access	RW							
HW Access	R							
Name	RANGE_LOW [15:8]							
Bits	23	22	21	20	19	18	17	16
SW Access	RW							
HW Access	R							
Name	RANGE_HIGH [23:16]							
Bits	31	30	29	28	27	26	25	24
SW Access	RW							
HW Access	R							
Name	RANGE_HIGH [31:24]							

Bits	Name	Description
31 : 16	RANGE_HIGH	High threshold for range detect. Default Value: 0
15 : 0	RANGE_LOW	Low threshold for range detect. Default Value: 0

17.1.6 SAR0_RANGE_COND

Global range detect mode register.

Address: 0x403A001C

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None							
HW Access	None							
Name	None [7:0]							
Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							
Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							
Bits	31	30	29	28	27	26	25	24
SW Access	RW		None					
HW Access	R		None					
Name	RANGE_COND [31:30]		None [29:24]					

Bits	Name	Description
31 : 30	RANGE_COND	Range condition select. Default Value: 0 0x0: BELOW : result < RANGE_LOW 0x1: INSIDE : RANGE_LOW <= result < RANGE_HIGH 0x2: ABOVE : RANGE_HIGH <= result 0x3: OUTSIDE : result < RANGE_LOW RANGE_HIGH <= result

17.1.7 SAR0_CHAN_EN

Enable bits for the channels

Address: 0x403A0020

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access	R							
Name	CHAN_EN [7:0]							
Bits	15	14	13	12	11	10	9	8
SW Access	RW							
HW Access	R							
Name	CHAN_EN [15:8]							
Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							
Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
15 : 0	CHAN_EN	Channel enable. - 0: the corresponding channel is disabled. - 1: the corresponding channel is enabled, it will be included in the next scan. Default Value: 0

17.1.8 SAR0_START_CTRL

Start control register (firmware trigger).

Address: 0x403A0024

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None							RW1S
HW Access	None							RW1C
Name	None [7:1]							FW_TRIGGER

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
0	FW_TRIGGER	When firmware writes a 1 here it will trigger the next scan of enabled channels, hardware clears this bit when the scan started with this trigger is completed. If scanning continuously the trigger is ignored and hardware clears this bit after the next scan is done. This bit is also cleared when the SAR is disabled. Default Value: 0

17.1.9 SAR0_CHAN_CONFIG0

Channel configuration register.

Address: 0x403A0080

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None	RW			None	RW		
HW Access	None	R			None	R		
Name	None	PORT_ADDR [6:4]			None	PIN_ADDR [2:0]		

Bits	15	14	13	12	11	10	9	8
SW Access	None		RW		None	RW	RW	RW
HW Access	None		R		None	R	R	R
Name	None [15:14]		SAMPLE_TIME_SEL [13:12]		None	AVG_EN	RESOLUTION	DIFFERENTIAL_EN

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	RW	None						
HW Access	R	None						
Name	DSI_OUT_EN	None [30:24]						

Bits	Name	Description
31	DSI_OUT_EN	DSI data output enable for this channel. - 0: the conversion result for this channel is only stored in the channel data register and the corresponding CHAN_DATA_VALID bit is set. - 1: the conversion result for this channel is stored in the channel data register and the corresponding CHAN_DATA_VALID bit is set. The same data (same formatting), together with the channel number, is sent out on the DSI communication channel for processing in UDBs. Default Value: 0
13 : 12	SAMPLE_TIME_SEL	Sample time select: select which of the 4 global sample times to use for this channel Default Value: 0
10	AVG_EN	Averaging enable for this channel. If set the AVG_CNT and AVG_SHIFT settings are used for sampling the addressed pin(s) Default Value: 0
9	RESOLUTION	Resolution for this channel. When AVG_EN is set this bit is ignored and always a 12-bit resolution (or highest resolution allowed by rounding) is used for this channel. Default Value: 0

17.1.9 SAR0_CHAN_CONFIG0 (continued)

0x0: MAXRES :

The maximum resolution is used for this channel (maximum resolution depends on wounding).

0x1: SUBRES :

The resolution specified by SUB_RESOLUTION in the SAR_SAMPLE_CTRL register is used for this channel.

8 DIFFERENTIAL_EN

Differential enable for this channel.

- 0: The voltage on the addressed pin is measured (Single-ended) and the resulting value is stored in the corresponding data register.

- 1: The differential voltage on the addressed pin pair is measured and the resulting value is stored in the corresponding data register. (PIN_ADDR[0] is ignored).

Default Value: 0

6 : 4 PORT_ADDR

Address of the port that contains the pin to be sampled by this channel.

Default Value: 0

0x0: SARMUX :

SARMUX pins.

0x1: CTB0 :

CTB0

0x2: CTB1 :

CTB1

0x3: CTB2 :

Reserved

0x4: CTB3 :

Reserved

0x5: AROUTE_VIRT2 :

Reserved

0x6: AROUTE_VIRT1 :

Reserved

0x7: SARMUX_VIRT :

SARMUX virtual port (VPORT0)

17.1.9 SAR0_CHAN_CONFIG0 (continued)

2 : 0	PIN_ADDR	Address of the pin to be sampled by this channel. If differential is enabled then PIN_ADDR[0] is ignored and considered to be 0, i.e. PIN_ADDR points to the even pin of a pin pair. For differential the even pin of the pair is connected to Vplus and the odd pin of the pair is connected to Vminus. Default Value: 0
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17.1.10 SAR0_CHAN_WORK0

Channel working data register

Address: 0x403A0100

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access	RW							
Name	WORK [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access	RW							
Name	WORK [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R	None						
HW Access	W	None						
Name	CHAN_WORK_VALID_MIR	None [30:24]						

Bits	Name	Description
31	CHAN_WORK_VALID_MIR	mirror bit of corresponding bit in SAR_CHAN_WORK_VALID register Default Value: 0
15 : 0	WORK	SAR conversion working data of the channel. The data is written here right after sampling this channel. Default Value: Undefined

17.1.11 SAR0_CHAN_RESULT0

Channel result data register

Address: 0x403A0180

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access	W							
Name	RESULT [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access	W							
Name	RESULT [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R	R	R	None				
HW Access	W	W	W	None				
Name	CHAN_RE-SULT_VALID_MIR	RANGE_INTR_MIR	SATURATE_INTR_MIR	None [28:24]				

Bits	Name	Description
31	CHAN_RE-SULT_VALID_MIR	mirror bit of corresponding bit in SAR_CHAN_RESULT_VALID register Default Value: 0
30	RANGE_INTR_MIR	mirror bit of corresponding bit in SAR_RANGE_INTR register Default Value: 0
29	SATURATE_INTR_MIR	mirror bit of corresponding bit in SAR_SATURATE_INTR register Default Value: 0
15 : 0	RESULT	SAR conversion result of the channel. The data is copied here from the WORK field after all enabled channels in this scan have been sampled. Default Value: Undefined

17.1.12 SAR0_CHAN_WORK_VALID

Channel working data register valid bits

Address: 0x403A0200

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access	RW							
Name	CHAN_WORK_VALID [7:0]							
Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access	RW							
Name	CHAN_WORK_VALID [15:8]							
Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							
Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
15 : 0	CHAN_WORK_VALID	If set the corresponding WORK data is valid, i.e. was already sampled during the current scan. Default Value: 0

17.1.13 SAR0_CHAN_RESULT_VALID

Channel result data register valid bits

Address: 0x403A0204

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access	RW							
Name	CHAN_RESULT_VALID [7:0]							
Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access	RW							
Name	CHAN_RESULT_VALID [15:8]							
Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							
Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
15 : 0	CHAN_RESULT_VALID	If set the corresponding RESULT data is valid, i.e. was sampled during the last scan. Default Value: 0

17.1.14 SAR0_STATUS

Current status of internal SAR registers (mostly for debug)

Address: 0x403A0208

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None			R				
HW Access	None			W				
Name	None [7:5]			CUR_CHAN [4:0]				

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R	R	None					
HW Access	W	W	None					
Name	BUSY	SW_VREF_NEG	None [29:24]					

Bits	Name	Description
31	BUSY	If high then the SAR is busy with a conversion. This bit is always high when CONTINUOUS is set. Firmware should wait for this bit to be low before putting the SAR in power down. Default Value: 0
30	SW_VREF_NEG	the current switch status, including DSI and sequencer controls, of the switch in the SARADC that shorts NEG with VREF input (see NEG_SEL). Default Value: 0
4 : 0	CUR_CHAN	current channel being sampled (channel 16 indicates the injection channel), only valid if BUSY. Default Value: 0

17.1.15 SAR0_AVG_STAT

Current averaging status (for debug)

Address: 0x403A020C

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access	W							
Name	CUR_AVG_ACCU [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access	W							
Name	CUR_AVG_ACCU [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None				R			
HW Access	None				W			
Name	None [23:20]				CUR_AVG_ACCU [19:16]			

Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access	W							
Name	CUR_AVG_CNT [31:24]							

Bits	Name	Description
31 : 24	CUR_AVG_CNT	the current value of the averaging counter. Note that the value shown is updated after the sampling time and therefore runs ahead of the accumulator update. Default Value: 0
19 : 0	CUR_AVG_ACCU	the current value of the averaging accumulator Default Value: 0

17.1.16 SAR0_INTR

Interrupt request register.

Address: 0x403A0210

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW1C	RW1C	RW1C	RW1C	RW1C	RW1C	RW1C	RW1C
HW Access	RW1S	RW1S	RW1S	RW1S	RW1S	RW1S	RW1S	RW1S
Name	INJ_COLLISION_INTR	INJ_RANGE_INTR	INJ_SATURATE_INTR	INJ_EOC_INTR	DSI_COLLISION_INTR	FW_COLLISION_INTR	OVERFLOW_INTR	EOS_INTR

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
7	INJ_COLLISION_INTR	Injection Collision Interrupt: hardware sets this interrupt when the injection trigger signal is asserted (INJ_START_EN==1 && INJ_TAILGATING==0) while the SAR is BUSY. Raising this interrupt is delayed to when the sampling of the injection channel has been completed, i.e. not when the preceeding scan with which this trigger collided is completed. When this interrupt is set it implies that the injection channel was sampled later than was intended. Write with '1' to clear bit. Default Value: 0
6	INJ_RANGE_INTR	Injection Range detect Interrupt: hardware sets this interrupt if the injection conversion result (after averaging) met the condition specified by the SAR_RANGE registers. Write with '1' to clear bit. Default Value: 0
5	INJ_SATURATE_INTR	Injection Saturation Interrupt: hardware sets this interrupt if an injection conversion result (before averaging) is either 0x000 or 0xFFFF (for 12-bit resolution), this is an indication that the ADC likely saturated. Write with '1' to clear bit. Default Value: 0

17.1.16 SAR0_INTR (continued)

4	INJ_EOC_INTR	Injection End of Conversion Interrupt: hardware sets this interrupt after completing the conversion for the injection channel (irrespective of if tailgating was used). Write with '1' to clear bit. Default Value: 0
3	DSI_COLLISION_INTR	This interrupt is set when a hardware trigger signal is asserted while the SAR is BUSY. Raising this interrupt is delayed to when the scan caused by the hardware trigger has been completed, i.e. not when the preceding scan with which this trigger collided is completed. When this interrupt is set it implies that the channels were sampled later than was intended (jitter). Write with '1' to clear bit. Default Value: 0
2	FW_COLLISION_INTR	Firmware Collision Interrupt: hardware sets this interrupt when FW_TRIGGER is asserted while the SAR is BUSY. Raising this interrupt is delayed to when the scan caused by the FW_TRIGGER has been completed, i.e. not when the preceding scan with which this trigger collided is completed. When this interrupt is set it implies that the channels were sampled later than was intended (jitter). Write with '1' to clear bit. Default Value: 0
1	OVERFLOW_INTR	Overflow Interrupt: hardware sets this interrupt when it sets a new EOS_INTR while that bit was not yet cleared by the firmware. Write with '1' to clear bit. Default Value: 0
0	EOS_INTR	End Of Scan Interrupt: hardware sets this interrupt after completing a scan of all the enabled channels. Write with '1' to clear bit. Default Value: 0

17.1.17 SAR0_INTR_SET

Interrupt set request register

Address: 0x403A0214

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW1S	RW1S	RW1S	RW1S	RW1S	RW1S	RW1S	RW1S
HW Access	A	A	A	A	A	A	A	A
Name	INJ_COLLISION_SET	INJ_RANGE_SET	INJ_SATURATE_SET	INJ_EOC_SET	DSI_COLLISION_SET	FW_COLLISION_SET	OVERFLOW_SET	EOS_SET

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
7	INJ_COLLISION_SET	Write with '1' to set corresponding bit in interrupt request register. Default Value: 0
6	INJ_RANGE_SET	Write with '1' to set corresponding bit in interrupt request register. Default Value: 0
5	INJ_SATURATE_SET	Write with '1' to set corresponding bit in interrupt request register. Default Value: 0
4	INJ_EOC_SET	Write with '1' to set corresponding bit in interrupt request register. Default Value: 0
3	DSI_COLLISION_SET	Write with '1' to set corresponding bit in interrupt request register. Default Value: 0
2	FW_COLLISION_SET	Write with '1' to set corresponding bit in interrupt request register. Default Value: 0
1	OVERFLOW_SET	Write with '1' to set corresponding bit in interrupt request register. Default Value: 0

17.1.17 SAR0_INTR_SET (continued)

0	EOS_SET	Write with '1' to set corresponding bit in interrupt request register. Default Value: 0
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17.1.18 SAR0_INTR_MASK

Interrupt mask register.

Address: 0x403A0218

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW	RW	RW	RW	RW	RW	RW	RW
HW Access	R	R	R	R	R	R	R	R
Name	INJ_COLLISION_MASK	INJ_RANGE_MASK	INJ_SATURATE_MASK	INJ_EOC_MASK	DSI_COLLISION_MASK	FW_COLLISION_MASK	OVERFLOW_MASK	EOS_MASK

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
7	INJ_COLLISION_MASK	Mask bit for corresponding bit in interrupt request register. Default Value: 0
6	INJ_RANGE_MASK	Mask bit for corresponding bit in interrupt request register. Default Value: 0
5	INJ_SATURATE_MASK	Mask bit for corresponding bit in interrupt request register. Default Value: 0
4	INJ_EOC_MASK	Mask bit for corresponding bit in interrupt request register. Default Value: 0
3	DSI_COLLISION_MASK	Mask bit for corresponding bit in interrupt request register. Default Value: 0
2	FW_COLLISION_MASK	Mask bit for corresponding bit in interrupt request register. Default Value: 0
1	OVERFLOW_MASK	Mask bit for corresponding bit in interrupt request register. Default Value: 0

17.1.18 SAR0_INTR_MASK (continued)

0	EOS_MASK	Mask bit for corresponding bit in interrupt request register. Default Value: 0
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17.1.19 SAR0_INTR_MASKED

Interrupt masked request register

Address: 0x403A021C

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R	R	R	R	R	R	R	R
HW Access	W	W	W	W	W	W	W	W
Name	INJ_COLLI- SION_MAS KED	IN- J_RANGE_ MASKED	INJ_SATU- RATE_MAS KED	IN- J_EOC_MA SKED	DSI_COLLI- SION_MAS KED	FW_COLLI- SION_MAS KED	OVER- FLOW_MA SKED	EO- S_MASKED

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
7	INJ_COLLI- SION_MASKED	Logical and of corresponding request and mask bits. Default Value: 0
6	INJ_RANGE_MASKED	Logical and of corresponding request and mask bits. Default Value: 0
5	INJ_SATU- RATE_MASKED	Logical and of corresponding request and mask bits. Default Value: 0
4	INJ_EOC_MASKED	Logical and of corresponding request and mask bits. Default Value: 0
3	DSI_COLLI- SION_MASKED	Logical and of corresponding request and mask bits. Default Value: 0
2	FW_COLLI- SION_MASKED	Logical and of corresponding request and mask bits. Default Value: 0
1	OVERFLOW_MASKED	Logical and of corresponding request and mask bits. Default Value: 0

17.1.19 SAR0_INTR_MASKED (continued)

0	EOS_MASKED	Logical and of corresponding request and mask bits. Default Value: 0
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17.1.20 SAR0_SATURATE_INTR

Saturate interrupt request register.

Address: 0x403A0220

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW1C							
HW Access	RW1S							
Name	SATURATE_INTR [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	RW1C							
HW Access	RW1S							
Name	SATURATE_INTR [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
15 : 0	SATURATE_INTR	Saturate Interrupt: hardware sets this interrupt for each channel if a conversion result (before averaging) of that channel is either 0x000 or 0xFFFF (for 12-bit resolution), this is an indication that the ADC likely saturated. Write with '1' to clear bit. Default Value: 0

17.1.21 SAR0_SATURATE_INTR_SET

Saturate interrupt set request register

Address: 0x403A0224

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW1S							
HW Access	A							
Name	SATURATE_SET [7:0]							
Bits	15	14	13	12	11	10	9	8
SW Access	RW1S							
HW Access	A							
Name	SATURATE_SET [15:8]							
Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							
Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
15 : 0	SATURATE_SET	Write with '1' to set corresponding bit in interrupt request register. Default Value: 0

17.1.22 SAR0_SATURATE_INTR_MASK

Saturate interrupt mask register.

Address: 0x403A0228

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access	R							
Name	SATURATE_MASK [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	RW							
HW Access	R							
Name	SATURATE_MASK [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
15 : 0	SATURATE_MASK	Mask bit for corresponding bit in interrupt request register. Default Value: 0

17.1.23 SAR0_SATURATE_INTR_MASKED

Saturate interrupt masked request register

Address: 0x403A022C

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access	W							
Name	SATURATE_MASKED [7:0]							
Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access	W							
Name	SATURATE_MASKED [15:8]							
Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							
Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
15 : 0	SATURATE_MASKED	Logical and of corresponding request and mask bits. Default Value: 0

17.1.24 SAR0_RANGE_INTR

Range detect interrupt request register.

Address: 0x403A0230

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW1C							
HW Access	RW1S							
Name	RANGE_INTR [7:0]							
Bits	15	14	13	12	11	10	9	8
SW Access	RW1C							
HW Access	RW1S							
Name	RANGE_INTR [15:8]							
Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							
Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
15 : 0	RANGE_INTR	Range detect Interrupt: hardware sets this interrupt for each channel if the conversion result (after averaging) of that channel met the condition specified by the SAR_RANGE registers. Write with '1' to clear bit. Default Value: 0

17.1.25 SAR0_RANGE_INTR_SET

Range detect interrupt set request register

Address: 0x403A0234

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW1S							
HW Access	A							
Name	RANGE_SET [7:0]							
Bits	15	14	13	12	11	10	9	8
SW Access	RW1S							
HW Access	A							
Name	RANGE_SET [15:8]							
Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							
Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
15 : 0	RANGE_SET	Write with '1' to set corresponding bit in interrupt request register. Default Value: 0

17.1.26 SAR0_RANGE_INTR_MASK

Range detect interrupt mask register.

Address: 0x403A0238

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access	R							
Name	RANGE_MASK [7:0]							
Bits	15	14	13	12	11	10	9	8
SW Access	RW							
HW Access	R							
Name	RANGE_MASK [15:8]							
Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							
Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
15 : 0	RANGE_MASK	Mask bit for corresponding bit in interrupt request register. Default Value: 0

17.1.27 SAR0_RANGE_INTR_MASKED

Range interrupt masked request register

Address: 0x403A023C

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access	W							
Name	RANGE_MASKED [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access	W							
Name	RANGE_MASKED [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
15 : 0	RANGE_MASKED	Logical and of corresponding request and mask bits. Default Value: 0

17.1.28 SAR0_INTR_CAUSE

Interrupt cause register

Address: 0x403A0240

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R	R	R	R	R	R	R	R
HW Access	W	W	W	W	W	W	W	W
Name	INJ_COLLISION_MASKED_MIR	INJ_RANGE_MASKED_MIR	INJ_SATURATION_MASKED_MIR	INJ_EOC_MASKED_MIR	DSI_COLLISION_MASKED_MIR	FW_COLLISION_MASKED_MIR	OVERFLOW_MASKED_MIR	EO-S_MASKED_MIR

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R	R	None					
HW Access	W	W	None					
Name	RANGE_MASKED_RED	SATURATION_MASKED_RED	None [29:24]					

Bits	Name	Description
31	RANGE_MASKED_RED	Reduction OR of all SAR_RANGE_INTR_MASKED bits Default Value: 0
30	SATURATION_MASKED_RED	Reduction OR of all SAR_SATURATION_INTR_MASKED bits Default Value: 0
7	INJ_COLLISION_MASKED_MIR	Mirror copy of corresponding bit in SAR_INTR_MASKED Default Value: 0
6	INJ_RANGE_MASKED_MIR	Mirror copy of corresponding bit in SAR_INTR_MASKED Default Value: 0
5	INJ_SATURATION_MASKED_MIR	Mirror copy of corresponding bit in SAR_INTR_MASKED Default Value: 0
4	INJ_EOC_MASKED_MIR	Mirror copy of corresponding bit in SAR_INTR_MASKED Default Value: 0

17.1.28 SAR0_INTR_CAUSE (continued)

3	DSI_COLLI- SION_MASKED_MIR	Mirror copy of corresponding bit in SAR_INTR_MASKED Default Value: 0
2	FW_COLLI- SION_MASKED_MIR	Mirror copy of corresponding bit in SAR_INTR_MASKED Default Value: 0
1	OVER- FLOW_MASKED_MIR	Mirror copy of corresponding bit in SAR_INTR_MASKED Default Value: 0
0	EOS_MASKED_MIR	Mirror copy of corresponding bit in SAR_INTR_MASKED Default Value: 0

17.1.29 SAR0_INJ_CHAN_CONFIG

Injection channel configuration register.

Address: 0x403A0280

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None	RW			None	RW		
HW Access	None	R			None	R		
Name	None	INJ_PORT_ADDR [6:4]			None	INJ_PIN_ADDR [2:0]		

Bits	15	14	13	12	11	10	9	8
SW Access	None		RW		None	RW	RW	RW
HW Access	None		R		None	R	R	R
Name	None [15:14]		INJ_SAMPLE_TIME_SEL [13:12]		None	IN-J_AVG_EN	INJ_RESOLUTION	INJ_DIFFERENTIAL_EN

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	RW1S	RW	None					
HW Access	RW1C	R	None					
Name	IN-J_START_EN	INJ_TAILGATING	None [29:24]					

Bits	Name	Description
31	INJ_START_EN	Set by firmware to enable the injection channel. If INJ_TAILGATING is not set this bit also functions as trigger for this channel. Cleared by hardware after this channel has been sampled (i.e. this channel is always one shot even if CONTINUOUS is set). Also cleared if the SAR is disabled. Default Value: 0
30	INJ_TAILGATING	Injection channel tailgating. - 0: no tailgating for this channel, SAR is immediately triggered when the INJ_START_EN bit is set if the SAR is not busy. If the SAR is busy, the INJ channel addressed pin is sampled at the end of the current scan. - 1: injection channel tailgating. The addressed pin is sampled after the next trigger and after all enabled channels have been scanned. Default Value: 0
13 : 12	INJ_SAMPLE_TIME_SEL	Injection sample time select: select which of the 4 global sample times to use for this channel Default Value: 0

17.1.29 SAR0_INJ_CHAN_CONFIG (continued)

10	INJ_AVG_EN	Averaging enable for this channel. If set the AVG_CNT and AVG_SHIFT settings are used for sampling the addressed pin(s) Default Value: 0
9	INJ_RESOLUTION	Resolution for this channel. Default Value: 0 0x0: 12B : 12-bit resolution is used for this channel. 0x1: SUBRES : The resolution specified by SUB_RESOLUTION in the SAR_SAMPLE_CTRL register is used for this channel.
8	INJ_DIFFERENTIAL_EN	Differential enable for this channel. - 0: The voltage on the addressed pin is measured (Single-ended) and the resulting value is stored in the corresponding data register. - 1: The differential voltage on the addressed pin pair is measured and the resulting value is stored in the corresponding data register. (INJ_PIN_ADDR[0] is ignored). Default Value: 0
6 : 4	INJ_PORT_ADDR	Address of the port that contains the pin to be sampled by this channel. Default Value: 0 0x0: SARMUX : SARMUX pins. 0x1: CTB0 : CTB0 0x2: CTB1 : CTB1 0x3: CTB2 : Reserved 0x4: CTB3 : Reserved 0x6: AROUTE_VIRT : Reserved 0x7: SARMUX_VIRT : SARMUX virtual port

17.1.29 SAR0_INJ_CHAN_CONFIG (continued)

2 : 0	INJ_PIN_ADDR	Address of the pin to be sampled by this injection channel. If differential is enabled then PIN_ADDR[0] is ignored and considered to be 0, i.e. PIN_ADDR points to the even pin of a pin pair. Default Value: 0
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17.1.30 SAR0_INJ_RESULT

Injection channel result register

Address: 0x403A0290

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access	W							
Name	INJ_RESULT [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access	W							
Name	INJ_RESULT [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R	R	R	R	None			
HW Access	W	W	W	W	None			
Name	IN- J_EOC_IN- TR_MIR	IN- J_RANGE_I NTR_MIR	INJ_SATU- RATE_IN- TR_MIR	INJ_COLLI- SION_IN- TR_MIR	None [27:24]			

Bits	Name	Description
31	INJ_EOC_INTR_MIR	mirror bit of corresponding bit in SAR_INTR register Default Value: 0
30	INJ_RANGE_INTR_MIR	mirror bit of corresponding bit in SAR_INTR register Default Value: 0
29	INJ_SATURATE_INTR_MIR	mirror bit of corresponding bit in SAR_INTR register Default Value: 0
28	INJ_COLLISION_INTR_MIR	mirror bit of corresponding bit in SAR_INTR register Default Value: 0
15 : 0	INJ_RESULT	SAR conversion result of the channel. Default Value: Undefined

17.1.31 SAR0_MUX_SWITCH0

SARMUX Firmware switch controls

Address: 0x403A0300

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW1S	RW1S	RW1S	RW1S	RW1S	RW1S	RW1S	RW1S
HW Access	RW1C	RW1C	RW1C	RW1C	RW1C	RW1C	RW1C	RW1C
Name	MUX_F-W_P7_VPLUS	MUX_F-W_P6_VPLUS	MUX_F-W_P5_VPLUS	MUX_F-W_P4_VPLUS	MUX_F-W_P3_VPLUS	MUX_F-W_P2_VPLUS	MUX_F-W_P1_VPLUS	MUX_F-W_P0_VPLUS

Bits	15	14	13	12	11	10	9	8
SW Access	RW1S	RW1S	RW1S	RW1S	RW1S	RW1S	RW1S	RW1S
HW Access	RW1C	RW1C	RW1C	RW1C	RW1C	RW1C	RW1C	RW1C
Name	MUX_F-W_P7_VMI-NUS	MUX_F-W_P6_VMI-NUS	MUX_F-W_P5_VMI-NUS	MUX_F-W_P4_VMI-NUS	MUX_F-W_P3_VMI-NUS	MUX_F-W_P2_VMI-NUS	MUX_F-W_P1_VMI-NUS	MUX_F-W_P0_VMI-NUS

Bits	23	22	21	20	19	18	17	16
SW Access	RW1S	RW1S	RW1S	RW1S	RW1S	RW1S	RW1S	RW1S
HW Access	RW1C	RW1C	RW1C	RW1C	RW1C	RW1C	RW1C	RW1C
Name	MUX_F-W_SAR-BUS1_VPLUS	MUX_F-W_SAR-BUS0_VPLUS	MUX_F-W_AMUX-BUSB_VMINUS	MUX_F-W_AMUX-BUSA_VMINUS	MUX_F-W_AMUX-BUSB_VPLUS	MUX_F-W_AMUX-BUSA_VPLUS	MUX_FW-TEMP_VPLUS	MUX_F-W_VS-SA_VMINUS

Bits	31	30	29	28	27	26	25	24
SW Access	None		RW1S	RW1S	RW1S	RW1S	RW1S	RW1S
HW Access	None		RW1C	RW1C	RW1C	RW1C	RW1C	RW1C
Name	None [31:30]		MUX_F-W_P7_COR EIO3	MUX_F-W_P6_COR EIO2	MUX_F-W_P5_COR EIO1	MUX_F-W_P4_COR EIO0	MUX_F-W_SAR-BUS1_VMINUS	MUX_F-W_SAR-BUS0_VMINUS

Bits	Name	Description
29	MUX_FW_P7_COREIO3	Reserved
28	MUX_FW_P6_COREIO2	Reserved
27	MUX_FW_P5_COREIO1	Reserved
26	MUX_FW_P4_COREIO0	Reserved

17.1.31 SAR0_MUX_SWITCH0 (continued)

25	MUX_FW_SAR-BUS1_VMINUS	Firmware control: 0=open, 1=close switch between sarbus1 and vminus signal. Write with '1' to set bit. Default Value: 0
24	MUX_FW_SAR-BUS0_VMINUS	Firmware control: 0=open, 1=close switch between sarbus0 and vminus signal. Write with '1' to set bit. Default Value: 0
23	MUX_FW_SAR-BUS1_VPLUS	Firmware control: 0=open, 1=close switch between sarbus1 and vplus signal. Write with '1' to set bit. Default Value: 0
22	MUX_FW_SAR-BUS0_VPLUS	Firmware control: 0=open, 1=close switch between sarbus0 and vplus signal. Write with '1' to set bit. Default Value: 0
21	MUX_FW_AMUXBUS-B_VMINUS	Firmware control: 0=open, 1=close switch between amuxbusb and vminus signal. Write with '1' to set bit. Default Value: 0
20	MUX_FW_AMUX-BUSA_VMINUS	Firmware control: 0=open, 1=close switch between amuxbusa and vminus signal. Write with '1' to set bit. Default Value: 0
19	MUX_FW_AMUXBUS-B_VPLUS	Firmware control: 0=open, 1=close switch between amuxbusb and vplus signal. Write with '1' to set bit. Default Value: 0
18	MUX_FW_AMUX-BUSA_VPLUS	Firmware control: 0=open, 1=close switch between amuxbusa and vplus signal. Write with '1' to set bit. Default Value: 0
17	MUX_FW_TEMP_VPLUS	Firmware control: 0=open, 1=close switch between temperature sensor and vplus signal, also powers on the temperature sensor. Write with '1' to set bit. Default Value: 0
16	MUX_FW_VSSA_VMINUS	Firmware control: 0=open, 1=close switch between vssa_kelvin and vminus signal. Write with '1' to set bit. Default Value: 0
15	MUX_FW_P7_VMINUS	Firmware control: 0=open, 1=close switch between pin P7 and vminus signal. Write with '1' to set bit. Default Value: 0
14	MUX_FW_P6_VMINUS	Firmware control: 0=open, 1=close switch between pin P6 and vminus signal. Write with '1' to set bit. Default Value: 0
13	MUX_FW_P5_VMINUS	Firmware control: 0=open, 1=close switch between pin P5 and vminus signal. Write with '1' to set bit. Default Value: 0
12	MUX_FW_P4_VMINUS	Firmware control: 0=open, 1=close switch between pin P4 and vminus signal. Write with '1' to set bit. Default Value: 0
11	MUX_FW_P3_VMINUS	Firmware control: 0=open, 1=close switch between pin P3 and vminus signal. Write with '1' to set bit. Default Value: 0
10	MUX_FW_P2_VMINUS	Firmware control: 0=open, 1=close switch between pin P2 and vminus signal. Write with '1' to set bit. Default Value: 0

17.1.31 SAR0_MUX_SWITCH0 (continued)

9	MUX_FW_P1_VMINUS	Firmware control: 0=open, 1=close switch between pin P1 and vminus signal. Write with '1' to set bit. Default Value: 0
8	MUX_FW_P0_VMINUS	Firmware control: 0=open, 1=close switch between pin P0 and vminus signal. Write with '1' to set bit. Default Value: 0
7	MUX_FW_P7_VPLUS	Firmware control: 0=open, 1=close switch between pin P7 and vplus signal. Write with '1' to set bit. Default Value: 0
6	MUX_FW_P6_VPLUS	Firmware control: 0=open, 1=close switch between pin P6 and vplus signal. Write with '1' to set bit. Default Value: 0
5	MUX_FW_P5_VPLUS	Firmware control: 0=open, 1=close switch between pin P5 and vplus signal. Write with '1' to set bit. Default Value: 0
4	MUX_FW_P4_VPLUS	Firmware control: 0=open, 1=close switch between pin P4 and vplus signal. Write with '1' to set bit. Default Value: 0
3	MUX_FW_P3_VPLUS	Firmware control: 0=open, 1=close switch between pin P3 and vplus signal. Write with '1' to set bit. Default Value: 0
2	MUX_FW_P2_VPLUS	Firmware control: 0=open, 1=close switch between pin P2 and vplus signal. Write with '1' to set bit. Default Value: 0
1	MUX_FW_P1_VPLUS	Firmware control: 0=open, 1=close switch between pin P1 and vplus signal. Write with '1' to set bit. Default Value: 0
0	MUX_FW_P0_VPLUS	Firmware control: 0=open, 1=close switch between pin P0 and vplus signal. Write with '1' to set bit. Default Value: 0

17.1.32 SAR0_MUX_SWITCH_CLEAR0

SARMUX Firmware switch control clear

Address: 0x403A0304

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW1C	RW1C	RW1C	RW1C	RW1C	RW1C	RW1C	RW1C
HW Access	A	A	A	A	A	A	A	A
Name	MUX_F-W_P7_VPLUS	MUX_F-W_P6_VPLUS	MUX_F-W_P5_VPLUS	MUX_F-W_P4_VPLUS	MUX_F-W_P3_VPLUS	MUX_F-W_P2_VPLUS	MUX_F-W_P1_VPLUS	MUX_F-W_P0_VPLUS

Bits	15	14	13	12	11	10	9	8
SW Access	RW1C	RW1C	RW1C	RW1C	RW1C	RW1C	RW1C	RW1C
HW Access	A	A	A	A	A	A	A	A
Name	MUX_F-W_P7_VMI-NUS	MUX_F-W_P6_VMI-NUS	MUX_F-W_P5_VMI-NUS	MUX_F-W_P4_VMI-NUS	MUX_F-W_P3_VMI-NUS	MUX_F-W_P2_VMI-NUS	MUX_F-W_P1_VMI-NUS	MUX_F-W_P0_VMI-NUS

Bits	23	22	21	20	19	18	17	16
SW Access	RW1C	RW1C	RW1C	RW1C	RW1C	RW1C	RW1C	RW1C
HW Access	A	A	A	A	A	A	A	A
Name	MUX_F-W_SAR-BUS1_VPLUS	MUX_F-W_SAR-BUS0_VPLUS	MUX_F-W_AMUX-BUSB_VMINUS	MUX_F-W_AMUX-BUSA_VMINUS	MUX_F-W_AMUX-BUSB_VPLUS	MUX_F-W_AMUX-BUSA_VPLUS	MUX_FW-TEMP_VPLUS	MUX_F-W_VS-SA_VMINUS

Bits	31	30	29	28	27	26	25	24
SW Access	None		RW1C	RW1C	RW1C	RW1C	RW1C	RW1C
HW Access	None		A	A	A	A	A	A
Name	None [31:30]		MUX_F-W_P7_COR EIO3	MUX_F-W_P6_COR EIO2	MUX_F-W_P5_COR EIO1	MUX_F-W_P4_COR EIO0	MUX_F-W_SAR-BUS1_VMINUS	MUX_F-W_SAR-BUS0_VMINUS

Bits	Name	Description
29	MUX_FW_P7_COREIO3	Reserved
28	MUX_FW_P6_COREIO2	Reserved
27	MUX_FW_P5_COREIO1	Reserved
26	MUX_FW_P4_COREIO0	Reserved

17.1.32 SAR0_MUX_SWITCH_CLEAR0 (continued)

25	MUX_FW_SAR-BUS1_VMINUS	Write '1' to clear corresponding bit in MUX_SWITCH0 Default Value: 0
24	MUX_FW_SAR-BUS0_VMINUS	Write '1' to clear corresponding bit in MUX_SWITCH0 Default Value: 0
23	MUX_FW_SAR-BUS1_VPLUS	Write '1' to clear corresponding bit in MUX_SWITCH0 Default Value: 0
22	MUX_FW_SAR-BUS0_VPLUS	Write '1' to clear corresponding bit in MUX_SWITCH0 Default Value: 0
21	MUX_FW_AMUXBUS-B_VMINUS	Write '1' to clear corresponding bit in MUX_SWITCH0 Default Value: 0
20	MUX_FW_AMUX-BUSA_VMINUS	Write '1' to clear corresponding bit in MUX_SWITCH0 Default Value: 0
19	MUX_FW_AMUXBUS-B_VPLUS	Write '1' to clear corresponding bit in MUX_SWITCH0 Default Value: 0
18	MUX_FW_AMUX-BUSA_VPLUS	Write '1' to clear corresponding bit in MUX_SWITCH0 Default Value: 0
17	MUX_FW_TEMP_VPLUS	Write '1' to clear corresponding bit in MUX_SWITCH0 Default Value: 0
16	MUX_FW_VSSA_VMINUS	Write '1' to clear corresponding bit in MUX_SWITCH0 Default Value: 0
15	MUX_FW_P7_VMINUS	Write '1' to clear corresponding bit in MUX_SWITCH0 Default Value: 0
14	MUX_FW_P6_VMINUS	Write '1' to clear corresponding bit in MUX_SWITCH0 Default Value: 0
13	MUX_FW_P5_VMINUS	Write '1' to clear corresponding bit in MUX_SWITCH0 Default Value: 0
12	MUX_FW_P4_VMINUS	Write '1' to clear corresponding bit in MUX_SWITCH0 Default Value: 0
11	MUX_FW_P3_VMINUS	Write '1' to clear corresponding bit in MUX_SWITCH0 Default Value: 0
10	MUX_FW_P2_VMINUS	Write '1' to clear corresponding bit in MUX_SWITCH0 Default Value: 0
9	MUX_FW_P1_VMINUS	Write '1' to clear corresponding bit in MUX_SWITCH0 Default Value: 0
8	MUX_FW_P0_VMINUS	Write '1' to clear corresponding bit in MUX_SWITCH0 Default Value: 0
7	MUX_FW_P7_VPLUS	Write '1' to clear corresponding bit in MUX_SWITCH0 Default Value: 0
6	MUX_FW_P6_VPLUS	Write '1' to clear corresponding bit in MUX_SWITCH0 Default Value: 0
5	MUX_FW_P5_VPLUS	Write '1' to clear corresponding bit in MUX_SWITCH0 Default Value: 0
4	MUX_FW_P4_VPLUS	Write '1' to clear corresponding bit in MUX_SWITCH0 Default Value: 0

17.1.32 SAR0_MUX_SWITCH_CLEAR0 (continued)

3	MUX_FW_P3_VPLUS	Write '1' to clear corresponding bit in MUX_SWITCH0 Default Value: 0
2	MUX_FW_P2_VPLUS	Write '1' to clear corresponding bit in MUX_SWITCH0 Default Value: 0
1	MUX_FW_P1_VPLUS	Write '1' to clear corresponding bit in MUX_SWITCH0 Default Value: 0
0	MUX_FW_P0_VPLUS	Write '1' to clear corresponding bit in MUX_SWITCH0 Default Value: 0

17.1.33 SAR0_MUX_SWITCH_HW_CTRL

SARMUX switch hardware control

Address: 0x403A0340

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW	RW	RW	RW	RW	RW	RW	RW
HW Access	R	R	R	R	R	R	R	R
Name	MUX- _HW_C- TRL_P7	MUX- _HW_C- TRL_P6	MUX- _HW_C- TRL_P5	MUX- _HW_C- TRL_P4	MUX- _HW_C- TRL_P3	MUX- _HW_C- TRL_P2	MUX- _HW_C- TRL_P1	MUX- _HW_C- TRL_P0

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	RW	RW	None		RW	RW	RW	RW
HW Access	R	R	None		R	R	R	R
Name	MUX- _HW_C- TRL_SARB US1	MUX- _HW_C- TRL_SARB US0	None [21:20]		MUX- _HW_C- TRL_AMUX BUSB	MUX- _HW_C- TRL_AMUX BUSA	MUX- _HW_C- TRL_TEMP	MUX- _HW_C- TRL_VSSA

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
23	MUX_HW_CTRL_SARBUS1	Hardware control: 0=only firmware control, 1=hardware control masked by firmware setting for sarbus1 switches. Default Value: 0
22	MUX_HW_CTRL_SARBUS0	Hardware control: 0=only firmware control, 1=hardware control masked by firmware setting for sarbus0 switches. Default Value: 0
19	MUX_HW_CTRL_AMUXBUSB	Hardware control: 0=only firmware control, 1=hardware control masked by firmware setting for amuxbusb switches. Default Value: 0
18	MUX_HW_CTRL_AMUXBUSA	Hardware control: 0=only firmware control, 1=hardware control masked by firmware setting for amuxbusa switches. Default Value: 0

17.1.33 SAR0_MUX_SWITCH_HW_CTRL (continued)

17	MUX_HW_CTRL_TEMP	Hardware control: 0=only firmware control, 1=hardware control masked by firmware setting for temp switch. Default Value: 0
16	MUX_HW_CTRL_VSSA	Hardware control: 0=only firmware control, 1=hardware control masked by firmware setting for vssa switch. Default Value: 0
7	MUX_HW_CTRL_P7	Hardware control: 0=only firmware control, 1=hardware control masked by firmware setting for pin P7 switches. Default Value: 0
6	MUX_HW_CTRL_P6	Hardware control: 0=only firmware control, 1=hardware control masked by firmware setting for pin P6 switches. Default Value: 0
5	MUX_HW_CTRL_P5	Hardware control: 0=only firmware control, 1=hardware control masked by firmware setting for pin P5 switches. Default Value: 0
4	MUX_HW_CTRL_P4	Hardware control: 0=only firmware control, 1=hardware control masked by firmware setting for pin P4 switches. Default Value: 0
3	MUX_HW_CTRL_P3	Hardware control: 0=only firmware control, 1=hardware control masked by firmware setting for pin P3 switches. Default Value: 0
2	MUX_HW_CTRL_P2	Hardware control: 0=only firmware control, 1=hardware control masked by firmware setting for pin P2 switches. Default Value: 0
1	MUX_HW_CTRL_P1	Hardware control: 0=only firmware control, 1=hardware control masked by firmware setting for pin P1 switches. Default Value: 0
0	MUX_HW_CTRL_P0	Hardware control: 0=only firmware control, 1=hardware control masked by firmware setting for pin P0 switches. Default Value: 0

17.1.34 SAR0_MUX_SWITCH_STATUS

SARMUX switch status

Address: 0x403A0348

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R	R	R	R	R	R	R	R
HW Access	W	W	W	W	W	W	W	W
Name	MUX_F-W_P7_VPLUS	MUX_F-W_P6_VPLUS	MUX_F-W_P5_VPLUS	MUX_F-W_P4_VPLUS	MUX_F-W_P3_VPLUS	MUX_F-W_P2_VPLUS	MUX_F-W_P1_VPLUS	MUX_F-W_P0_VPLUS

Bits	15	14	13	12	11	10	9	8
SW Access	R	R	R	R	R	R	R	R
HW Access	W	W	W	W	W	W	W	W
Name	MUX_F-W_P7_VMINUS	MUX_F-W_P6_VMINUS	MUX_F-W_P5_VMINUS	MUX_F-W_P4_VMINUS	MUX_F-W_P3_VMINUS	MUX_F-W_P2_VMINUS	MUX_F-W_P1_VMINUS	MUX_F-W_P0_VMINUS

Bits	23	22	21	20	19	18	17	16
SW Access	R	R	R	R	R	R	R	R
HW Access	W	W	W	W	W	W	W	W
Name	MUX_F-W_SAR-BUS1_VPLUS	MUX_F-W_SAR-BUS0_VPLUS	MUX_F-W_AMUX-BUSB_VMINUS	MUX_F-W_AMUX-BUSA_VMINUS	MUX_F-W_AMUX-BUSB_VPLUS	MUX_F-W_AMUX-BUSA_VPLUS	MUX_FW-TEMP_VPLUS	MUX_F-W_VS-SA_VMINUS

Bits	31	30	29	28	27	26	25	24
SW Access	None						R	R
HW Access	None						W	W
Name	None [31:26]						MUX_F-W_SAR-BUS1_VMINUS	MUX_F-W_SAR-BUS0_VMINUS

Bits	Name	Description
25	MUX_FW_SAR-BUS1_VMINUS	switch status of corresponding bit in MUX_SWITCH0 Default Value: 0
24	MUX_FW_SAR-BUS0_VMINUS	switch status of corresponding bit in MUX_SWITCH0 Default Value: 0
23	MUX_FW_SAR-BUS1_VPLUS	switch status of corresponding bit in MUX_SWITCH0 Default Value: 0
22	MUX_FW_SAR-BUS0_VPLUS	switch status of corresponding bit in MUX_SWITCH0 Default Value: 0

17.1.34 SAR0_MUX_SWITCH_STATUS (continued)

21	MUX_FW_AMUXBUS- B_VMINUS	switch status of corresponding bit in MUX_SWITCH0 Default Value: 0
20	MUX_FW_AMUX- BUSA_VMINUS	switch status of corresponding bit in MUX_SWITCH0 Default Value: 0
19	MUX_FW_AMUXBUS- B_VPLUS	switch status of corresponding bit in MUX_SWITCH0 Default Value: 0
18	MUX_FW_AMUX- BUSA_VPLUS	switch status of corresponding bit in MUX_SWITCH0 Default Value: 0
17	MUX_FW_TEMP_VPLUS	switch status of corresponding bit in MUX_SWITCH0 Default Value: 0
16	MUX_FW_VSSA_VMI- NUS	switch status of corresponding bit in MUX_SWITCH0 Default Value: 0
15	MUX_FW_P7_VMINUS	switch status of corresponding bit in MUX_SWITCH0 Default Value: 0
14	MUX_FW_P6_VMINUS	switch status of corresponding bit in MUX_SWITCH0 Default Value: 0
13	MUX_FW_P5_VMINUS	switch status of corresponding bit in MUX_SWITCH0 Default Value: 0
12	MUX_FW_P4_VMINUS	switch status of corresponding bit in MUX_SWITCH0 Default Value: 0
11	MUX_FW_P3_VMINUS	switch status of corresponding bit in MUX_SWITCH0 Default Value: 0
10	MUX_FW_P2_VMINUS	switch status of corresponding bit in MUX_SWITCH0 Default Value: 0
9	MUX_FW_P1_VMINUS	switch status of corresponding bit in MUX_SWITCH0 Default Value: 0
8	MUX_FW_P0_VMINUS	switch status of corresponding bit in MUX_SWITCH0 Default Value: 0
7	MUX_FW_P7_VPLUS	switch status of corresponding bit in MUX_SWITCH0 Default Value: 0
6	MUX_FW_P6_VPLUS	switch status of corresponding bit in MUX_SWITCH0 Default Value: 0
5	MUX_FW_P5_VPLUS	switch status of corresponding bit in MUX_SWITCH0 Default Value: 0
4	MUX_FW_P4_VPLUS	switch status of corresponding bit in MUX_SWITCH0 Default Value: 0
3	MUX_FW_P3_VPLUS	switch status of corresponding bit in MUX_SWITCH0 Default Value: 0
2	MUX_FW_P2_VPLUS	switch status of corresponding bit in MUX_SWITCH0 Default Value: 0
1	MUX_FW_P1_VPLUS	switch status of corresponding bit in MUX_SWITCH0 Default Value: 0
0	MUX_FW_P0_VPLUS	switch status of corresponding bit in MUX_SWITCH0 Default Value: 0

17.1.35 SAR0_PUMP_CTRL

Switch pump control

Address: 0x403A0380

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None							RW
HW Access	None							R
Name	None [7:1]							CLOCK_-SEL

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	RW	None						
HW Access	R	None						
Name	ENABLED	None [30:24]						

Bits	Name	Description
31	ENABLED	0=disabled: pump output is VDDA_PUMP, 1=enabled: pump output is boosted. Default Value: 0
0	CLOCK_SEL	Clock select: 0=external clock, 1=internal clock (deprecated). Default Value: 0

18 PASS MMIO Registers



This section discusses the PASS MMIO registers. It lists all the registers in mapping tables, in address order.

18.1 Register Details

Register	Address	Description
PASS0_INTR_CAUSE	0x403F0000	Interrupt cause register
PASS0_DFT_CTRL	0x403F0030	DFT control register
PASS0_PASS_CTRL	0x403F0108	PASS Control
PASS0_DSAB_DSAB_CTRL	0x403F0E00	global DSAB control
PASS0_DSAB_DSAB_DFT	0x403F0E04	DFT bits
PASS0_DSAB_TRIM	0x403F0F00	DSAB Trim bits

18.1.1 PASS0_INTR_CAUSE

Interrupt cause register

Address: 0x403F0000

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None							R
HW Access	None							W
Name	None [7:1]							CTB0_INT

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
0	CTB0_INT	CTB0 interrupt pending Default Value: 0

18.1.2 PASS0_DFT_CTRL

DFT control register

Address: 0x403F0030

Retention: Not Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None							RW
HW Access	None							R
Name	None [7:1]							DSAB_AD- FT_RES_E N

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
0	DSAB_ADFT_RES_EN	Close the switch to connect the DSAB ADFT resistor to the AMUXBUS Default Value: 0

18.1.3 PASS0_PASS_CTRL

PASS Control

Address: 0x403F0108

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None						RW	RW
HW Access	None						R	R
Name	None [7:2]						PMP-CLK_SRC	PMP-CLK_BYP

Bits	15	14	13	12	11	10	9	8
SW Access	RW							
HW Access	R							
Name	RMB_BITS [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
15 : 8	RMB_BITS	Risk mitigation bits Default Value: 0
1	PMPCLK_SRC	- 0: Pump clk is clk_hf - 1: Pump clk is direct from SRSS Default Value: 0
0	PMPCLK_BYP	- 0: Pump clk is clk_hf/2 - 1: Pump clk is selected from PMPCLK_SRC Default Value: 0

18.1.4 PASS0_DSAB_DSAB_CTRL

global DSAB control

Address: 0x403F0E00

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None		RW					
HW Access	None		R					
Name	None [7:6]		CURRENT_SEL [5:0]					
Bits	15	14	13	12	11	10	9	8
SW Access	None				RW			
HW Access	None				R			
Name	None [15:12]				SEL_OUT [11:8]			
Bits	23	22	21	20	19	18	17	16
SW Access	None				RW			
HW Access	None				R			
Name	None [23:20]				REF_SWAP_EN [19:16]			
Bits	31	30	29	28	27	26	25	24
SW Access	RW	None		RW	None			RW
HW Access	R	None		R	None			R
Name	ENABLED	None [30:29]		START-UP_RM	None [27:25]			BYPASS_MODE_EN

Bits	Name	Description
31	ENABLED	This field (along with SEL_OUT and REF_SWAP_EN) provides bitwise selection of the current sources that drive the DSAB ZTC and PTAT outputs. See SEL_OUT field for truth tables. In SRSSLT devices, in active mode, this bit is overridden to '1', that is - it is always enabled in active mode. Default Value: 0
28	STARTUP_RM	Risk mitigation control 1 - Force start the startup circuit Default Value: 0
24	BYPASS_MODE_EN	0 - DSAB PTAT generator is powered from DSAB regulator: VDDA must be at least 2.4V 1 - DSAB PTAT generator is powered directly from VDDA: VDDA cannot exceed 4.0V Default Value: 0
19 : 16	REF_SWAP_EN	This field (along with SEL_OUT and ENABLED) provides bitwise selection of the current sources that drive the DSAB ZTC and PTAT outputs. See SEL_OUT field for truth tables. Default Value: 0

18.1.4 PASS0_DSAB_DSAB_CTRL (continued)

11 : 8 SEL_OUT

This field (along with REF_SWAP_EN and ENABLED) provides bitwise selection of the current sources that drive the DSAB ZTC and PTAT outputs. The available current sources are a function of the type of SRSS. The SRSS current sources are disabled in chip DeepSleep mode which is indicated by () entries in the truth table. CTB(m) factory trim has SRSS-ZTC and either SRSS-PTAT (SRSSv2) or DSAB_PTAT (SRSS-LITE) enabled. If different settings are used, then a periodic re-trim of CTB(m) offset should be performed.

Truth Table with SRSSv2 (backwards compatible with PSoC4A family)

ENABLED	SEL_OUT	REF_SWAP_EN	"Chip Power Mode"	ZTC Output	PTAT Output
0	X	X	Active	SRSS-ZTC	SRSS-PTAT
0	X	X	DeepSleep	(SRSS-ZTC)	(SRSS-PTAT)
1	0	0	Active	SRSS-ZTC	SRSS-PTAT
1	0	0	DeepSleep	(SRSS-ZTC)	(SRSS-PTAT)
1	0	1	Active	SRSS-PTAT	SRSS-ZTC
1	0	1	DeepSleep	(SRSS-PTAT)	(SRSS-ZTC)
1	1	0	Active	DSAB-PTAT	SRSS-PTAT
1	1	0	DeepSleep	DSAB-PTAT	(SRSS-PTAT)
1	1	1	Active	SRSS-PTAT	DSAB-PTAT
1	1	1	DeepSleep	(SRSS-PTAT)	DSAB-PTAT

Truth Table with SRSS-LITE

ENABLED	SEL_OUT	REF_SWAP_EN	"Chip Power Mode"	ZTC Output	PTAT Output
0	X	X	Active	SRSS-ZTC	DSAB-PTAT
0	X	X	DeepSleep	(SRSS-ZTC)	--
1	0	0	Active	SRSS-ZTC	DSAB-PTAT
1	0	0	DeepSleep	(SRSS-ZTC)	--
1	0	1	Active	DSAB-PTAT	SRSS-ZTC
1	0	1	DeepSleep	DSAB-PTAT	(SRSS-ZTC)
1	1	0	Active	DSAB-PTAT	--
1	1	0	DeepSleep	DSAB-PTAT	--
1	1	1	Active	--	DSAB-PTAT
1	1	1	DeepSleep	--	DSAB-PTAT

Default Value: 0

5 : 0 CURRENT_SEL

DSAB DAC control field

Nominal DSAB Output Current = CURRENT_SEL * 0.075 uA

In products with SRSS-LITE, this setting impacts the CTB(m) offset. A value of 0x20 is used during factory trim and is required to maintain low offsets across temperature variation. If a different setting is used then a periodic re-trim of CTB(m) offset should be performed.
 Default Value: 0

18.1.5 PASS0_DSAB_DSAB_DFT

DFT bits

Address: 0x403F0E04

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None				RW			
HW Access	None				R			
Name	None [7:4]				EN_DFT [3:0]			

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
3 : 0	EN_DFT	- 0: DSAB DFT disabled - 1: DSAB DFT enabled (connect output to amuxbus)
		0001 - PTAT<0> 0010 - PTAT<1> 0011 - PTAT<1:0> 0100 - PTAT<2> 0111 - PTAT<2:0> 1000 - PTAT<3> 1111 - PTAT<3:0> 1001 - DSAB Reg Out Default Value: 0

18.1.6 PASS0_DSAB_TRIM

DSAB Trim bits

Address: 0x403F0F00

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None		RW		RW			
HW Access	None		R		R			
Name	None [7:6]		DSAB_RMB_BITS [5:4]		IBIAS_TRIM [3:0]			

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
5 : 4	DSAB_RMB_BITS	Risk mitigation bits Default Value: 0
3 : 0	IBIAS_TRIM	1111=lowest, 0000=highest Default Value: 0

19 Cortex M0+ (CM0P) Registers



This section discusses the CM0P registers. It lists all the registers in mapping tables, in address order.

19.1 Register Details

Register	Address	Description
CM0P_DWT_PID4	0xE0001FD0	Watchpoint Unit CoreSight ROM Table Peripheral ID #4
CM0P_DWT_PID0	0xE0001FE0	Watchpoint Unit CoreSight ROM Table Peripheral ID #0
CM0P_DWT_PID1	0xE0001FE4	Watchpoint Unit CoreSight ROM Table Peripheral ID #1
CM0P_DWT_PID2	0xE0001FE8	Watchpoint Unit CoreSight ROM Table Peripheral ID #2
CM0P_DWT_PID3	0xE0001FEC	Watchpoint Unit CoreSight ROM Table Peripheral ID #3
CM0P_DWT_CID0	0xE0001FF0	Watchpoint Unit CoreSight ROM Table Component ID #0
CM0P_DWT_CID1	0xE0001FF4	Watchpoint Unit CoreSight ROM Table Component ID #1
CM0P_DWT_CID2	0xE0001FF8	Watchpoint Unit CoreSight ROM Table Component ID #2
CM0P_DWT_CID3	0xE0001FFC	Watchpoint Unit CoreSight ROM Table Component ID #3
CM0P_BP_PID4	0xE0002FD0	Breakpoint Unit CoreSight ROM Table Peripheral ID #4
CM0P_BP_PID0	0xE0002FE0	Breakpoint Unit CoreSight ROM Table Peripheral ID #0
CM0P_BP_PID1	0xE0002FE4	Breakpoint Unit CoreSight ROM Table Peripheral ID #1
CM0P_BP_PID2	0xE0002FE8	Breakpoint Unit CoreSight ROM Table Peripheral ID #2
CM0P_BP_PID3	0xE0002FEC	Breakpoint Unit CoreSight ROM Table Peripheral ID #3
CM0P_BP_CID0	0xE0002FF0	Breakpoint Unit CoreSight ROM Table Component ID #0
CM0P_BP_CID1	0xE0002FF4	Breakpoint Unit CoreSight ROM Table Component ID #1
CM0P_BP_CID2	0xE0002FF8	Breakpoint Unit CoreSight ROM Table Component ID #2
CM0P_BP_CID3	0xE0002FFC	Breakpoint Unit CoreSight ROM Table Component ID #3
CM0P_SYST_CSR	0xE000E010	SysTick Control & Status
CM0P_SYST_RVR	0xE000E014	SysTick Reload Value
CM0P_SYST_CVR	0xE000E018	SysTick Current Value
CM0P_SYST_CALIB	0xE000E01C	SysTick Calibration Value
CM0P_ISER	0xE000E100	Interrupt Set-Enable Register
CM0P_ICER	0xE000E180	Interrupt Clear Enable Register
CM0P_ISPR	0xE000E200	Interrupt Set-Pending Register
CM0P_ICPR	0xE000E280	Interrupt Clear-Pending Register
CM0P_IPR0	0xE000E400	Interrupt Priority Registers

Register	Address	Description
CM0P_IPR1	0xE000E404	Interrupt Priority Registers. See CM0P_IPR0 for the details of bit fields.
CM0P_IPR2	0xE000E408	Interrupt Priority Registers. See CM0P_IPR0 for the details of bit fields.
CM0P_IPR3	0xE000E40C	Interrupt Priority Registers. See CM0P_IPR0 for the details of bit fields.
CM0P_IPR4	0xE000E410	Interrupt Priority Registers. See CM0P_IPR0 for the details of bit fields.
CM0P_IPR5	0xE000E414	Interrupt Priority Registers. See CM0P_IPR0 for the details of bit fields.
CM0P_IPR6	0xE000E418	Interrupt Priority Registers. See CM0P_IPR0 for the details of bit fields.
CM0P_IPR7	0xE000E41C	Interrupt Priority Registers. See CM0P_IPR0 for the details of bit fields.
CM0P_CPUID	0xE000ED00	CPUID Register
CM0P_ICSR	0xE000ED04	Interrupt Control State Register
CM0P_VTOR	0xE000ED08	Vector Table Offset Register
CM0P_AIRCR	0xE000ED0C	Application Interrupt and Reset Control Register
CM0P_SCR	0xE000ED10	System Control Register
CM0P_CCR	0xE000ED14	Configuration and Control Register
CM0P_SHPR2	0xE000ED1C	System Handler Priority Register 2
CM0P_SHPR3	0xE000ED20	System Handler Priority Register 3
CM0P_SHCSR	0xE000ED24	System Handler Control and State Register
CM0P_SCS_PID4	0xE000EFD0	System Control Space ROM Table Peripheral ID #4
CM0P_SCS_PID0	0xE000EFE0	System Control Space ROM Table Peripheral ID #0
CM0P_SCS_PID1	0xE000EFE4	System Control Space ROM Table Peripheral ID #1
CM0P_SCS_PID2	0xE000EFE8	System Control Space ROM Table Peripheral ID #2
CM0P_SCS_PID3	0xE000EFEC	System Control Space ROM Table Peripheral ID #3
CM0P_SCS_CID0	0xE000EFF0	System Control Space ROM Table Component ID #0
CM0P_SCS_CID1	0xE000EFF4	System Control Space ROM Table Component ID #1
CM0P_SCS_CID2	0xE000EFF8	System Control Space ROM Table Component ID #2
CM0P_SCS_CID3	0xE000EFFC	System Control Space ROM Table Component ID #3
CM0P_ROM_SCS	0xE00FF000	CM0+ CoreSight ROM Table Peripheral #0
CM0P_ROM_DWT	0xE00FF004	CM0+ CoreSight ROM Table Peripheral #1
CM0P_ROM_BPU	0xE00FF008	CM0+ CoreSight ROM Table Peripheral #2
CM0P_ROM_END	0xE00FF00C	CM0+ CoreSight ROM Table End Marker
CM0P_ROM_CSMT	0xE00FF0CC	CM0+ CoreSight ROM Table Memory Type
CM0P_ROM_PID4	0xE00FFFD0	CM0+ CoreSight ROM Table Peripheral ID #4
CM0P_ROM_PID0	0xE00FFFE0	CM0+ CoreSight ROM Table Peripheral ID #0
CM0P_ROM_PID1	0xE00FFFE4	CM0+ CoreSight ROM Table Peripheral ID #1
CM0P_ROM_PID2	0xE00FFFE8	CM0+ CoreSight ROM Table Peripheral ID #2
CM0P_ROM_PID3	0xE00FF FEC	CM0+ CoreSight ROM Table Peripheral ID #3
CM0P_ROM_CID0	0xE00FFFF0	CM0+ CoreSight ROM Table Component ID #0
CM0P_ROM_CID1	0xE00FFFF4	CM0+ CoreSight ROM Table Component ID #1
CM0P_ROM_CID2	0xE00FFFF8	CM0+ CoreSight ROM Table Component ID #2
CM0P_ROM_CID3	0xE00FFFFC	CM0+ CoreSight ROM Table Component ID #3

19.1.1 CM0P_DWT_PID4

Watchpoint Unit CoreSight ROM Table Peripheral ID #4

Address: 0xE0001FD0

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access								
Name	VALUE [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access								
Name	VALUE [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access								
Name	VALUE [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access								
Name	VALUE [31:24]							

Bits	Name	Description
31 : 0	VALUE	Peripheral ID #4 Default Value: 4

19.1.2 CM0P_DWT_PID0

Watchpoint Unit CoreSight ROM Table Peripheral ID #0

Address: 0xE0001FE0

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access								
Name	VALUE [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access								
Name	VALUE [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access								
Name	VALUE [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access								
Name	VALUE [31:24]							

Bits	Name	Description
31 : 0	VALUE	Peripheral ID #0 Default Value: 10

19.1.3 CM0P_DWT_PID1

Watchpoint Unit CoreSight ROM Table Peripheral ID #1

Address: 0xE0001FE4

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access								
Name	VALUE [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access								
Name	VALUE [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access								
Name	VALUE [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access								
Name	VALUE [31:24]							

Bits	Name	Description
31 : 0	VALUE	Peripheral ID #1 Default Value: 176

19.1.4 CM0P_DWT_PID2

Watchpoint Unit CoreSight ROM Table Peripheral ID #2

Address: 0xE0001FE8

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access								
Name	VALUE [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access								
Name	VALUE [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access								
Name	VALUE [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access								
Name	VALUE [31:24]							

Bits	Name	Description
31 : 0	VALUE	Peripheral ID #2 Default Value: 11

19.1.5 CM0P_DWT_PID3

Watchpoint Unit CoreSight ROM Table Peripheral ID #3

Address: 0xE0001FEC

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access								
Name	VALUE [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access								
Name	VALUE [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access								
Name	VALUE [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access								
Name	VALUE [31:24]							

Bits	Name	Description
31 : 0	VALUE	Peripheral ID #3 Default Value: 0

19.1.6 CM0P_DWT_CID0

Watchpoint Unit CoreSight ROM Table Component ID #0

Address: 0xE0001FF0

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access								
Name	VALUE [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access								
Name	VALUE [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access								
Name	VALUE [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access								
Name	VALUE [31:24]							

Bits	Name	Description
31 : 0	VALUE	Component ID #0 Default Value: 13

19.1.7 CM0P_DWT_CID1

Watchpoint Unit CoreSight ROM Table Component ID #1

Address: 0xE0001FF4

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access								
Name	VALUE [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access								
Name	VALUE [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access								
Name	VALUE [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access								
Name	VALUE [31:24]							

Bits	Name	Description
31 : 0	VALUE	Component ID #1 Default Value: 224

19.1.8 CM0P_DWT_CID2

Watchpoint Unit CoreSight ROM Table Component ID #2

Address: 0xE0001FF8

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access								
Name	VALUE [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access								
Name	VALUE [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access								
Name	VALUE [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access								
Name	VALUE [31:24]							

Bits	Name	Description
31 : 0	VALUE	Component ID #2 Default Value: 5

19.1.9 CM0P_DWT_CID3

Watchpoint Unit CoreSight ROM Table Component ID #3

Address: 0xE0001FFC

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access								
Name	VALUE [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access								
Name	VALUE [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access								
Name	VALUE [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access								
Name	VALUE [31:24]							

Bits	Name	Description
31 : 0	VALUE	Component ID #3 Default Value: 177

19.1.10 CM0P_BP_PID4

Breakpoint Unit CoreSight ROM Table Peripheral ID #4

Address: 0xE0002FD0

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access								
Name	VALUE [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access								
Name	VALUE [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access								
Name	VALUE [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access								
Name	VALUE [31:24]							

Bits	Name	Description
31 : 0	VALUE	Peripheral ID #4 Default Value: 4

19.1.11 CM0P_BP_PID0

Breakpoint Unit CoreSight ROM Table Peripheral ID #0

Address: 0xE0002FE0

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access								
Name	VALUE [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access								
Name	VALUE [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access								
Name	VALUE [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access								
Name	VALUE [31:24]							

Bits	Name	Description
31 : 0	VALUE	Peripheral ID #0 Default Value: 11

19.1.12 CM0P_BP_PID1

Breakpoint Unit CoreSight ROM Table Peripheral ID #1

Address: 0xE0002FE4

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access								
Name	VALUE [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access								
Name	VALUE [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access								
Name	VALUE [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access								
Name	VALUE [31:24]							

Bits	Name	Description
31 : 0	VALUE	Peripheral ID #1 Default Value: 176

19.1.13 CM0P_BP_PID2

Breakpoint Unit CoreSight ROM Table Peripheral ID #2

Address: 0xE0002FE8

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access								
Name	VALUE [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access								
Name	VALUE [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access								
Name	VALUE [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access								
Name	VALUE [31:24]							

Bits	Name	Description
31 : 0	VALUE	Peripheral ID #2 Default Value: 11

19.1.14 CM0P_BP_PID3

Breakpoint Unit CoreSight ROM Table Peripheral ID #3

Address: 0xE0002FEC

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access								
Name	VALUE [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access								
Name	VALUE [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access								
Name	VALUE [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access								
Name	VALUE [31:24]							

Bits	Name	Description
31 : 0	VALUE	Peripheral ID #3 Default Value: 0

19.1.15 CM0P_BP_CID0

Breakpoint Unit CoreSight ROM Table Component ID #0

Address: 0xE0002FF0

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access								
Name	VALUE [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access								
Name	VALUE [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access								
Name	VALUE [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access								
Name	VALUE [31:24]							

Bits	Name	Description
31 : 0	VALUE	Component ID #0 Default Value: 13

19.1.16 CM0P_BP_CID1

Breakpoint Unit CoreSight ROM Table Component ID #1

Address: 0xE0002FF4

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access								
Name	VALUE [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access								
Name	VALUE [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access								
Name	VALUE [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access								
Name	VALUE [31:24]							

Bits	Name	Description
31 : 0	VALUE	Component ID #1 Default Value: 224

19.1.17 CM0P_BP_CID2

Breakpoint Unit CoreSight ROM Table Component ID #2

Address: 0xE0002FF8

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access								
Name	VALUE [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access								
Name	VALUE [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access								
Name	VALUE [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access								
Name	VALUE [31:24]							

Bits	Name	Description
31 : 0	VALUE	Component ID #2 Default Value: 5

19.1.18 CM0P_BP_CID3

Breakpoint Unit CoreSight ROM Table Component ID #3

Address: 0xE0002FFC

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access								
Name	VALUE [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access								
Name	VALUE [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access								
Name	VALUE [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access								
Name	VALUE [31:24]							

Bits	Name	Description
31 : 0	VALUE	Component ID #3 Default Value: 177

19.1.19 CM0P_SYST_CSR

SysTick Control & Status

Address: 0xE000E010

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None					RW	RW	RW
HW Access	None					R	R	R
Name	None [7:3]					CLK-SOURCE	TICKINT	ENABLE

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							R
HW Access	None							RW
Name	None [23:17]							COUNT-FLAG

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
16	COUNTFLAG	Indicates whether the counter has counted to "0" since the last read of this register: '0': counter has not counted to "0". '1': counter has counted to "0". COUNTFLAG is set to '1' by a count transition from "1" to "0". COUNTFLAG is cleared to '0' by a read of this register, and by any write to the SYST_CVR register. Default Value: 0

19.1.19 CM0P_SYST_CSR (continued)

2	CLKSOURCE	Indicates the SysTick counter clock source: '0': SysTick uses the low frequency clock "clk_lf". For this mode to function, "clk_lf" should be less than half the frequency of "clk_sys". Note that "clk_lf" is generated by a low accuracy ILO (Internal Low power Oscillator), with a target frequency of 32.768 kHz (frequency can be as low as 15 KHz and as high as 60 kHz). '1': SysTick uses the system/processor clock "clk_sys". In PSoC4A-BLE (and later products), SysTick counter functionality on the low frequency clock is provided. in SF, TSG6M products, this functionality is not provided. For these products, this field should be set to '1', such that SysTick uses the system clock "clk_sys". Default Value: 0
1	TICKINT	Indicates whether counting to "0" causes the status of the SysTick exception to change to pending: '0': count to "0" does not affect the SysTick exception status. '1': count to "0" changes the SysTick exception status to pending. Changing the value of the counter to "0" by writing zero to the SYST_CVR register to "0" never changes the status of the SysTick exception. Default Value: 0
0	ENABLE	Indicates the enabled status of the SysTick counter: '0': counter is disabled. '1': counter is operating. Default Value: 0

19.1.20 CM0P_SYST_RVR

SysTick Reload Value

Address: 0xE000E014

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access	R							
Name	RELOAD [7:0]							
Bits	15	14	13	12	11	10	9	8
SW Access	RW							
HW Access	R							
Name	RELOAD [15:8]							
Bits	23	22	21	20	19	18	17	16
SW Access	RW							
HW Access	R							
Name	RELOAD [23:16]							
Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
23 : 0	RELOAD	The value to load into the SYST_CVR register when the counter reaches 0. Default Value: X

19.1.21 CM0P_SYST_CVR

SysTick Current Value

Address: 0xE000E018

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW							
HW Access	R							
Name	CURRENT [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	RW							
HW Access	R							
Name	CURRENT [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	RW							
HW Access	R							
Name	CURRENT [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
23 : 0	CURRENT	Current counter value. This is the value of the counter at the time it is sampled. Default Value: X

19.1.22 CM0P_SYST_CALIB

SysTick Calibration Value

Address: 0xE000E01C

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access	RW							
Name	TENMS [7:0]							
Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access	RW							
Name	TENMS [15:8]							
Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access	RW							
Name	TENMS [23:16]							
Bits	31	30	29	28	27	26	25	24
SW Access	R	R	None					
HW Access		RW	None					
Name	NOREF	SKEW	None [29:24]					

Bits	Name	Description
31	NOREF	Indicates whether a implementation defined reference clock is provided: '0': the reference clock is provided. '1': the reference clock is not provided. When this bit is '1', the SYST_CSR.CLKSOURCE is forced to '1' and cannot be cleared to '0'. In PSoC4A-BLE (and later products), SysTick counter functionality on the low frequency clock is provided and this field is '0'. In earlier products, SysTick counter functionality on the low frequency clock is NOT provided and this field is '1'. Default Value: 0
30	SKEW	Indicates whether the 10ms calibration value is exact: '0': 10ms calibration value is exact. '1': 10ms calibration value is inexact, because of the clock frequency. In PSoC4A-BLE (and later products), SysTick counter functionality on the low frequency clock is provided and this field is '1' (due to the low accuracy ILO). In earlier products, SysTick counter functionality on the low frequency clock is NOT provided and this field is '0'. Default Value: X

19.1.22 CM0P_SYST_CALIB (continued)

23 : 0 TENMS

Optionally, holds a reload value to be used for 10ms (100Hz) timing, subject to system clock skew errors. If this field is "0", the calibration value is not known.

In PSoC4A-BLE (and later products), SysTick counter functionality on the low frequency clock is provided and this field is 0x00:00147. In earlier products, SysTick counter functionality on the low frequency clock is NOT provided and this field is 0x00:0000.
 Default Value: X

19.1.23 CM0P_ISER

Interrupt Set-Enable Register

Address: 0xE000E100

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW1S							
HW Access	R							
Name	SETENA [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	RW1S							
HW Access	R							
Name	SETENA [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	RW1S							
HW Access	R							
Name	SETENA [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	RW1S							
HW Access	R							
Name	SETENA [31:24]							

Bits	Name	Description
31 : 0	SETENA	Enables, or reads the enabled state of one or more interrupts. Each bit corresponds to the same numbered interrupt. Default Value: 0

19.1.24 CM0P_ICER

Interrupt Clear Enable Register

Address: 0xE000E180

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW1C							
HW Access	R							
Name	CLRENA [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	RW1C							
HW Access	R							
Name	CLRENA [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	RW1C							
HW Access	R							
Name	CLRENA [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	RW1C							
HW Access	R							
Name	CLRENA [31:24]							

Bits	Name	Description
31 : 0	CLRENA	Disables, or reads the enabled state of one or more interrupts. Each bit corresponds to the same numbered interrupt. Default Value: 0

19.1.25 CM0P_ISPR

Interrupt Set-Pending Register

Address: 0xE000E200

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW1S							
HW Access	R							
Name	SETPEND [7:0]							
Bits	15	14	13	12	11	10	9	8
SW Access	RW1S							
HW Access	R							
Name	SETPEND [15:8]							
Bits	23	22	21	20	19	18	17	16
SW Access	RW1S							
HW Access	R							
Name	SETPEND [23:16]							
Bits	31	30	29	28	27	26	25	24
SW Access	RW1S							
HW Access	R							
Name	SETPEND [31:24]							

Bits	Name	Description
31 : 0	SETPEND	Changes the state of one or more interrupts to pending. Each bit corresponds to the same numbered interrupt. Default Value: 0

19.1.26 CM0P_ICPR

Interrupt Clear-Pending Register

Address: 0xE000E280

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW1C							
HW Access	R							
Name	CLRPEND [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	RW1C							
HW Access	R							
Name	CLRPEND [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	RW1C							
HW Access	R							
Name	CLRPEND [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	RW1C							
HW Access	R							
Name	CLRPEND [31:24]							

Bits	Name	Description
31 : 0	CLRPEND	Changes the state of one or more interrupts to not pending. Each bit corresponds to the same numbered interrupt. Default Value: 0

19.1.27 CM0P_IPR0

Interrupt Priority Registers

Address: 0xE000E400

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	RW		None					
HW Access	R		None					
Name	PRI_N0 [7:6]		None [5:0]					

Bits	15	14	13	12	11	10	9	8
SW Access	RW		None					
HW Access	R		None					
Name	PRI_N1 [15:14]		None [13:8]					

Bits	23	22	21	20	19	18	17	16
SW Access	RW		None					
HW Access	R		None					
Name	PRI_N2 [23:22]		None [21:16]					

Bits	31	30	29	28	27	26	25	24
SW Access	RW		None					
HW Access	R		None					
Name	PRI_N3 [31:30]		None [29:24]					

Bits	Name	Description
31 : 30	PRI_N3	Priority of interrupt number N+3. Default Value: 0
23 : 22	PRI_N2	Priority of interrupt number N+2. Default Value: 0
15 : 14	PRI_N1	Priority of interrupt number N+1. Default Value: 0
7 : 6	PRI_N0	Priority of interrupt number N. Default Value: 0

19.1.28 CM0P_CPUID

CPUID Register

Address: 0xE000ED00

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R				R			
HW Access								
Name	PARTNO [7:4]				REVISION [3:0]			

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access								
Name	PARTNO [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	R				R			
HW Access								
Name	VARIANT [23:20]				CONSTANT [19:16]			

Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access								
Name	IMPLEMENTER [31:24]							

Bits	Name	Description
31 : 24	IMPLEMENTER	Implementer code for ARM. Default Value: 65
23 : 20	VARIANT	Implementation defined. In ARM implementations this is the major revision number n in the rn part of the rnpn revision status, Product revision status on page xii. Default Value: 0
19 : 16	CONSTANT	Indicates the architecture, ARMv6-M Default Value: 12
15 : 4	PARTNO	Indicates part number, Cortex-M0+ Default Value: 3168
3 : 0	REVISION	Indicates revision. In ARM implementations this is the minor revision number n in the pn part of the rnpn revision status, see Product revision status on page xii. For release r0p1. Default Value: 1

19.1.29 CM0P_ICSR

Interrupt Control State Register

Address: 0xE000ED04

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access	RW							
Name	VECTACTIVE [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R				None			R
HW Access	RW				None			RW
Name	VECTPENDING [15:12]				None [11:9]			VECTACTIVE

Bits	23	22	21	20	19	18	17	16
SW Access	R	R	None	R				
HW Access	RW	RW	None	RW				
Name	ISRPRE-EMPT	ISRPENDING	None	VECTPENDING [20:16]				

Bits	31	30	29	28	27	26	25	24
SW Access	RW1S	None		RW1S	RW1C	RW1S	RW1C	None
HW Access	RW	None		RW	R	RW	R	None
Name	NMIPENDSET	None [30:29]		PENDSVSET	PENDSVCLR	PENDSTSETb	PENDSTCLR	None

Bits	Name	Description
31	NMIPENDSET	Activates an NMI exception or reads back the current state. Because NMI is the highest priority exception, it activates as soon as it is registered. Default Value: 0
28	PENDSVSET	Sets a pending PendSV interrupt or reads back the current state. Use this normally to request a context switch. Writing PENDSVSET and PENDSVCLR to '1' concurrently is UNPREDICTABLE. Default Value: 0
27	PENDSVCLR	Clears a pending PendSV interrupt. Default Value: 0
26	PENDSTSETb	Sets a pending SysTick or reads back the current state. Writing PENDSTSET and PENDSTCLR to '1' concurrently is UNPREDICTABLE. Default Value: 0
25	PENDSTCLR	Clears a pending SysTick, whether set here or by the timer hardware. Default Value: 0
23	ISRPREEMPT	Indicates whether a pending exception will be serviced on exit from debug halt state. Default Value: 0

19.1.29 CM0P_ICSR (continued)

22	ISRPENDING	Indicates if an external configurable, NVIC generated, interrupt is pending. Default Value: 0
20 : 12	VECTPENDING	The exception number for the highest priority pending exception. 0= No pending exceptions. The pending state includes the effect of memory-mapped enable and mask registers. It does not include the PRIMASK special-purpose register qualifier. Default Value: 0
8 : 0	VECTACTIVE	The exception number for the current executing exception. 0= Thread mode. This is the same value as IPSR[8:0] Default Value: 0

19.1.30 CM0P_VTOR

Vector Table Offset Register

Address: 0xE000ED08

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None							
HW Access	None							
Name	None [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	RW							
HW Access	R							
Name	TBLOFF [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	RW							
HW Access	R							
Name	TBLOFF [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	RW							
HW Access	R							
Name	TBLOFF [31:24]							

Bits	Name	Description
31 : 8	TBLOFF	Table offset address. All bits of the Vector table address that are not defined by the VTOR are zero. Default Value: 0

19.1.31 CM0P_AIRCR

Application Interrupt and Reset Control Register

Address: 0xE000ED0C

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None					RW1S	RW1C	None
HW Access	None					R	R	None
Name	None [7:3]					SYSRESE- TREQ	VECTCL- RACTIVE	None

Bits	15	14	13	12	11	10	9	8
SW Access	R	None						
HW Access		None						
Name	ENDIAN- NESS	None [14:8]						

Bits	23	22	21	20	19	18	17	16
SW Access	RW							
HW Access	R							
Name	VECTKEY [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	RW							
HW Access	R							
Name	VECTKEY [31:24]							

Bits	Name	Description
31 : 16	VECTKEY	Vector Key. The value 0x05FA must be written to this register, otherwise the register write is UNPREDICTABLE. Readback value is UNKNOWN. Default Value: X
15	ENDIANNESS	Indicates the memory system data endianness: 0 little endian 1 big endian. See Endian support on page A3-44 for more information. Default Value: 0
2	SYSRESETREQ	System Reset Request. Writing 1 to this bit asserts a signal to request a reset by the external system. This will cause a full system reset of the CPU and all other components in the device. See Reset management on page B1-240 for more information. Default Value: 0
1	VECTCLRACTIVE	Clears all active state information for fixed and configurable exceptions. The effect of writing a 1 to this bit if the processor is not halted in Debug state is UNPREDICTABLE. Default Value: 0

19.1.32 CM0P_SCR

System Control Register

Address: 0xE000ED10

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None			RW	None	RW	RW	None
HW Access	None			R	None	R	R	None
Name	None [7:5]			SEVON- PEND	None	SLEEP- DEEP	SLEEPON- EXIT	None

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
4	SEVONPEND	Determines whether an interrupt transition from inactive state to pending state is a wakeup event: 0: transitions from inactive to pending are not wakeup events. 1: transitions from inactive to pending are wakeup events. See WFE on page A6-197 for more information. Default Value: 0
2	SLEEPDEEP	An implementation can use this bit to select DeepSleep/Hibernate power modes upon execution of WFI/WFE: 0: Select Sleep mode 1: Select DeepSleep/Hibernate (depends on PWR_CONTROL.HIBERNATE) Default Value: 0
1	SLEEPONEXIT	Determines whether, on an exit from an ISR that returns to the base level of execution priority, the processor enters a sleep state: 0 do not enter sleep state. 1 enter sleep state. See Power management on page B1-240 for more information. Default Value: 0

19.1.33 CM0P_CCR

Configuration and Control Register

Address: 0xE00ED14

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None				R	None		
HW Access	None					None		
Name	None [7:4]				UN-ALIGN_TRP	None		

Bits	15	14	13	12	11	10	9	8
SW Access	None						R	None
HW Access	None							None
Name	None [15:10]						STKALIGN	None

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
9	STKALIGN	1: On exception entry, the SP used prior to the exception is adjusted to be 8-byte aligned and the context to restore it is saved. The SP is restored on the associated exception return. Default Value: 1
3	UNALIGN_TRP	1: unaligned word and halfword accesses generate a HardFault exception. Default Value: 1

19.1.34 CM0P_SHPR2

System Handler Priority Register 2

Address: 0xE000ED1C

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None							
HW Access	None							
Name	None [7:0]							
Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							
Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							
Bits	31	30	29	28	27	26	25	24
SW Access	RW		None					
HW Access	R		None					
Name	PRI_11 [31:30]		None [29:24]					

Bits	Name	Description
31 : 30	PRI_11	Priority of system handler 11, SVCall Default Value: 0

19.1.35 CM0P_SHPR3

System Handler Priority Register 3

Address: 0xE000ED20

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None							
HW Access	None							
Name	None [7:0]							
Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							
Bits	23	22	21	20	19	18	17	16
SW Access	RW		None					
HW Access	R		None					
Name	PRI_14 [23:22]		None [21:16]					
Bits	31	30	29	28	27	26	25	24
SW Access	RW		None					
HW Access	R		None					
Name	PRI_15 [31:30]		None [29:24]					

Bits	Name	Description
31 : 30	PRI_15	Priority of system handler 15, SysTick Default Value: 0
23 : 22	PRI_14	Priority of system handler 14, PendSV Default Value: 0

19.1.36 CM0P_SHCSR

System Handler Control and State Register

Address: 0xE000ED24

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None							
HW Access	None							
Name	None [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	RW	None						
HW Access	RW	None						
Name	SVCALL- PENDEDED	None						

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
15	SVCALLPENDEDED	0 SVCAll is not pending. 1 SVCAll is pending. This bit reflects the pending state on a read, and updates the pending state, to the value written, on a write. (Pending state bits are set to 1 when an exception occurs, and are cleared to 0 when an exception becomes active.) Default Value: 0

19.1.37 CM0P_SCS_PID4

System Control Space ROM Table Peripheral ID #4

Address: 0xE000EFD0

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access								
Name	VALUE [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access								
Name	VALUE [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access								
Name	VALUE [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access								
Name	VALUE [31:24]							

Bits	Name	Description
31 : 0	VALUE	Peripheral ID #4 Default Value: 4

19.1.38 CM0P_SCS_PID0

System Control Space ROM Table Peripheral ID #0

Address: 0xE000EFE0

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access								
Name	VALUE [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access								
Name	VALUE [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access								
Name	VALUE [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access								
Name	VALUE [31:24]							

Bits	Name	Description
31 : 0	VALUE	Peripheral ID #0 Default Value: 8

19.1.39 CM0P_SCS_PID1

System Control Space ROM Table Peripheral ID #1

Address: 0xE000EFE4

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access								
Name	VALUE [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access								
Name	VALUE [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access								
Name	VALUE [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access								
Name	VALUE [31:24]							

Bits	Name	Description
31 : 0	VALUE	Peripheral ID #1 Default Value: 176

19.1.40 CM0P_SCS_PID2

System Control Space ROM Table Peripheral ID #2

Address: 0xE000EFE8

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access								
Name	VALUE [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access								
Name	VALUE [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access								
Name	VALUE [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access								
Name	VALUE [31:24]							

Bits	Name	Description
31 : 0	VALUE	Peripheral ID #2 Default Value: 11

19.1.41 CM0P_SCS_PID3

System Control Space ROM Table Peripheral ID #3

Address: 0xE000EFEC

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access								
Name	VALUE [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access								
Name	VALUE [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access								
Name	VALUE [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access								
Name	VALUE [31:24]							

Bits	Name	Description
31 : 0	VALUE	Peripheral ID #3 Default Value: 0

19.1.42 CM0P_SCS_CID0

System Control Space ROM Table Component ID #0

Address: 0xE000EFF0

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access								
Name	VALUE [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access								
Name	VALUE [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access								
Name	VALUE [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access								
Name	VALUE [31:24]							

Bits	Name	Description
31 : 0	VALUE	Component ID #0 Default Value: 13

19.1.43 CM0P_SCS_CID1

System Control Space ROM Table Component ID #1

Address: 0xE000EFF4

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access								
Name	VALUE [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access								
Name	VALUE [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access								
Name	VALUE [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access								
Name	VALUE [31:24]							

Bits	Name	Description
31 : 0	VALUE	Component ID #1 Default Value: 224

19.1.44 CM0P_SCS_CID2

System Control Space ROM Table Component ID #2

Address: 0xE000EFF8

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access								
Name	VALUE [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access								
Name	VALUE [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access								
Name	VALUE [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access								
Name	VALUE [31:24]							

Bits	Name	Description
31 : 0	VALUE	Component ID #2 Default Value: 5

19.1.45 CM0P_SCS_CID3

System Control Space ROM Table Component ID #3

Address: 0xE000EFFF

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access								
Name	VALUE [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access								
Name	VALUE [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access								
Name	VALUE [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access								
Name	VALUE [31:24]							

Bits	Name	Description
31 : 0	VALUE	Component ID #3 Default Value: 177

19.1.46 CM0P_ROM_SCS

CM0+ CoreSight ROM Table Peripheral #0

Address: 0xE00FF000

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access								
Name	VALUE [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access								
Name	VALUE [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access								
Name	VALUE [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access								
Name	VALUE [31:24]							

Bits	Name	Description
31 : 0	VALUE	Offset to SCS ROM Table Default Value: 4293980163

19.1.47 CM0P_ROM_DWT

CM0+ CoreSight ROM Table Peripheral #1

Address: 0xE00FF004

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access								
Name	VALUE [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access								
Name	VALUE [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access								
Name	VALUE [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access								
Name	VALUE [31:24]							

Bits	Name	Description
31 : 0	VALUE	Offset to DWT ROM Table Default Value: 4293926915

19.1.48 CM0P_ROM_BPU

CM0+ CoreSight ROM Table Peripheral #2

Address: 0xE00FF008

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access								
Name	VALUE [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access								
Name	VALUE [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access								
Name	VALUE [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access								
Name	VALUE [31:24]							

Bits	Name	Description
31 : 0	VALUE	Offset to BPU ROM Table Default Value: 4293931011

19.1.49 CM0P_ROM_END

CM0+ CoreSight ROM Table End Marker

Address: 0xE00FF00C

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access								
Name	VALUE [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access								
Name	VALUE [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access								
Name	VALUE [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access								
Name	VALUE [31:24]							

Bits	Name	Description
31 : 0	VALUE	End marker in peripheral list Default Value: 0

19.1.50 CM0P_ROM_CSMT

CM0+ CoreSight ROM Table Memory Type

Address: 0xE00FFCC

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access								
Name	VALUE [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access								
Name	VALUE [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access								
Name	VALUE [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access								
Name	VALUE [31:24]							

Bits	Name	Description
31 : 0	VALUE	Memory Type Default Value: 1

19.1.51 CM0P_ROM_PID4

CM0+ CoreSight ROM Table Peripheral ID #4

Address: 0xE00FFFD0

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access								
Name	VALUE [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access								
Name	VALUE [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access								
Name	VALUE [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access								
Name	VALUE [31:24]							

Bits	Name	Description
31 : 0	VALUE	Peripheral ID #4 Default Value: 4

19.1.52 CM0P_ROM_PID0

CM0+ CoreSight ROM Table Peripheral ID #0

Address: 0xE00FFE0

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access								
Name	VALUE [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access								
Name	VALUE [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access								
Name	VALUE [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access								
Name	VALUE [31:24]							

Bits	Name	Description
31 : 0	VALUE	Peripheral ID #0 Default Value: 192

19.1.53 CM0P_ROM_PID1

CM0+ CoreSight ROM Table Peripheral ID #1

Address: 0xE00FFE4

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access								
Name	VALUE [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access								
Name	VALUE [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access								
Name	VALUE [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access								
Name	VALUE [31:24]							

Bits	Name	Description
31 : 0	VALUE	Peripheral ID #1 Default Value: 180

19.1.54 CM0P_ROM_PID2

CM0+ CoreSight ROM Table Peripheral ID #2

Address: 0xE00FFE8

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access								
Name	VALUE [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access								
Name	VALUE [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access								
Name	VALUE [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access								
Name	VALUE [31:24]							

Bits	Name	Description
31 : 0	VALUE	Peripheral ID #2 Default Value: 11

19.1.55 CM0P_ROM_PID3

CM0+ CoreSight ROM Table Peripheral ID #3

Address: 0xE00FFEC

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access								
Name	VALUE [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access								
Name	VALUE [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access								
Name	VALUE [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access								
Name	VALUE [31:24]							

Bits	Name	Description
31 : 0	VALUE	Peripheral ID #3 Default Value: 0

19.1.56 CM0P_ROM_CID0

CM0+ CoreSight ROM Table Component ID #0

Address: 0xE00FFF0

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access								
Name	VALUE [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access								
Name	VALUE [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access								
Name	VALUE [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access								
Name	VALUE [31:24]							

Bits	Name	Description
31 : 0	VALUE	Component ID #0 Default Value: 13

19.1.57 CM0P_ROM_CID1

CM0+ CoreSight ROM Table Component ID #1

Address: 0xE00FFF4

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access								
Name	VALUE [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access								
Name	VALUE [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access								
Name	VALUE [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access								
Name	VALUE [31:24]							

Bits	Name	Description
31 : 0	VALUE	Component ID #1 Default Value: 16

19.1.58 CM0P_ROM_CID2

CM0+ CoreSight ROM Table Component ID #2

Address: 0xE00FFF8

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access								
Name	VALUE [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access								
Name	VALUE [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access								
Name	VALUE [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access								
Name	VALUE [31:24]							

Bits	Name	Description
31 : 0	VALUE	Component ID #2 Default Value: 5

19.1.59 CM0P_ROM_CID3

CM0+ CoreSight ROM Table Component ID #3

Address: 0xE00FFFC

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access								
Name	VALUE [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access								
Name	VALUE [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access								
Name	VALUE [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access								
Name	VALUE [31:24]							

Bits	Name	Description
31 : 0	VALUE	Component ID #3 Default Value: 177

20 ROM Table Registers



This section discusses the ROM Table registers. It lists all the registers in mapping tables, in address order.

20.1 Register Details

Register	Address	Description
ROMTABLE_ADDR	0xF0000000	Link to Cortex M0 ROM Table.
ROMTABLE_DID	0xF0000FCC	Device Type Identifier register.
ROMTABLE_PID4	0xF0000FD0	Peripheral Identification Register 4.
ROMTABLE_PID5	0xF0000FD4	Peripheral Identification Register 5.
ROMTABLE_PID6	0xF0000FD8	Peripheral Identification Register 6.
ROMTABLE_PID7	0xF0000FDC	Peripheral Identification Register 7.
ROMTABLE_PID0	0xF0000FE0	Peripheral Identification Register 0.
ROMTABLE_PID1	0xF0000FE4	Peripheral Identification Register 1.
ROMTABLE_PID2	0xF0000FE8	Peripheral Identification Register 2.
ROMTABLE_PID3	0xF0000FEC	Peripheral Identification Register 3.
ROMTABLE_CID0	0xF0000FF0	Component Identification Register 0.
ROMTABLE_CID1	0xF0000FF4	Component Identification Register 1.
ROMTABLE_CID2	0xF0000FF8	Component Identification Register 2.
ROMTABLE_CID3	0xF0000FFC	Component Identification Register 3.

20.1.1 ROMTABLE_ADDR

Link to Cortex M0 ROM Table.

Address: 0xF0000000

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	None						R	R
HW Access	None						R	R
Name	None [7:2]						FOR-MAT_32BIT	PRESENT

Bits	15	14	13	12	11	10	9	8
SW Access	R				None			
HW Access	R				None			
Name	ADDR_OFFSET [15:12]				None [11:8]			

Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access	R							
Name	ADDR_OFFSET [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access	R							
Name	ADDR_OFFSET [31:24]							

Bits	Name	Description
31 : 12	ADDR_OFFSET	Address offset of the Cortex-M0 ROM Table base address (0xe00f:f000) wrt. Cypress chip specific ROM Table base address (0xf000:0000). ADDR_OFFSET[19:0] = 0xe00f:f - 0xf000:0 = 0xf00f:f. Default Value: 983295
1	FORMAT_32BIT	ROM Table format: '0': 8-bit format. '1': 32-bit format. Default Value: 1
0	PRESENT	Entry present. Default Value: 1

20.1.2 ROMTABLE_DID

Device Type Identifier register.

Address: 0xF0000FCC

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access	R							
Name	VALUE [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access	R							
Name	VALUE [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access	R							
Name	VALUE [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access	R							
Name	VALUE [31:24]							

Bits	Name	Description
31 : 0	VALUE	. Default Value: 1

20.1.3 ROMTABLE_PID4

Peripheral Identification Register 4.

Address: 0xF000FD0

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R				R			
HW Access	R				R			
Name	COUNT [7:4]				JEP_CONTINUATION [3:0]			

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
7 : 4	COUNT	Size of ROM Table is $2^{\text{COUNT}} \times 4$ KByte. Default Value: 0
3 : 0	JEP_CONTINUATION	JEP106 continuation code. This value is product specific and specified as part of the product definition in the CPUSS.JEPCONTINUATION parameter. Default Value: Undefined

20.1.4 ROMTABLE_PID5

Peripheral Identification Register 5.

Address: 0xF0000FD4

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access	R							
Name	VALUE [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access	R							
Name	VALUE [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access	R							
Name	VALUE [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access	R							
Name	VALUE [31:24]							

Bits	Name	Description
31 : 0	VALUE	. Default Value: 0

20.1.5 ROMTABLE_PID6

Peripheral Identification Register 6.

Address: 0xF0000FD8

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access	R							
Name	VALUE [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access	R							
Name	VALUE [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access	R							
Name	VALUE [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access	R							
Name	VALUE [31:24]							

Bits	Name	Description
31 : 0	VALUE	. Default Value: 0

20.1.6 ROMTABLE_PID7

Peripheral Identification Register 7.

Address: 0xF0000FDC

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access	R							
Name	VALUE [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access	R							
Name	VALUE [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access	R							
Name	VALUE [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access	R							
Name	VALUE [31:24]							

Bits	Name	Description
31 : 0	VALUE	. Default Value: 0

20.1.7 ROMTABLE_PID0

Peripheral Identification Register 0.

Address: 0xF0000FE0

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access	R							
Name	PN_MIN [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
7 : 0	PN_MIN	JEP106 part number. 4 lsbs of CPUSS.PARTNUMBER parameter. Default Value: Undefined

20.1.8 ROMTABLE_PID1

Peripheral Identification Register 1.

Address: 0xF0000FE4

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R				R			
HW Access	R				R			
Name	JEPID_MIN [7:4]				PN_MAJ [3:0]			

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
7 : 4	JEPID_MIN	JEP106 vendor id. 4 lsbs of CPUSS.JEPID parameter. Default Value: Undefined
3 : 0	PN_MAJ	JEP106 part number. 4 msbs of CPUSS.PARTNUMBER parameter. Default Value: Undefined

20.1.9 ROMTABLE_PID2

Peripheral Identification Register 2.

Address: 0xF0000FE8

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R				None	R		
HW Access	R				None	R		
Name	REV [7:4]				None	JEPID_MAJ [2:0]		

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
7 : 4	REV	Major REVision number (chip specific). Identifies the design iteration of the component. For first tape out: 0x1. This field is incremented on subsequent tape outs. Default Value: Undefined
2 : 0	JEPID_MAJ	JEP106 vendor id. 4 msbs of CPUSS.JEPID parameter. Default Value: Undefined

20.1.10 ROMTABLE_PID3

Peripheral Identification Register 3.

Address: 0xF0000FEC

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R				R			
HW Access	R				R			
Name	REV_AND [7:4]				CM [3:0]			

Bits	15	14	13	12	11	10	9	8
SW Access	None							
HW Access	None							
Name	None [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	None							
HW Access	None							
Name	None [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	None							
HW Access	None							
Name	None [31:24]							

Bits	Name	Description
7 : 4	REV_AND	Minor REVision number (chip specific). For first tape out: 0x1. This field is incremented on subsequent tape outs. Default Value: Undefined
3 : 0	CM	Customer modified field. This field is used to track modifications to the original component design as a result of component IP reuse. Default Value: 0

20.1.11 ROMTABLE_CID0

Component Identification Register 0.

Address: 0xF0000FF0

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access	R							
Name	VALUE [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access	R							
Name	VALUE [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access	R							
Name	VALUE [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access	R							
Name	VALUE [31:24]							

Bits	Name	Description
31 : 0	VALUE	Component identification byte 0 of 4-byte component identification 0xB105:100D. Default Value: 13

20.1.12 ROMTABLE_CID1

Component Identification Register 1.

Address: 0xF0000FF4

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access	R							
Name	VALUE [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access	R							
Name	VALUE [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access	R							
Name	VALUE [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access	R							
Name	VALUE [31:24]							

Bits	Name	Description
31 : 0	VALUE	Component identification byte 1 of 4-byte component identification 0xB105:100D. Component class: "ROM Table". Default Value: 16

20.1.13 ROMTABLE_CID2

Component Identification Register 2.

Address: 0xF0000FF8

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access	R							
Name	VALUE [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access	R							
Name	VALUE [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access	R							
Name	VALUE [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access	R							
Name	VALUE [31:24]							

Bits	Name	Description
31 : 0	VALUE	Component identification byte 2 of 4-byte component identification 0xB105:100D. Default Value: 5

20.1.14 ROMTABLE_CID3

Component Identification Register 3.

Address: 0xF0000FFC

Retention: Retained

Bits	7	6	5	4	3	2	1	0
SW Access	R							
HW Access	R							
Name	VALUE [7:0]							

Bits	15	14	13	12	11	10	9	8
SW Access	R							
HW Access	R							
Name	VALUE [15:8]							

Bits	23	22	21	20	19	18	17	16
SW Access	R							
HW Access	R							
Name	VALUE [23:16]							

Bits	31	30	29	28	27	26	25	24
SW Access	R							
HW Access	R							
Name	VALUE [31:24]							

Bits	Name	Description
31 : 0	VALUE	Component identification byte 3 of 4-byte component identification 0xB105:100D. Default Value: 177

Revision History



Revision History

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Document Number: 002-27800			
Revision	ECN#	Issue Date	Description of Change
**	6771730	03/11/2020	Initial release for PSoC 4100S Plus 256KB PSoC 4 Registers TRM.
*A	8057737	07/29/2024	Fixed broken links.